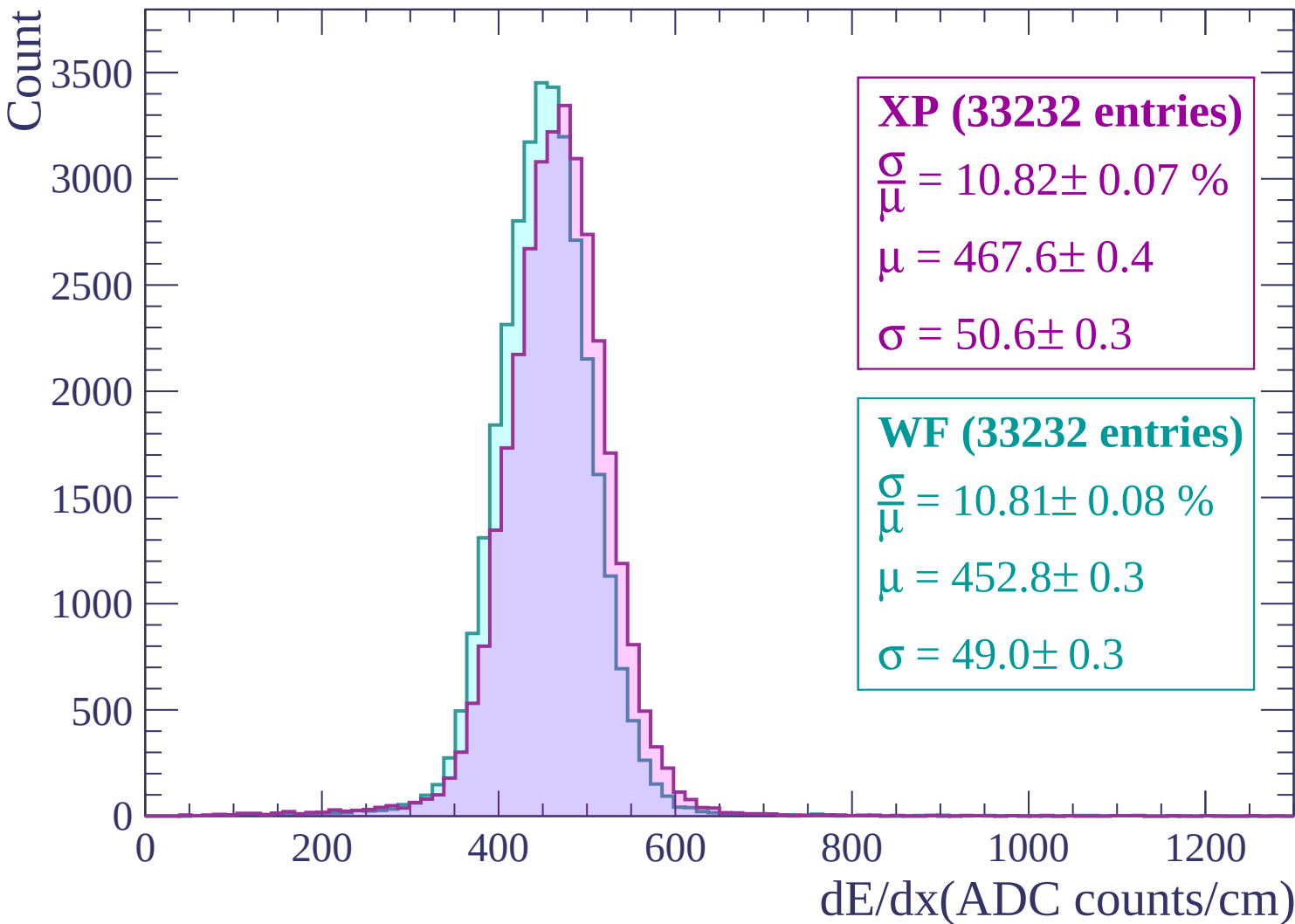
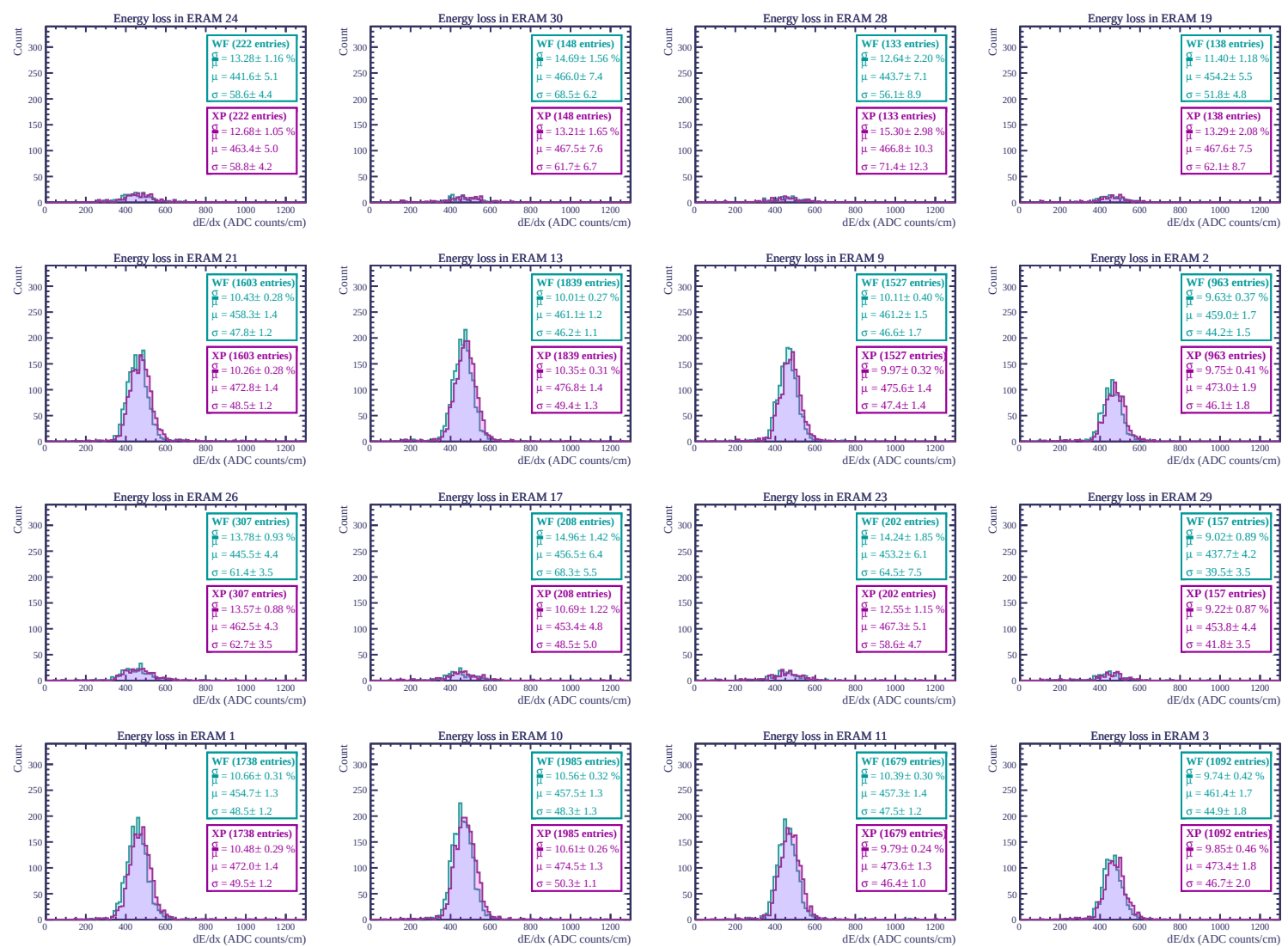
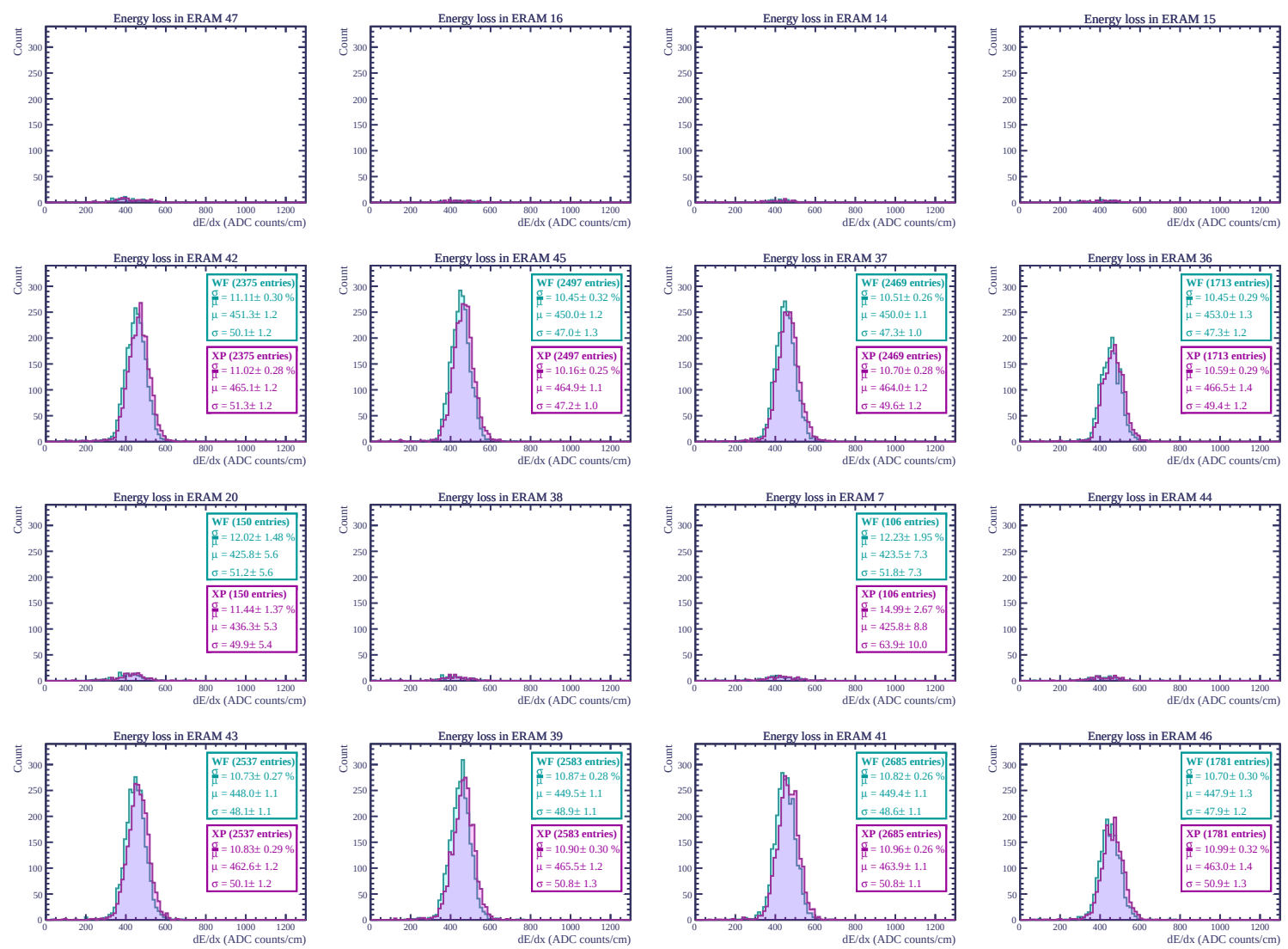
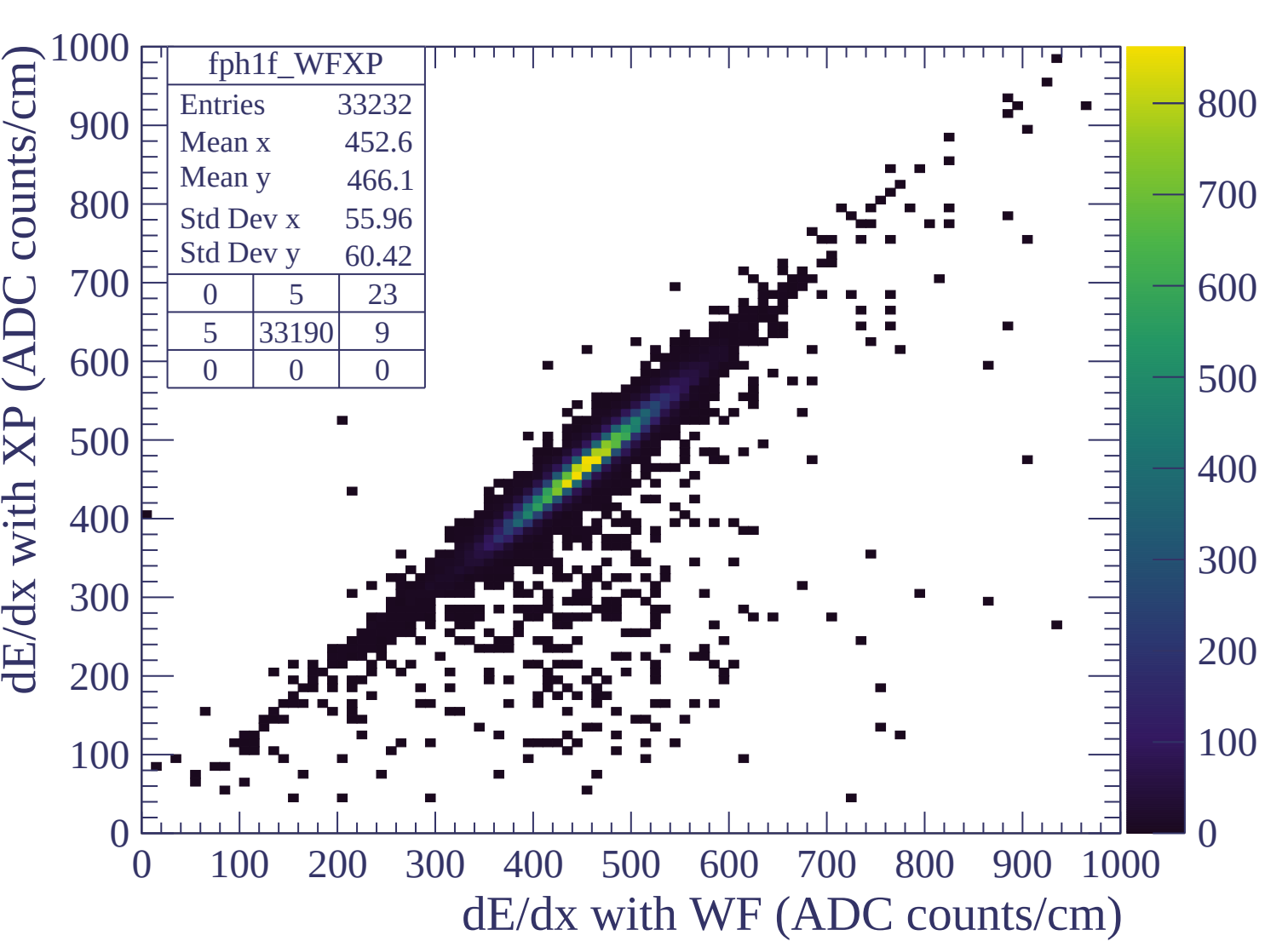
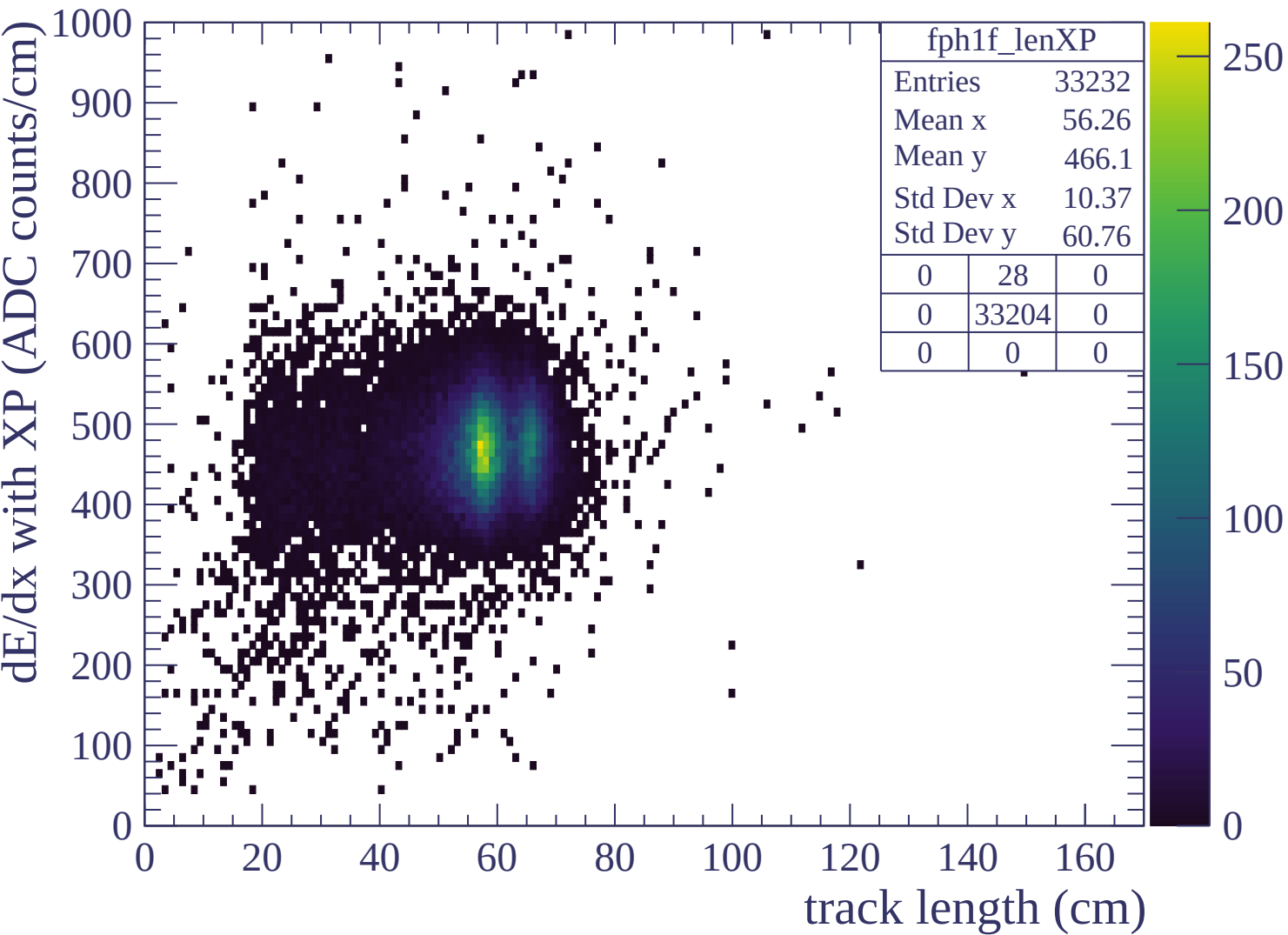


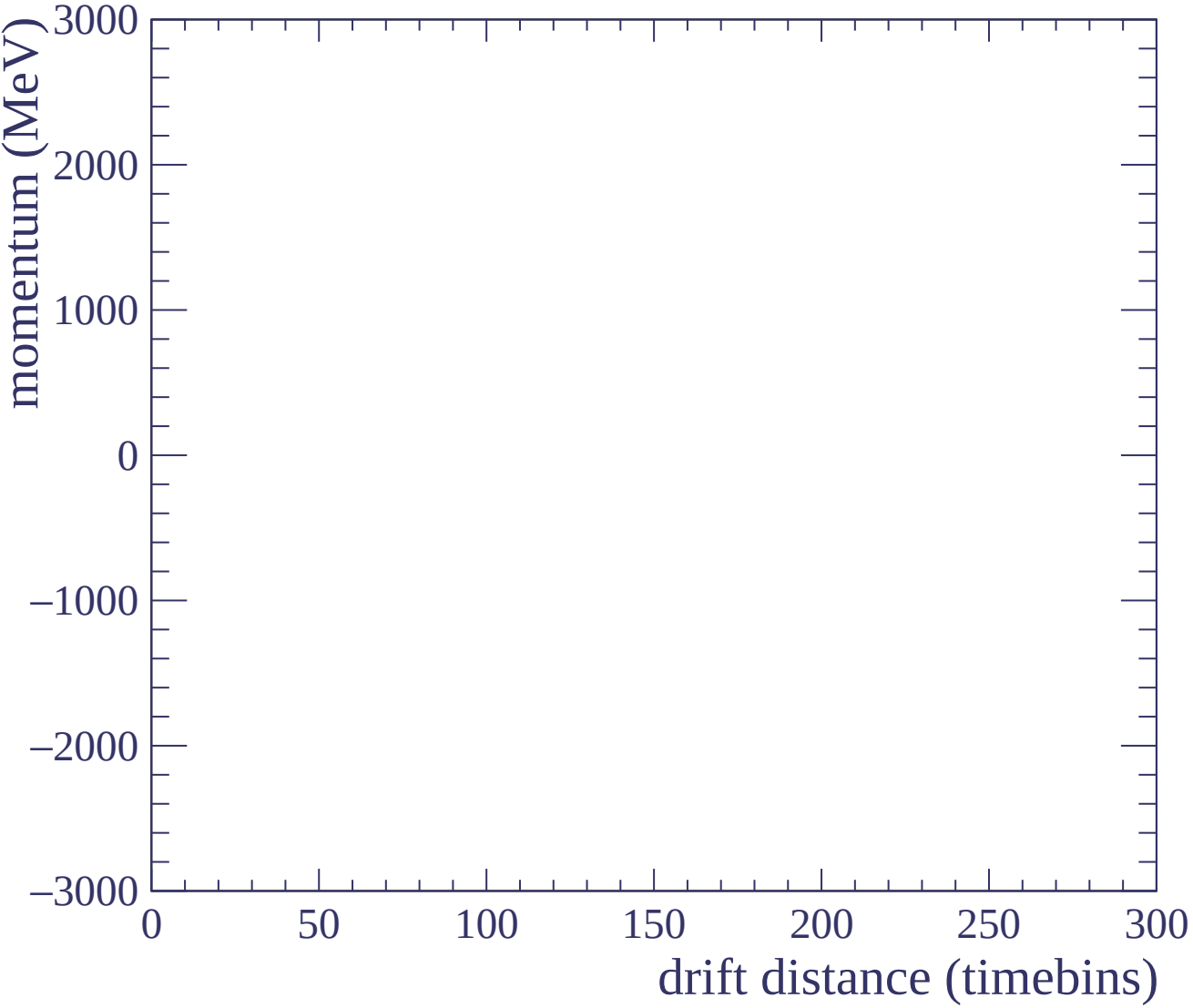


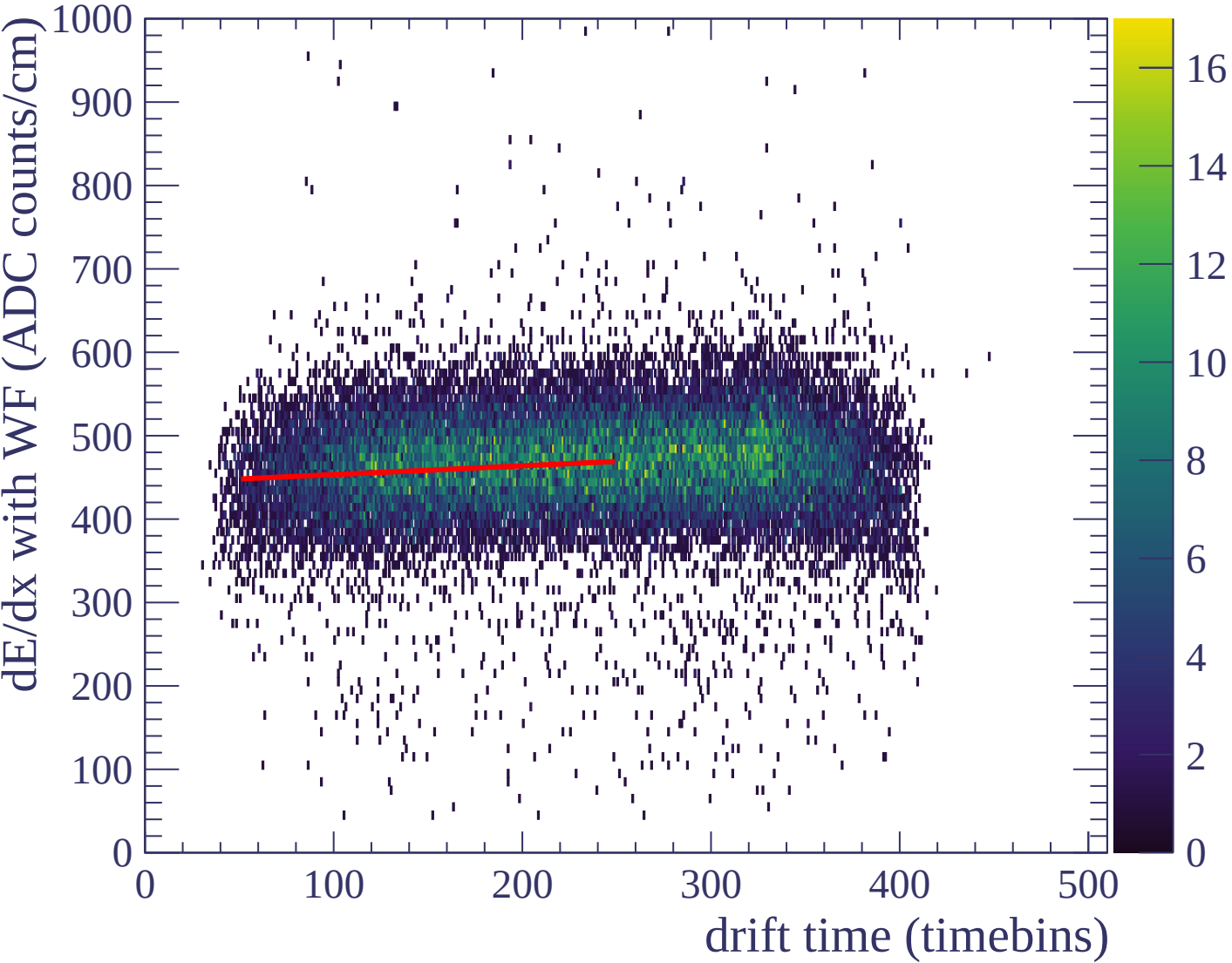


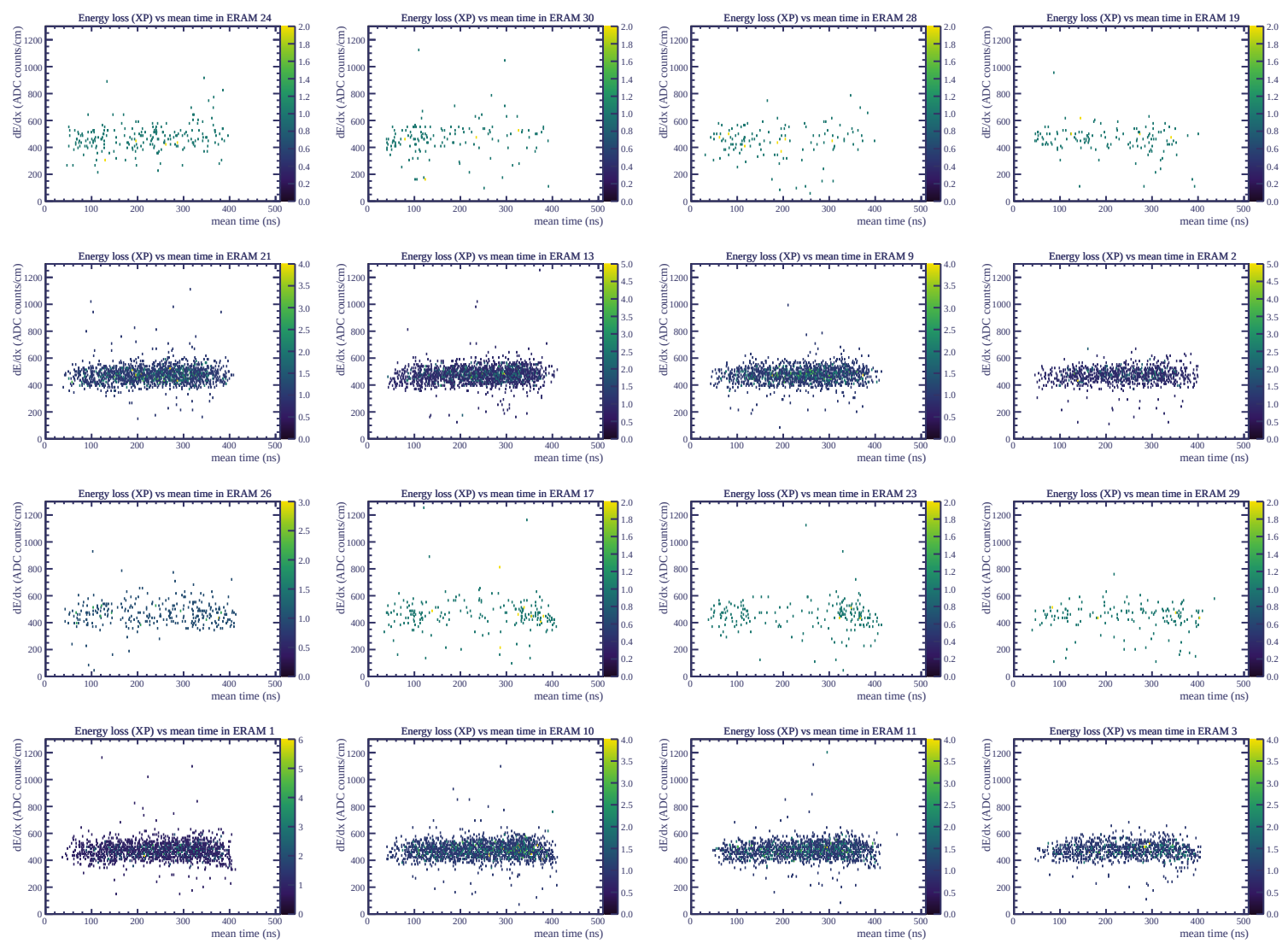


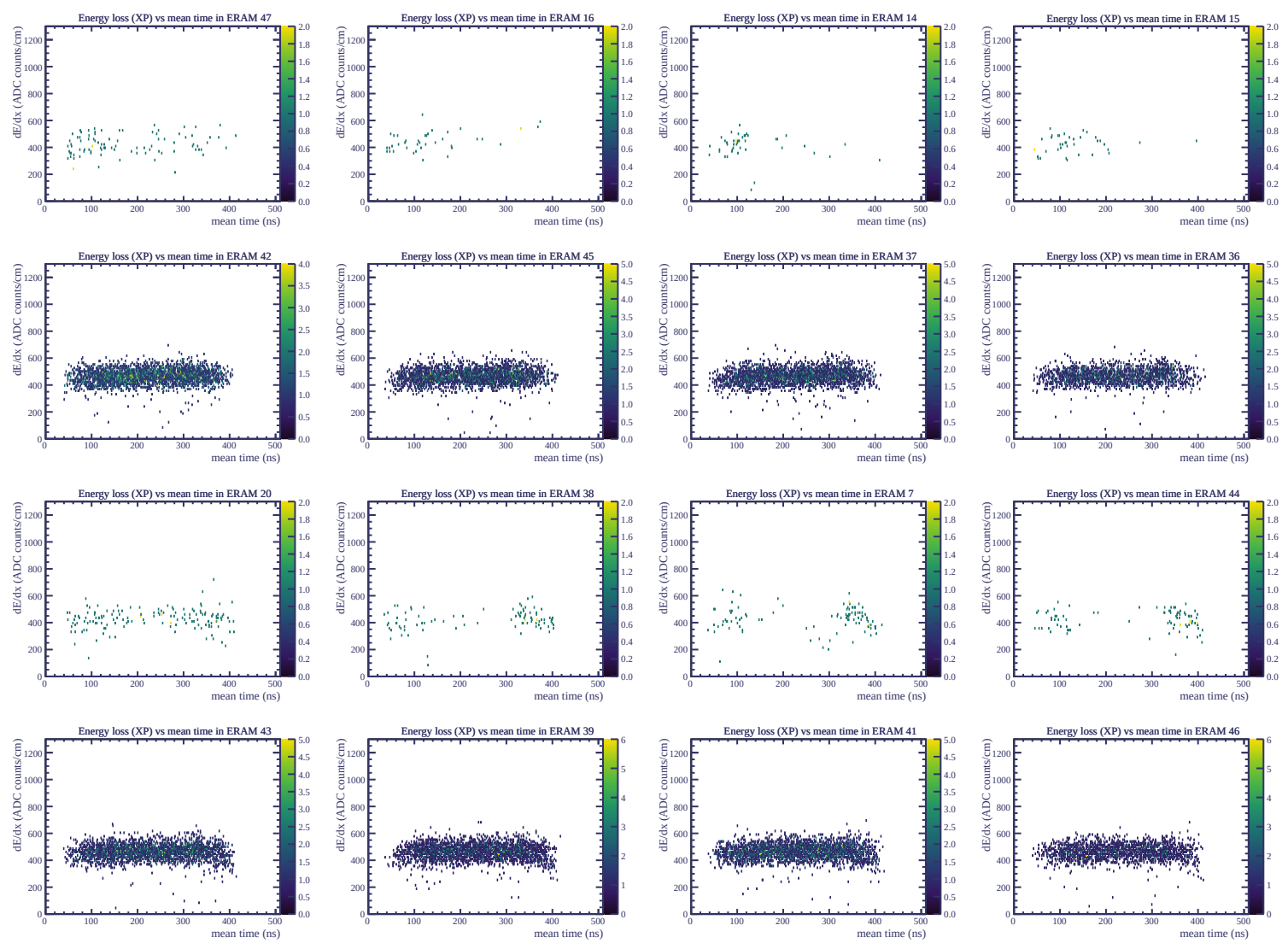


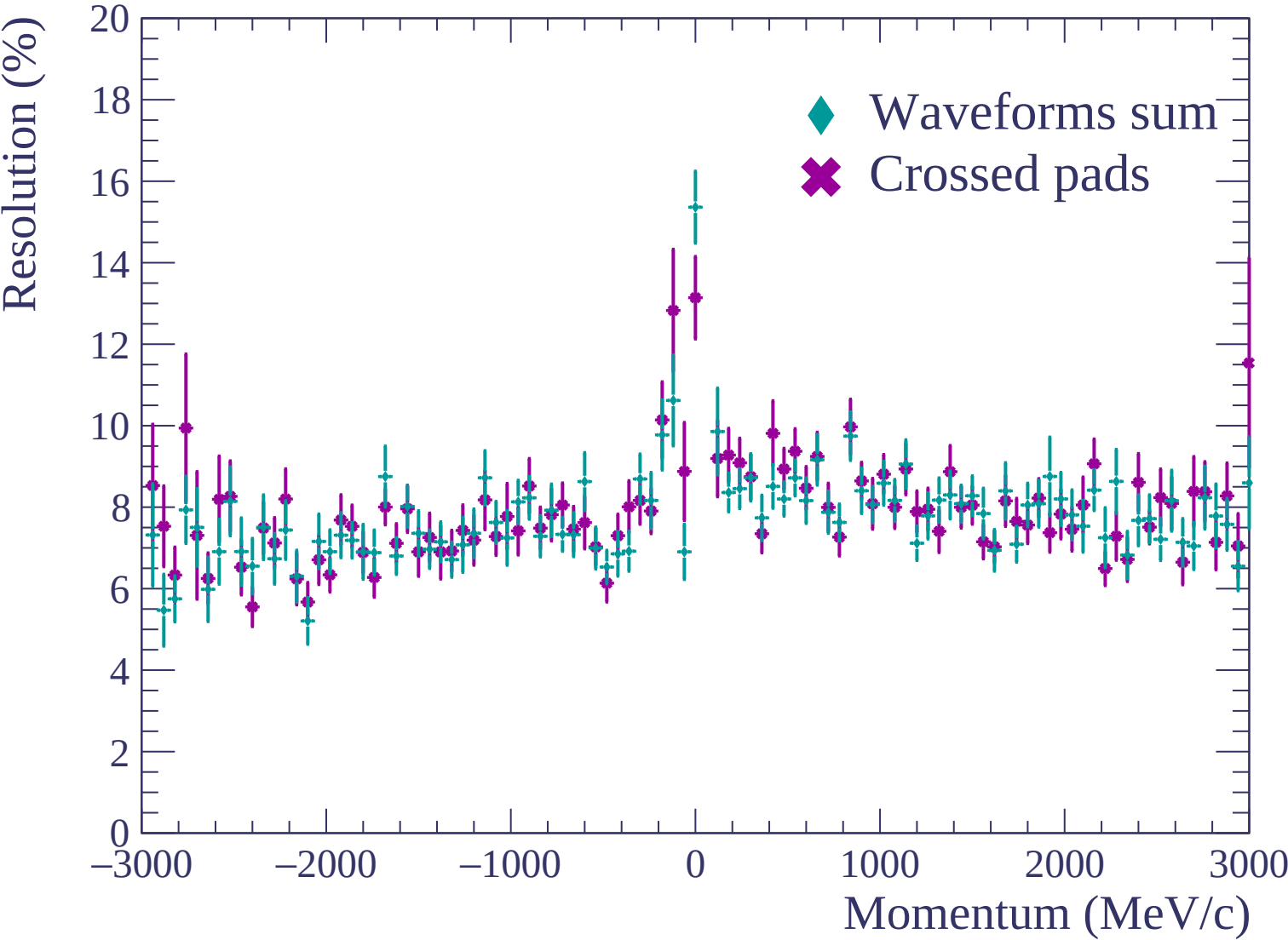


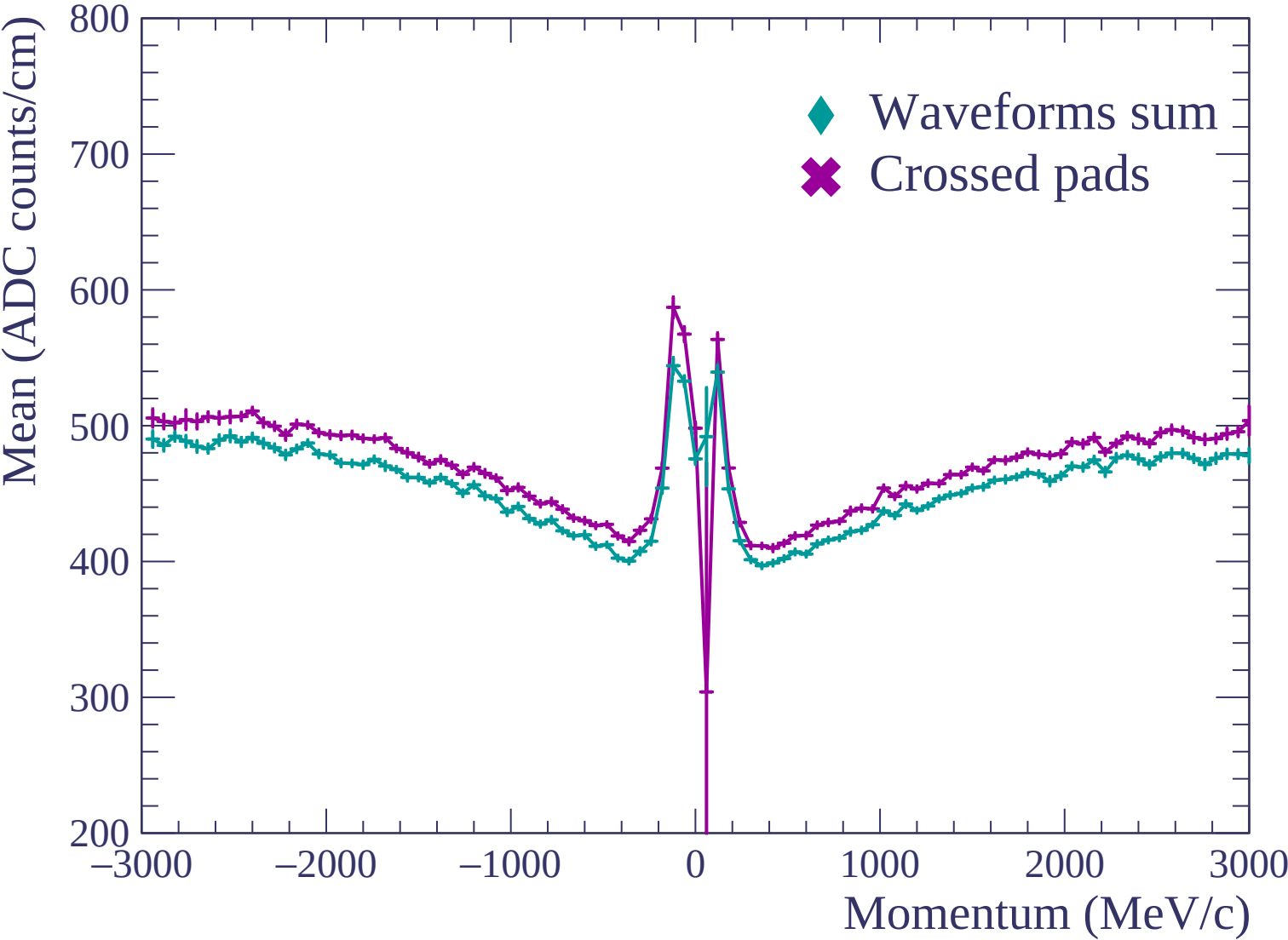


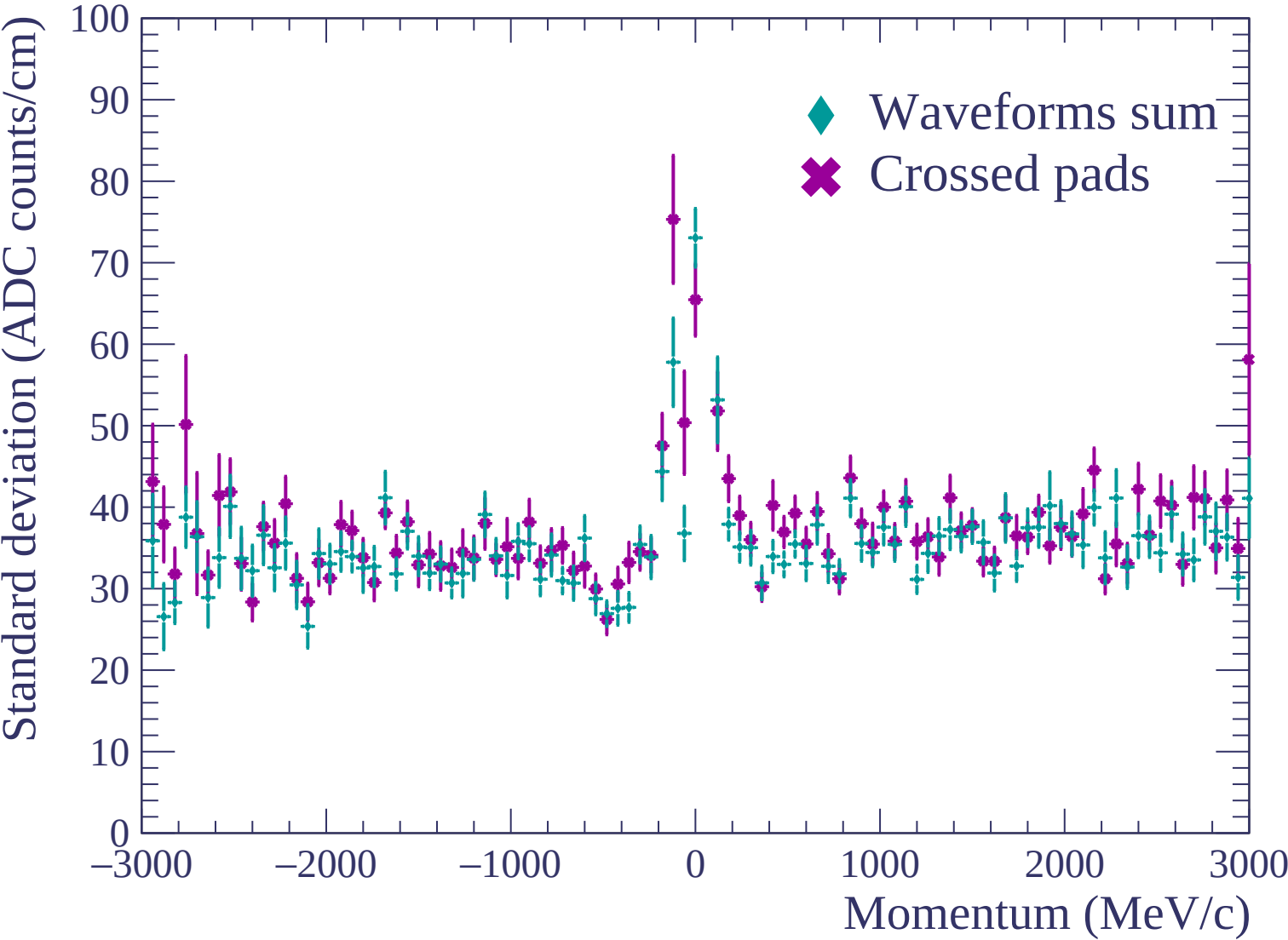


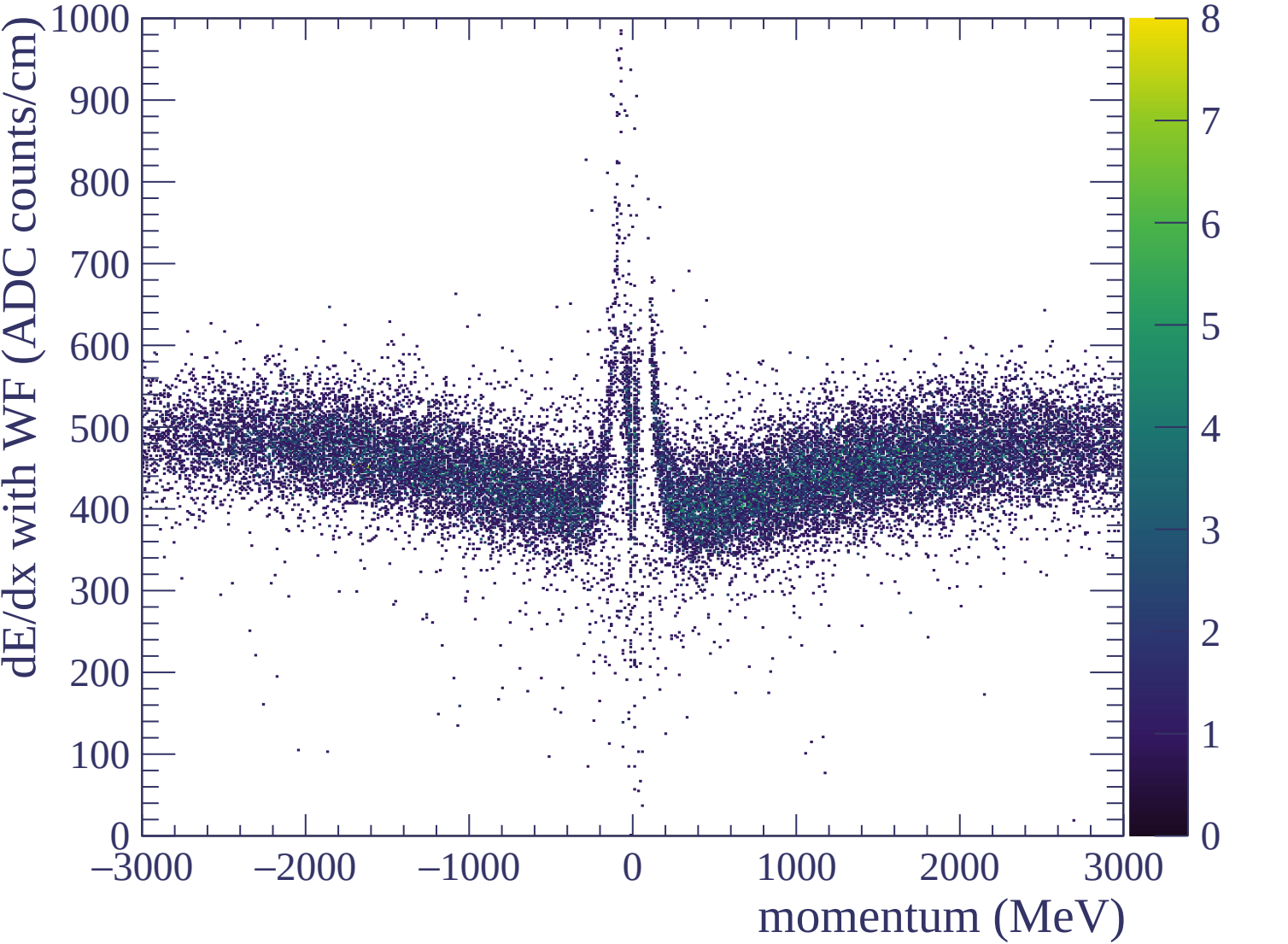


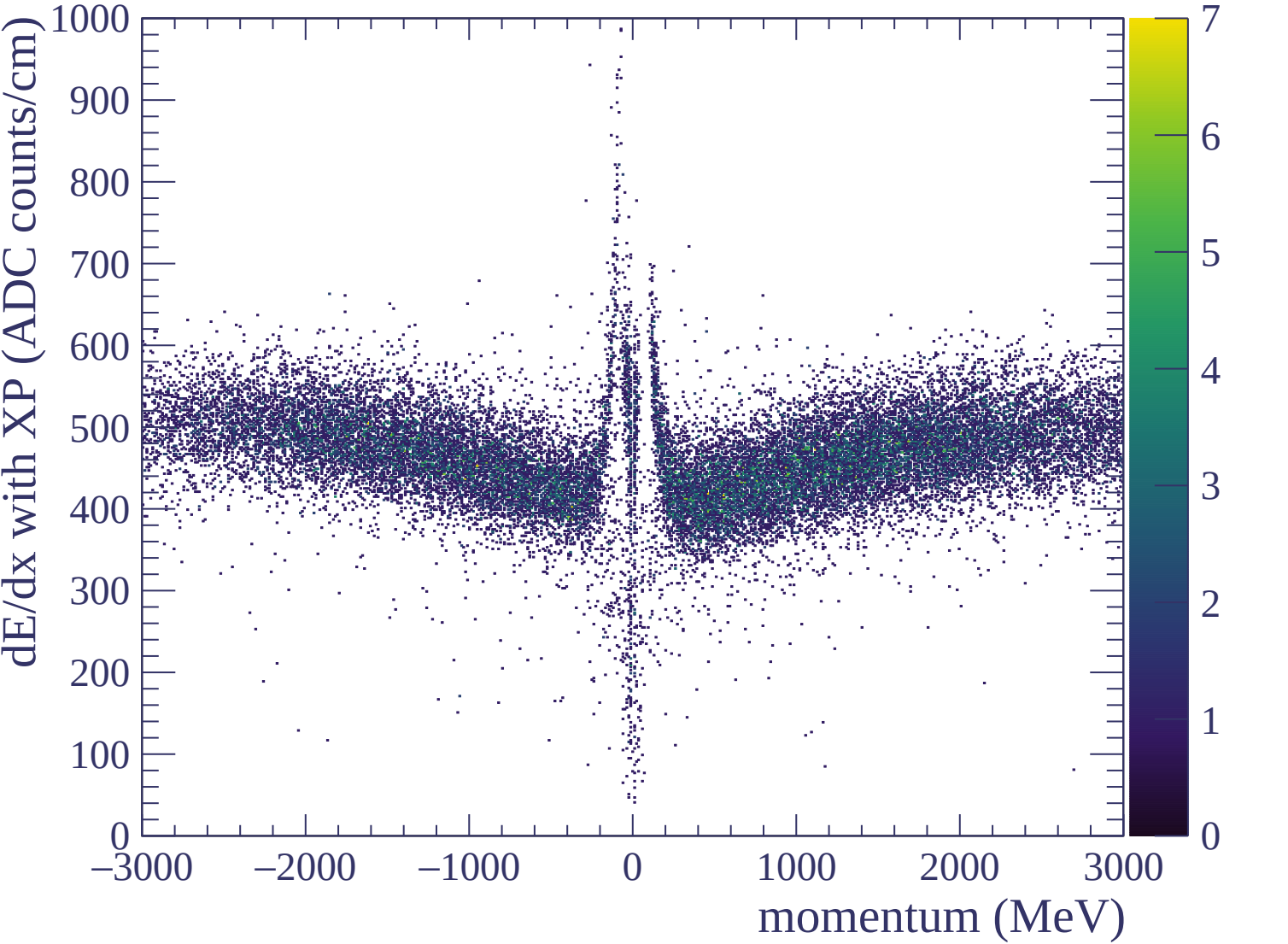


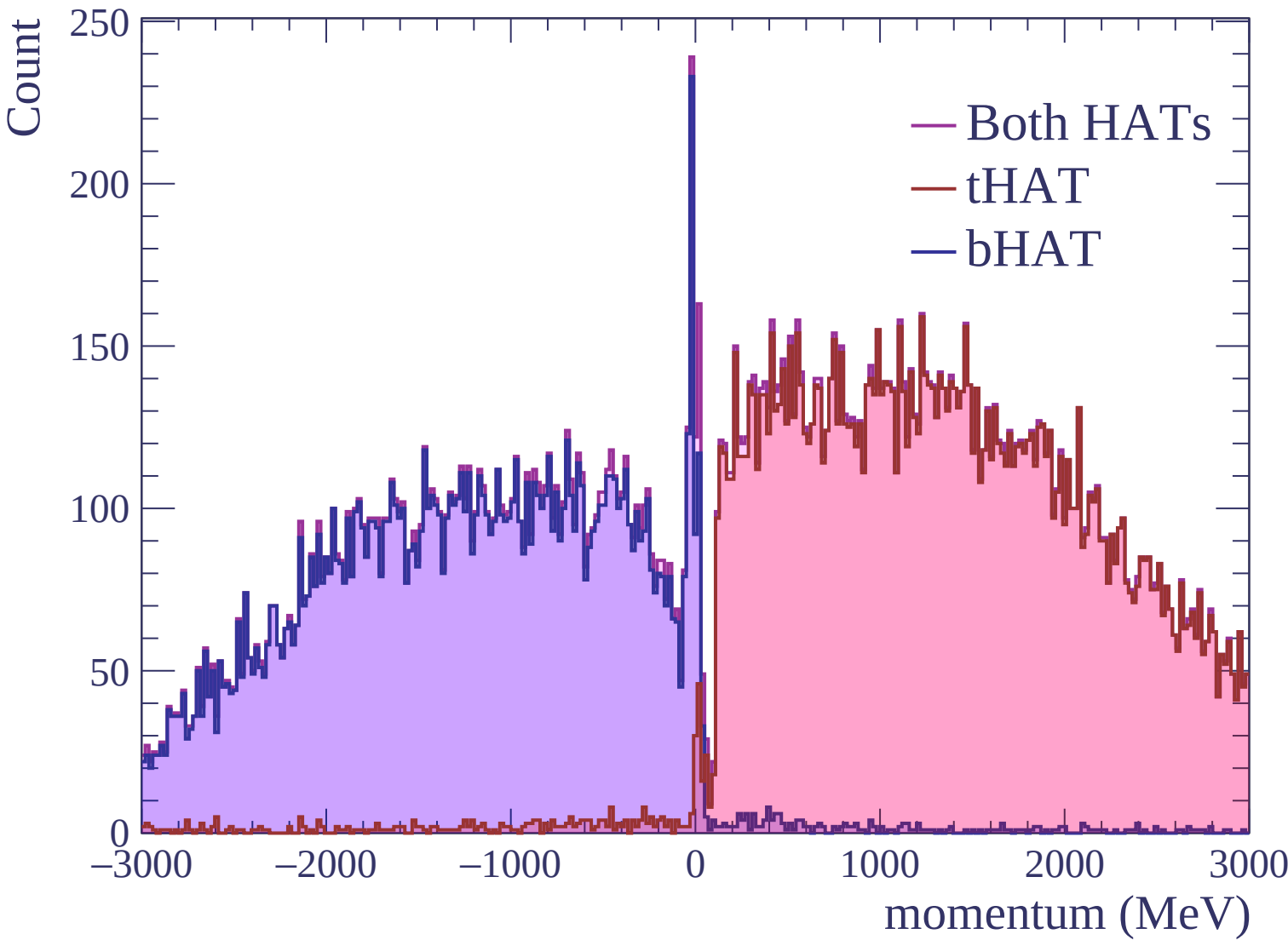


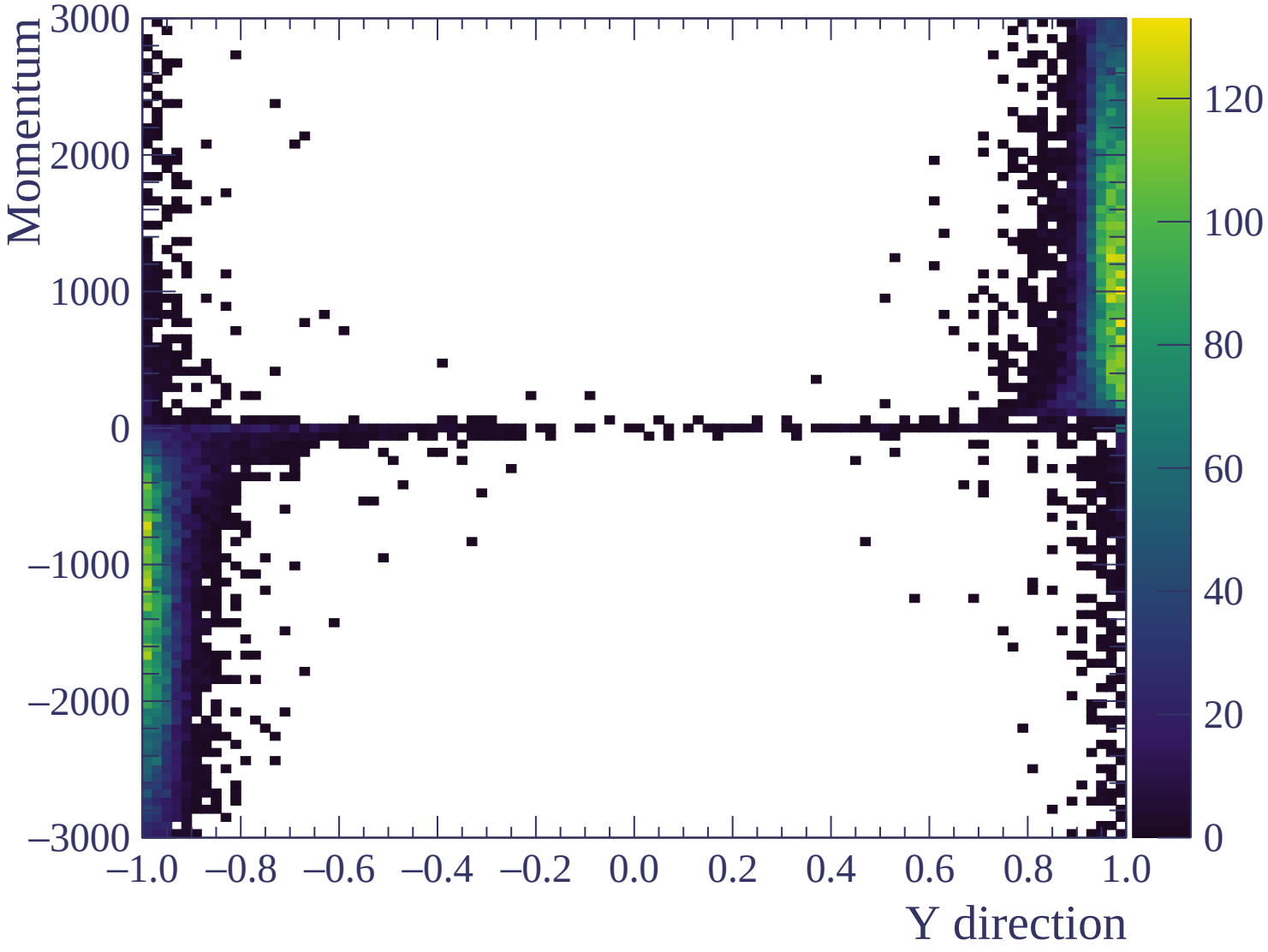


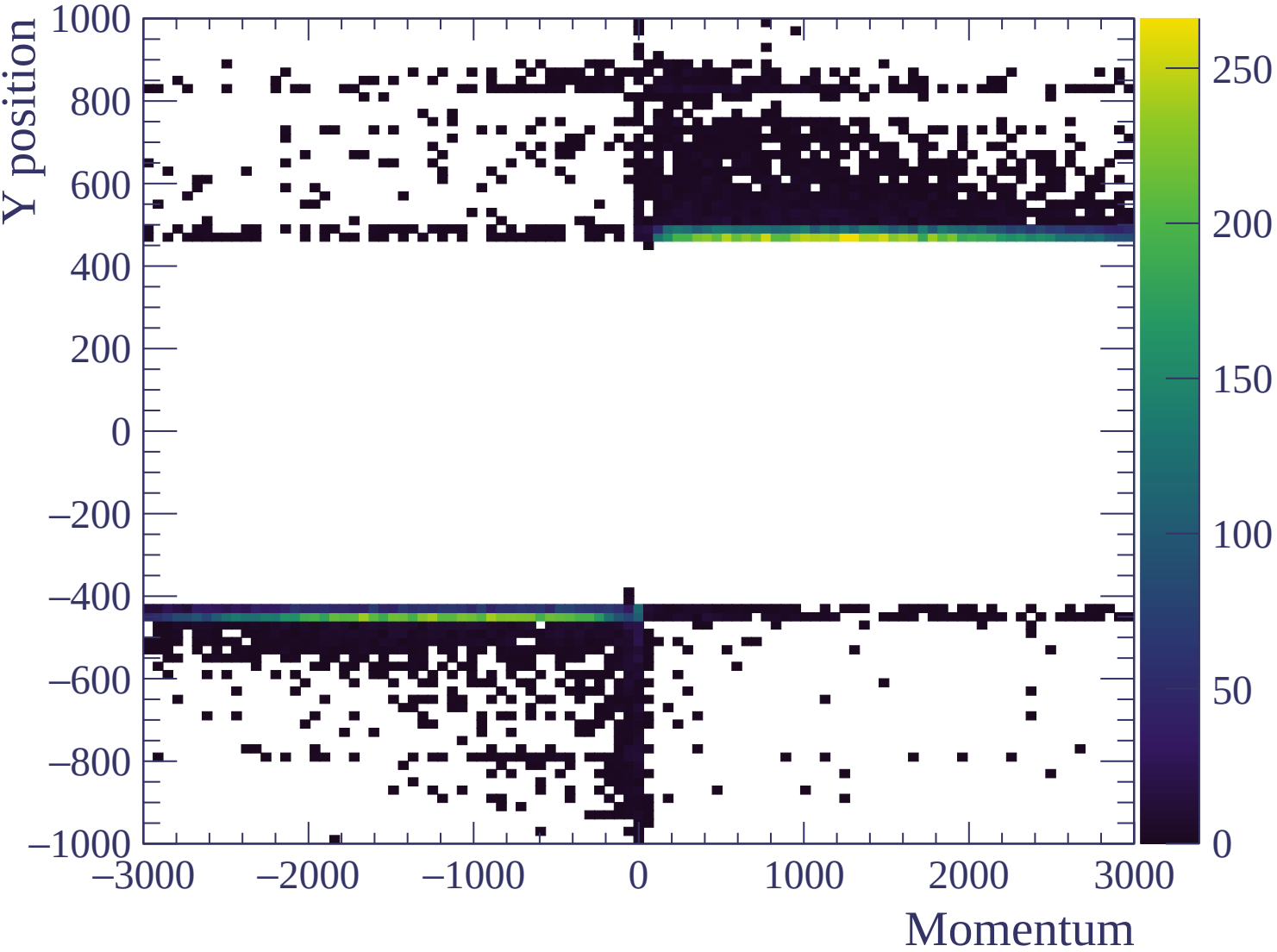


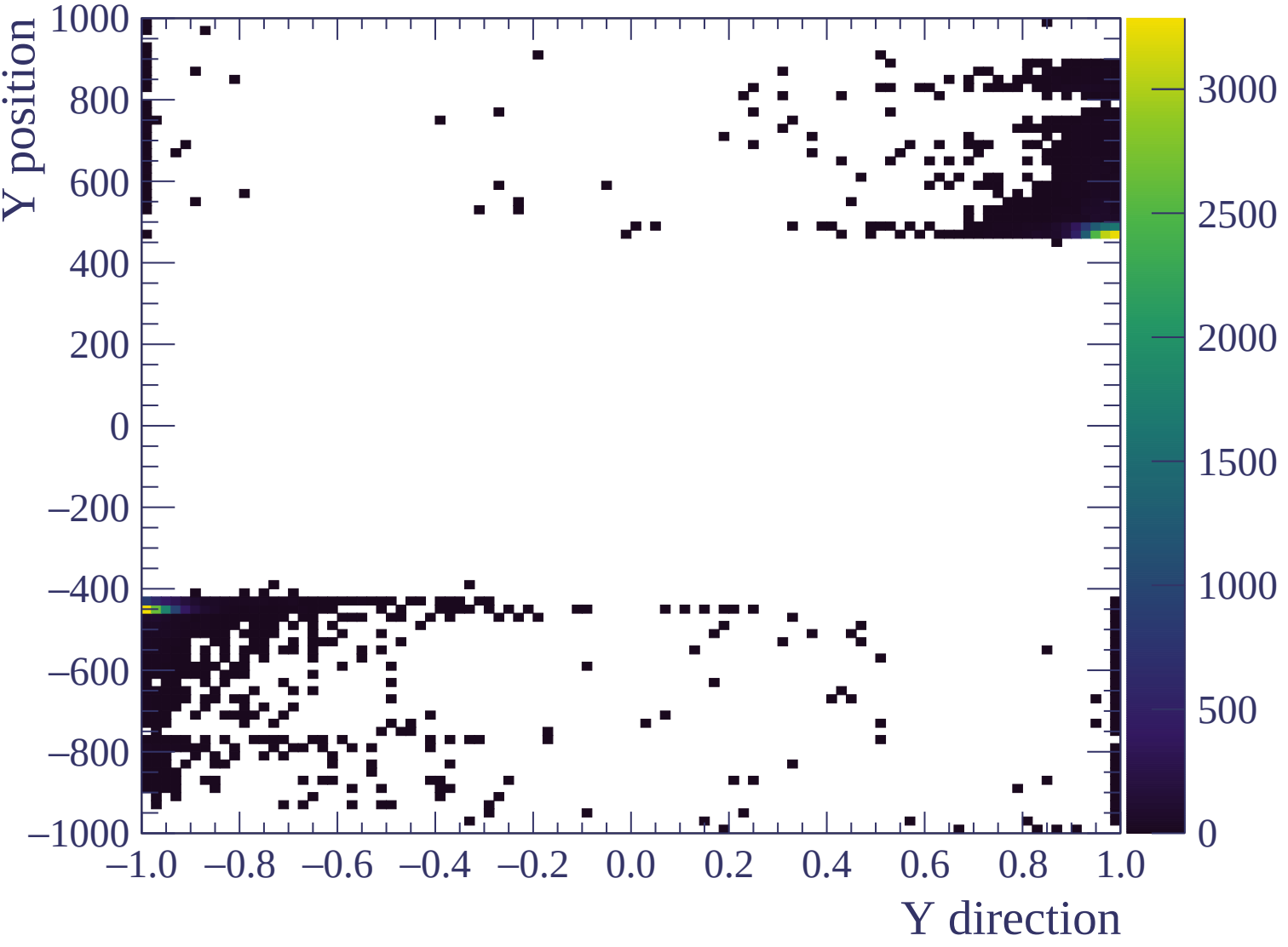


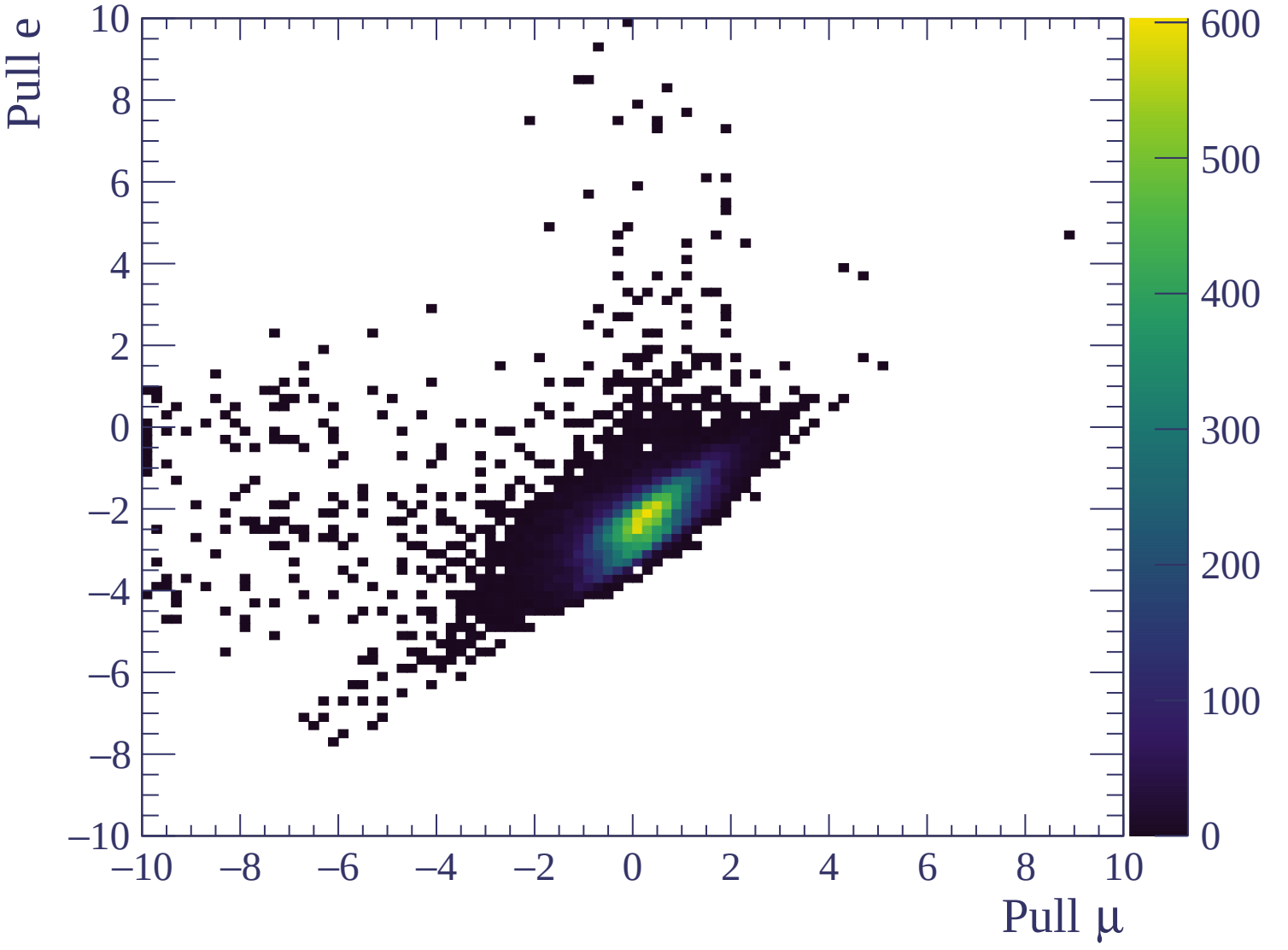






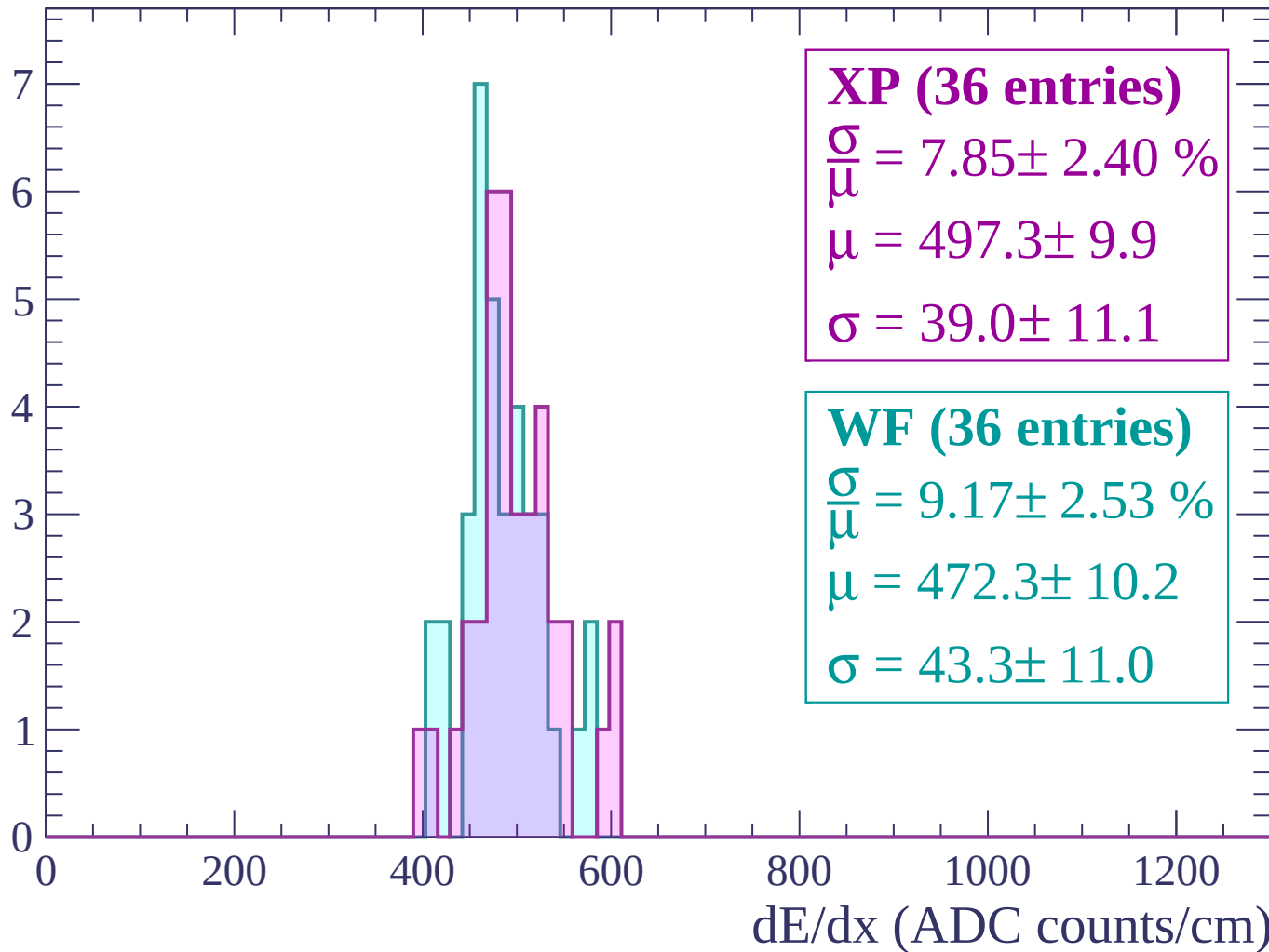






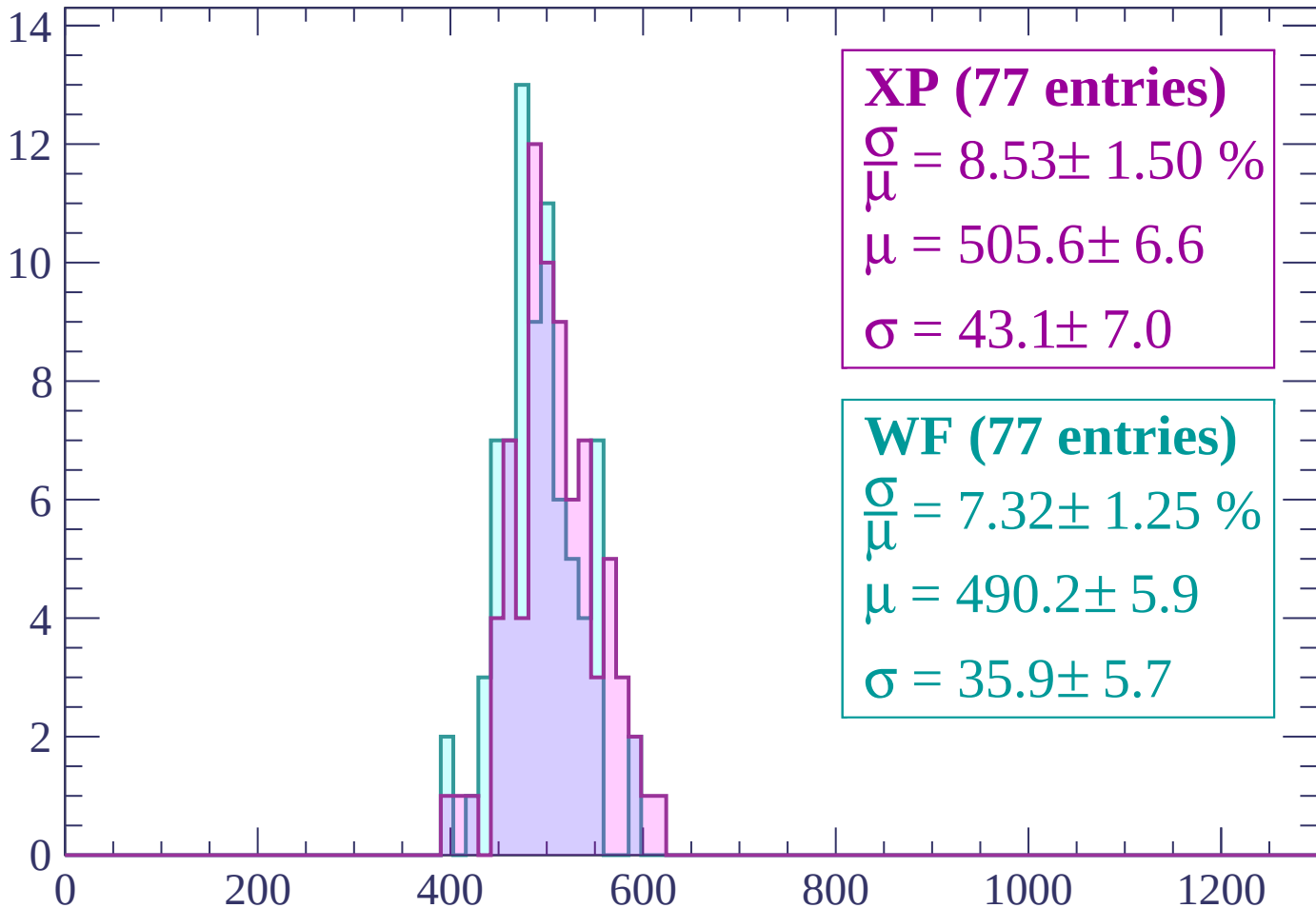
Energy loss | -3000 < p < -2940

Count



Energy loss | $-2940 < p < -2880$

Count



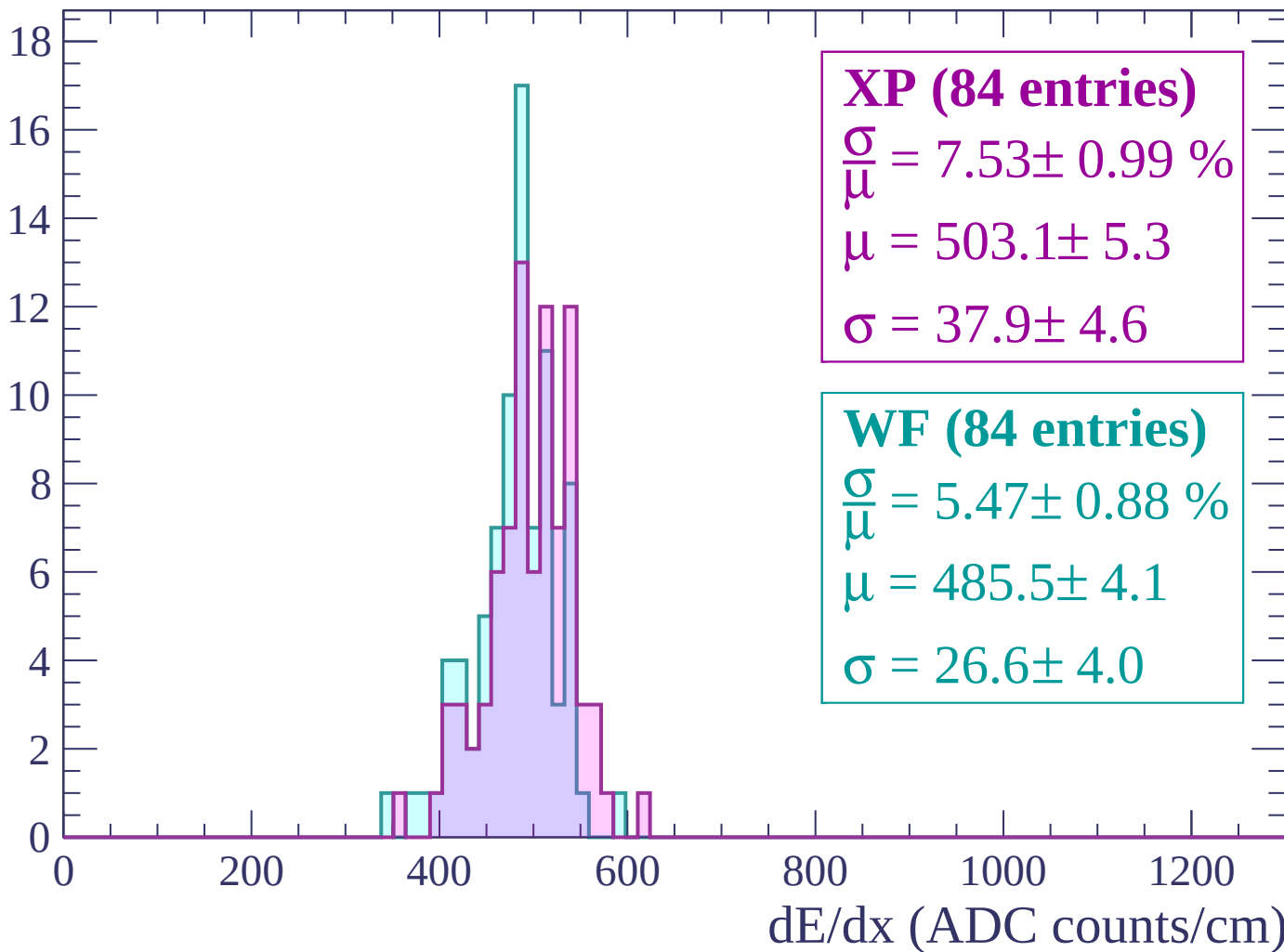
XP (77 entries)
 $\frac{\sigma}{\mu} = 8.53 \pm 1.50 \%$
 $\mu = 505.6 \pm 6.6$
 $\sigma = 43.1 \pm 7.0$

WF (77 entries)
 $\frac{\sigma}{\mu} = 7.32 \pm 1.25 \%$
 $\mu = 490.2 \pm 5.9$
 $\sigma = 35.9 \pm 5.7$

dE/dx (ADC counts/cm)

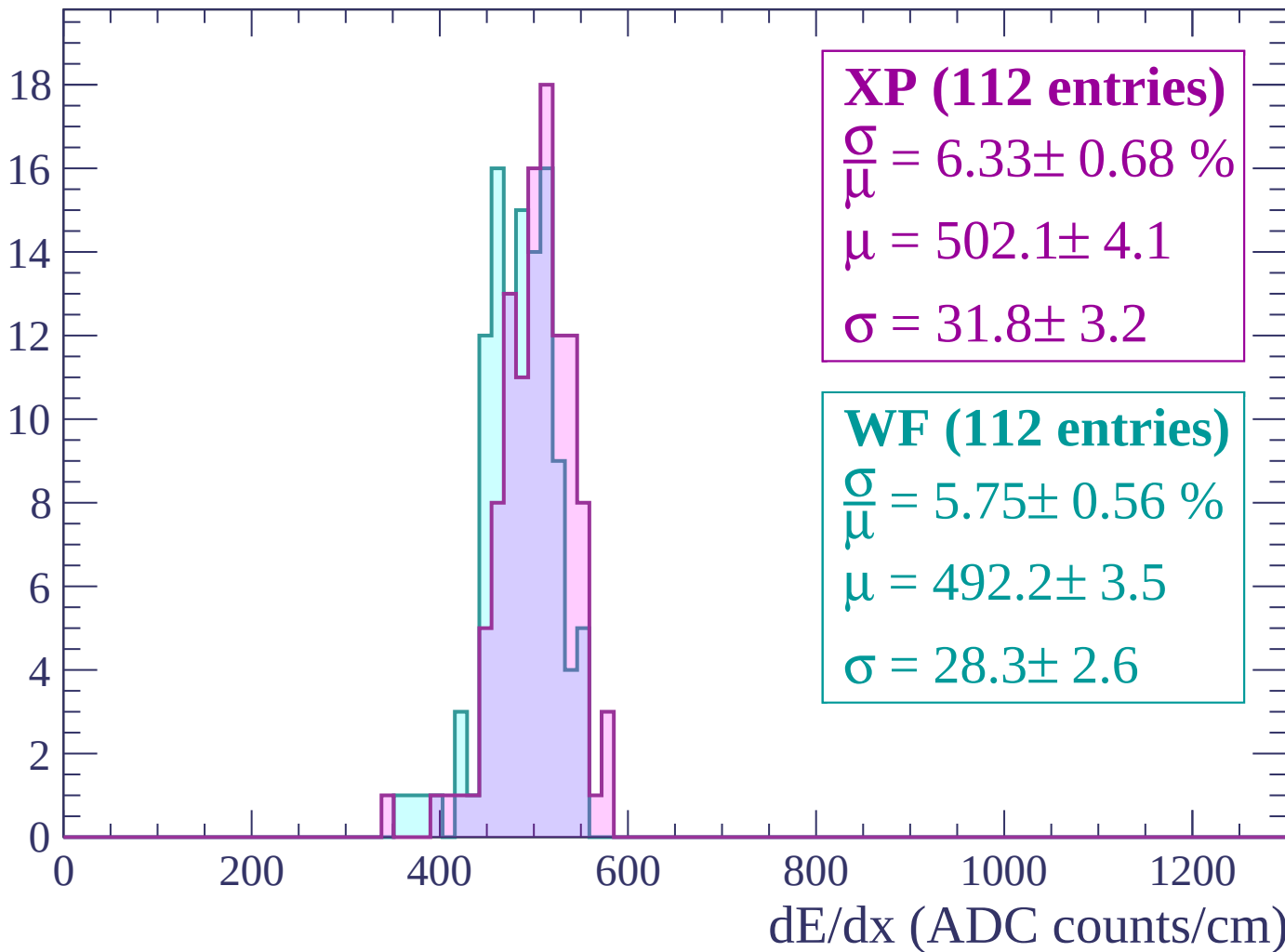
Energy loss | $-2880 < p < -2820$

Count



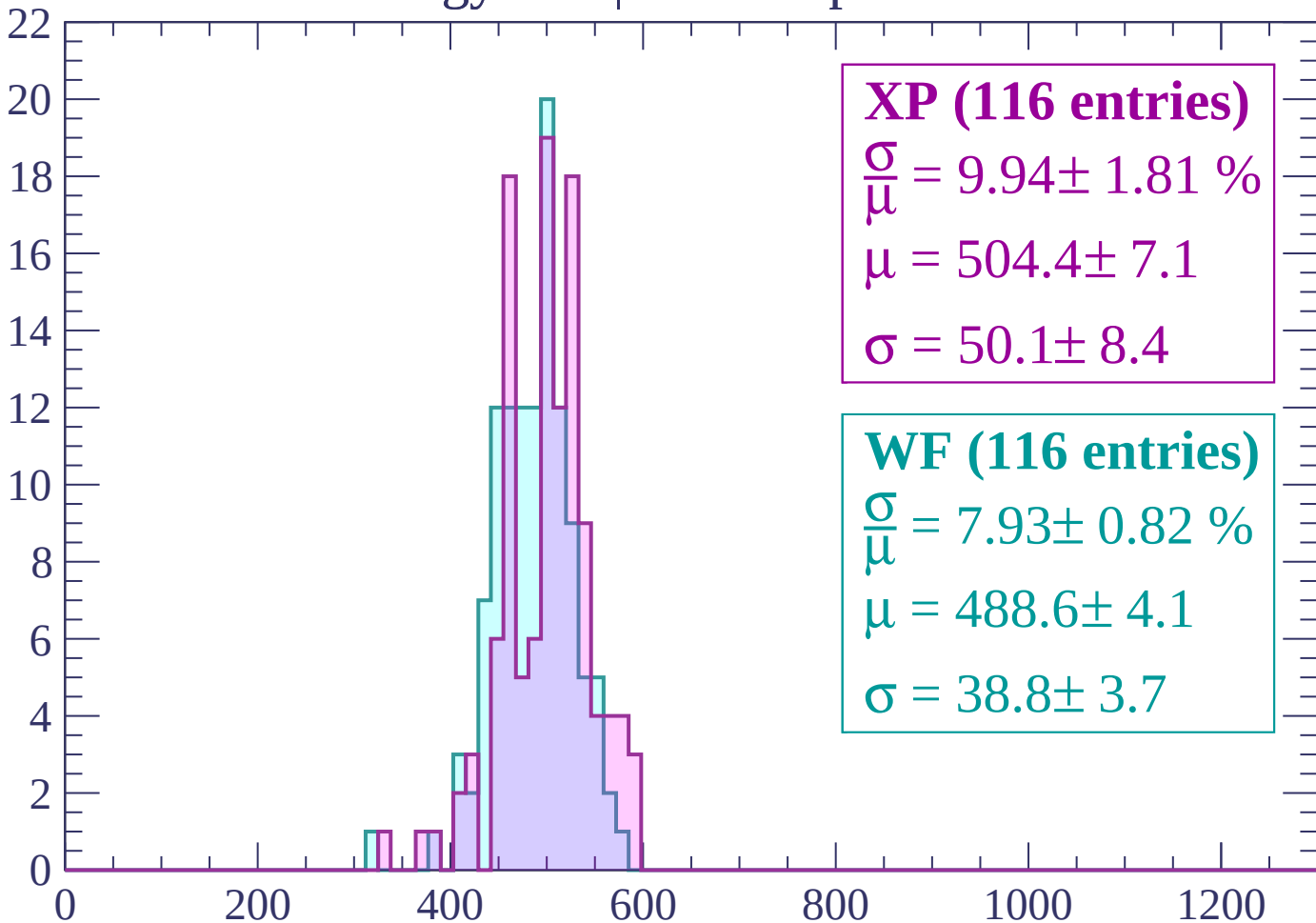
Energy loss | $-2820 < p < -2760$

Count



Energy loss | $-2760 < p < -2700$

Count



XP (116 entries)

$\frac{\sigma}{\mu} = 9.94 \pm 1.81 \%$

$\mu = 504.4 \pm 7.1$

$\sigma = 50.1 \pm 8.4$

WF (116 entries)

$\frac{\sigma}{\mu} = 7.93 \pm 0.82 \%$

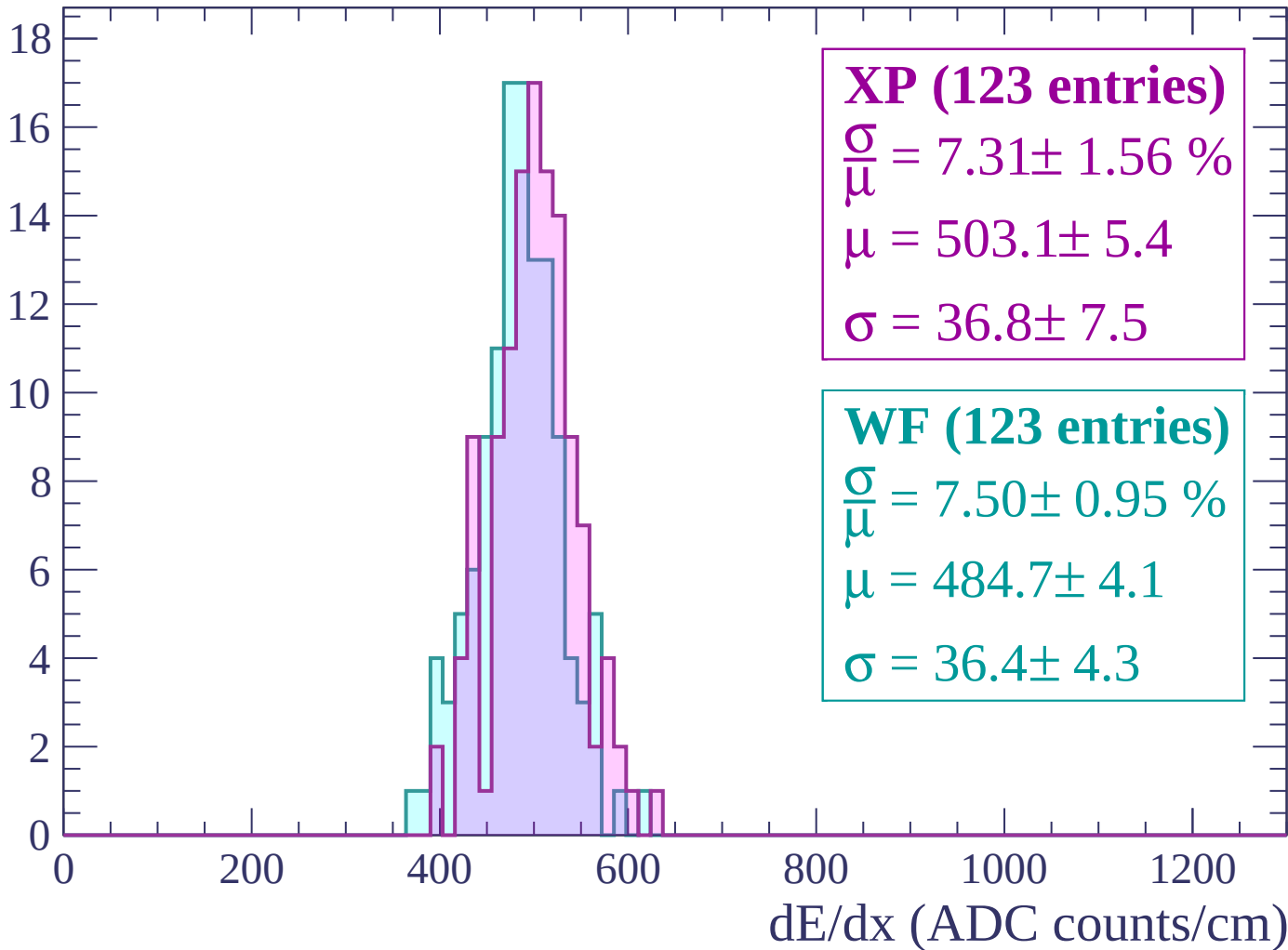
$\mu = 488.6 \pm 4.1$

$\sigma = 38.8 \pm 3.7$

dE/dx (ADC counts/cm)

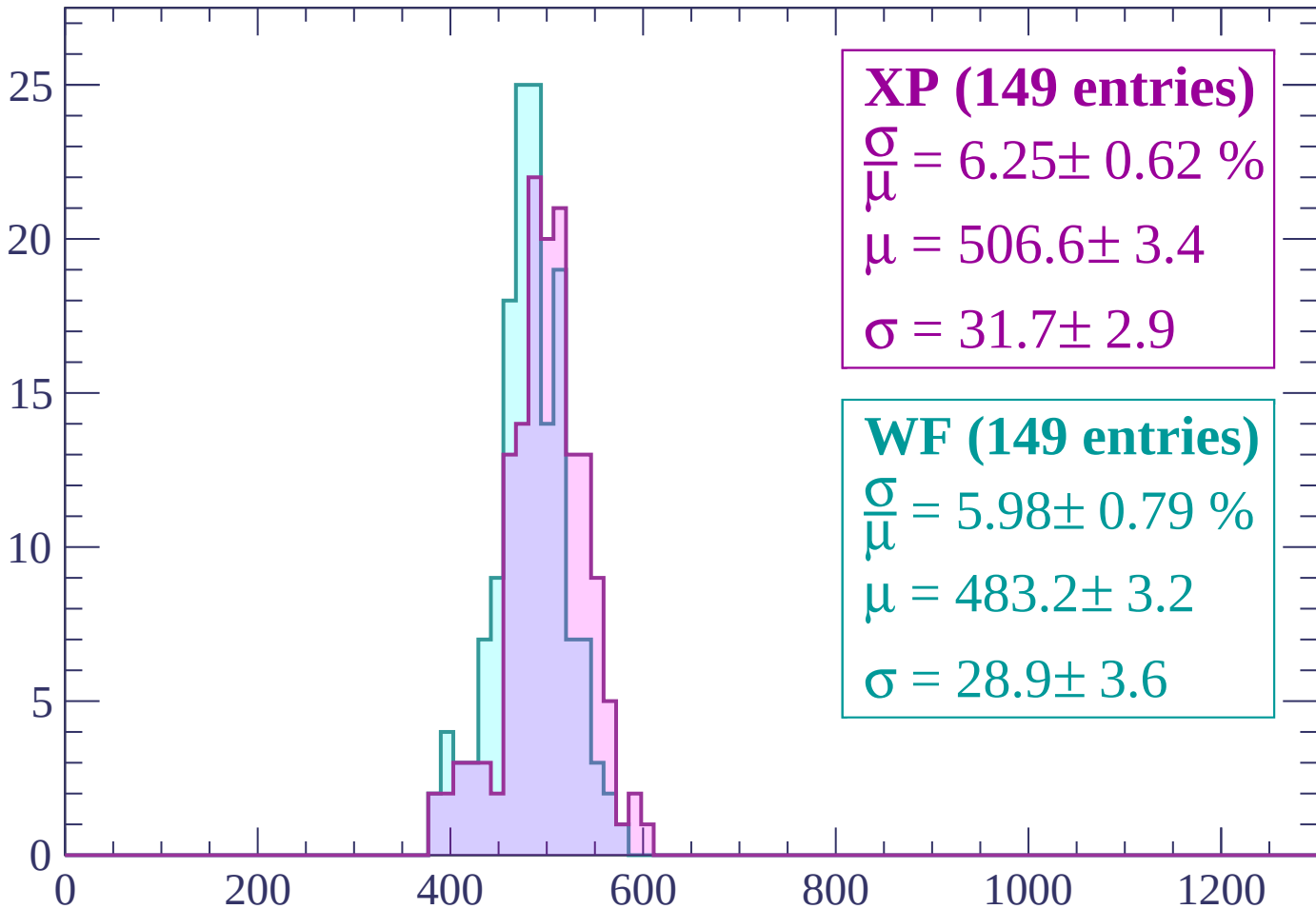
Energy loss | -2700 < p < -2640

Count



Energy loss | $-2640 < p < -2580$

Count



XP (149 entries)

$$\frac{\sigma}{\mu} = 6.25 \pm 0.62 \%$$

$$\mu = 506.6 \pm 3.4$$

$$\sigma = 31.7 \pm 2.9$$

WF (149 entries)

$$\frac{\sigma}{\mu} = 5.98 \pm 0.79 \%$$

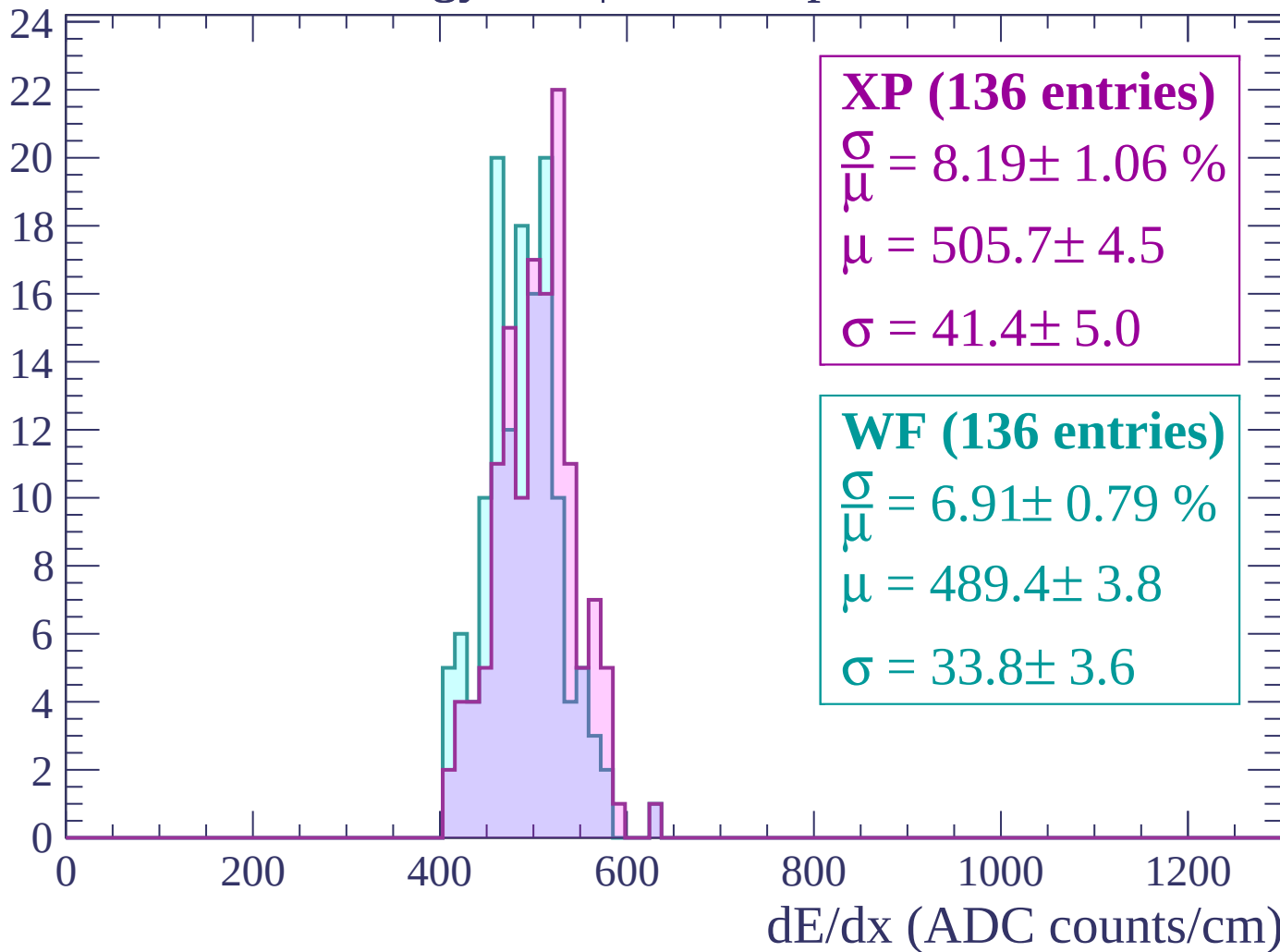
$$\mu = 483.2 \pm 3.2$$

$$\sigma = 28.9 \pm 3.6$$

dE/dx (ADC counts/cm)

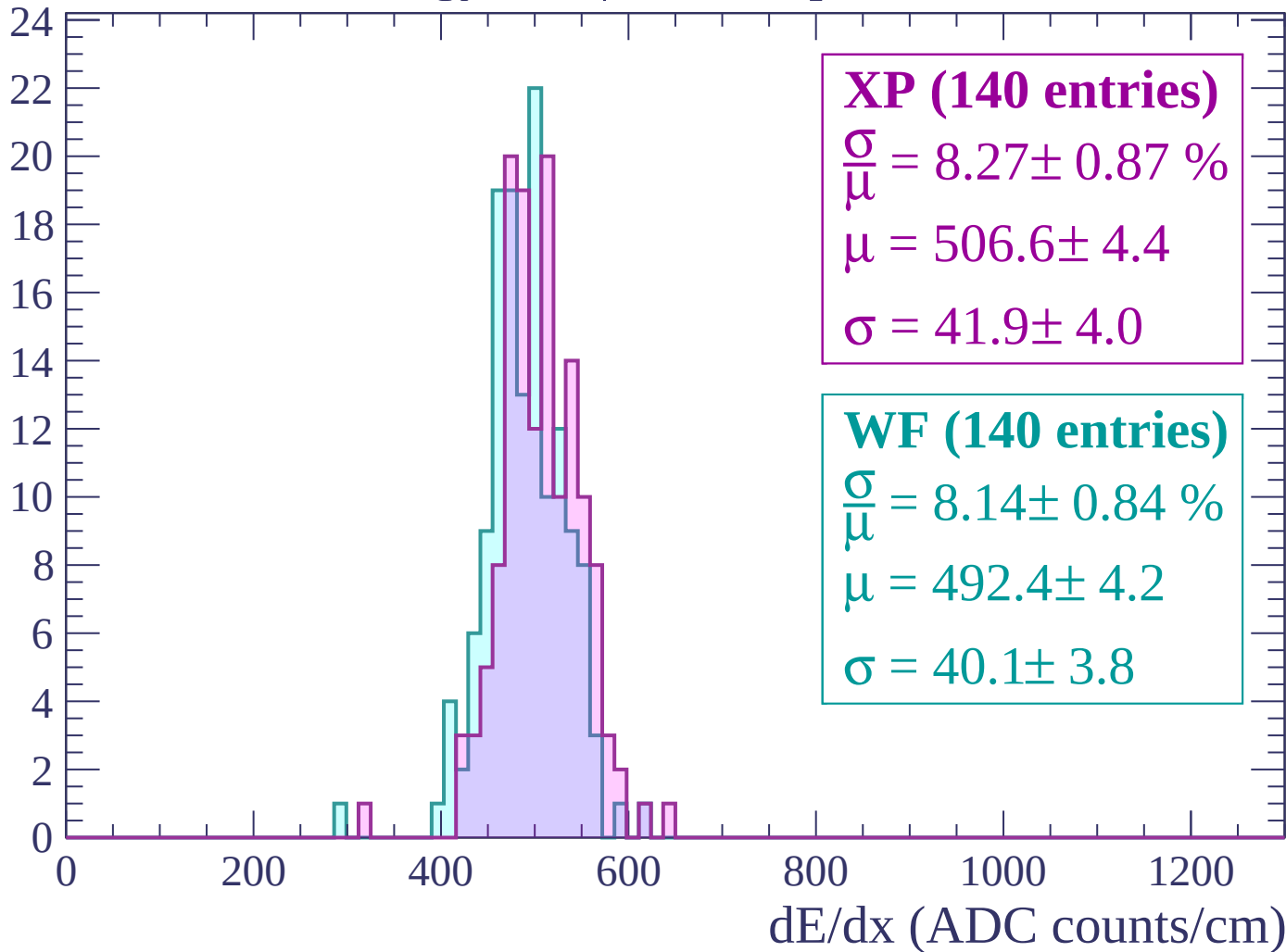
Energy loss | $-2580 < p < -2520$

Count



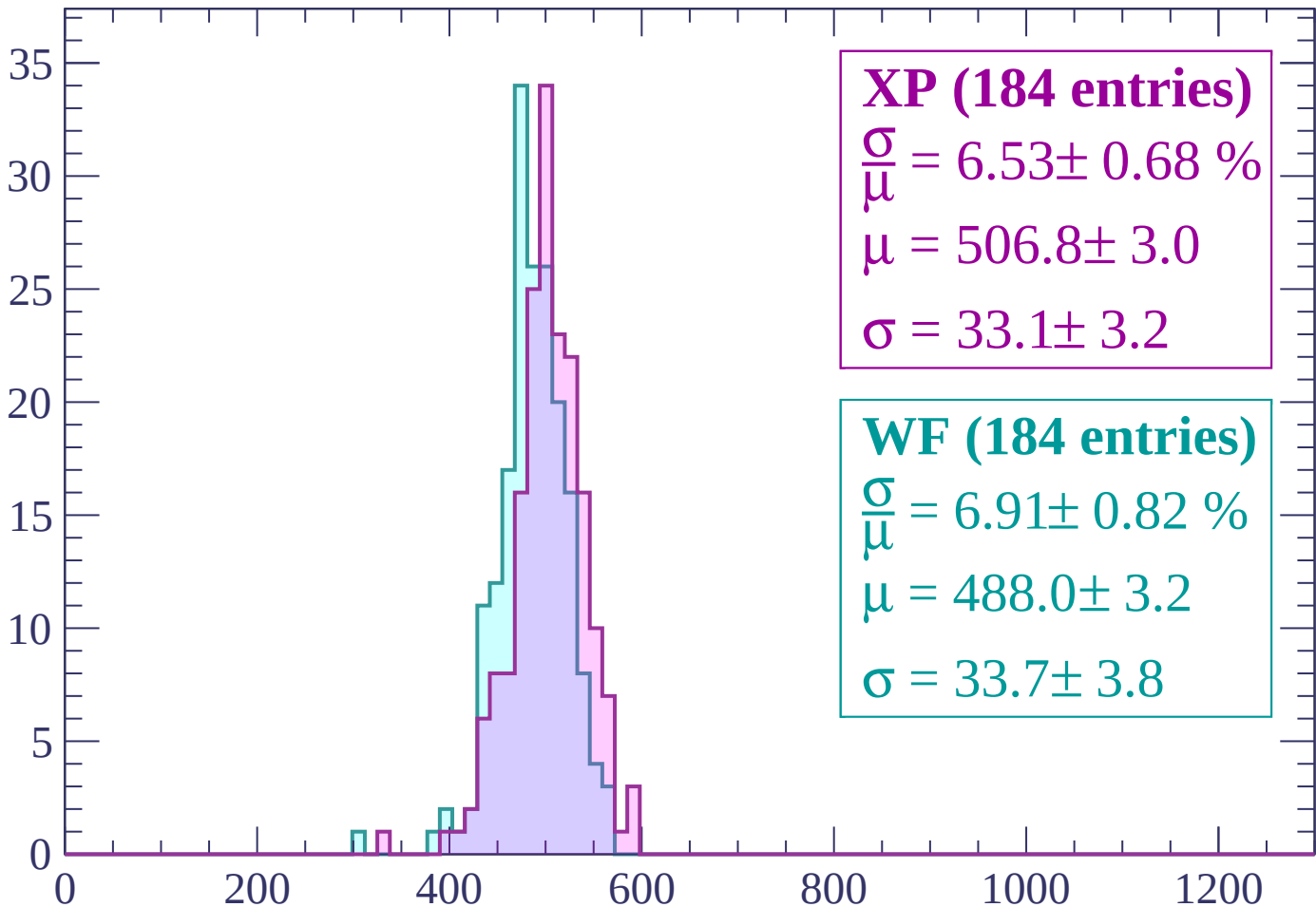
Energy loss | -2520 < p < -2460

Count



Energy loss | $-2460 < p < -2400$

Count



XP (184 entries)

$\frac{\sigma}{\mu} = 6.53 \pm 0.68 \%$

$\mu = 506.8 \pm 3.0$

$\sigma = 33.1 \pm 3.2$

WF (184 entries)

$\frac{\sigma}{\mu} = 6.91 \pm 0.82 \%$

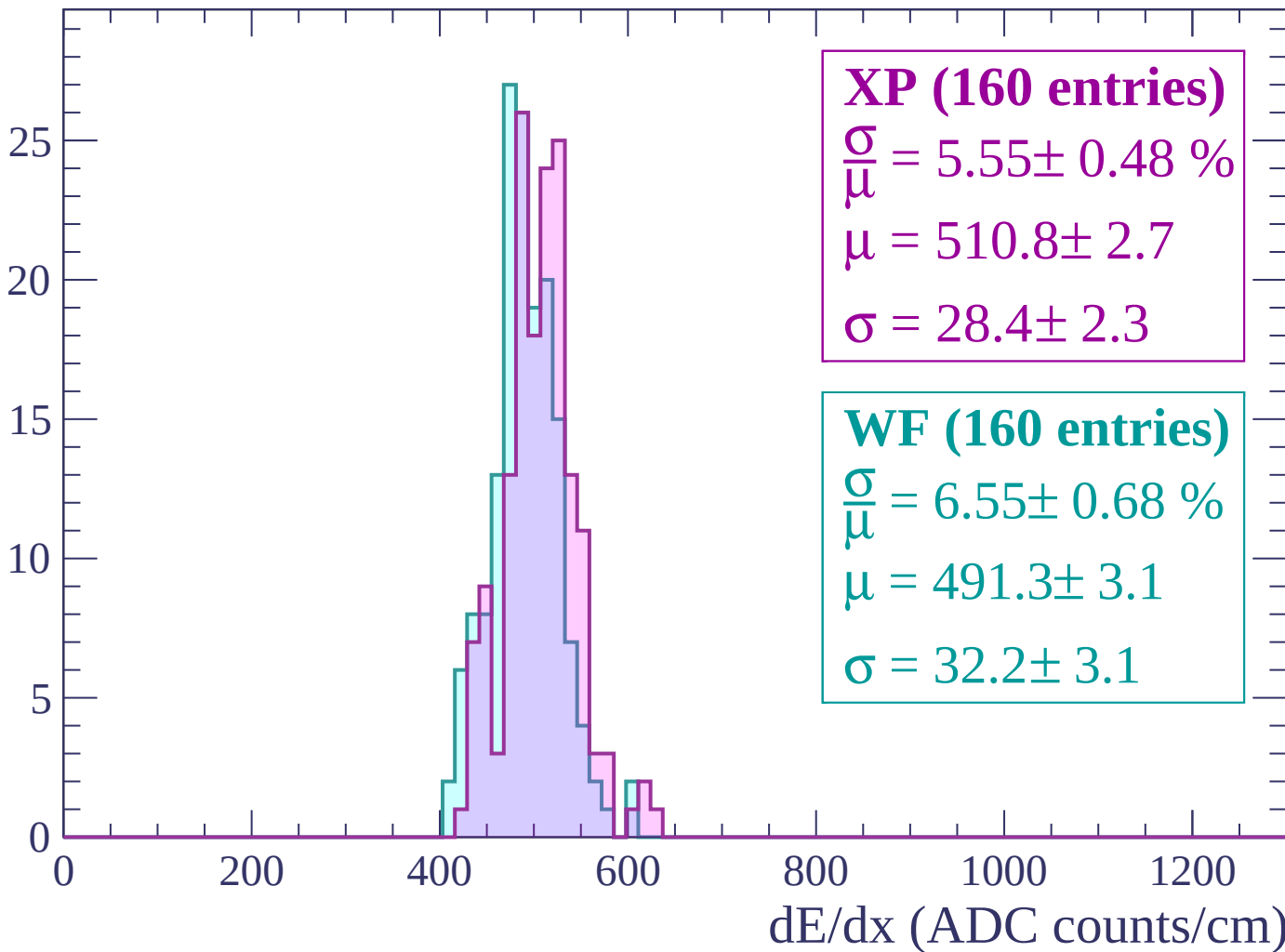
$\mu = 488.0 \pm 3.2$

$\sigma = 33.7 \pm 3.8$

dE/dx (ADC counts/cm)

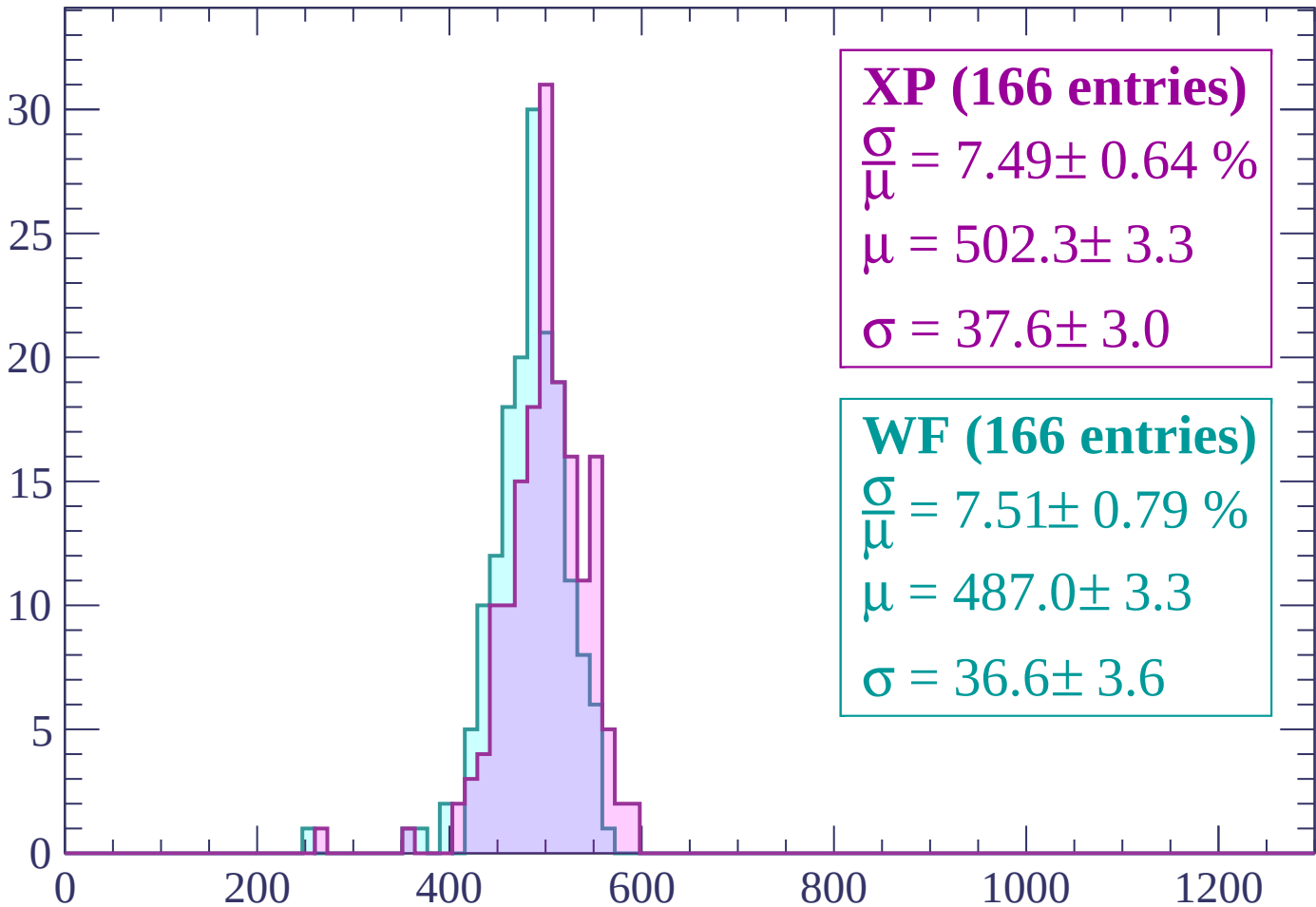
Energy loss | $-2400 < p < -2340$

Count



Energy loss | $-2340 < p < -2280$

Count



XP (166 entries)

$\frac{\sigma}{\mu} = 7.49 \pm 0.64 \%$

$\mu = 502.3 \pm 3.3$

$\sigma = 37.6 \pm 3.0$

WF (166 entries)

$\frac{\sigma}{\mu} = 7.51 \pm 0.79 \%$

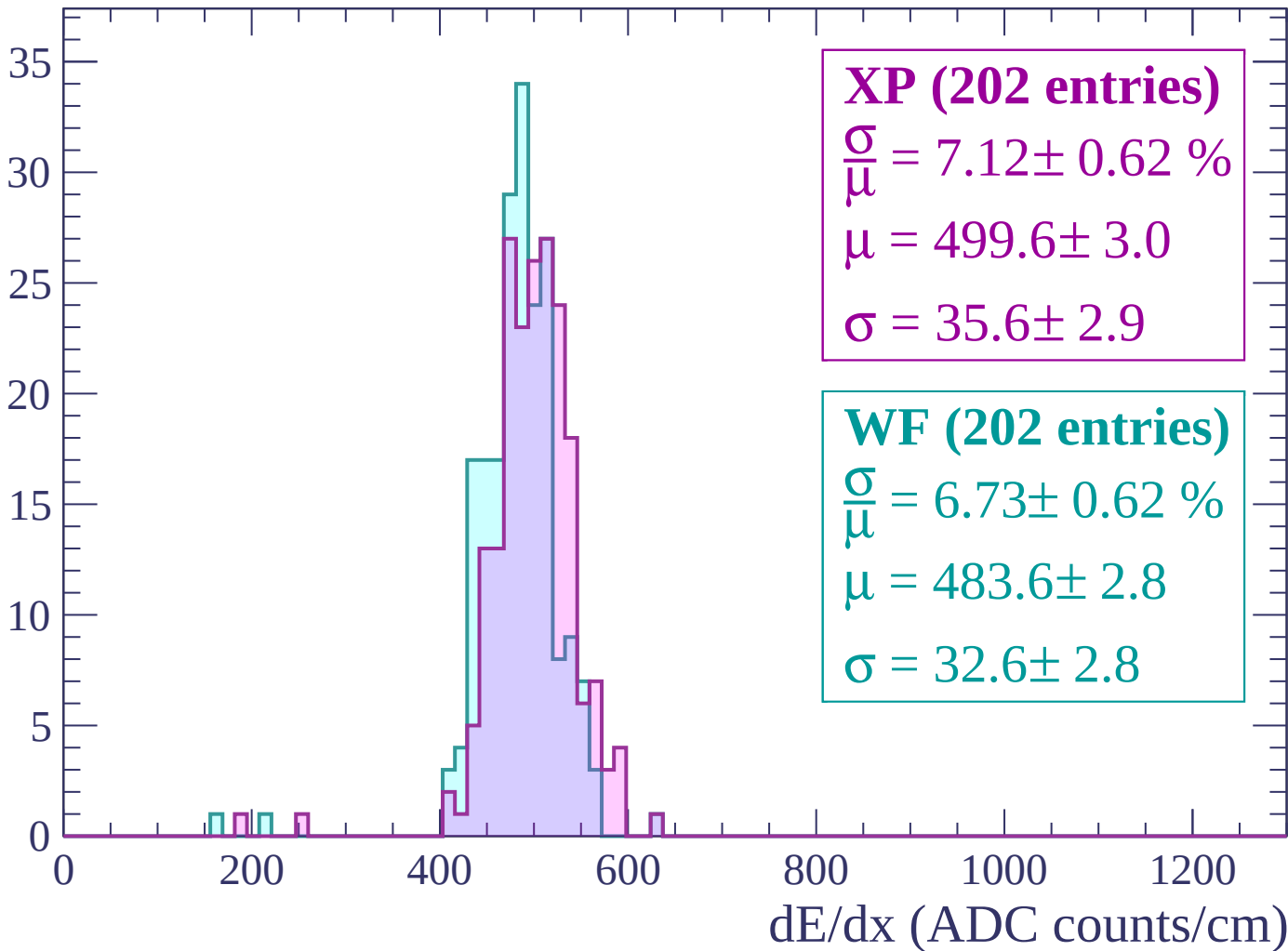
$\mu = 487.0 \pm 3.3$

$\sigma = 36.6 \pm 3.6$

dE/dx (ADC counts/cm)

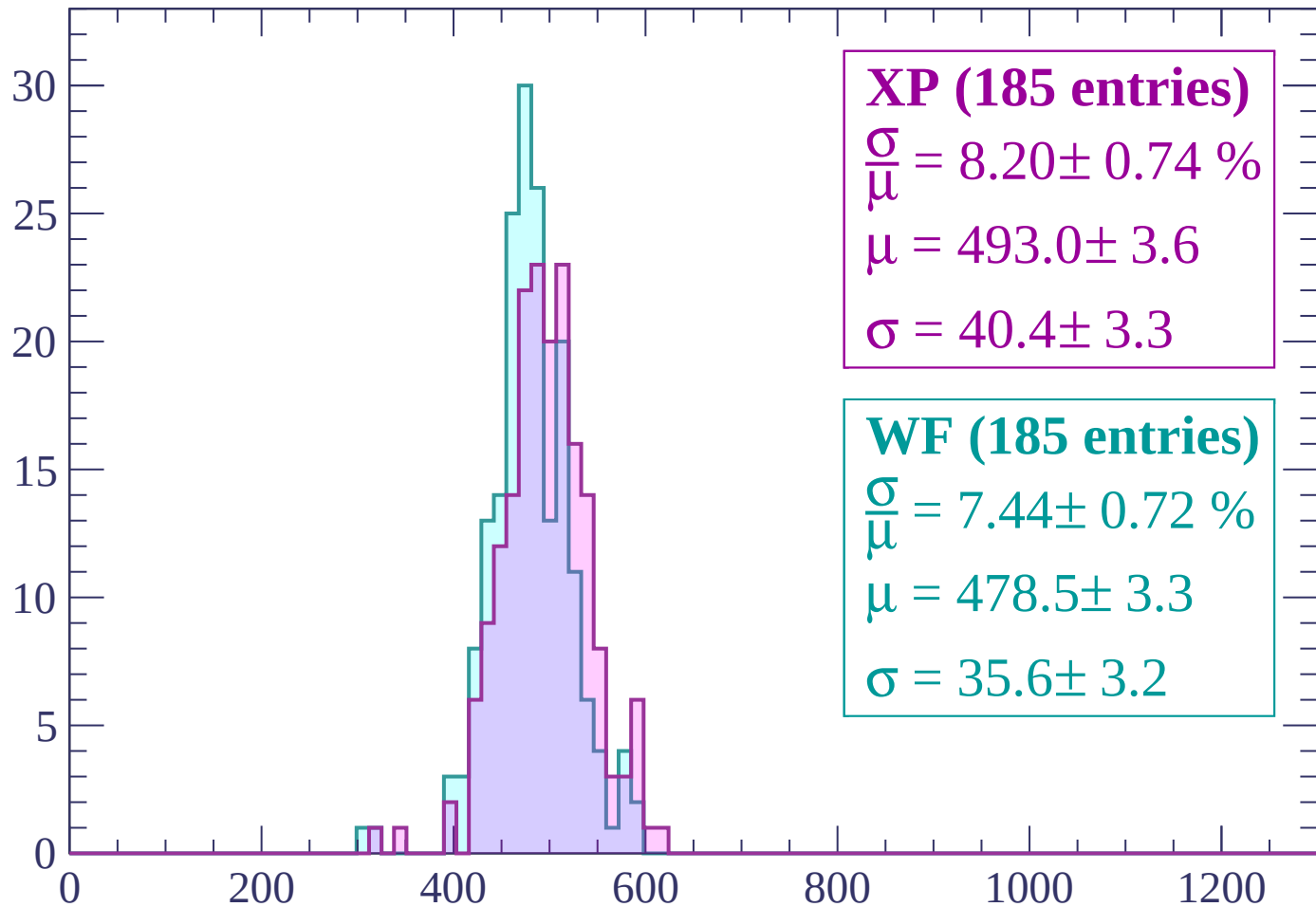
Energy loss | $-2280 < p < -2220$

Count



Energy loss | $-2220 < p < -2160$

Count



XP (185 entries)

$$\frac{\sigma}{\mu} = 8.20 \pm 0.74 \%$$

$$\mu = 493.0 \pm 3.6$$

$$\sigma = 40.4 \pm 3.3$$

WF (185 entries)

$$\frac{\sigma}{\mu} = 7.44 \pm 0.72 \%$$

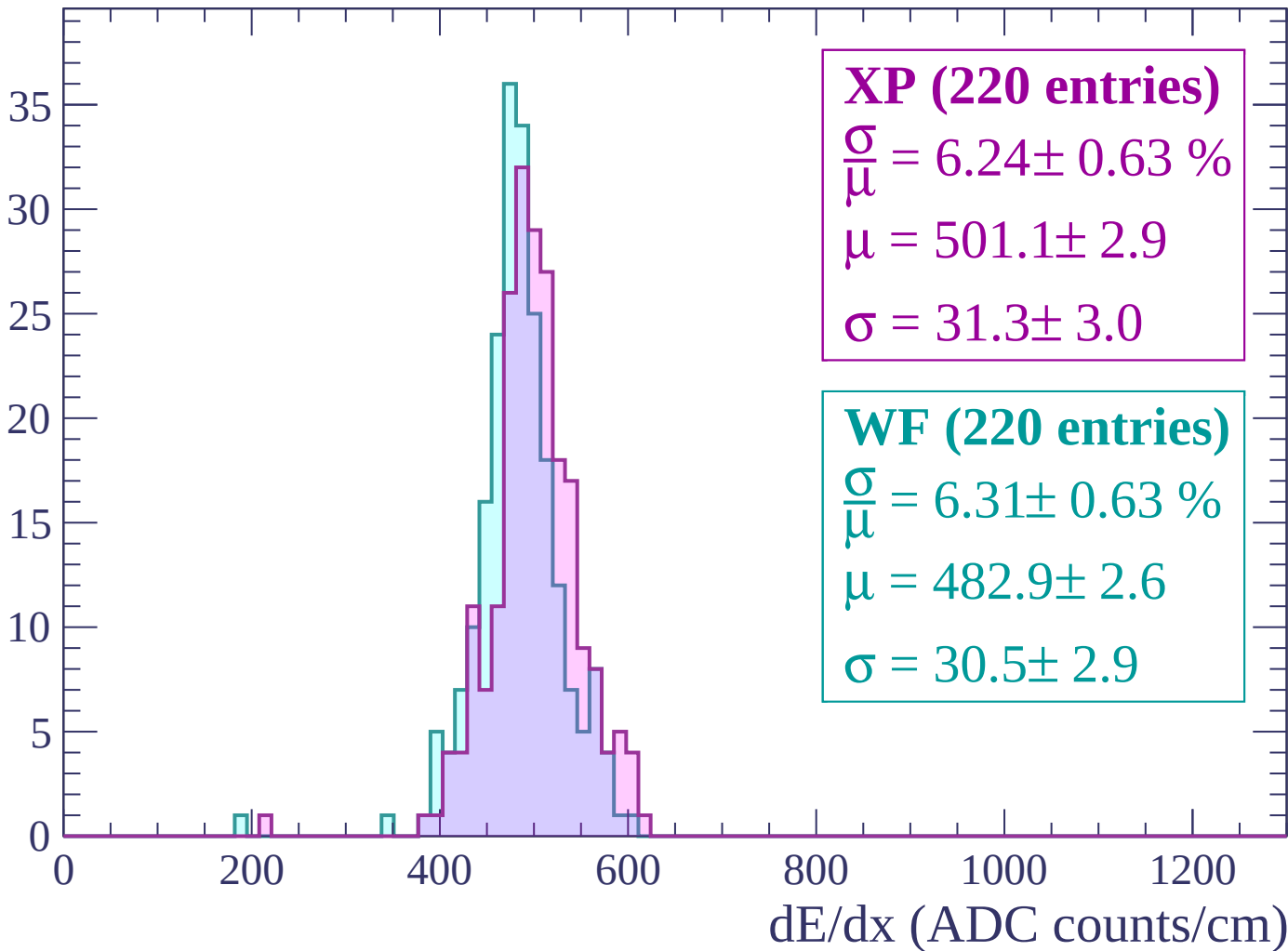
$$\mu = 478.5 \pm 3.3$$

$$\sigma = 35.6 \pm 3.2$$

dE/dx (ADC counts/cm)

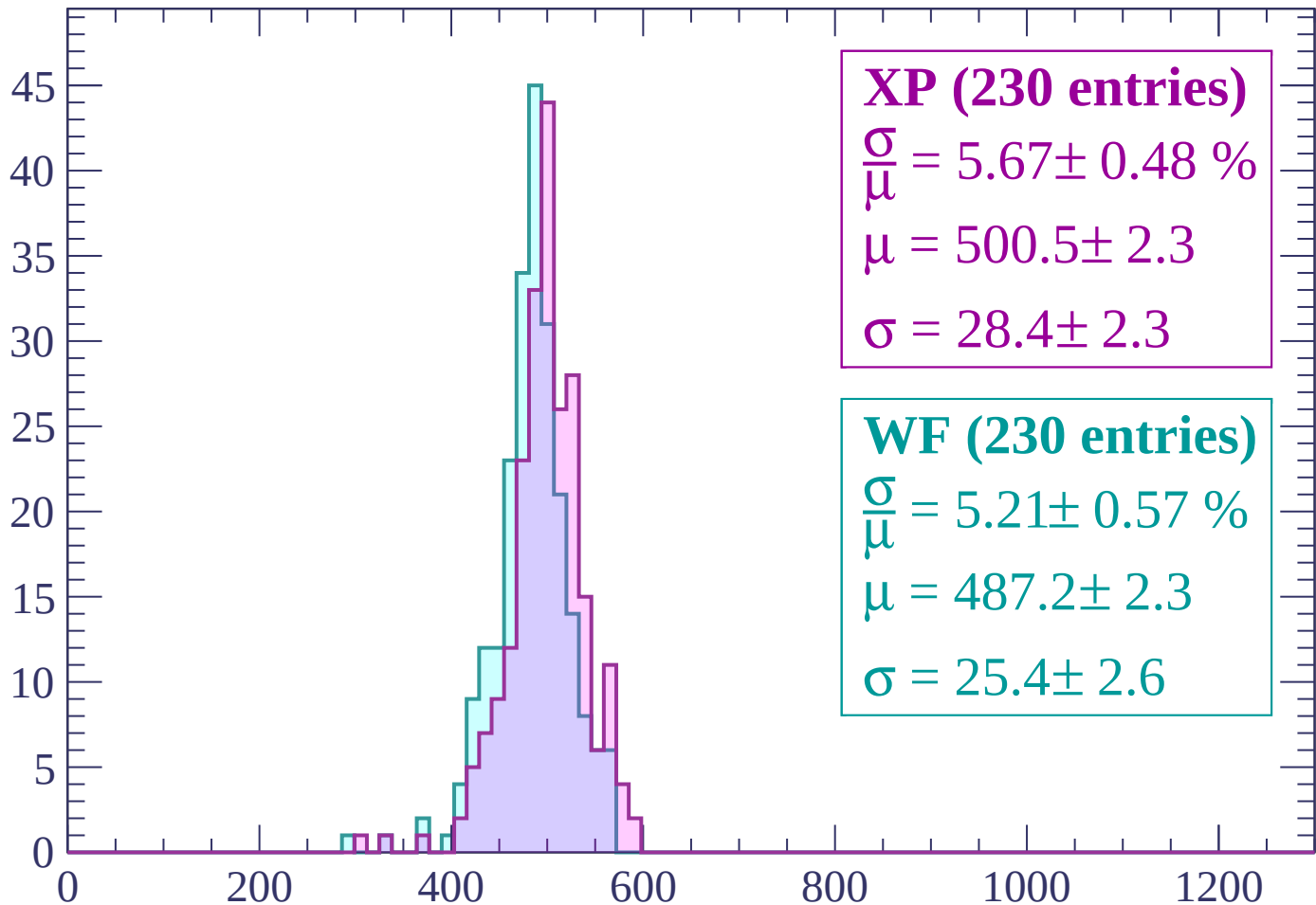
Energy loss | $-2160 < p < -2100$

Count



Energy loss | -2100 < p < -2040

Count



XP (230 entries)

$\frac{\sigma}{\mu} = 5.67 \pm 0.48 \%$

$\mu = 500.5 \pm 2.3$

$\sigma = 28.4 \pm 2.3$

WF (230 entries)

$\frac{\sigma}{\mu} = 5.21 \pm 0.57 \%$

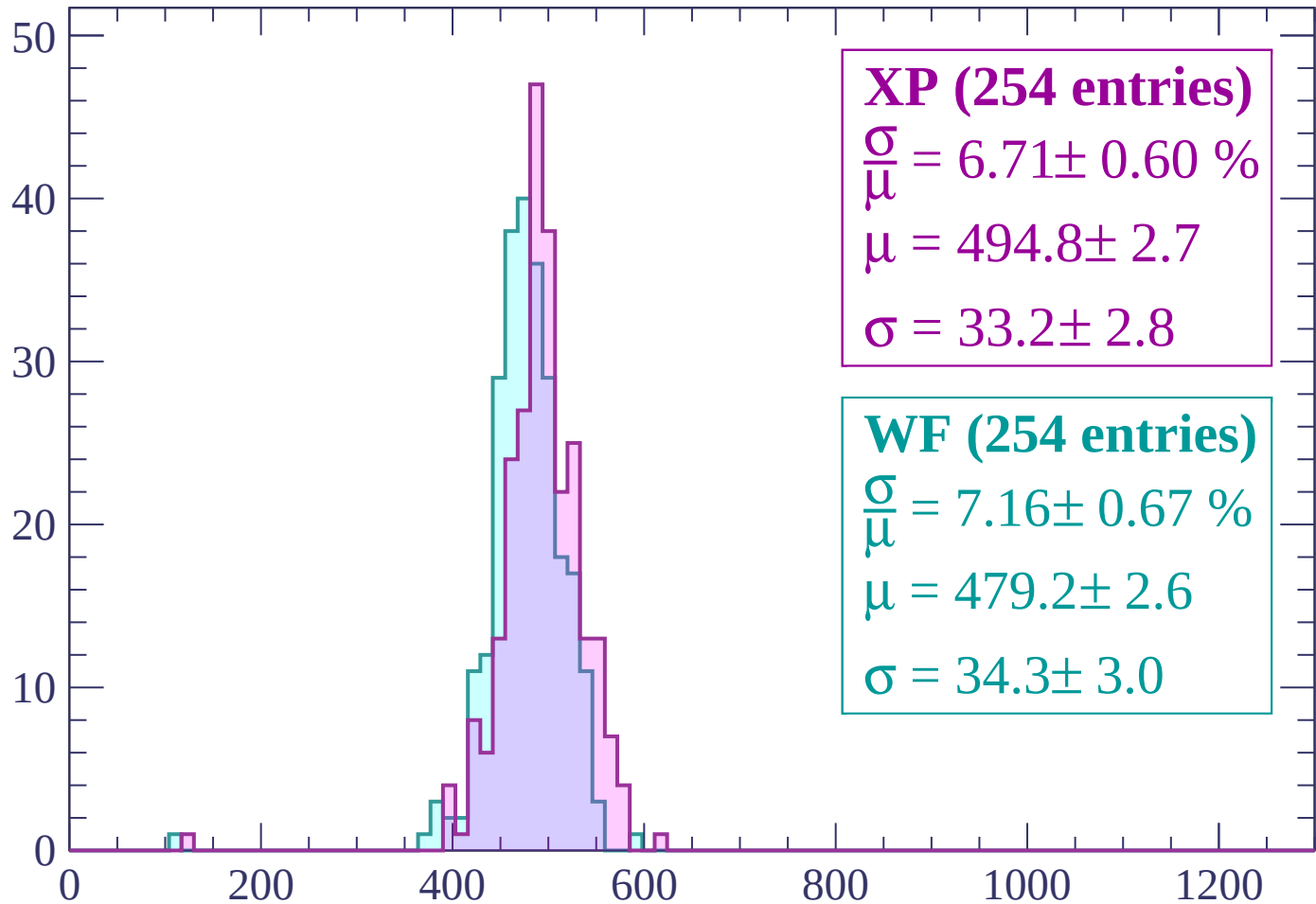
$\mu = 487.2 \pm 2.3$

$\sigma = 25.4 \pm 2.6$

dE/dx (ADC counts/cm)

Energy loss | -2040 < p < -1980

Count



XP (254 entries)

$\frac{\sigma}{\mu} = 6.71 \pm 0.60 \%$

$\mu = 494.8 \pm 2.7$

$\sigma = 33.2 \pm 2.8$

WF (254 entries)

$\frac{\sigma}{\mu} = 7.16 \pm 0.67 \%$

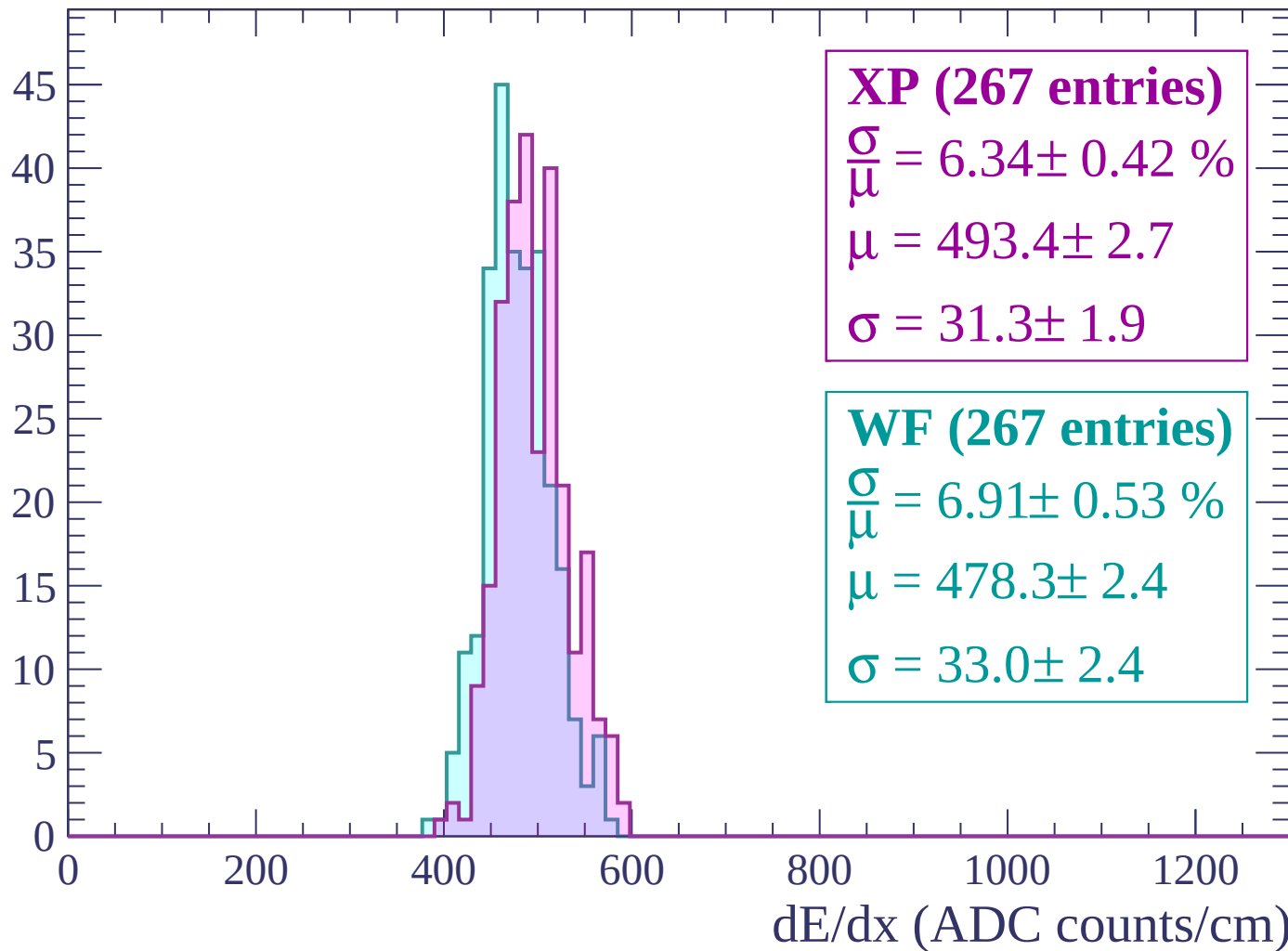
$\mu = 479.2 \pm 2.6$

$\sigma = 34.3 \pm 3.0$

dE/dx (ADC counts/cm)

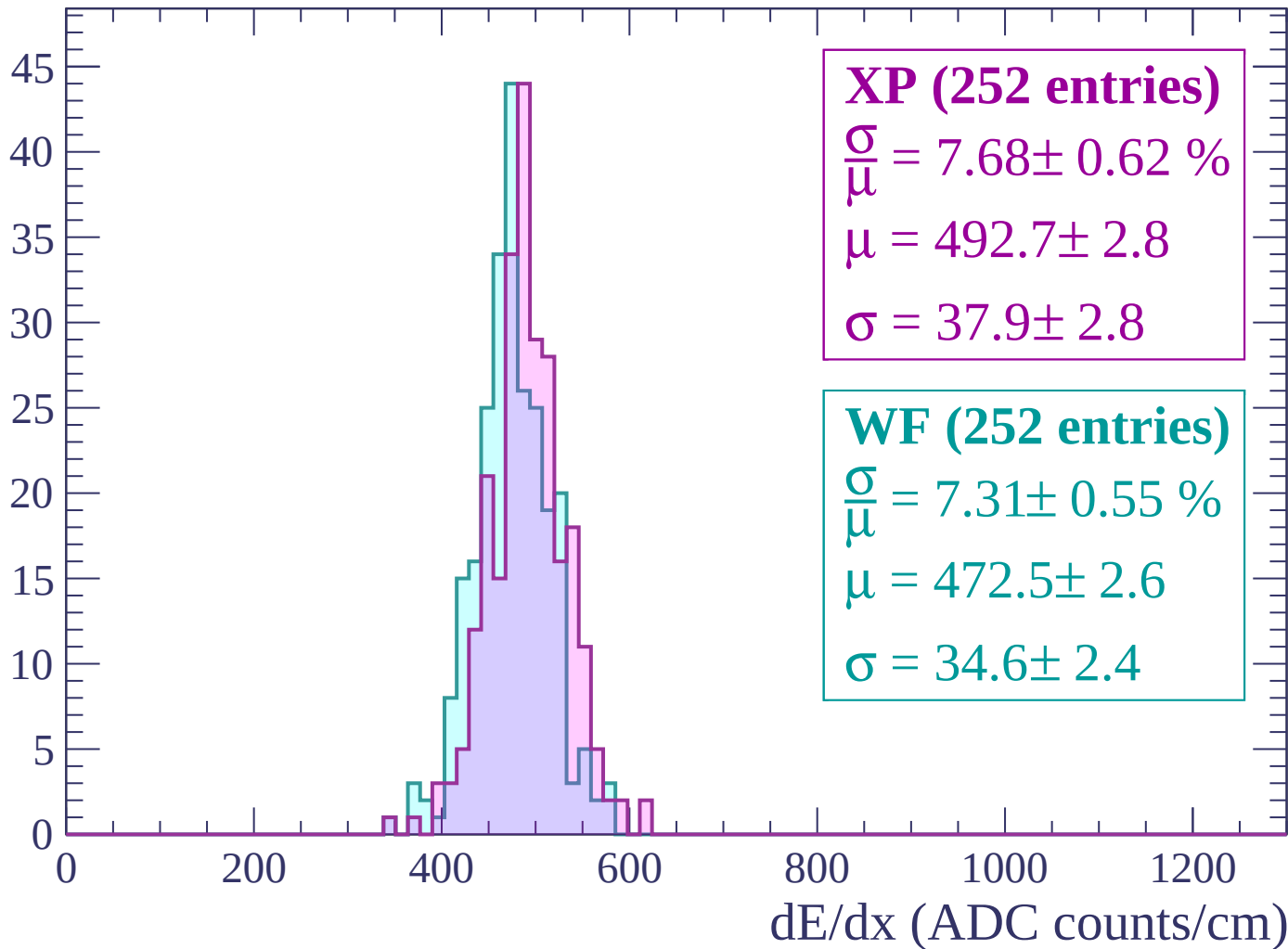
Energy loss | -1980 < p < -1920

Count



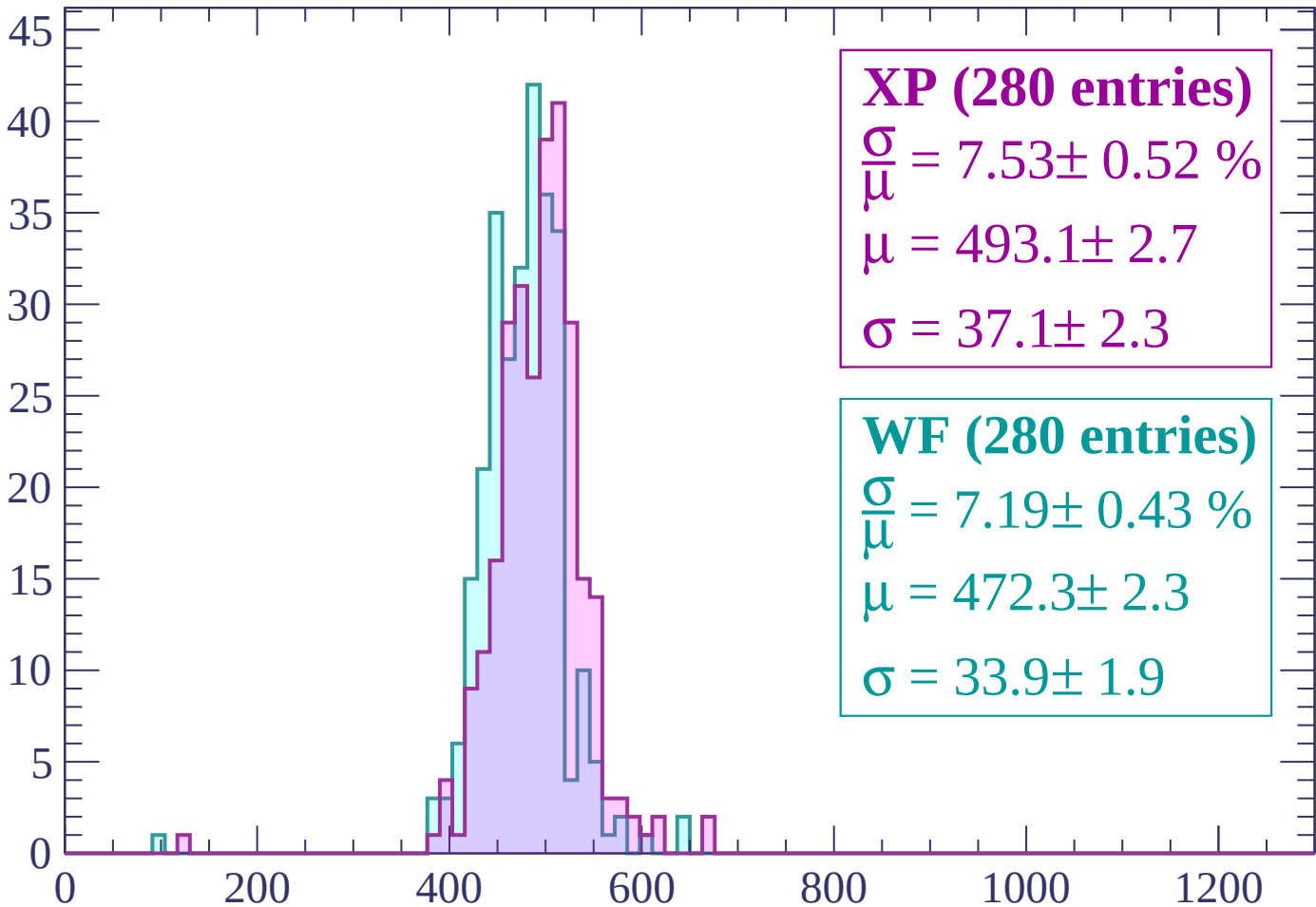
Energy loss | -1920 < p < -1860

Count



Energy loss | -1860 < p < -1800

Count



XP (280 entries)

$$\frac{\sigma}{\mu} = 7.53 \pm 0.52 \%$$

$$\mu = 493.1 \pm 2.7$$

$$\sigma = 37.1 \pm 2.3$$

WF (280 entries)

$$\frac{\sigma}{\mu} = 7.19 \pm 0.43 \%$$

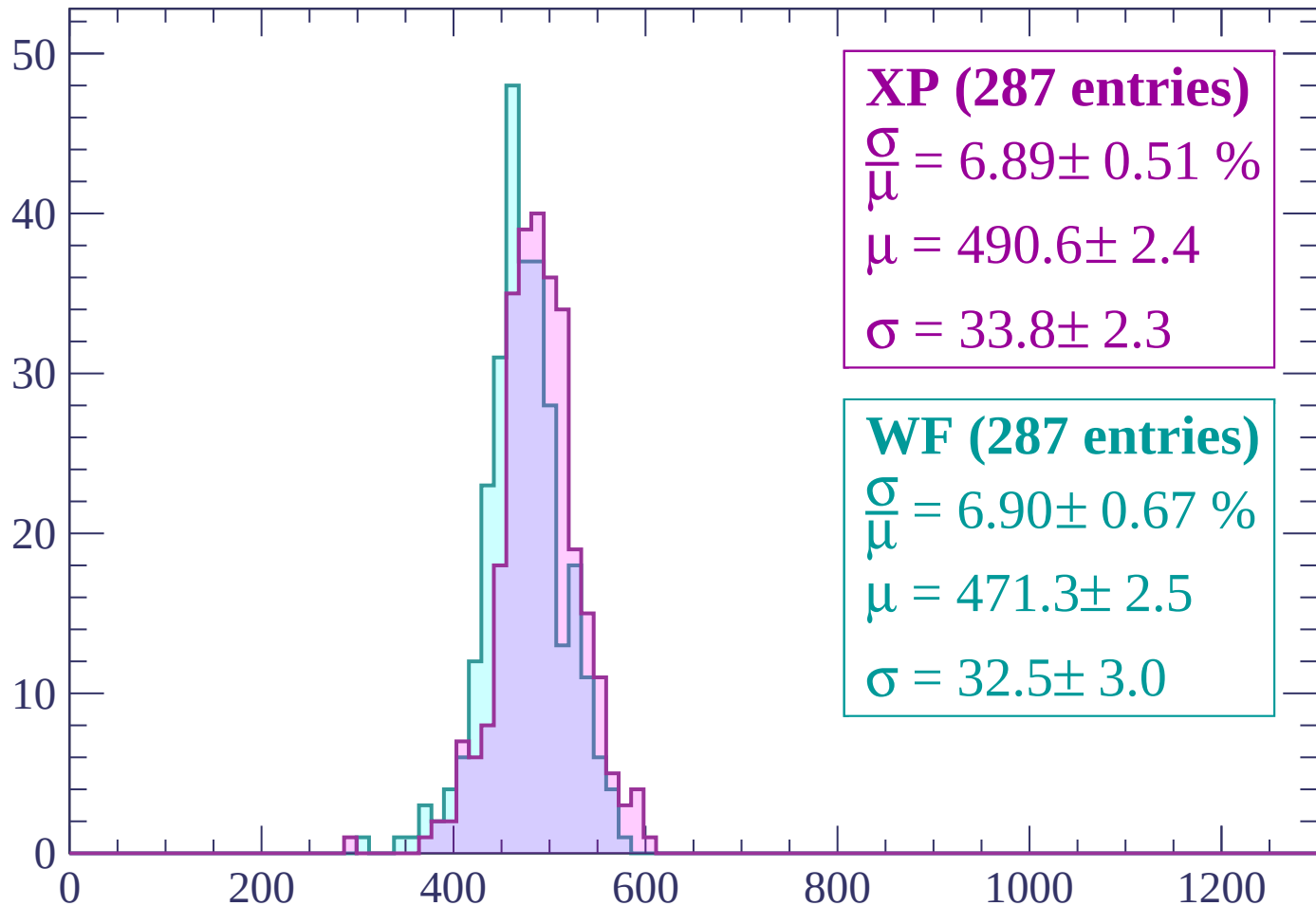
$$\mu = 472.3 \pm 2.3$$

$$\sigma = 33.9 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1740$

Count



XP (287 entries)

$$\frac{\sigma}{\mu} = 6.89 \pm 0.51 \%$$

$$\mu = 490.6 \pm 2.4$$

$$\sigma = 33.8 \pm 2.3$$

WF (287 entries)

$$\frac{\sigma}{\mu} = 6.90 \pm 0.67 \%$$

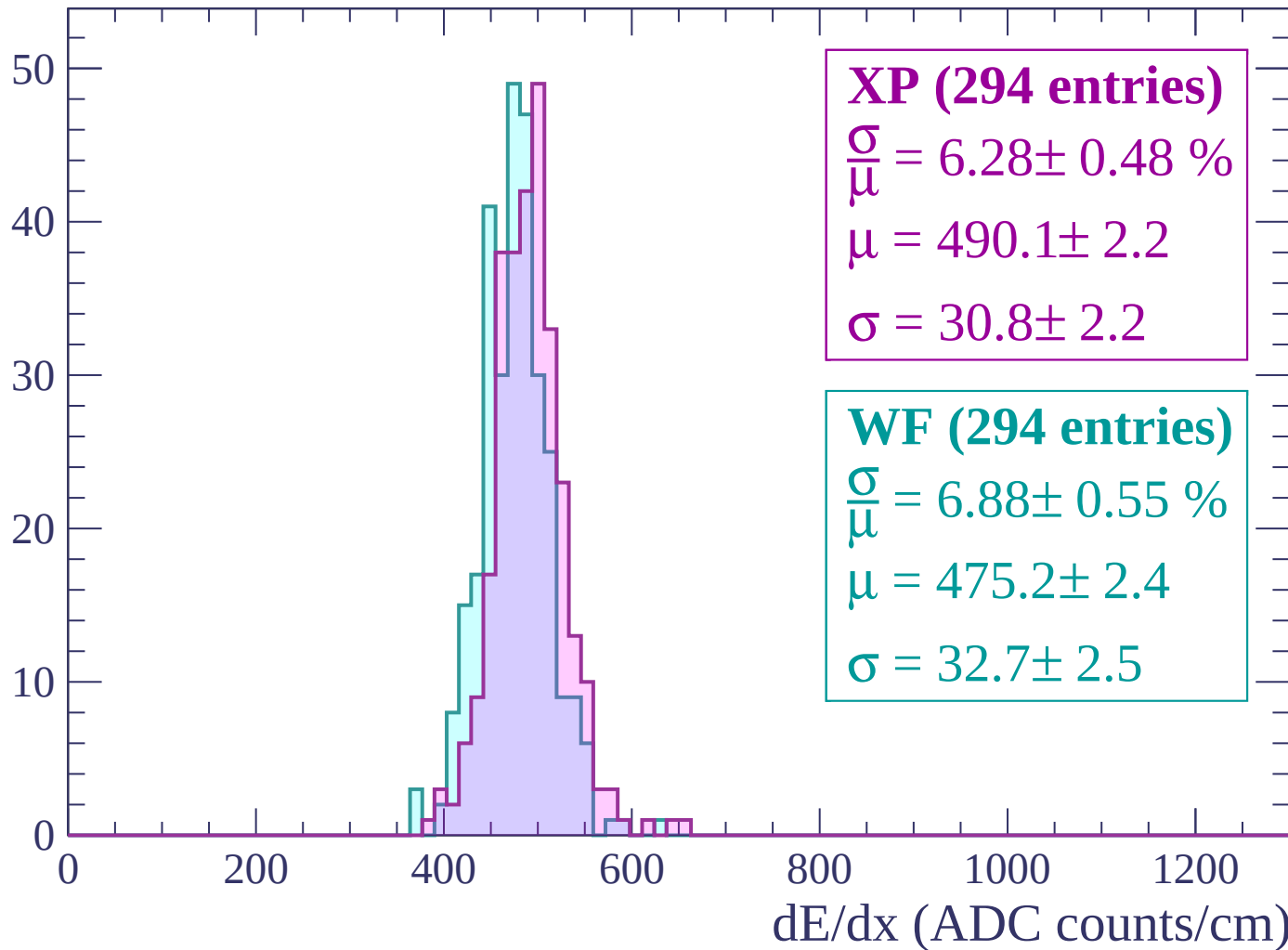
$$\mu = 471.3 \pm 2.5$$

$$\sigma = 32.5 \pm 3.0$$

dE/dx (ADC counts/cm)

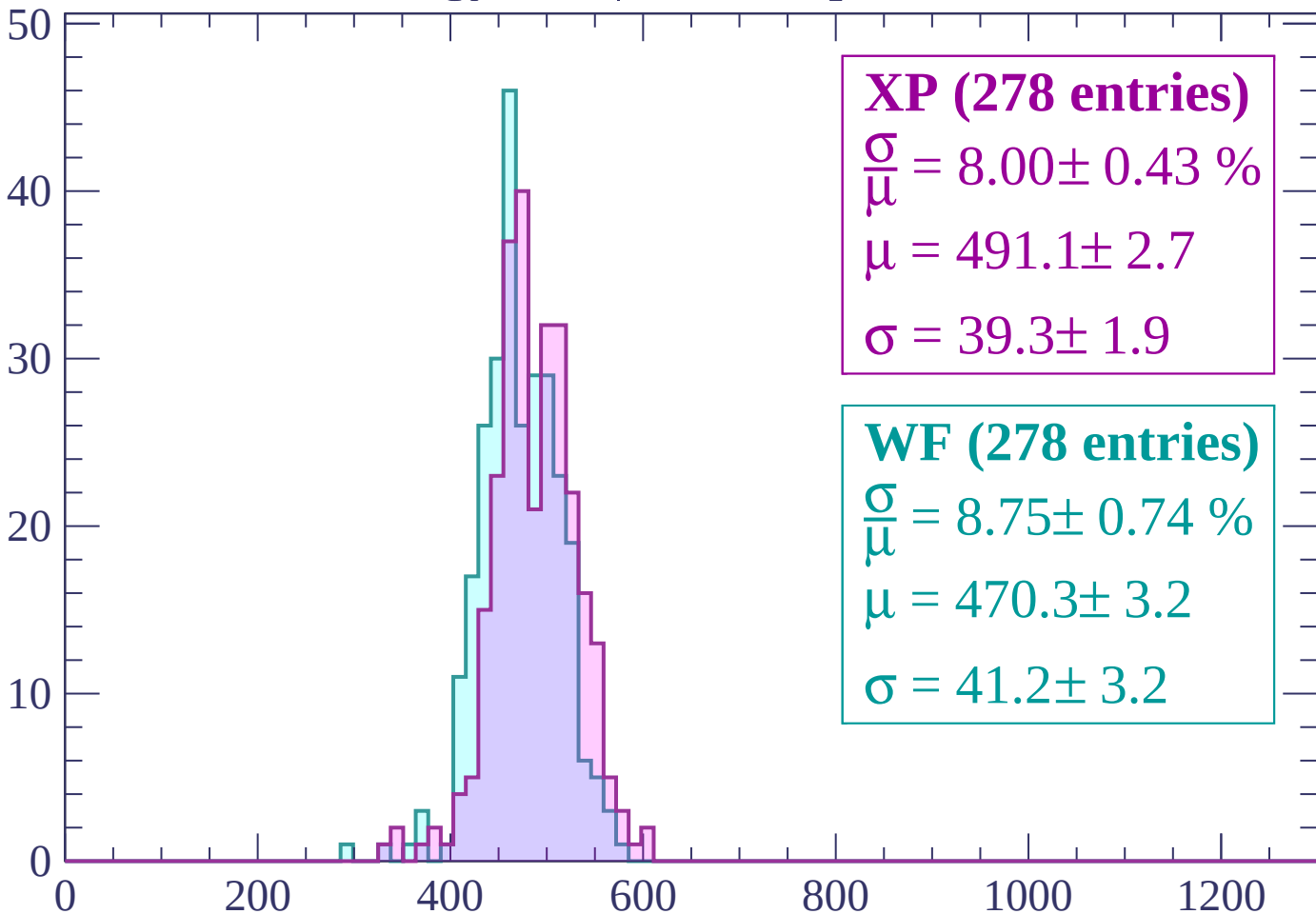
Energy loss | $-1740 < p < -1680$

Count



Energy loss | $-1680 < p < -1620$

Count



XP (278 entries)

$\frac{\sigma}{\mu} = 8.00 \pm 0.43 \%$

$\mu = 491.1 \pm 2.7$

$\sigma = 39.3 \pm 1.9$

WF (278 entries)

$\frac{\sigma}{\mu} = 8.75 \pm 0.74 \%$

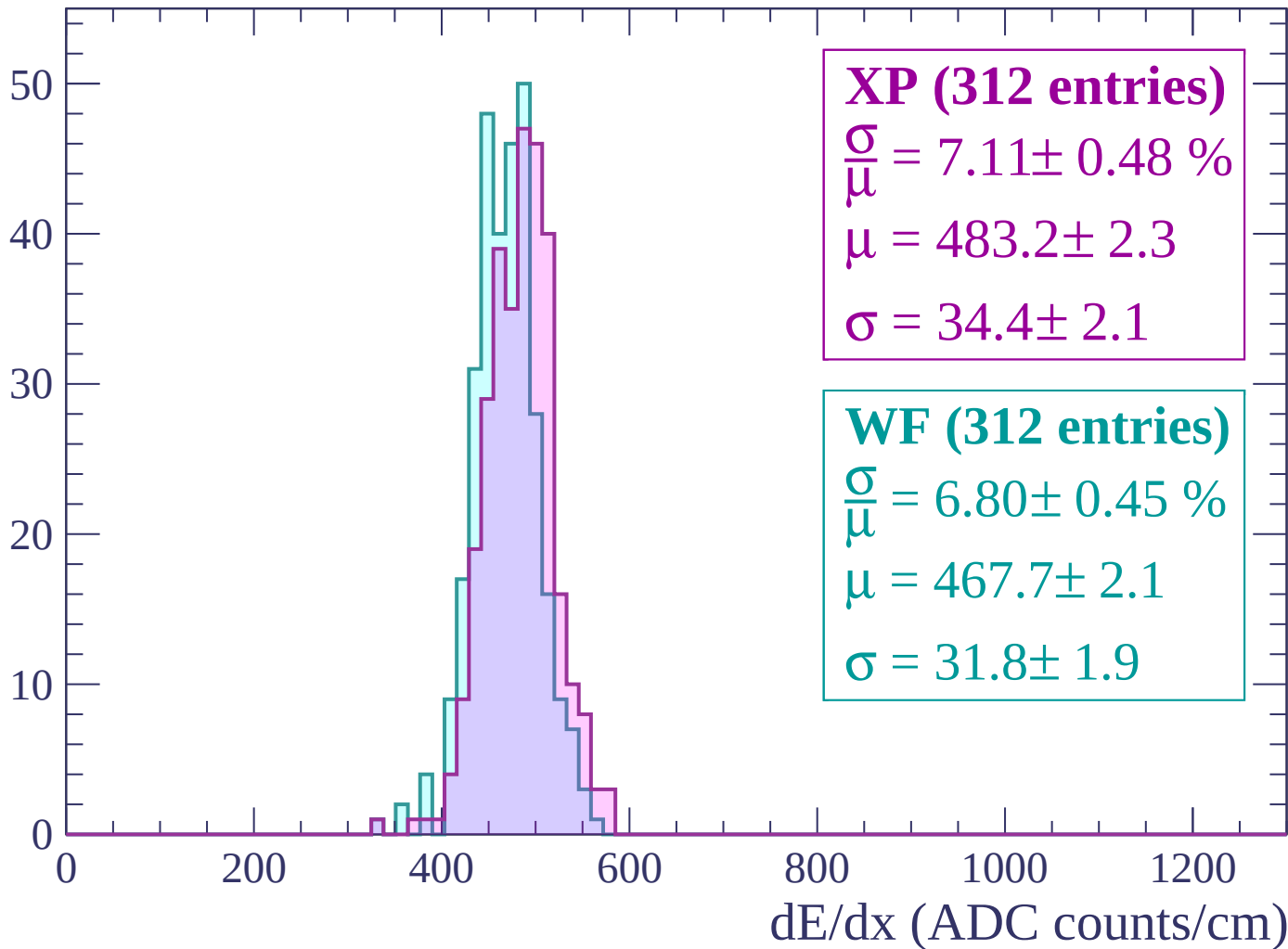
$\mu = 470.3 \pm 3.2$

$\sigma = 41.2 \pm 3.2$

dE/dx (ADC counts/cm)

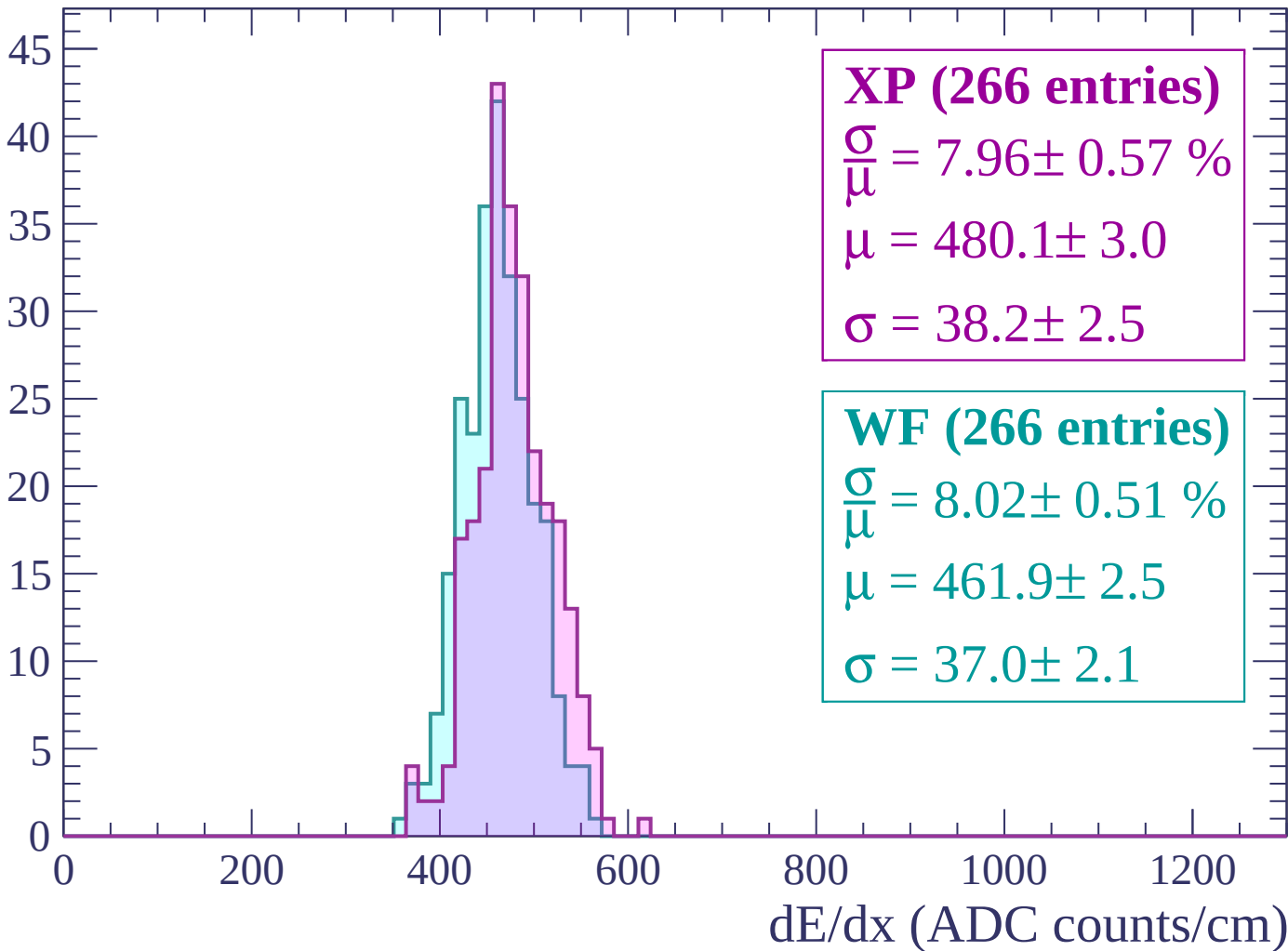
Energy loss | $-1620 < p < -1560$

Count



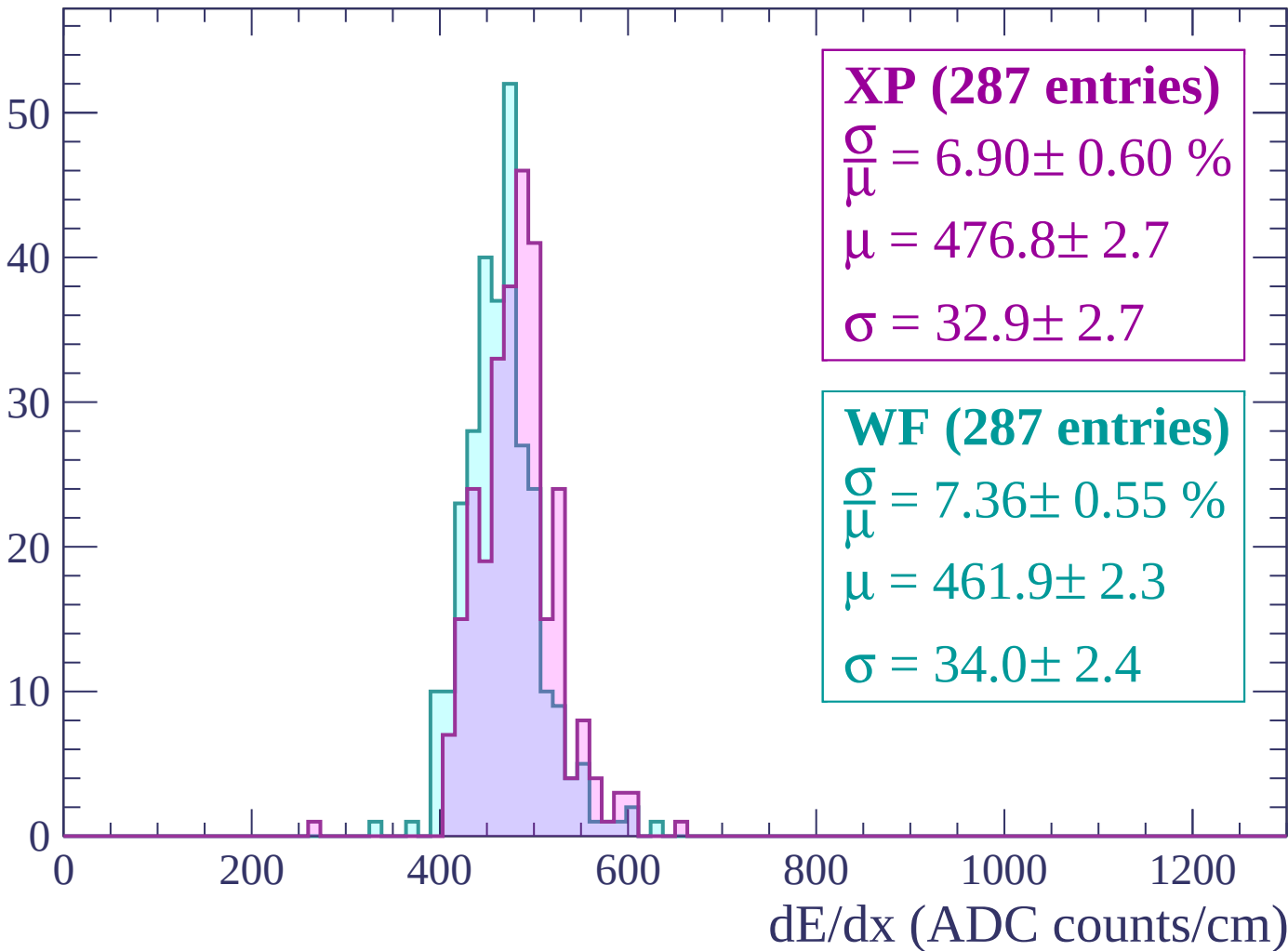
Energy loss | $-1560 < p < -1500$

Count



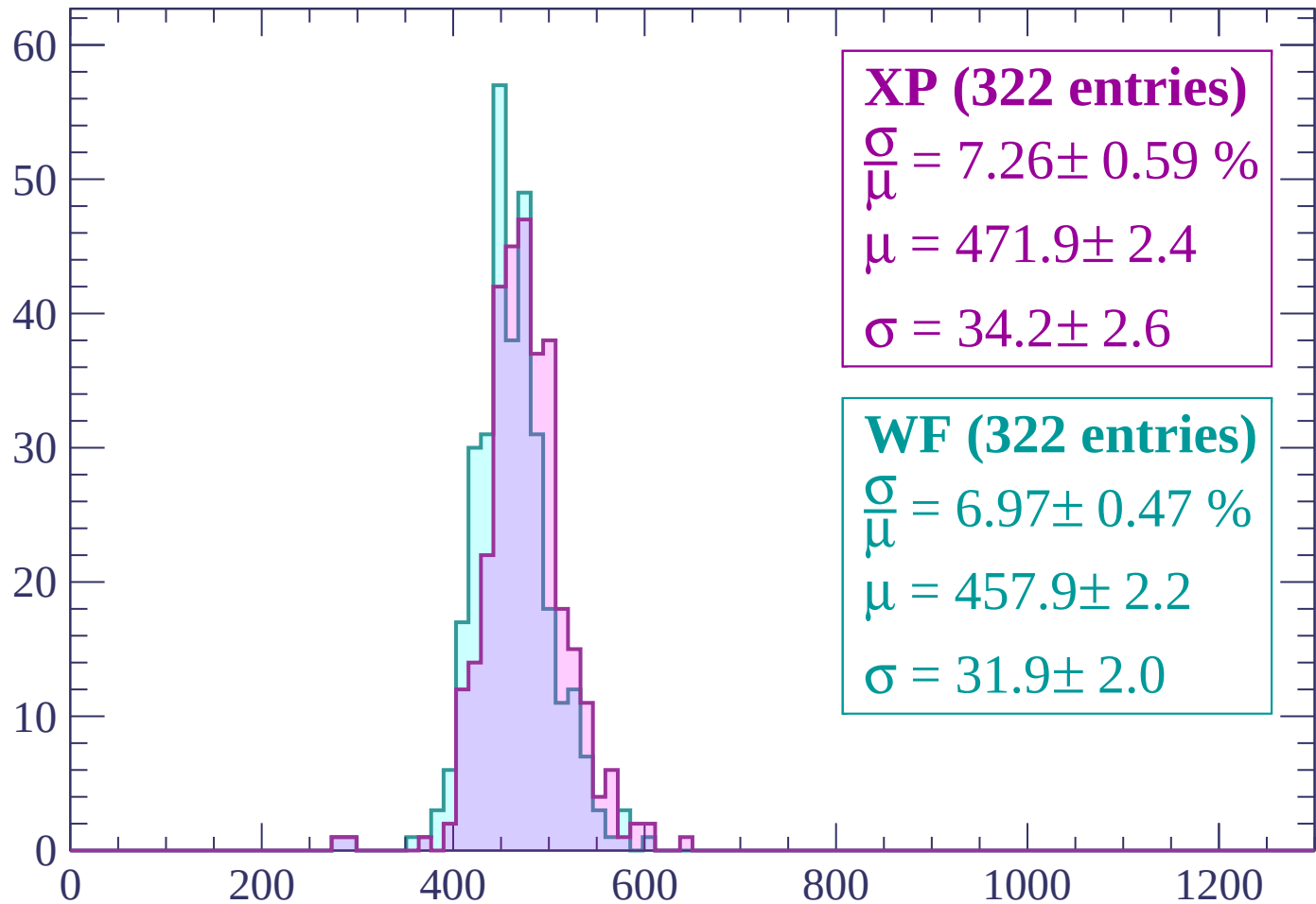
Energy loss | $-1500 < p < -1440$

Count



Energy loss | $-1440 < p < -1380$

Count



XP (322 entries)

$\frac{\sigma}{\mu} = 7.26 \pm 0.59 \%$

$\mu = 471.9 \pm 2.4$

$\sigma = 34.2 \pm 2.6$

WF (322 entries)

$\frac{\sigma}{\mu} = 6.97 \pm 0.47 \%$

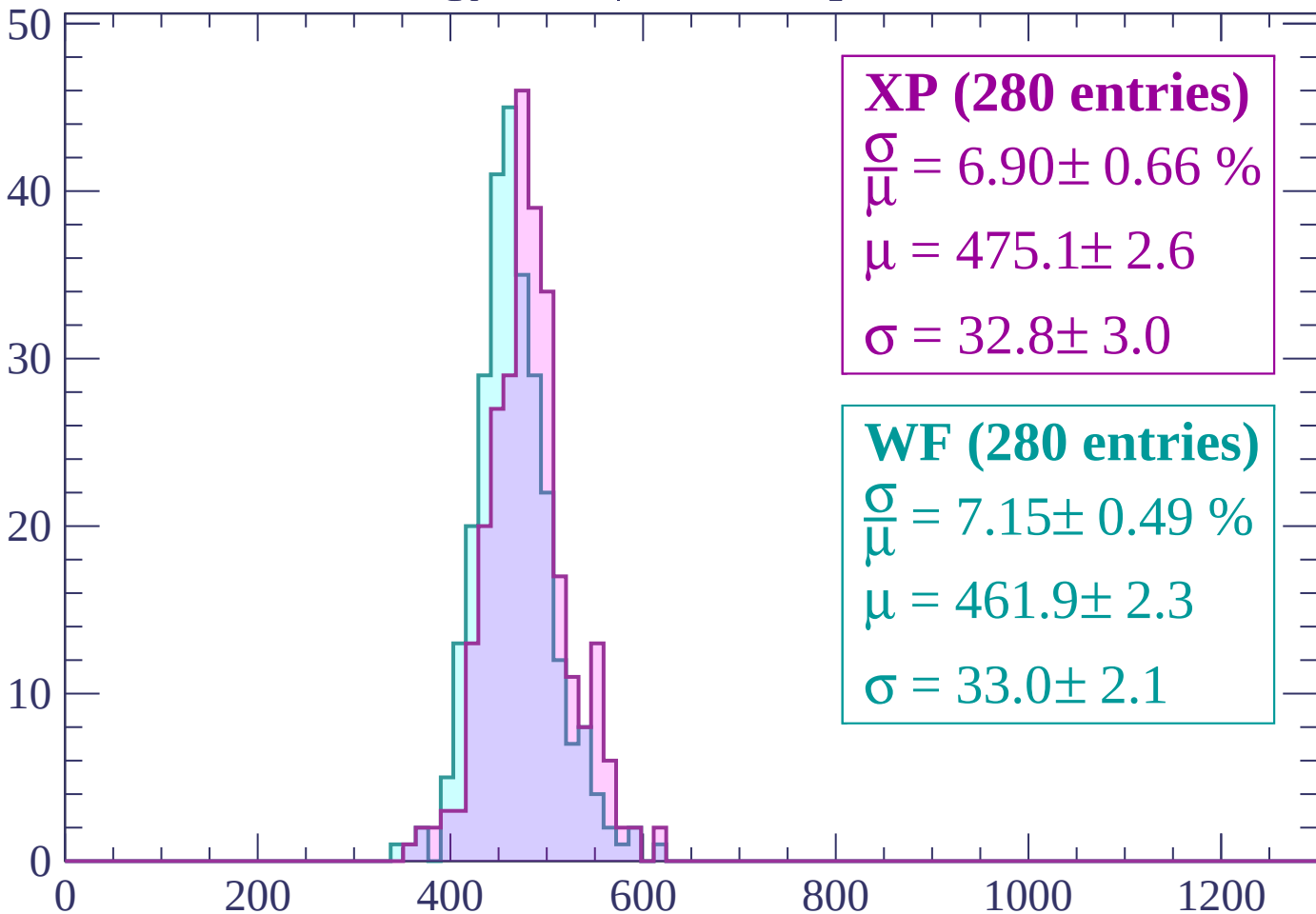
$\mu = 457.9 \pm 2.2$

$\sigma = 31.9 \pm 2.0$

dE/dx (ADC counts/cm)

Energy loss | $-1380 < p < -1320$

Count



XP (280 entries)

$\frac{\sigma}{\mu} = 6.90 \pm 0.66 \%$

$\mu = 475.1 \pm 2.6$

$\sigma = 32.8 \pm 3.0$

WF (280 entries)

$\frac{\sigma}{\mu} = 7.15 \pm 0.49 \%$

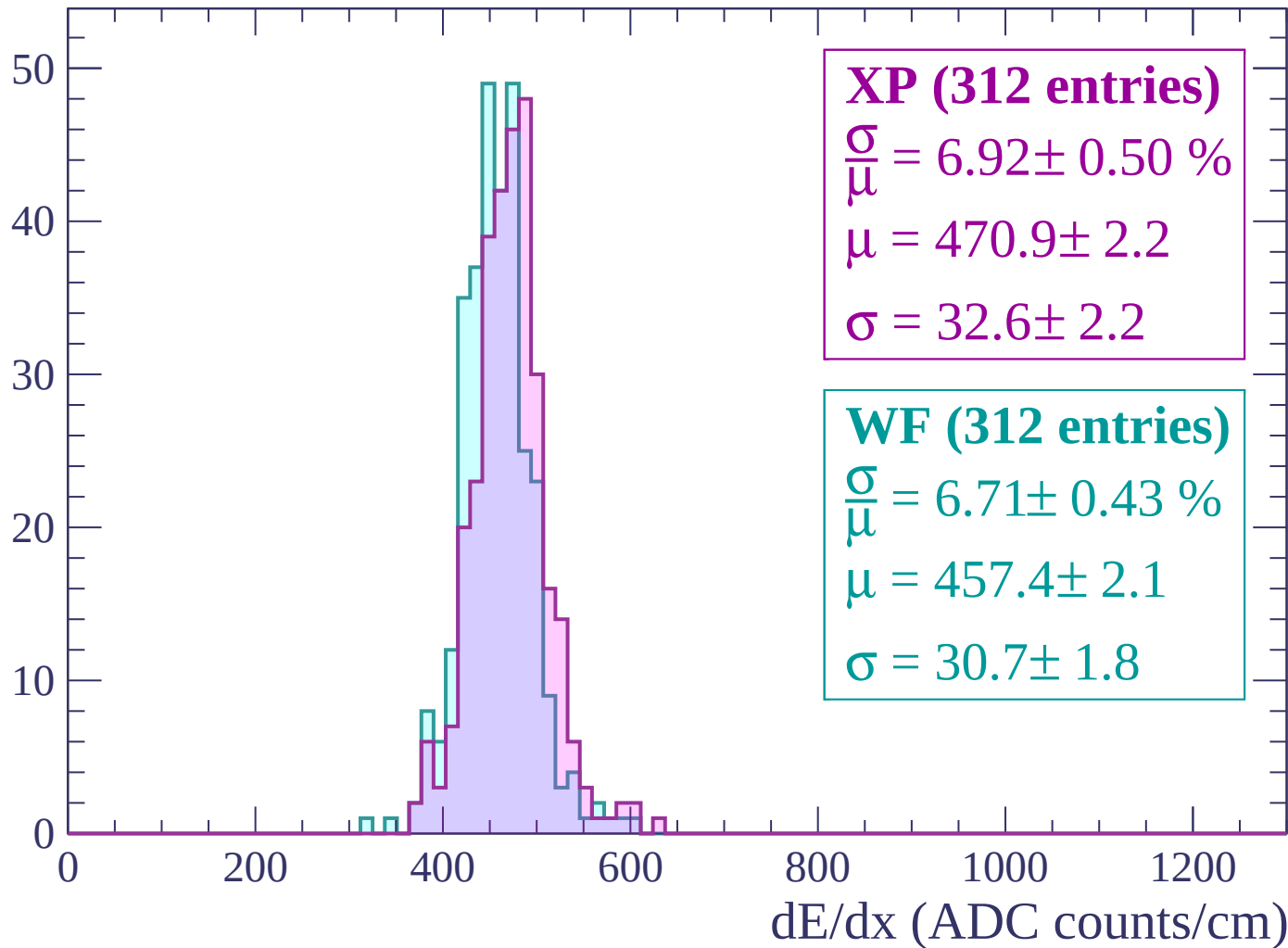
$\mu = 461.9 \pm 2.3$

$\sigma = 33.0 \pm 2.1$

dE/dx (ADC counts/cm)

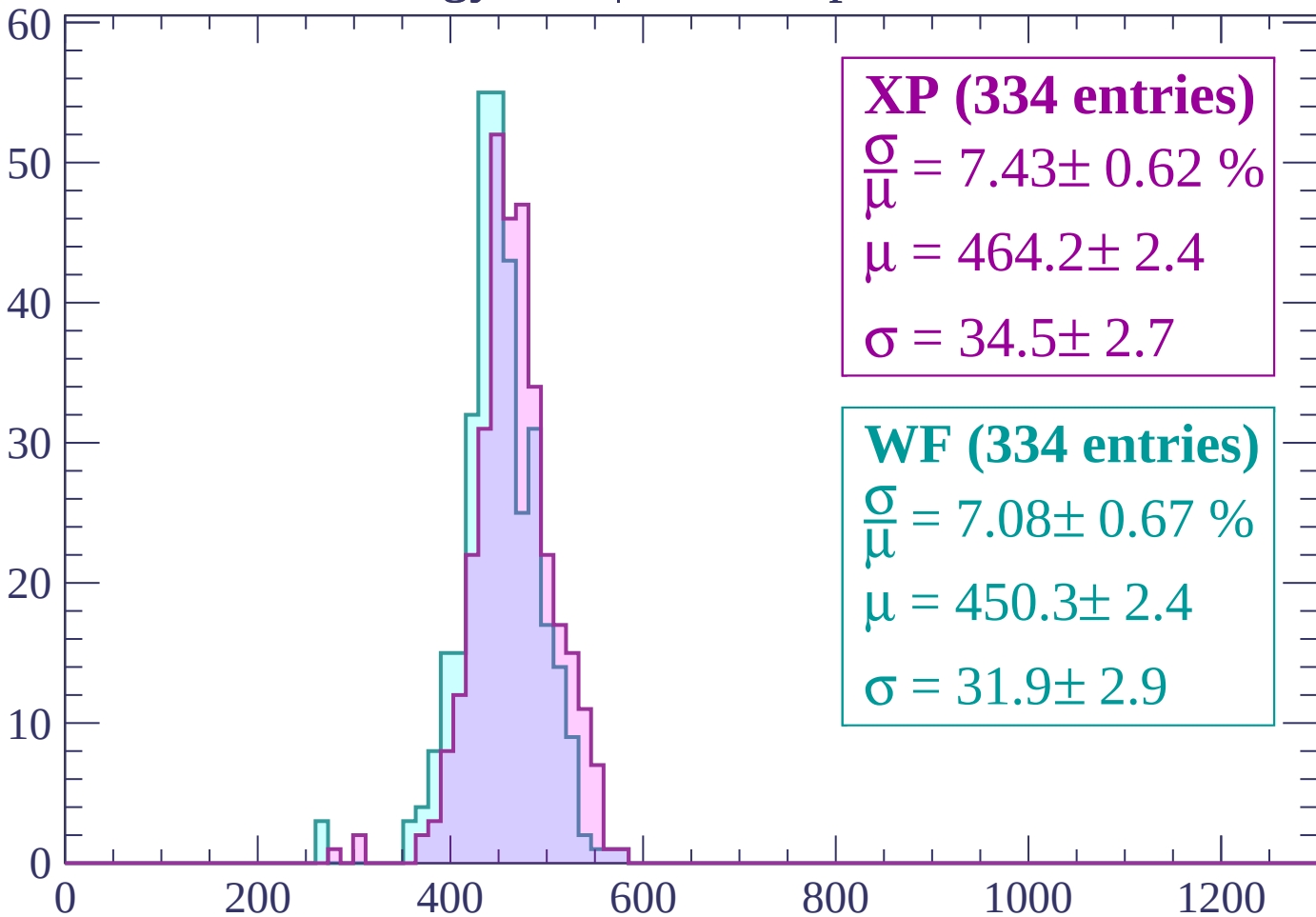
Energy loss | $-1320 < p < -1260$

Count



Energy loss | $-1260 < p < -1200$

Count



XP (334 entries)

$$\frac{\sigma}{\mu} = 7.43 \pm 0.62 \%$$

$$\mu = 464.2 \pm 2.4$$

$$\sigma = 34.5 \pm 2.7$$

WF (334 entries)

$$\frac{\sigma}{\mu} = 7.08 \pm 0.67 \%$$

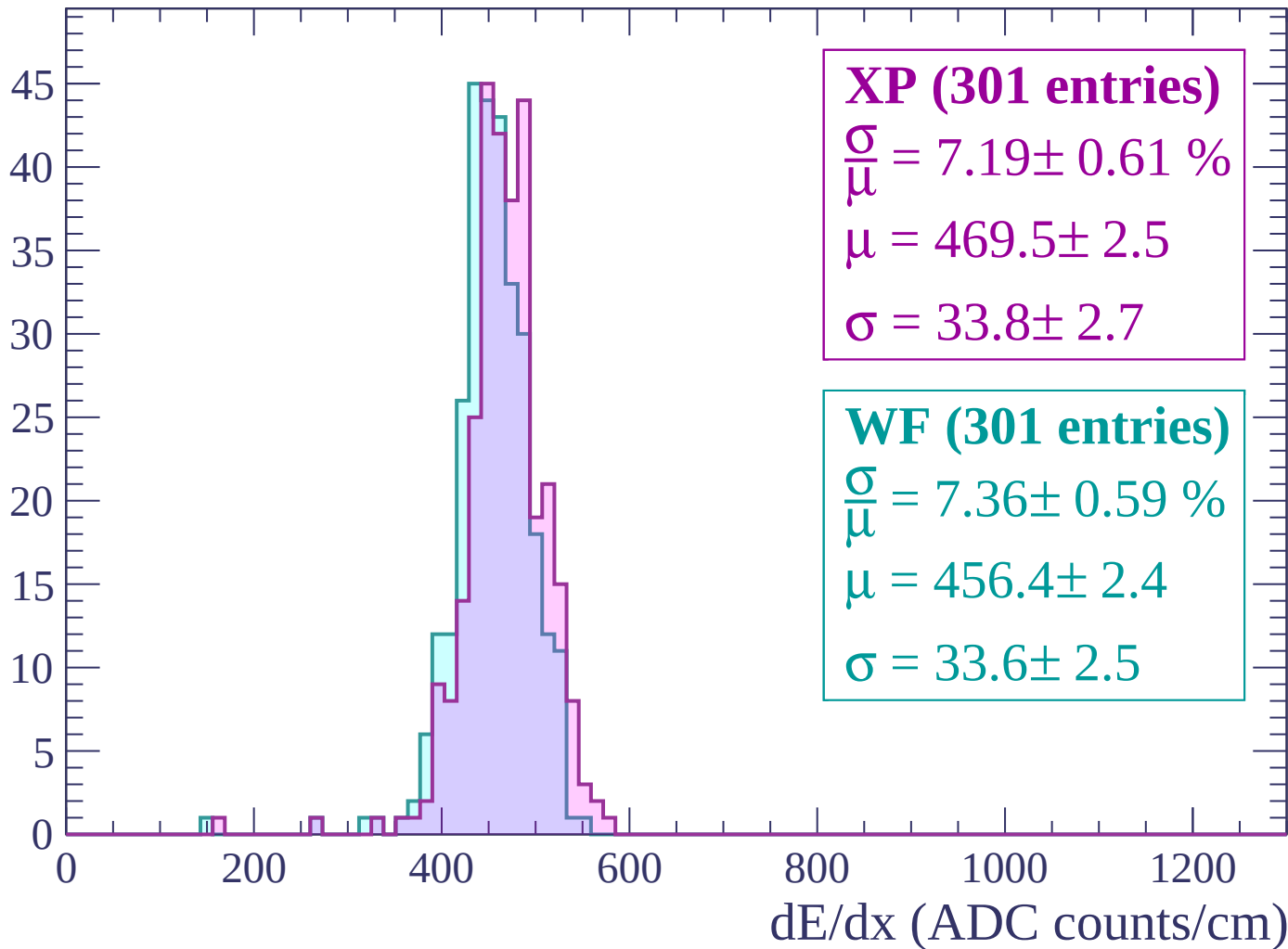
$$\mu = 450.3 \pm 2.4$$

$$\sigma = 31.9 \pm 2.9$$

dE/dx (ADC counts/cm)

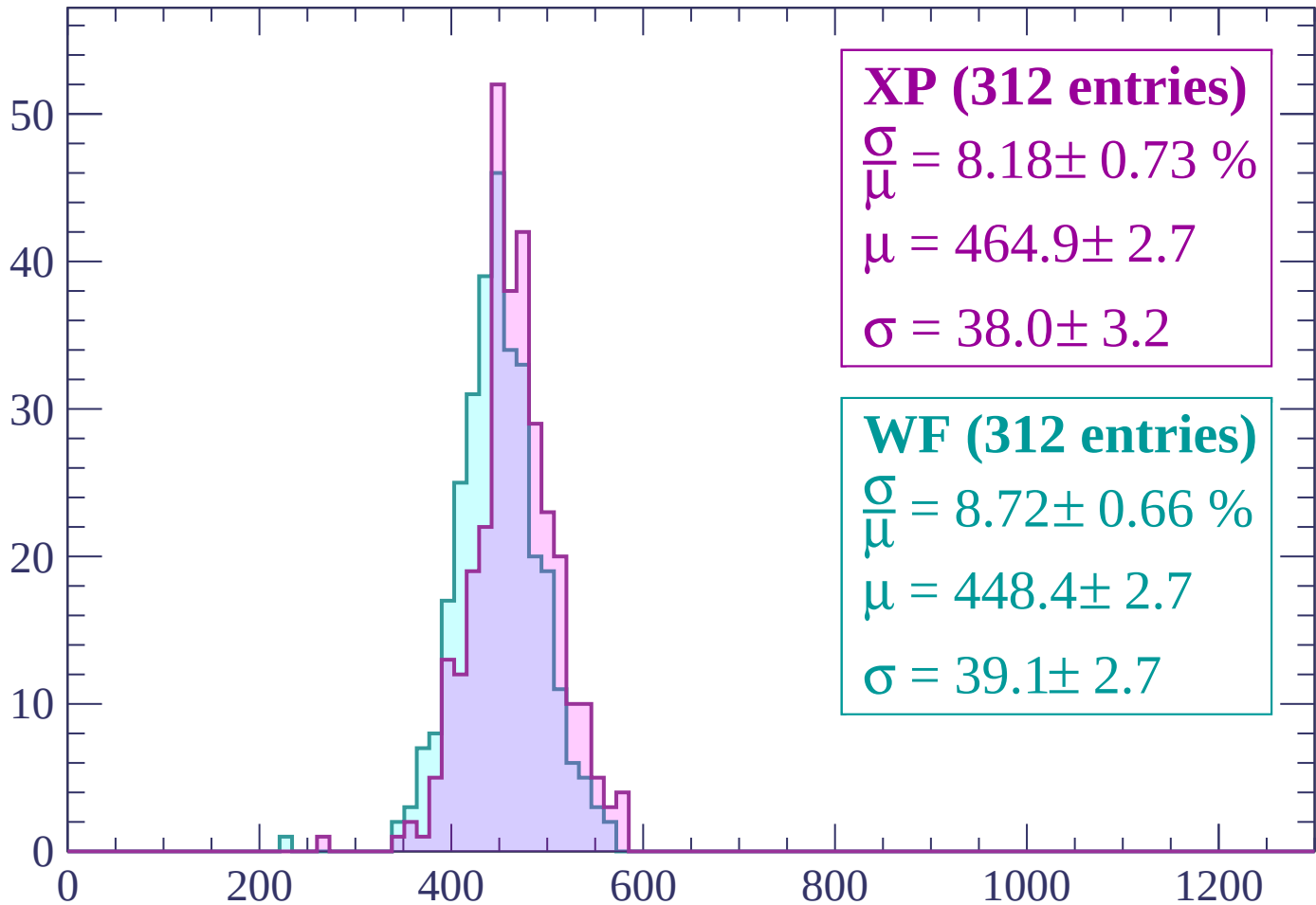
Energy loss | $-1200 < p < -1140$

Count



Energy loss | $-1140 < p < -1080$

Count



XP (312 entries)

$\frac{\sigma}{\mu} = 8.18 \pm 0.73 \%$

$\mu = 464.9 \pm 2.7$

$\sigma = 38.0 \pm 3.2$

WF (312 entries)

$\frac{\sigma}{\mu} = 8.72 \pm 0.66 \%$

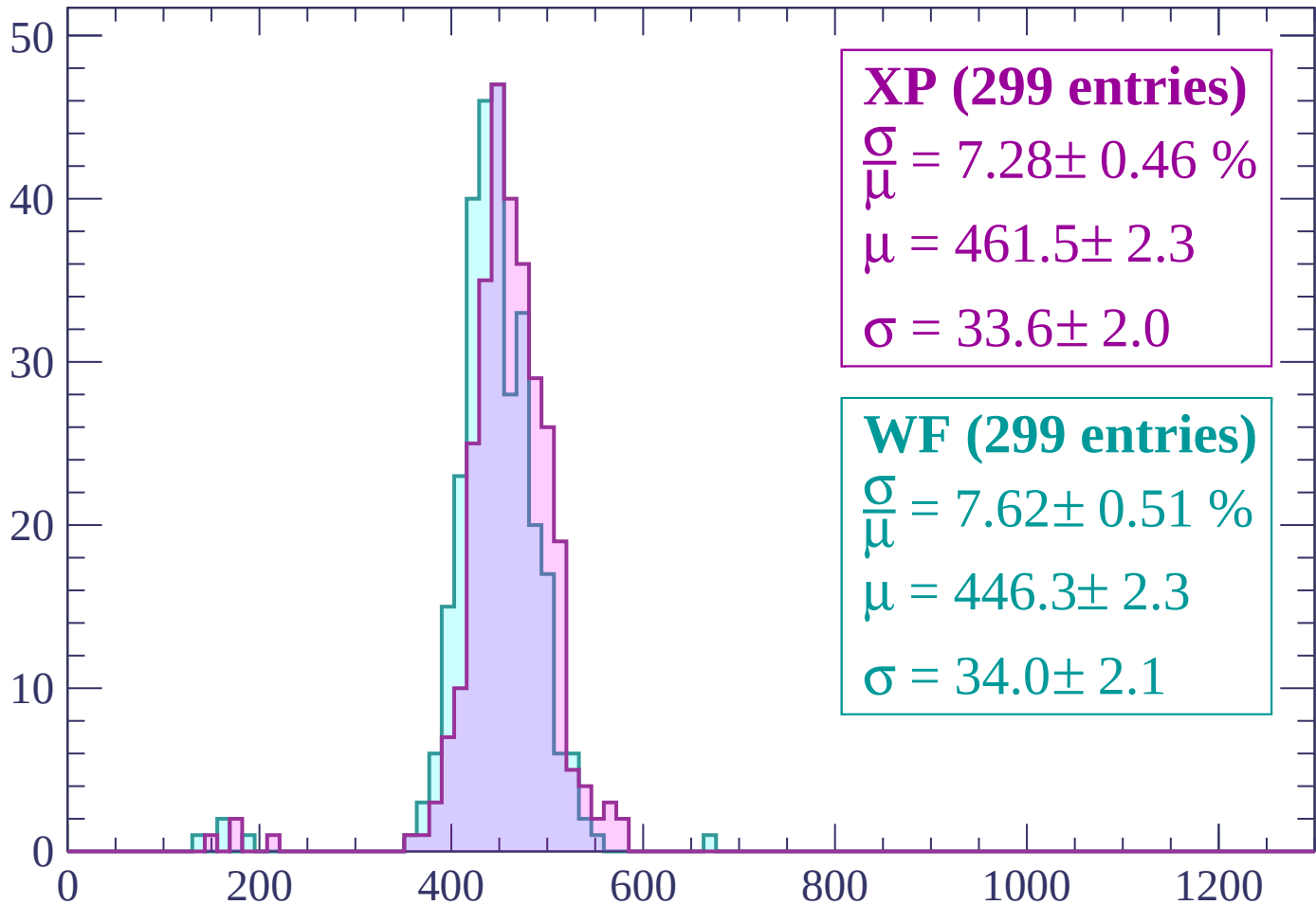
$\mu = 448.4 \pm 2.7$

$\sigma = 39.1 \pm 2.7$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1020$

Count



XP (299 entries)

$$\frac{\sigma}{\mu} = 7.28 \pm 0.46 \%$$

$$\mu = 461.5 \pm 2.3$$

$$\sigma = 33.6 \pm 2.0$$

WF (299 entries)

$$\frac{\sigma}{\mu} = 7.62 \pm 0.51 \%$$

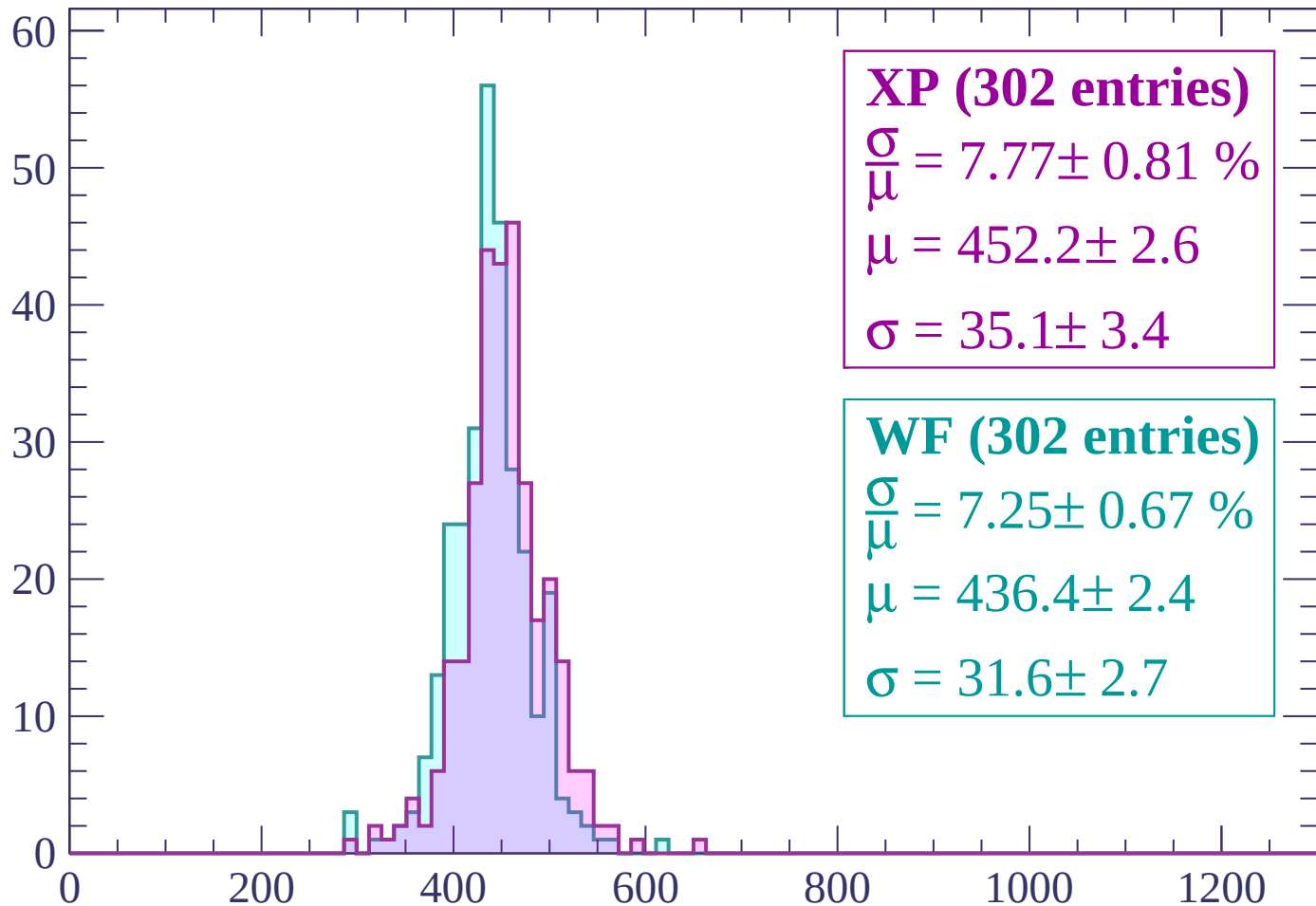
$$\mu = 446.3 \pm 2.3$$

$$\sigma = 34.0 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$

Count



XP (302 entries)

$$\frac{\sigma}{\mu} = 7.77 \pm 0.81 \%$$

$$\mu = 452.2 \pm 2.6$$

$$\sigma = 35.1 \pm 3.4$$

WF (302 entries)

$$\frac{\sigma}{\mu} = 7.25 \pm 0.67 \%$$

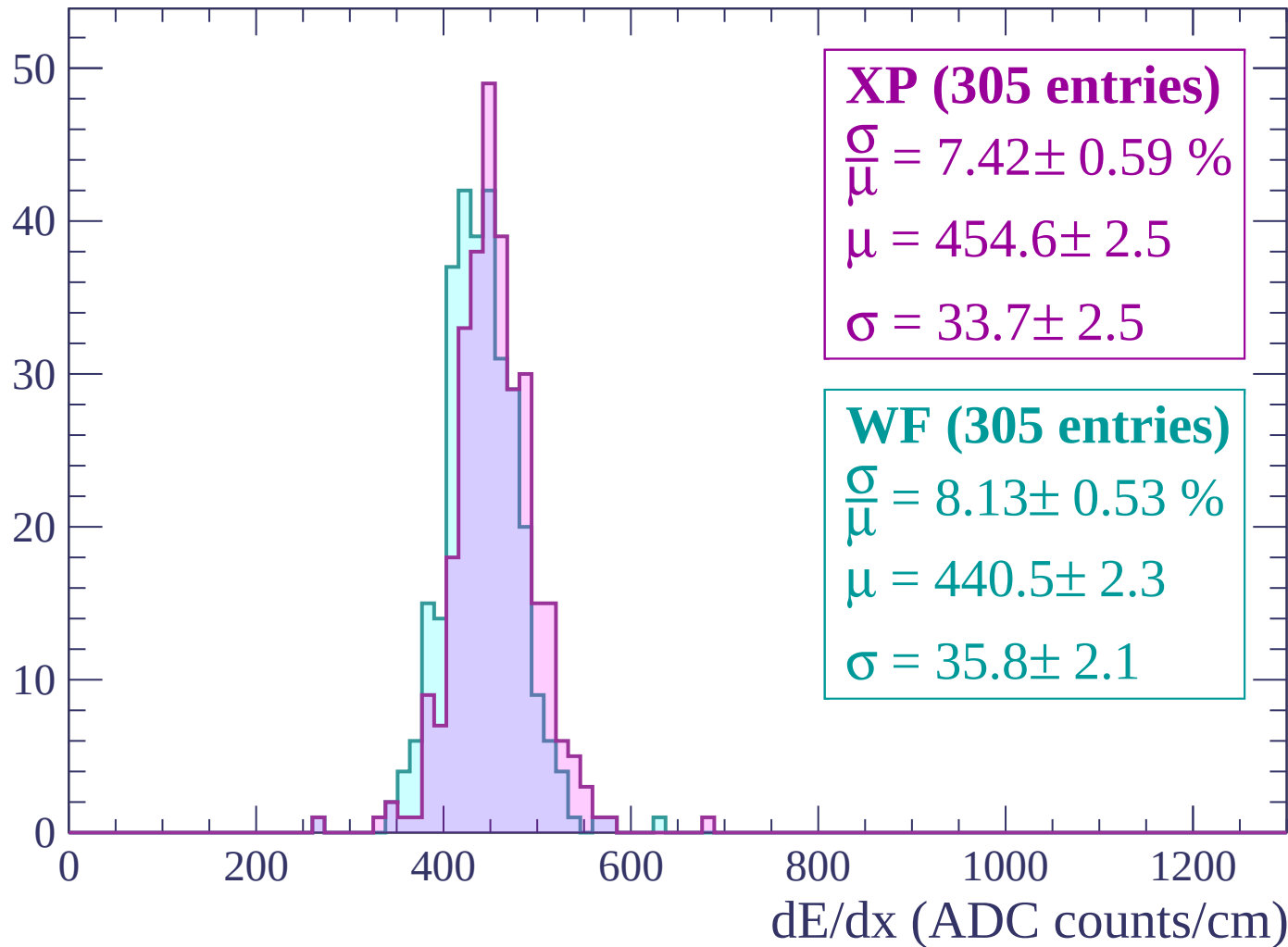
$$\mu = 436.4 \pm 2.4$$

$$\sigma = 31.6 \pm 2.7$$

dE/dx (ADC counts/cm)

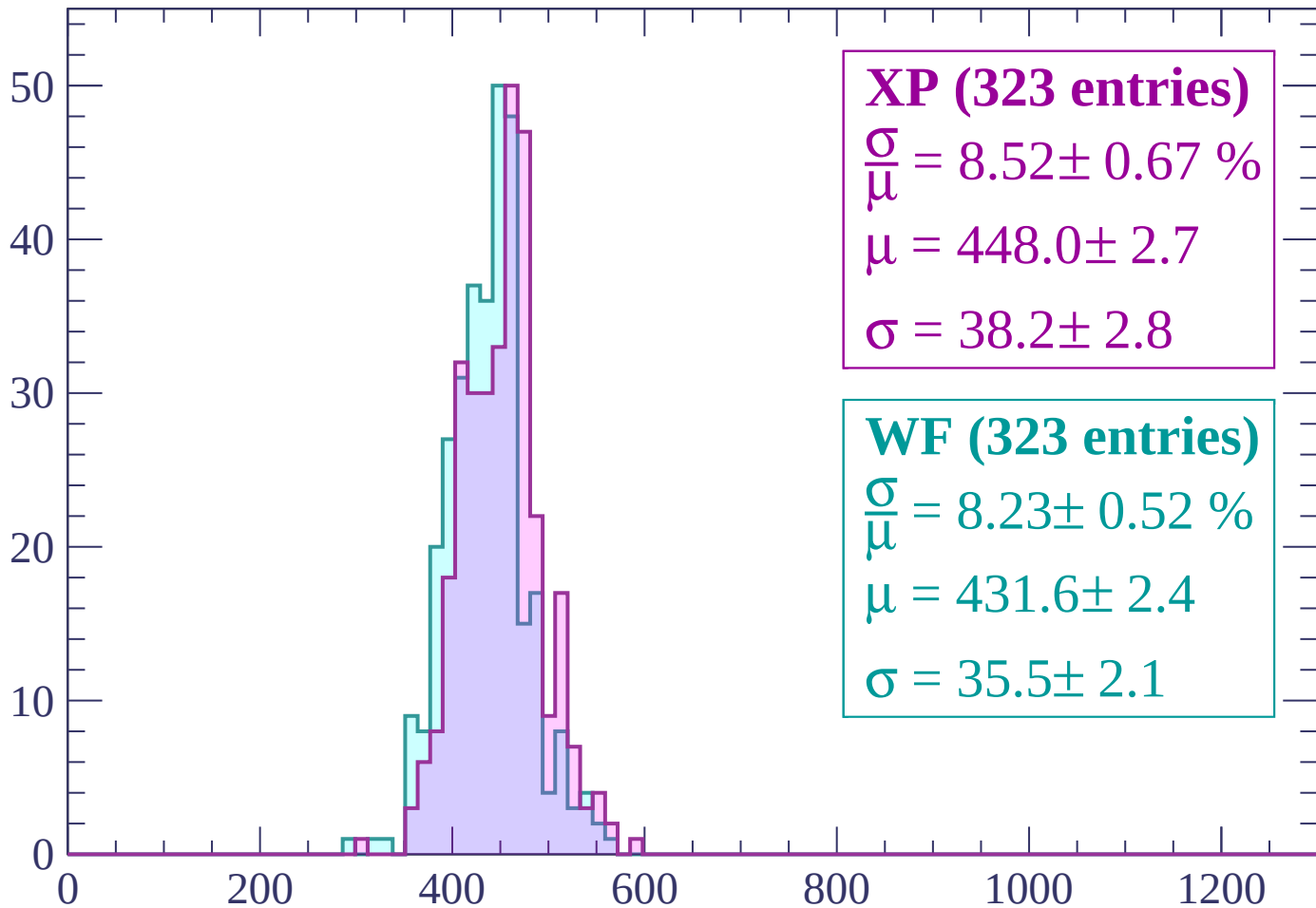
Energy loss | $-960 < p < -900$

Count



Energy loss | $-900 < p < -840$

Count



XP (323 entries)

$$\frac{\sigma}{\mu} = 8.52 \pm 0.67 \%$$

$$\mu = 448.0 \pm 2.7$$

$$\sigma = 38.2 \pm 2.8$$

WF (323 entries)

$$\frac{\sigma}{\mu} = 8.23 \pm 0.52 \%$$

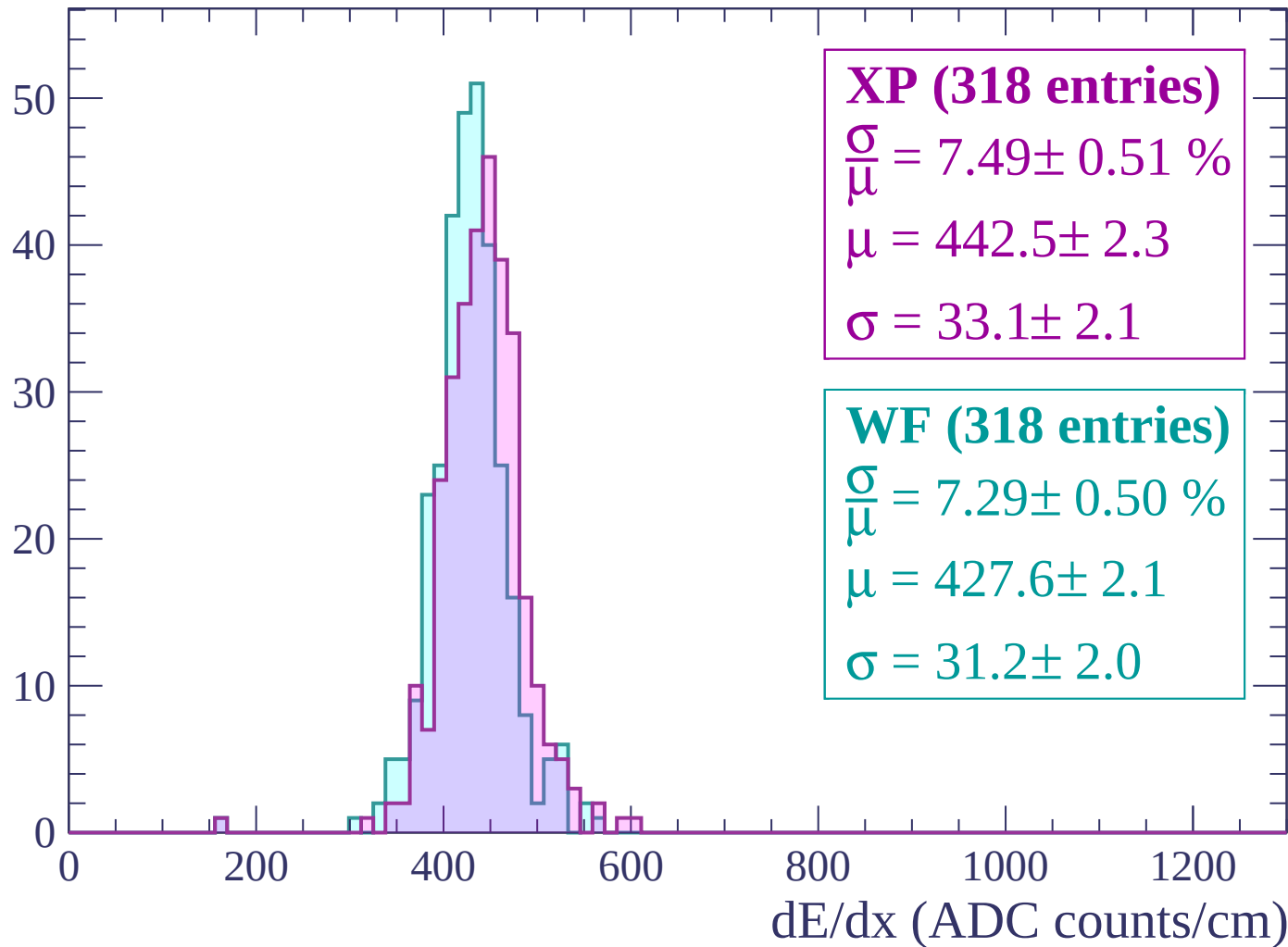
$$\mu = 431.6 \pm 2.4$$

$$\sigma = 35.5 \pm 2.1$$

dE/dx (ADC counts/cm)

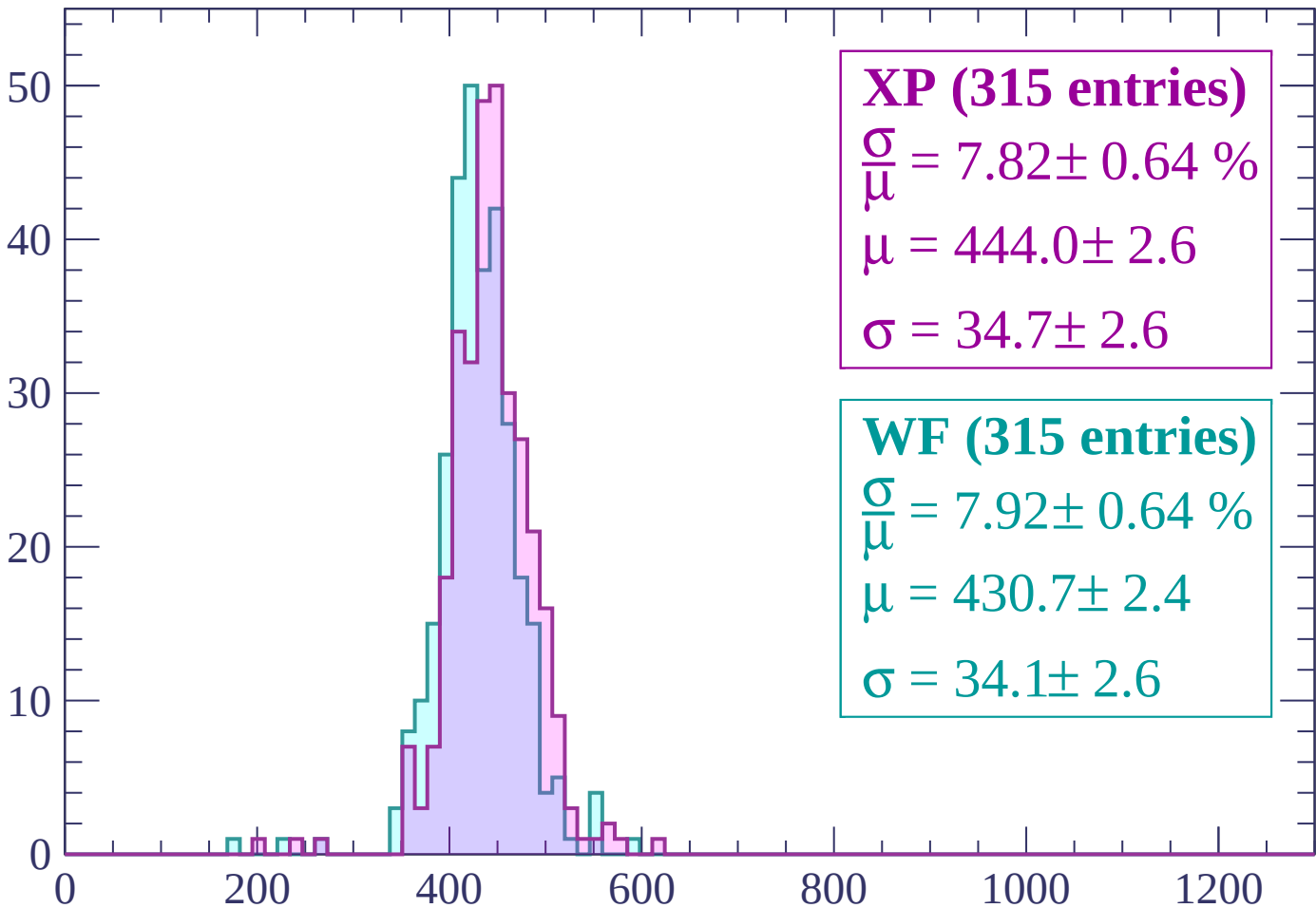
Energy loss | $-840 < p < -780$

Count



Energy loss | $-780 < p < -720$

Count



XP (315 entries)

$\frac{\sigma}{\mu} = 7.82 \pm 0.64 \%$

$\mu = 444.0 \pm 2.6$

$\sigma = 34.7 \pm 2.6$

WF (315 entries)

$\frac{\sigma}{\mu} = 7.92 \pm 0.64 \%$

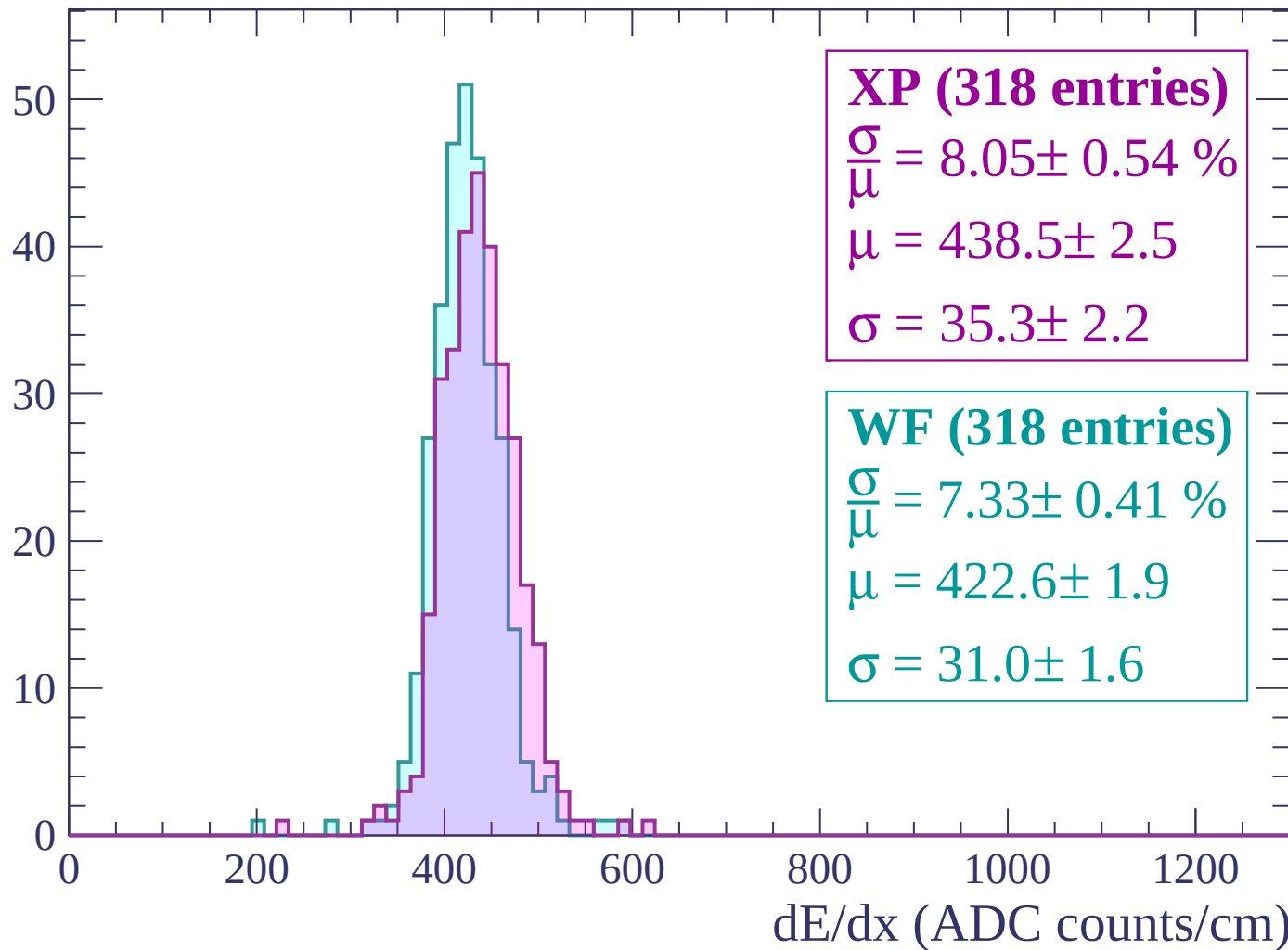
$\mu = 430.7 \pm 2.4$

$\sigma = 34.1 \pm 2.6$

dE/dx (ADC counts/cm)

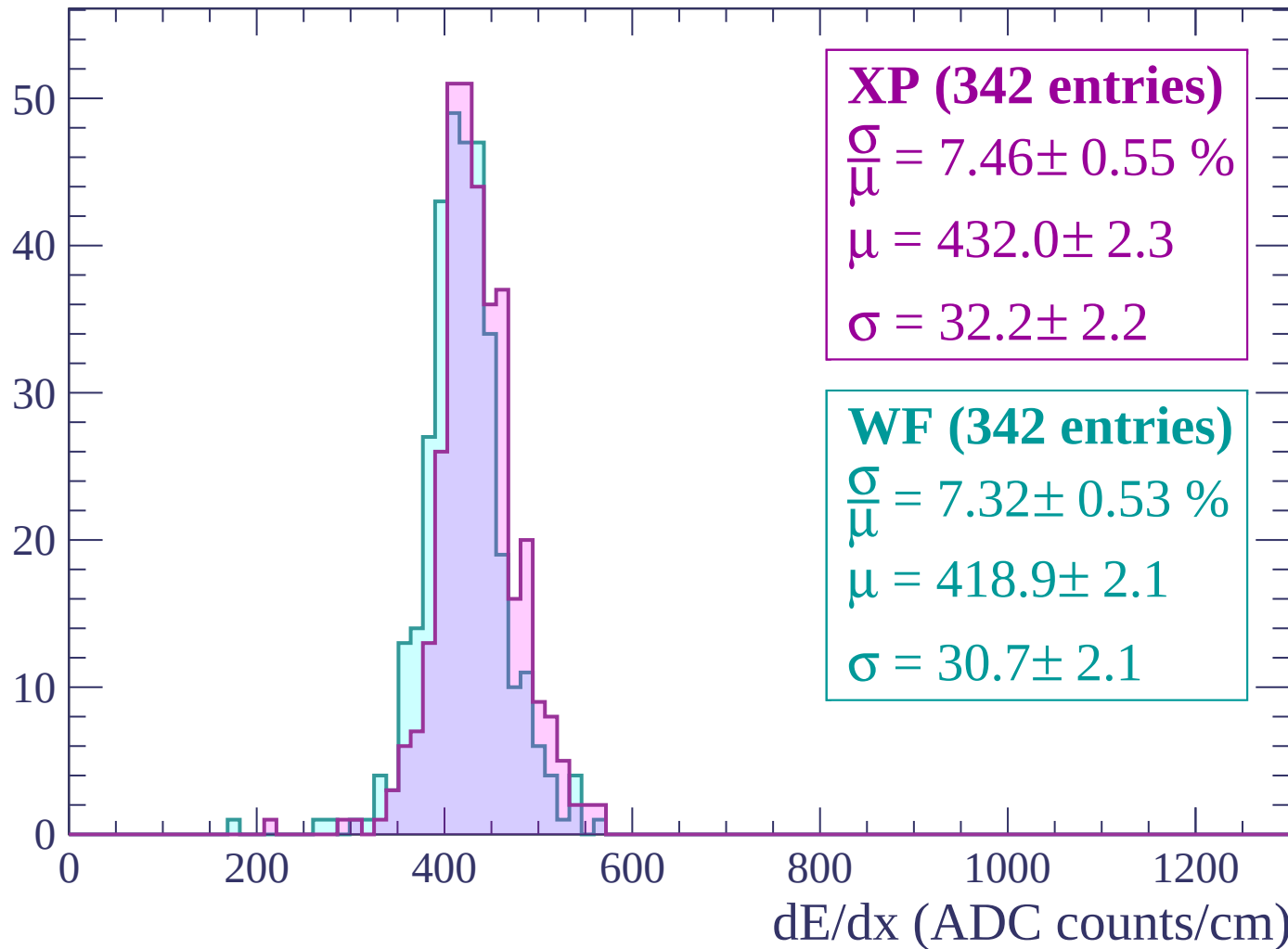
Energy loss | $-720 < p < -660$

Count



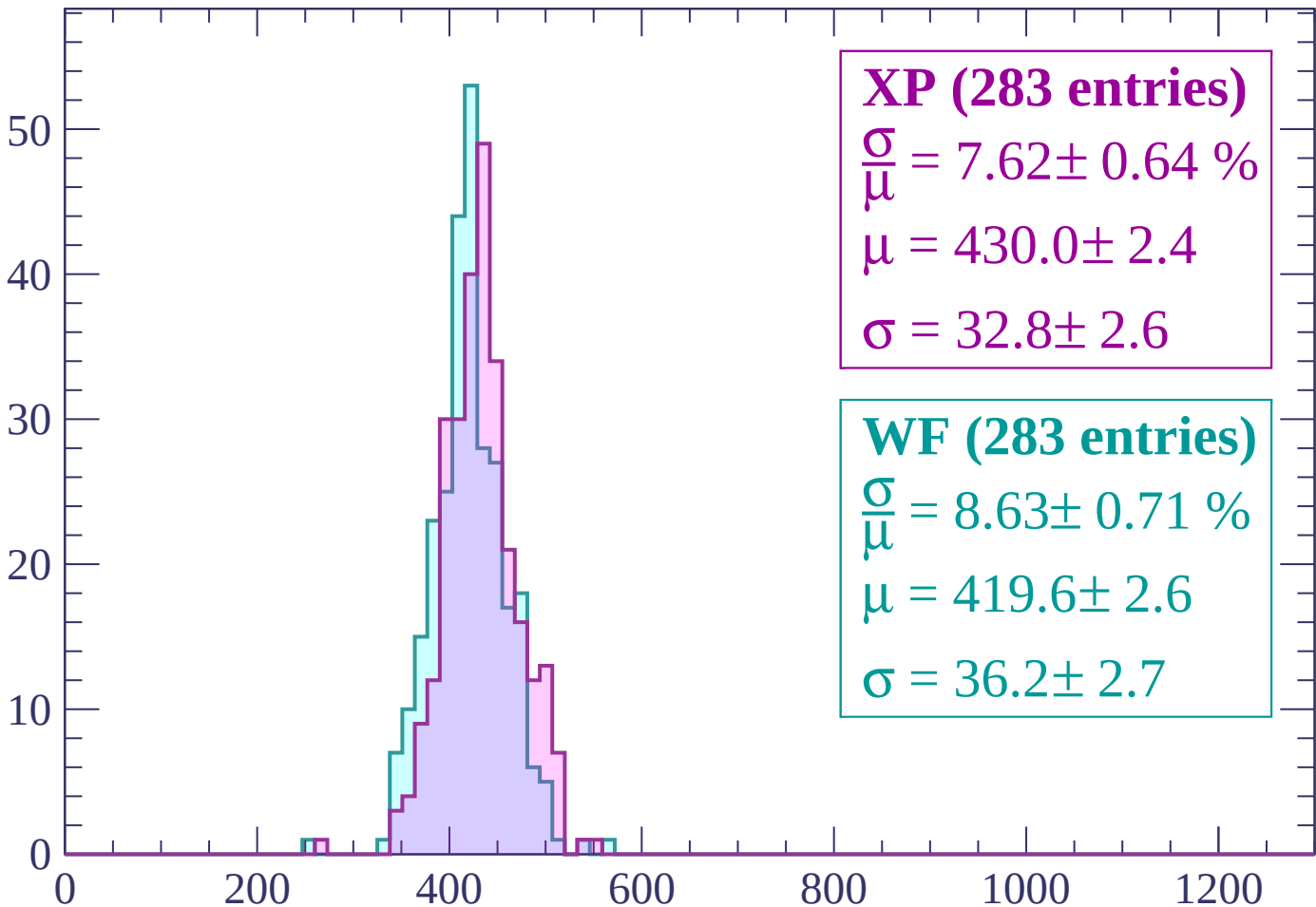
Energy loss | $-660 < p < -600$

Count



Energy loss | $-600 < p < -540$

Count



XP (283 entries)

$\frac{\sigma}{\mu} = 7.62 \pm 0.64 \%$

$\mu = 430.0 \pm 2.4$

$\sigma = 32.8 \pm 2.6$

WF (283 entries)

$\frac{\sigma}{\mu} = 8.63 \pm 0.71 \%$

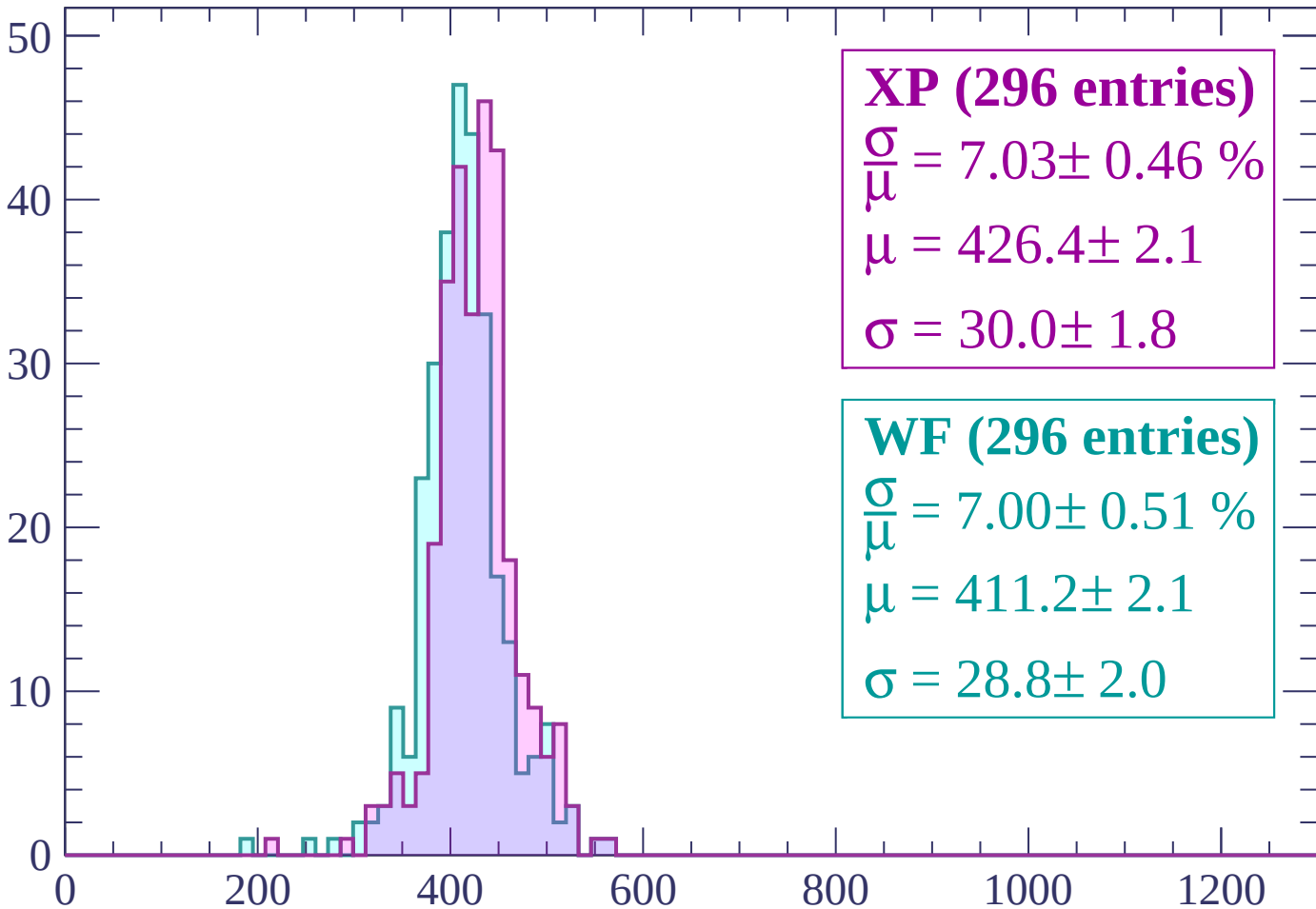
$\mu = 419.6 \pm 2.6$

$\sigma = 36.2 \pm 2.7$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count



XP (296 entries)

$$\frac{\sigma}{\mu} = 7.03 \pm 0.46 \%$$

$$\mu = 426.4 \pm 2.1$$

$$\sigma = 30.0 \pm 1.8$$

WF (296 entries)

$$\frac{\sigma}{\mu} = 7.00 \pm 0.51 \%$$

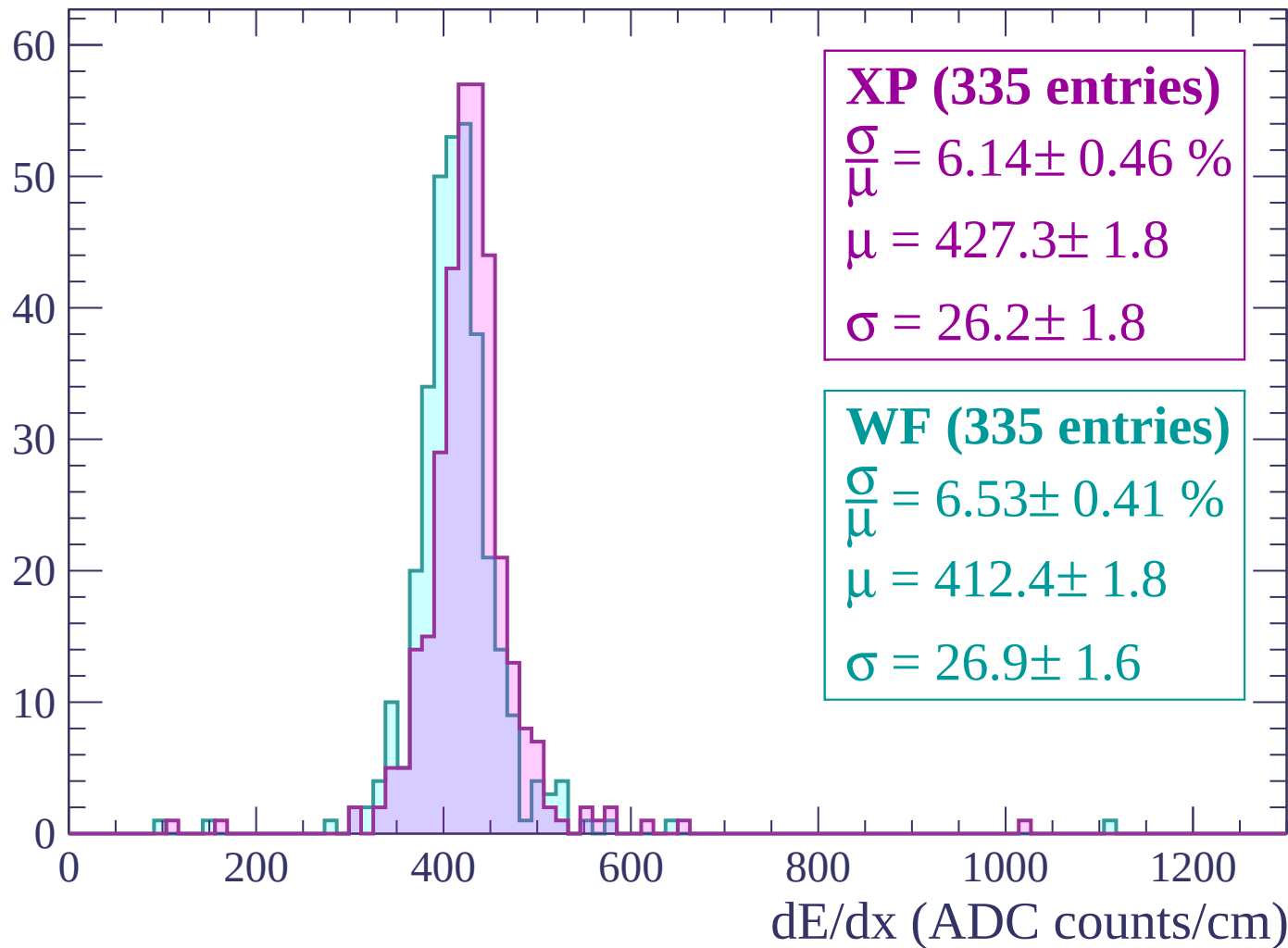
$$\mu = 411.2 \pm 2.1$$

$$\sigma = 28.8 \pm 2.0$$

dE/dx (ADC counts/cm)

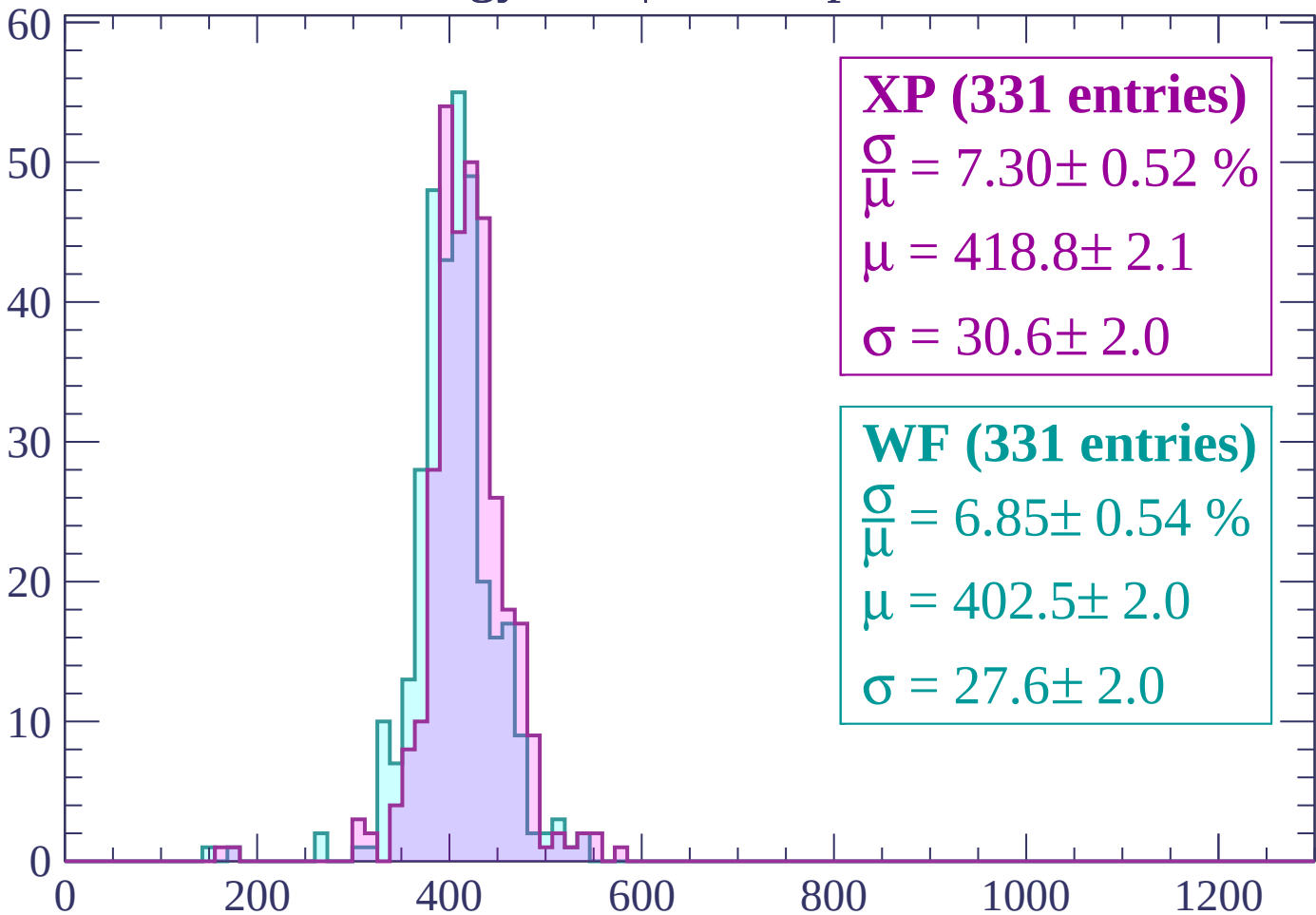
Energy loss | $-480 < p < -420$

Count



Energy loss | $-420 < p < -360$

Count



XP (331 entries)

$$\frac{\sigma}{\mu} = 7.30 \pm 0.52 \%$$

$$\mu = 418.8 \pm 2.1$$

$$\sigma = 30.6 \pm 2.0$$

WF (331 entries)

$$\frac{\sigma}{\mu} = 6.85 \pm 0.54 \%$$

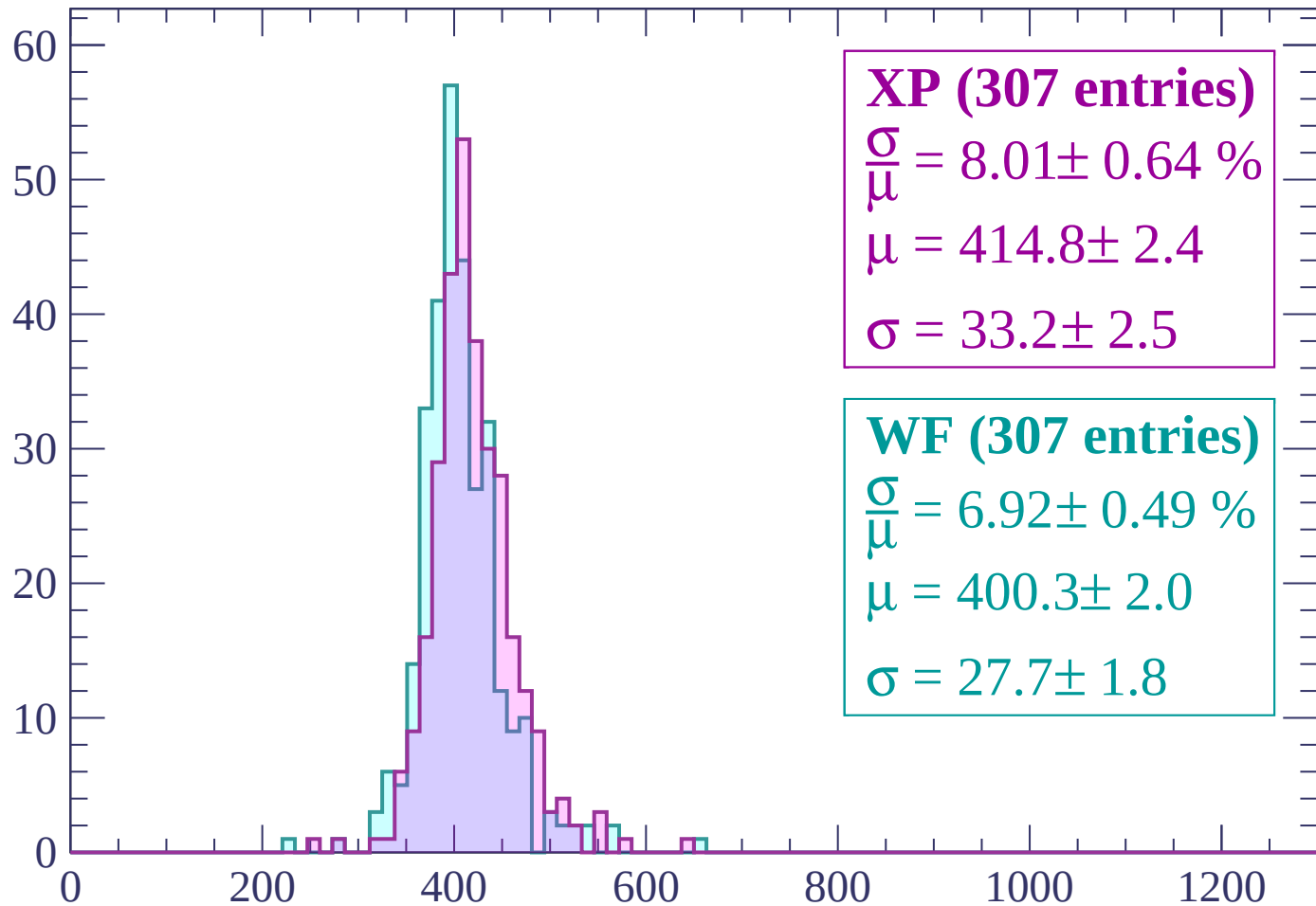
$$\mu = 402.5 \pm 2.0$$

$$\sigma = 27.6 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$

Count



XP (307 entries)

$\frac{\sigma}{\mu} = 8.01 \pm 0.64 \%$

$\mu = 414.8 \pm 2.4$

$\sigma = 33.2 \pm 2.5$

WF (307 entries)

$\frac{\sigma}{\mu} = 6.92 \pm 0.49 \%$

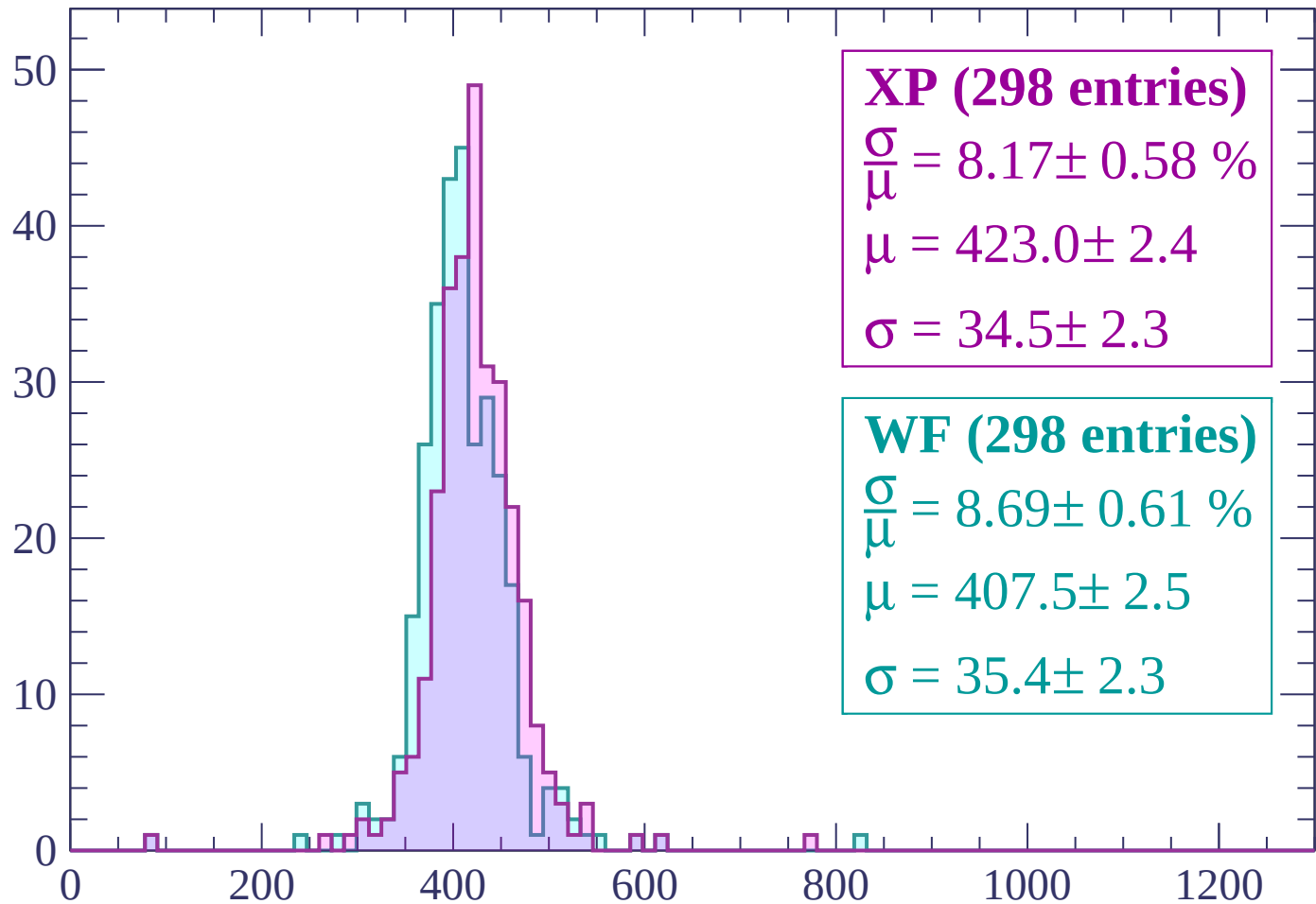
$\mu = 400.3 \pm 2.0$

$\sigma = 27.7 \pm 1.8$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP (298 entries)

$\frac{\sigma}{\mu} = 8.17 \pm 0.58 \%$

$\mu = 423.0 \pm 2.4$

$\sigma = 34.5 \pm 2.3$

WF (298 entries)

$\frac{\sigma}{\mu} = 8.69 \pm 0.61 \%$

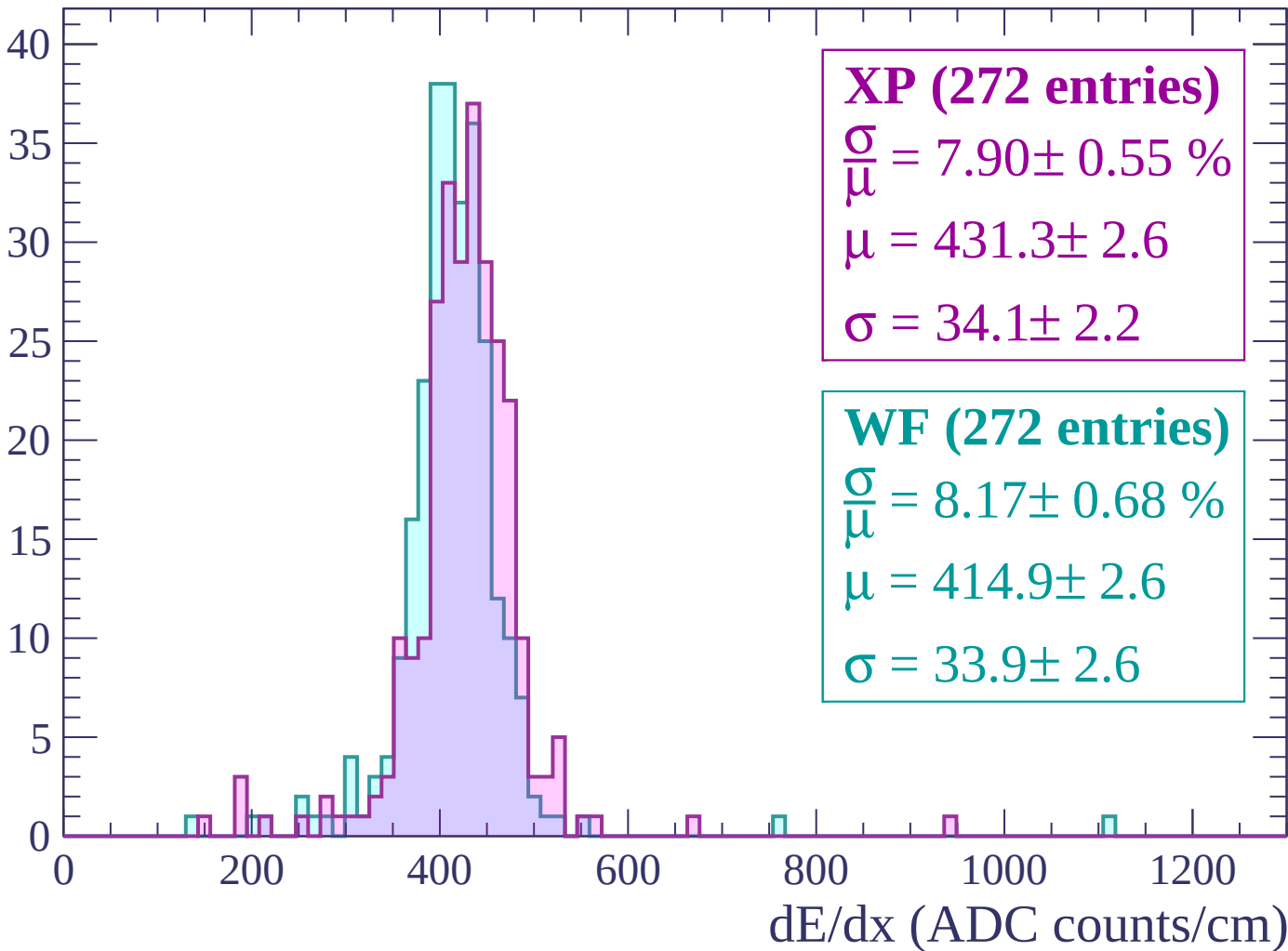
$\mu = 407.5 \pm 2.5$

$\sigma = 35.4 \pm 2.3$

dE/dx (ADC counts/cm)

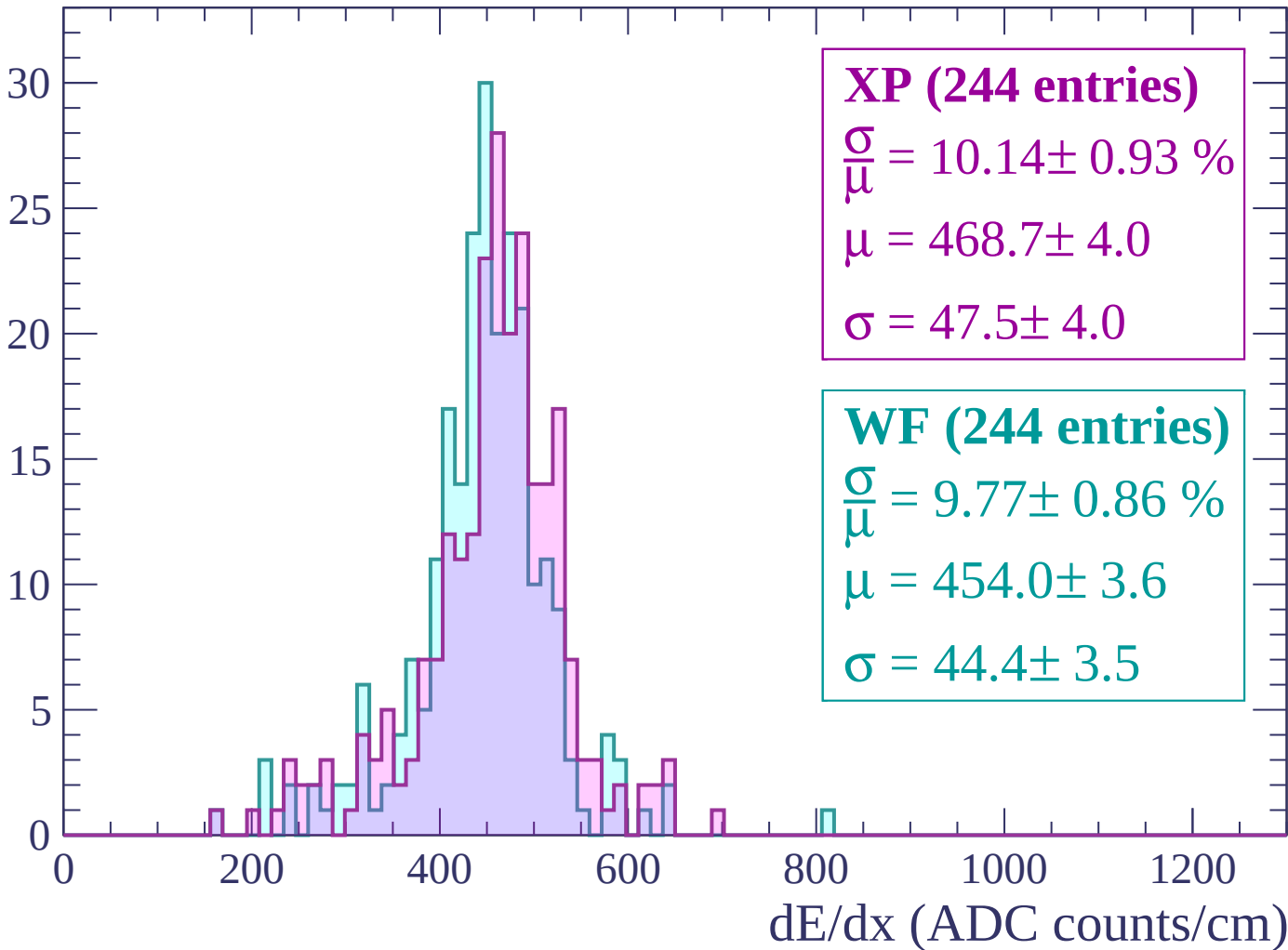
Energy loss | $-240 < p < -180$

Count



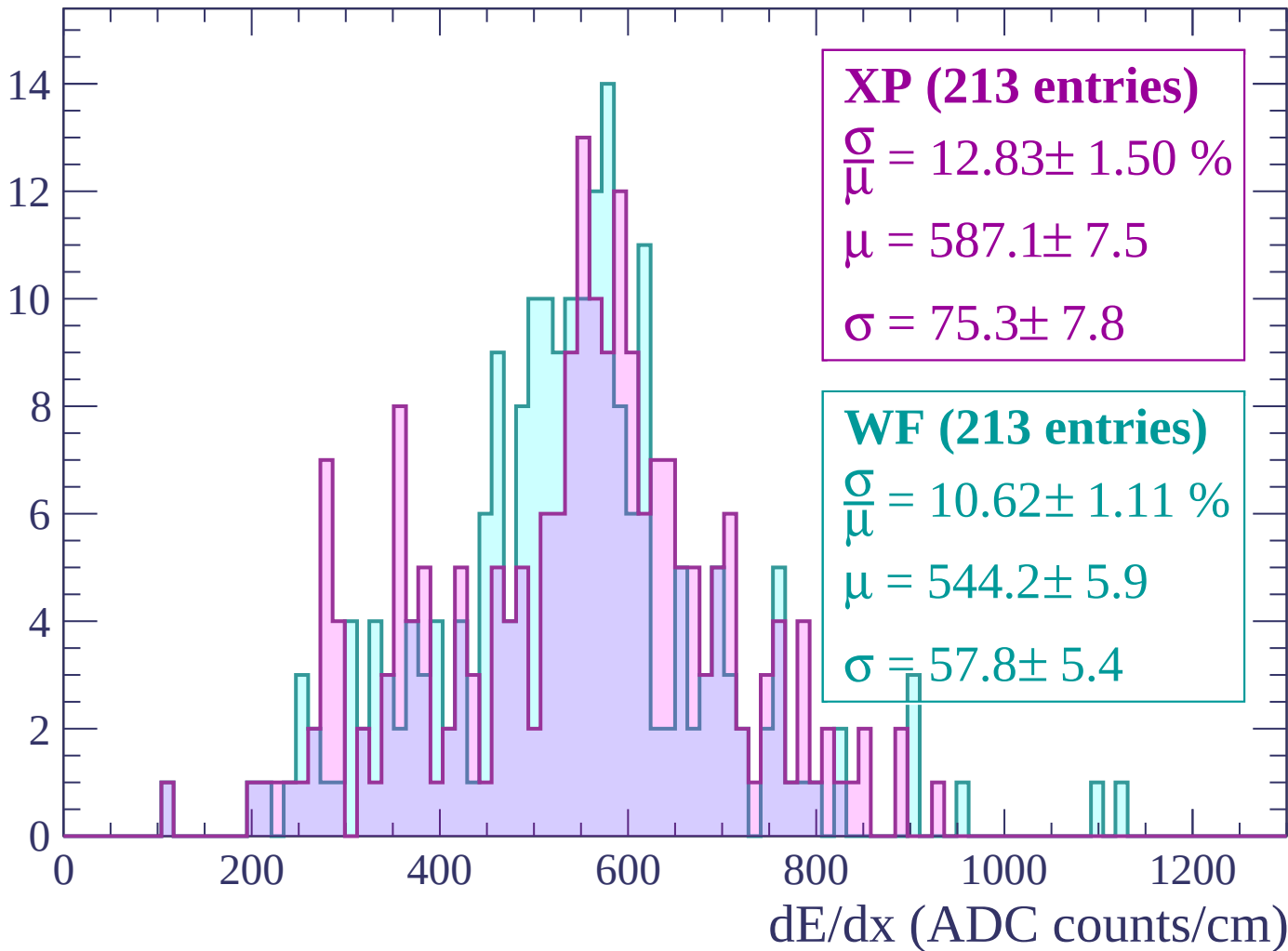
Energy loss | $-180 < p < -120$

Count



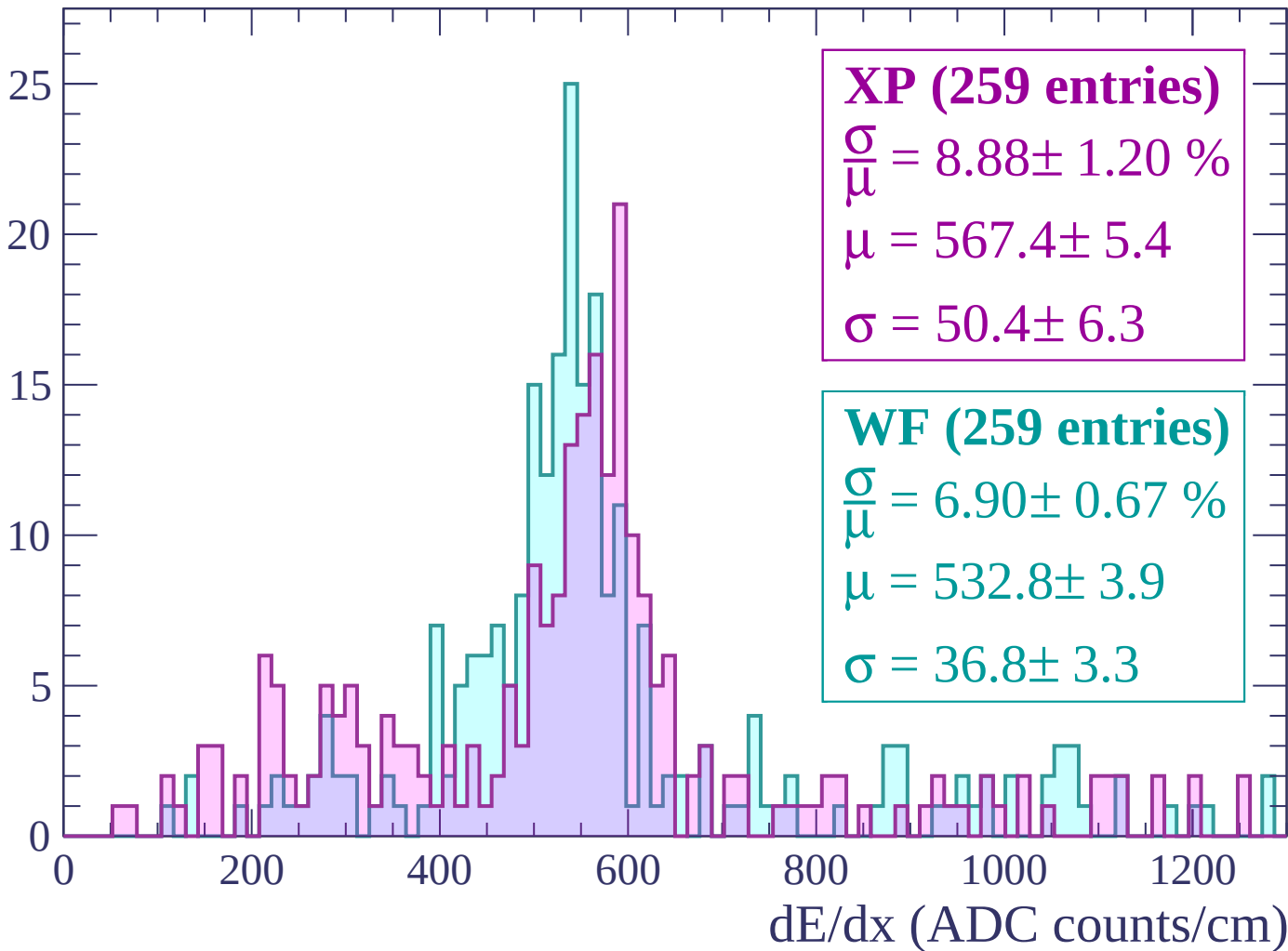
Energy loss | $-120 < p < -60$

Count



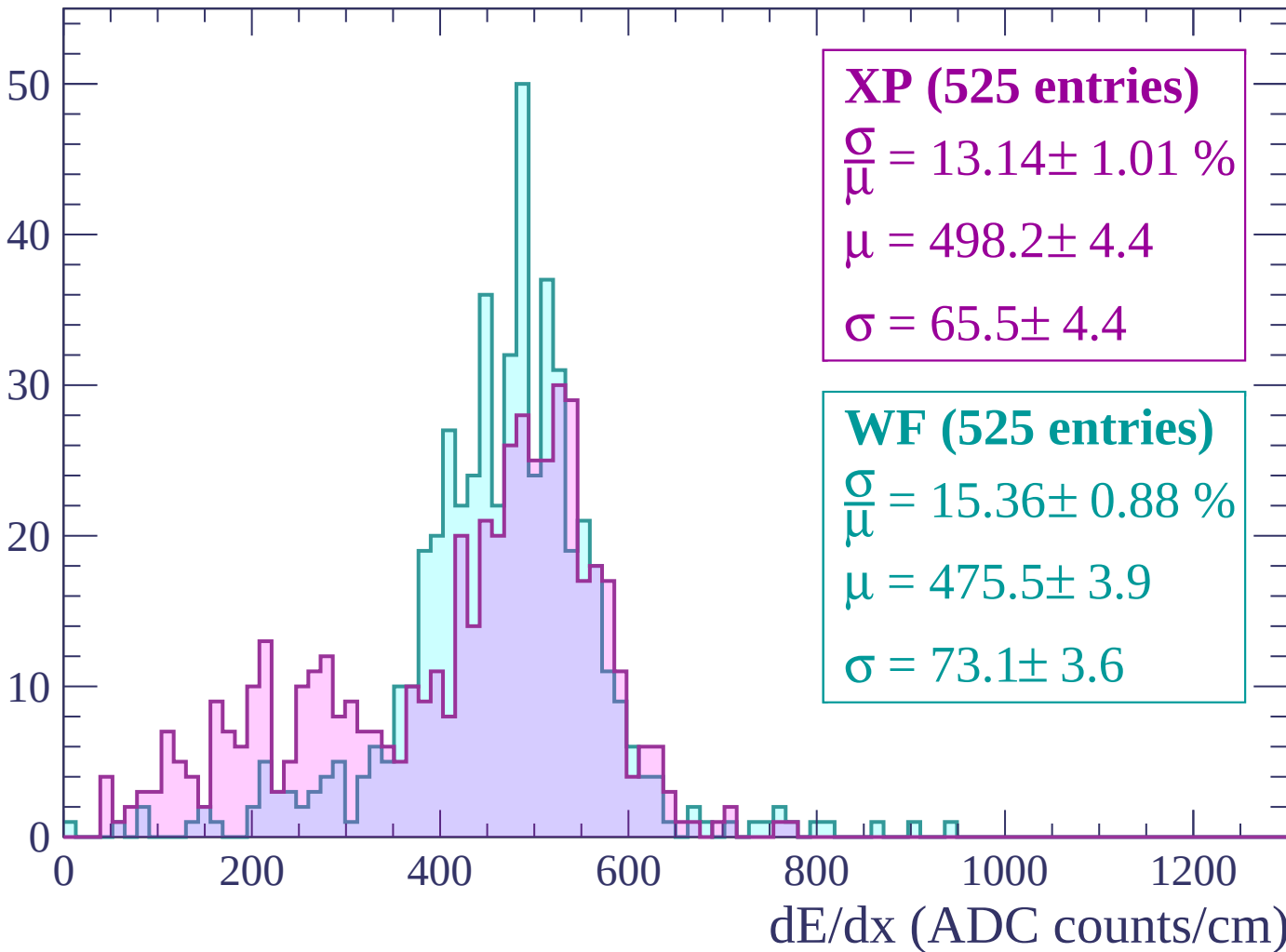
Energy loss | $-60 < p < 0$

Count



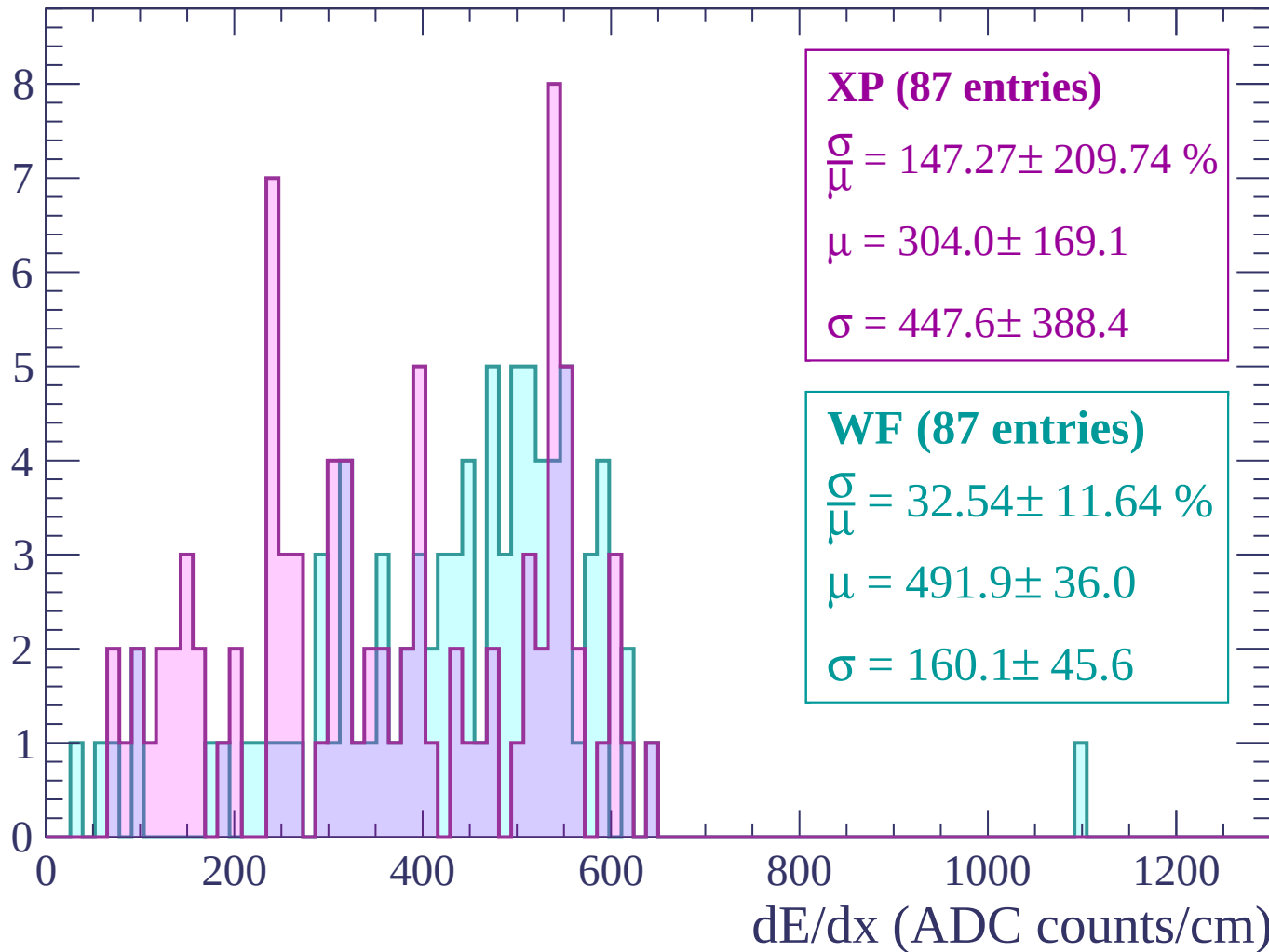
Energy loss | $0 < p < 60$

Count



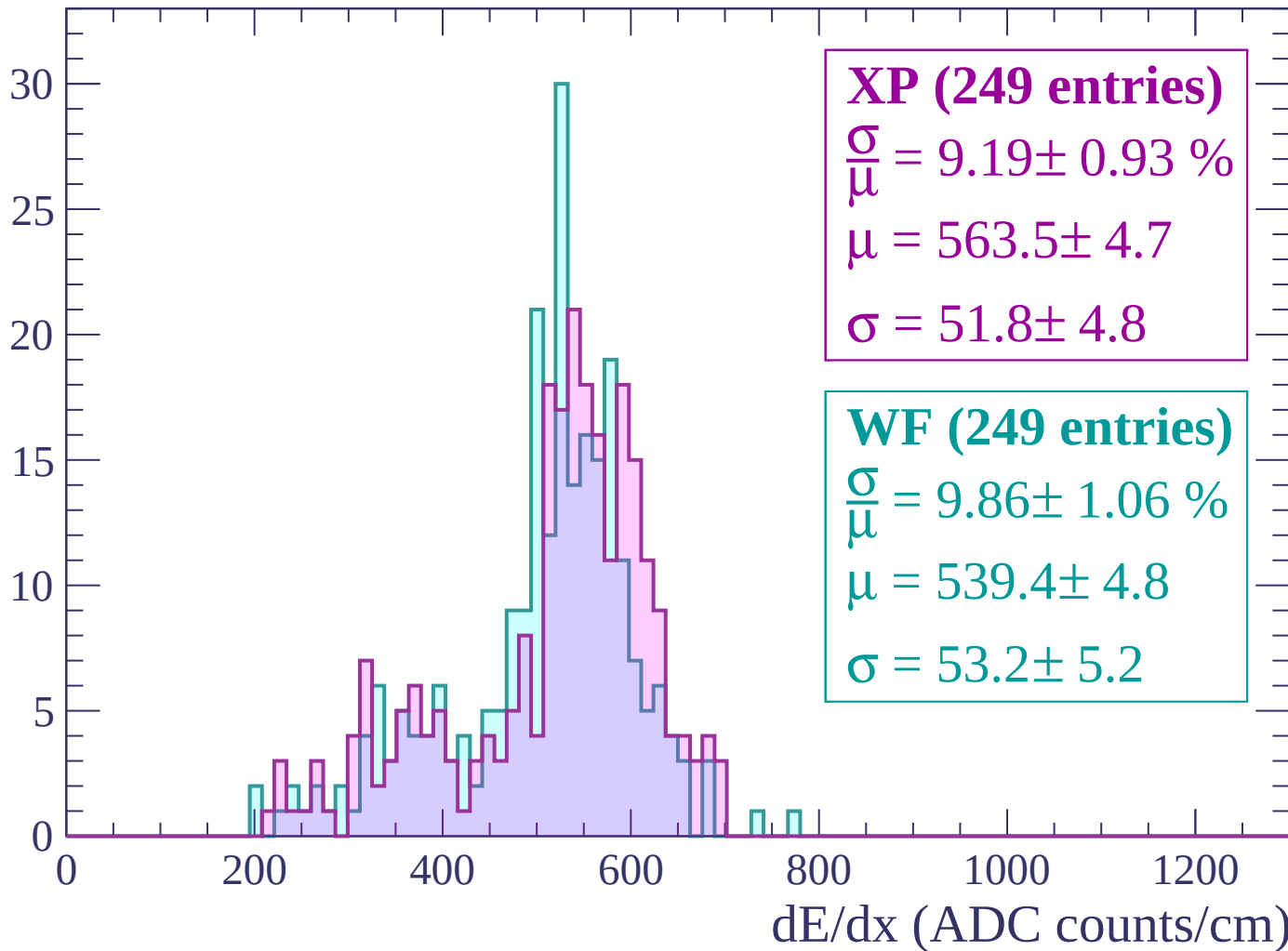
Energy loss | $60 < p < 120$

Count



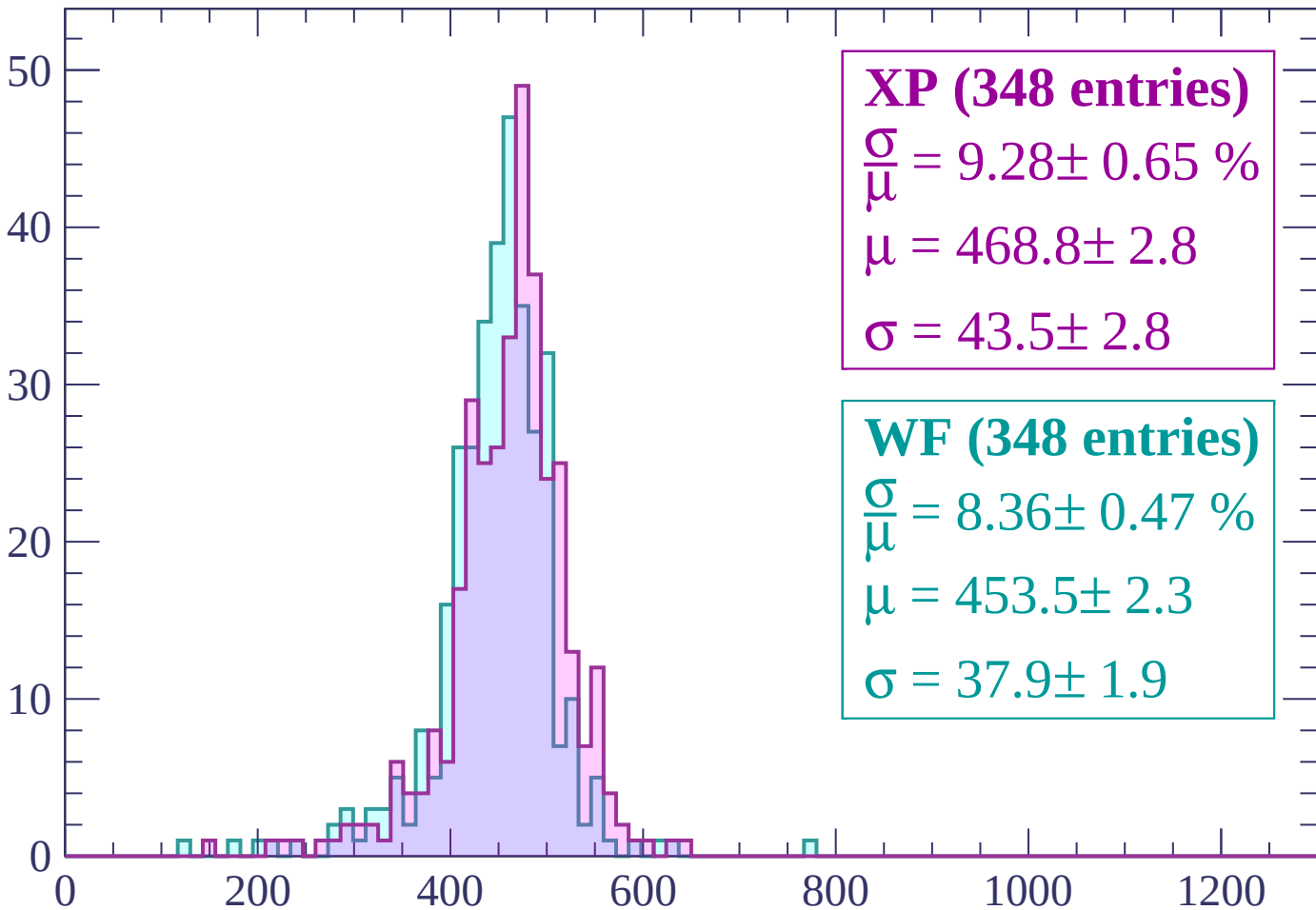
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (348 entries)

$$\frac{\sigma}{\mu} = 9.28 \pm 0.65 \%$$

$$\mu = 468.8 \pm 2.8$$

$$\sigma = 43.5 \pm 2.8$$

WF (348 entries)

$$\frac{\sigma}{\mu} = 8.36 \pm 0.47 \%$$

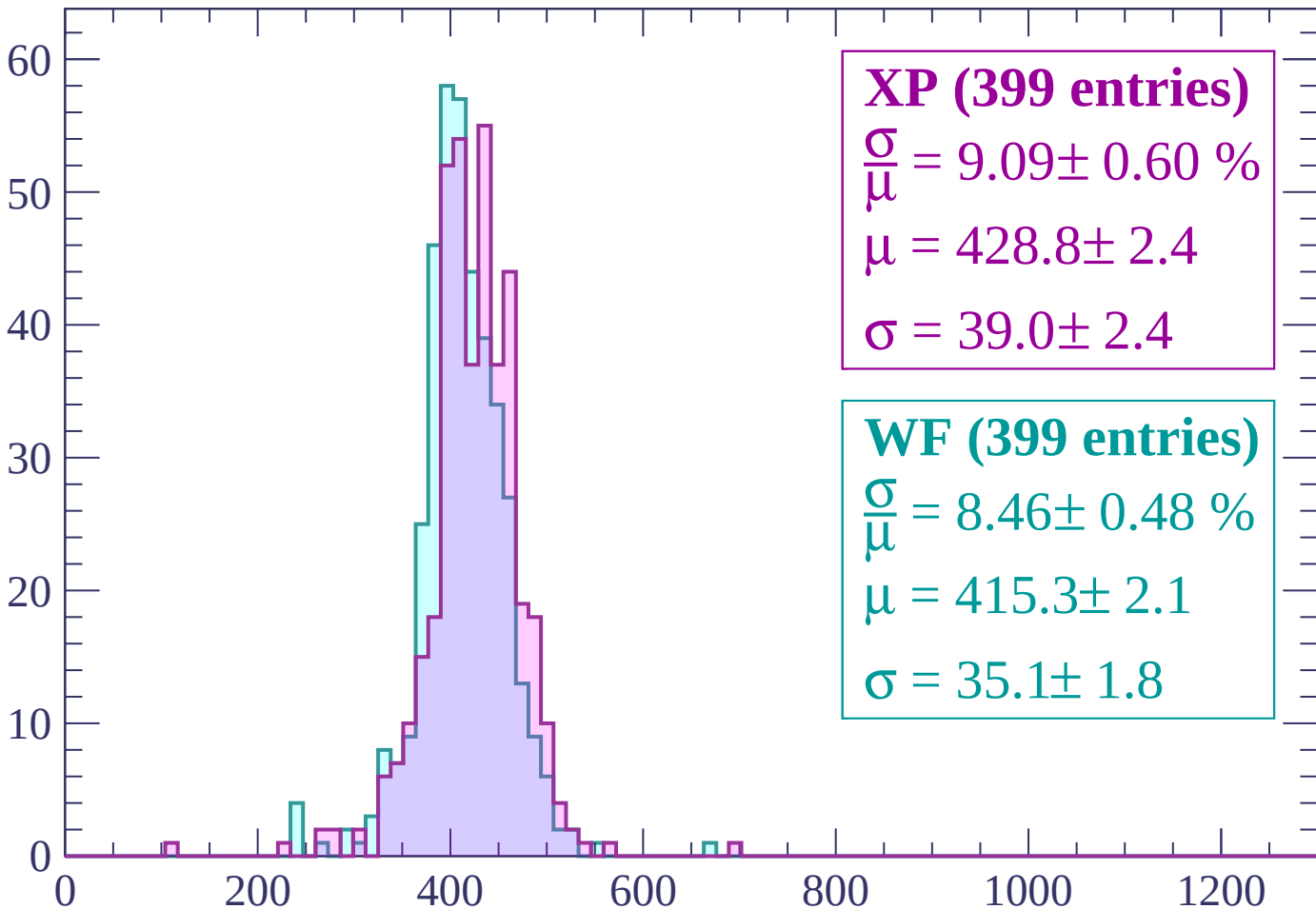
$$\mu = 453.5 \pm 2.3$$

$$\sigma = 37.9 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP (399 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.60 \%$$

$$\mu = 428.8 \pm 2.4$$

$$\sigma = 39.0 \pm 2.4$$

WF (399 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.48 \%$$

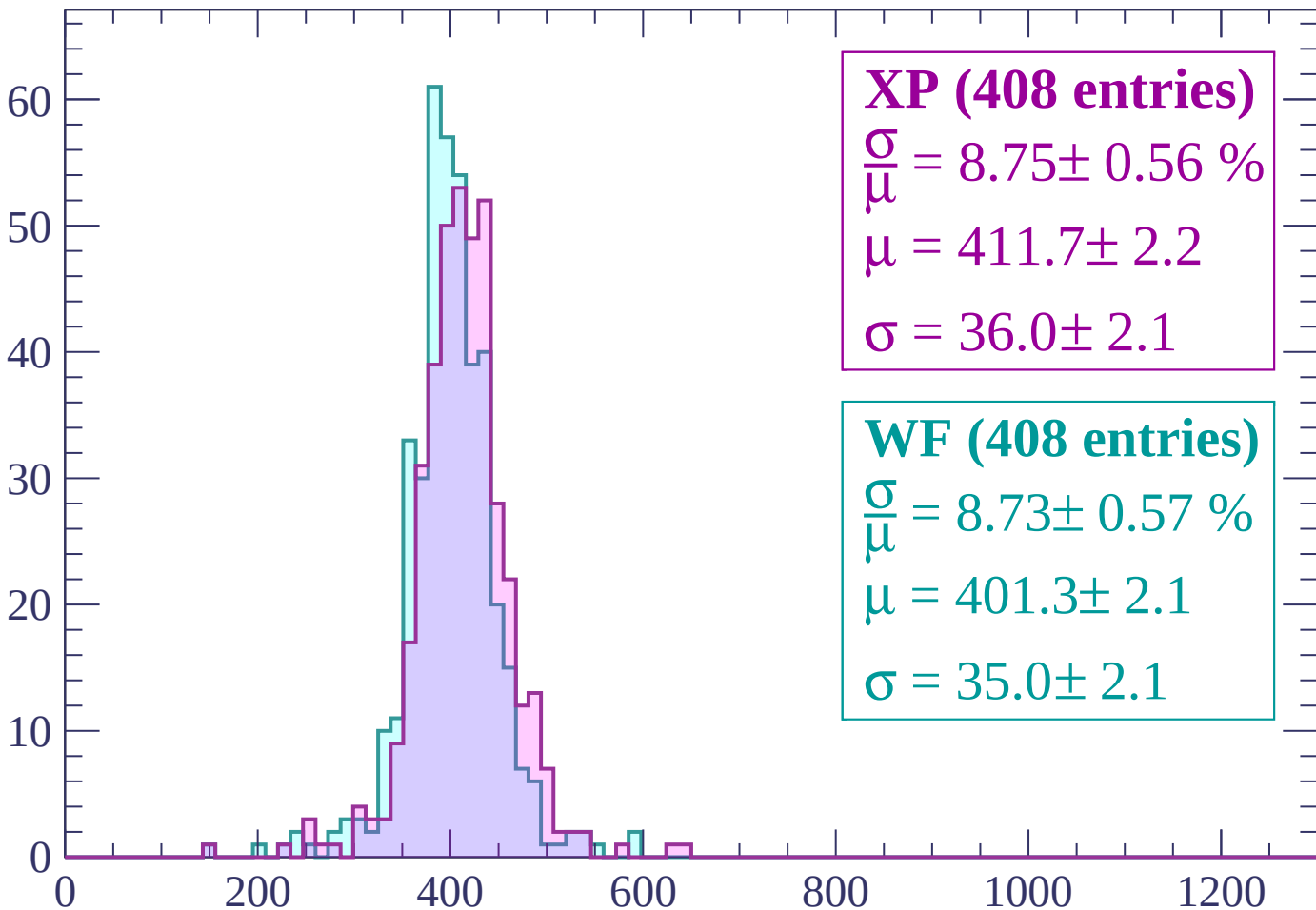
$$\mu = 415.3 \pm 2.1$$

$$\sigma = 35.1 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (408 entries)

$$\frac{\sigma}{\mu} = 8.75 \pm 0.56 \%$$

$$\mu = 411.7 \pm 2.2$$

$$\sigma = 36.0 \pm 2.1$$

WF (408 entries)

$$\frac{\sigma}{\mu} = 8.73 \pm 0.57 \%$$

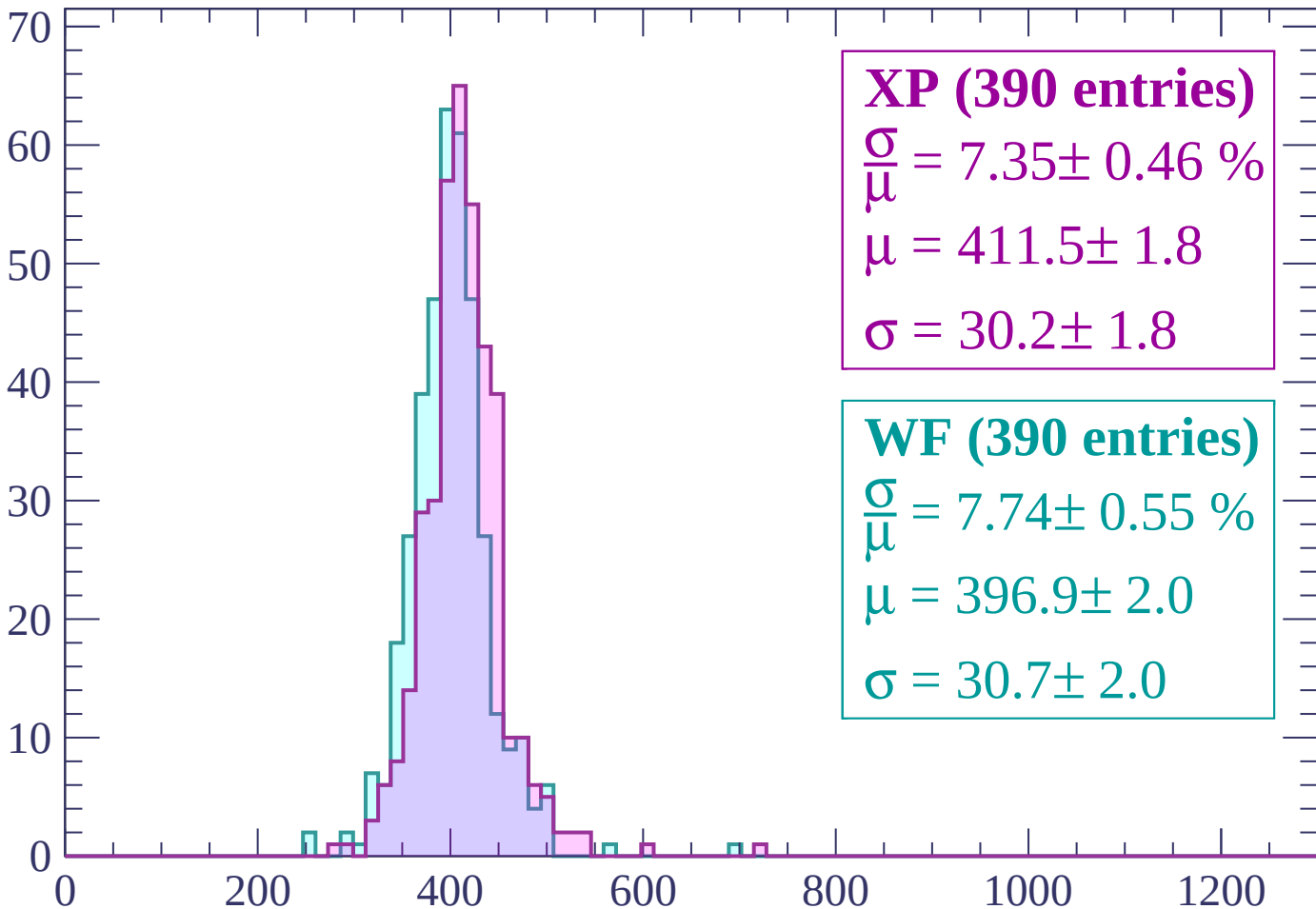
$$\mu = 401.3 \pm 2.1$$

$$\sigma = 35.0 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (390 entries)

$\frac{\sigma}{\mu} = 7.35 \pm 0.46 \%$

$\mu = 411.5 \pm 1.8$

$\sigma = 30.2 \pm 1.8$

WF (390 entries)

$\frac{\sigma}{\mu} = 7.74 \pm 0.55 \%$

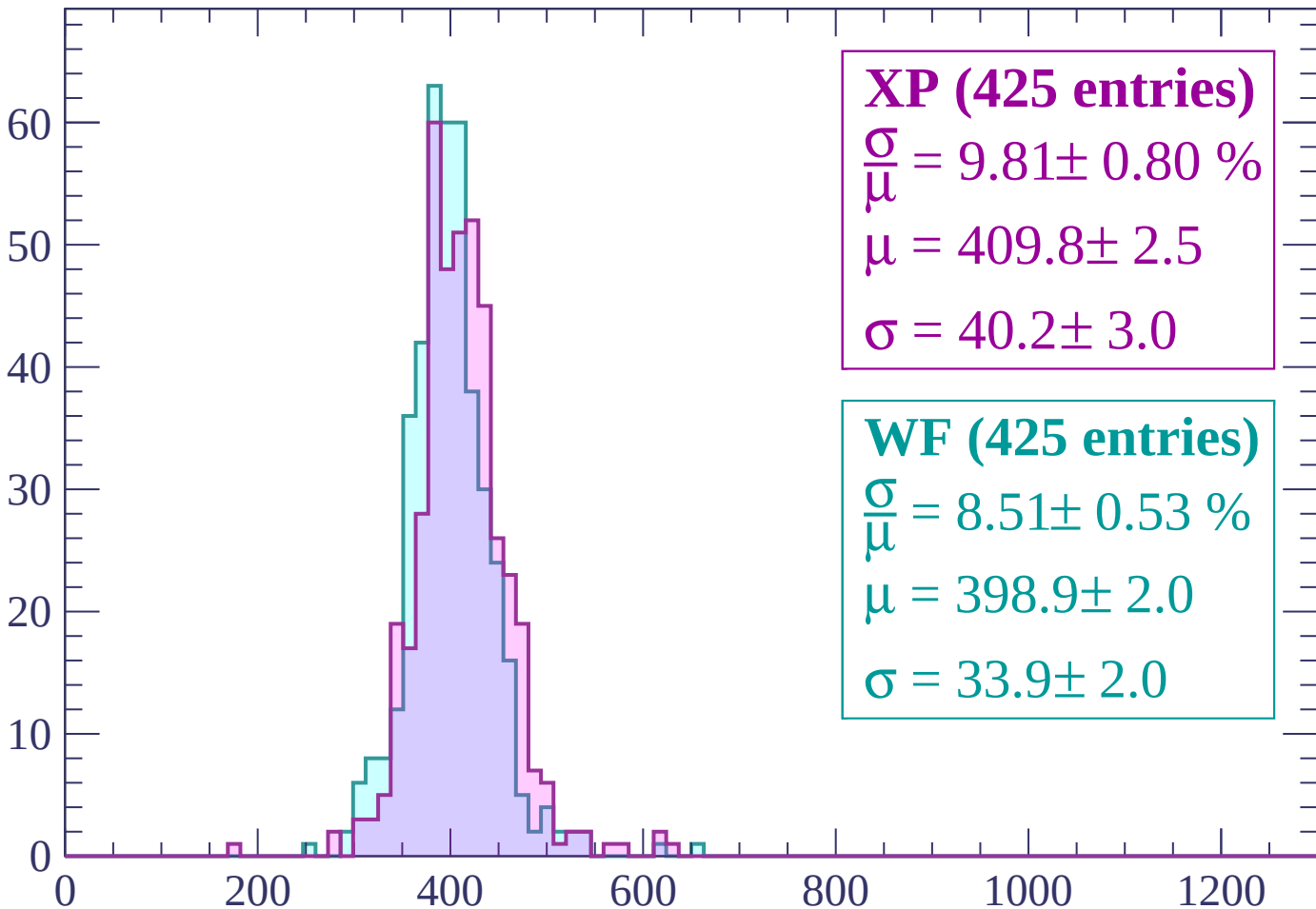
$\mu = 396.9 \pm 2.0$

$\sigma = 30.7 \pm 2.0$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (425 entries)

$$\frac{\sigma}{\mu} = 9.81 \pm 0.80 \%$$

$$\mu = 409.8 \pm 2.5$$

$$\sigma = 40.2 \pm 3.0$$

WF (425 entries)

$$\frac{\sigma}{\mu} = 8.51 \pm 0.53 \%$$

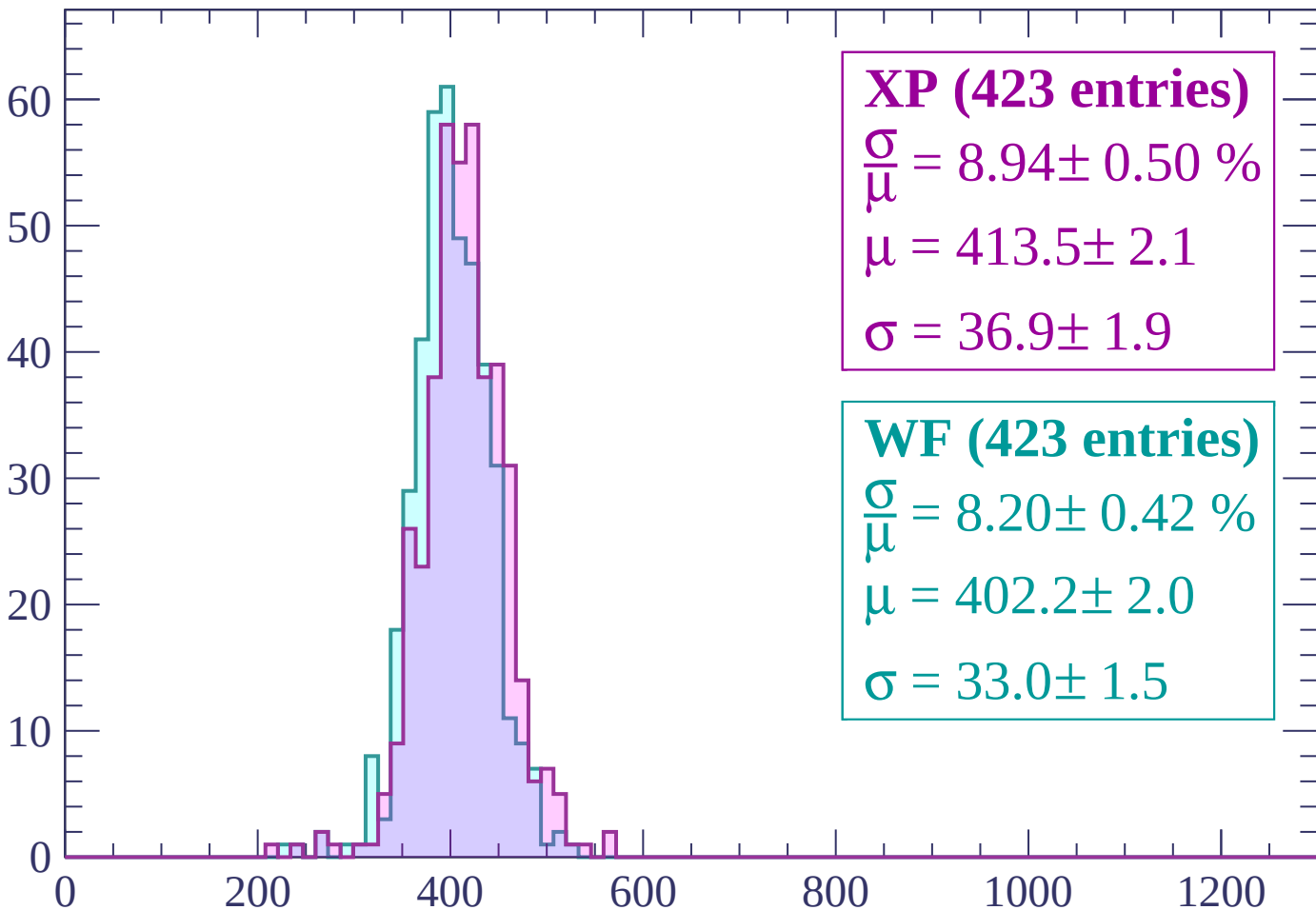
$$\mu = 398.9 \pm 2.0$$

$$\sigma = 33.9 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (423 entries)

$\frac{\sigma}{\mu} = 8.94 \pm 0.50 \%$

$\mu = 413.5 \pm 2.1$

$\sigma = 36.9 \pm 1.9$

WF (423 entries)

$\frac{\sigma}{\mu} = 8.20 \pm 0.42 \%$

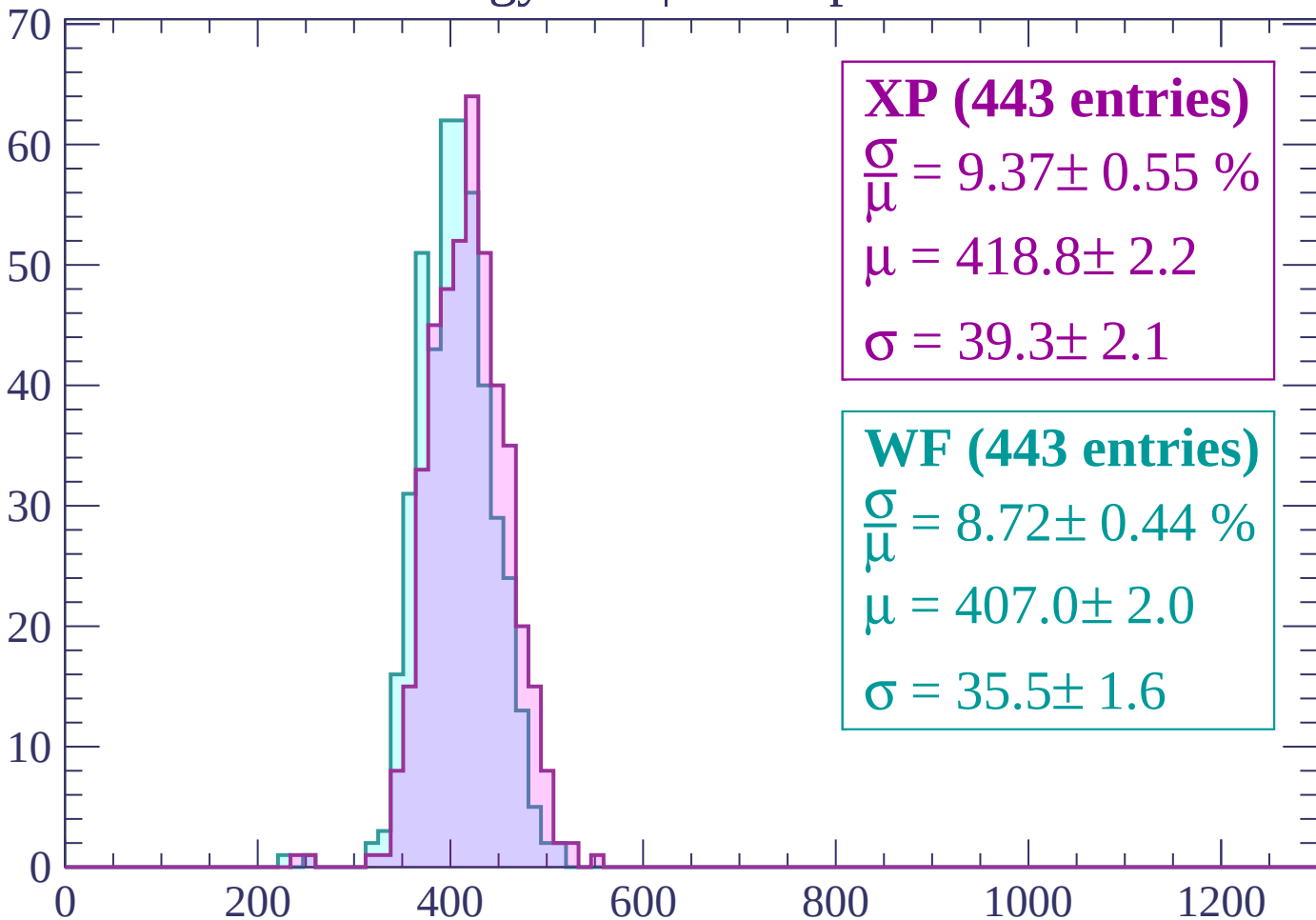
$\mu = 402.2 \pm 2.0$

$\sigma = 33.0 \pm 1.5$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (443 entries)

$$\frac{\sigma}{\mu} = 9.37 \pm 0.55 \%$$

$$\mu = 418.8 \pm 2.2$$

$$\sigma = 39.3 \pm 2.1$$

WF (443 entries)

$$\frac{\sigma}{\mu} = 8.72 \pm 0.44 \%$$

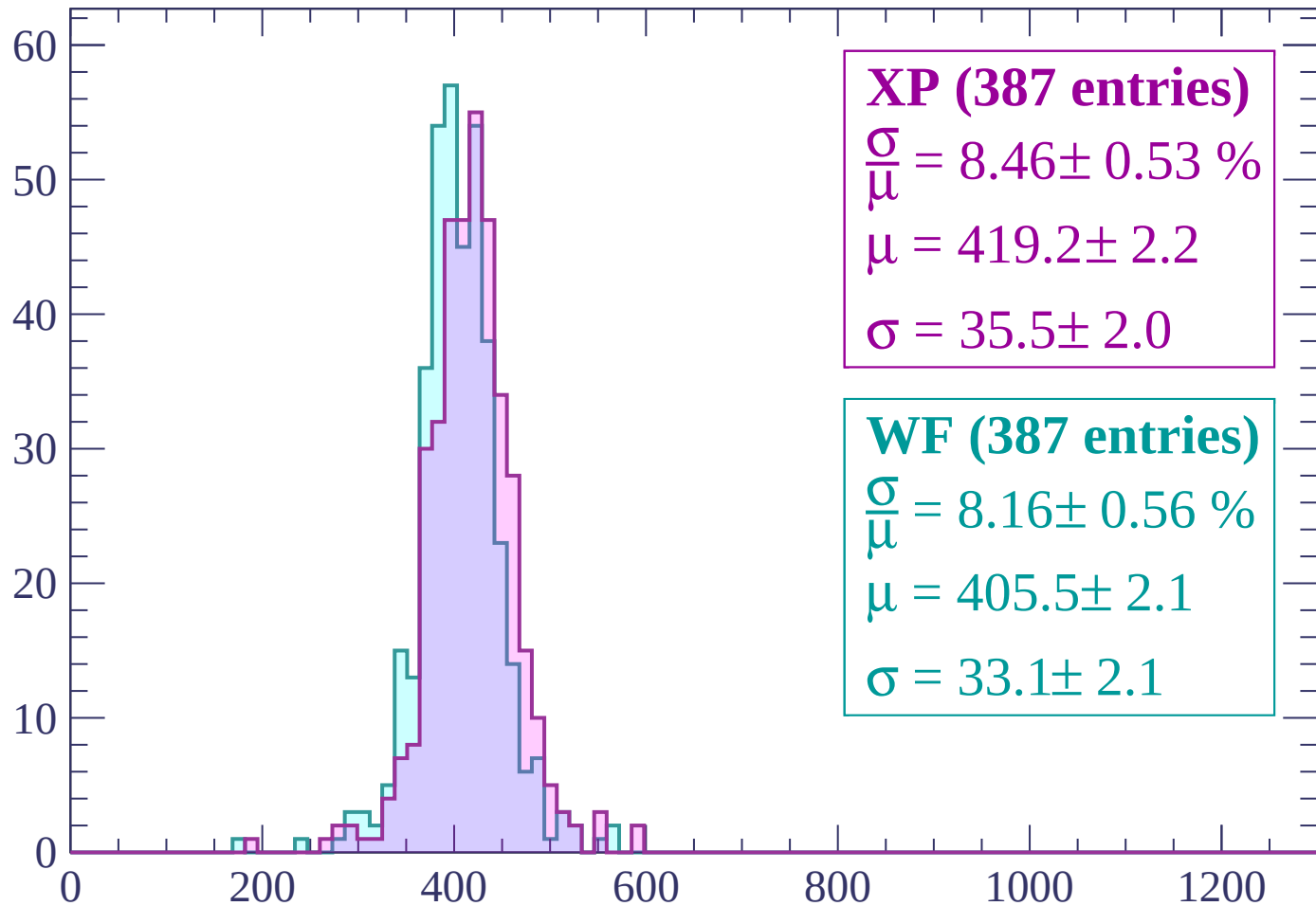
$$\mu = 407.0 \pm 2.0$$

$$\sigma = 35.5 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (387 entries)

$\frac{\sigma}{\mu} = 8.46 \pm 0.53 \%$

$\mu = 419.2 \pm 2.2$

$\sigma = 35.5 \pm 2.0$

WF (387 entries)

$\frac{\sigma}{\mu} = 8.16 \pm 0.56 \%$

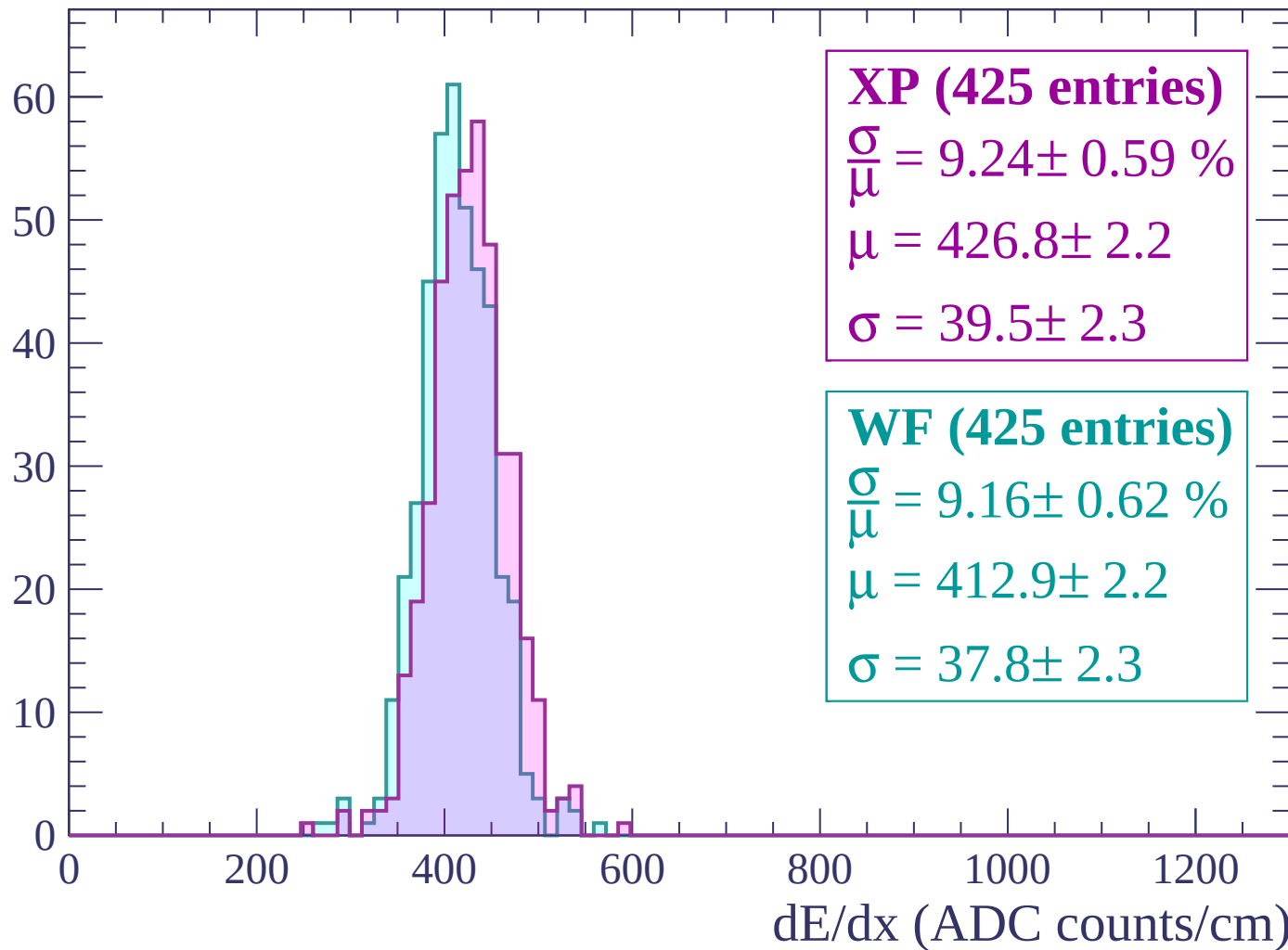
$\mu = 405.5 \pm 2.1$

$\sigma = 33.1 \pm 2.1$

dE/dx (ADC counts/cm)

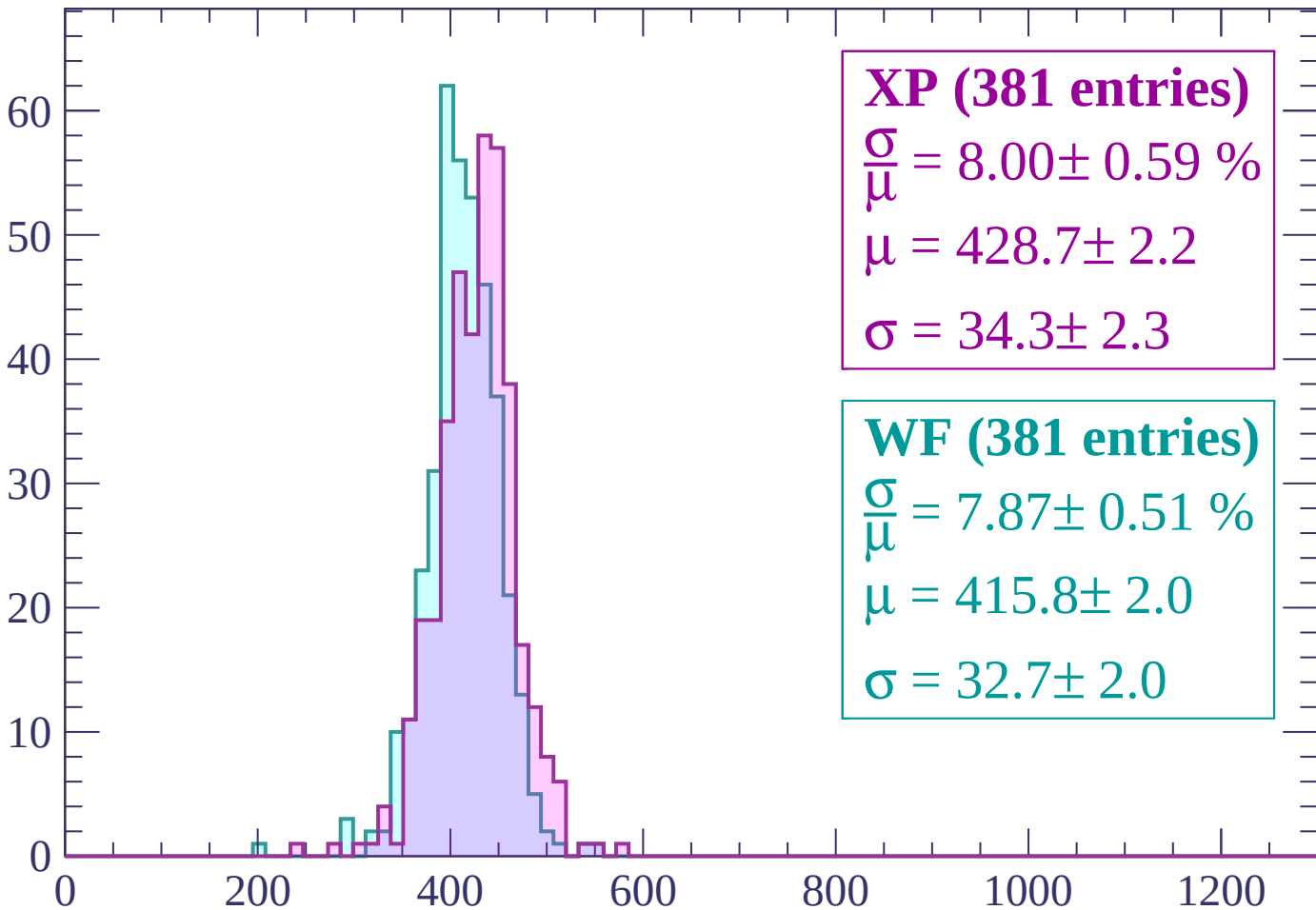
Energy loss | $660 < p < 720$

Count



Energy loss | $720 < p < 780$

Count



XP (381 entries)

$$\frac{\sigma}{\mu} = 8.00 \pm 0.59 \%$$

$$\mu = 428.7 \pm 2.2$$

$$\sigma = 34.3 \pm 2.3$$

WF (381 entries)

$$\frac{\sigma}{\mu} = 7.87 \pm 0.51 \%$$

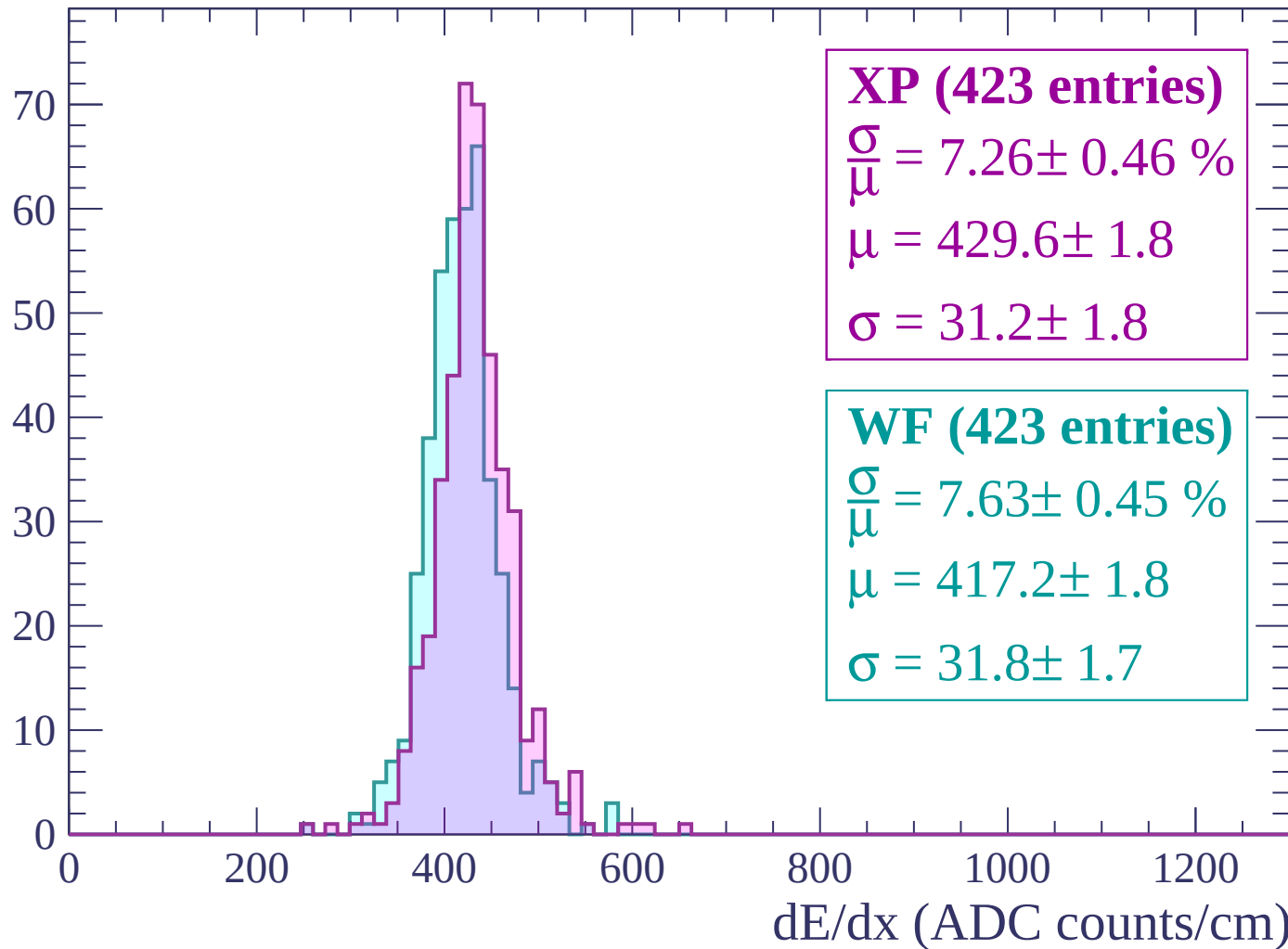
$$\mu = 415.8 \pm 2.0$$

$$\sigma = 32.7 \pm 2.0$$

dE/dx (ADC counts/cm)

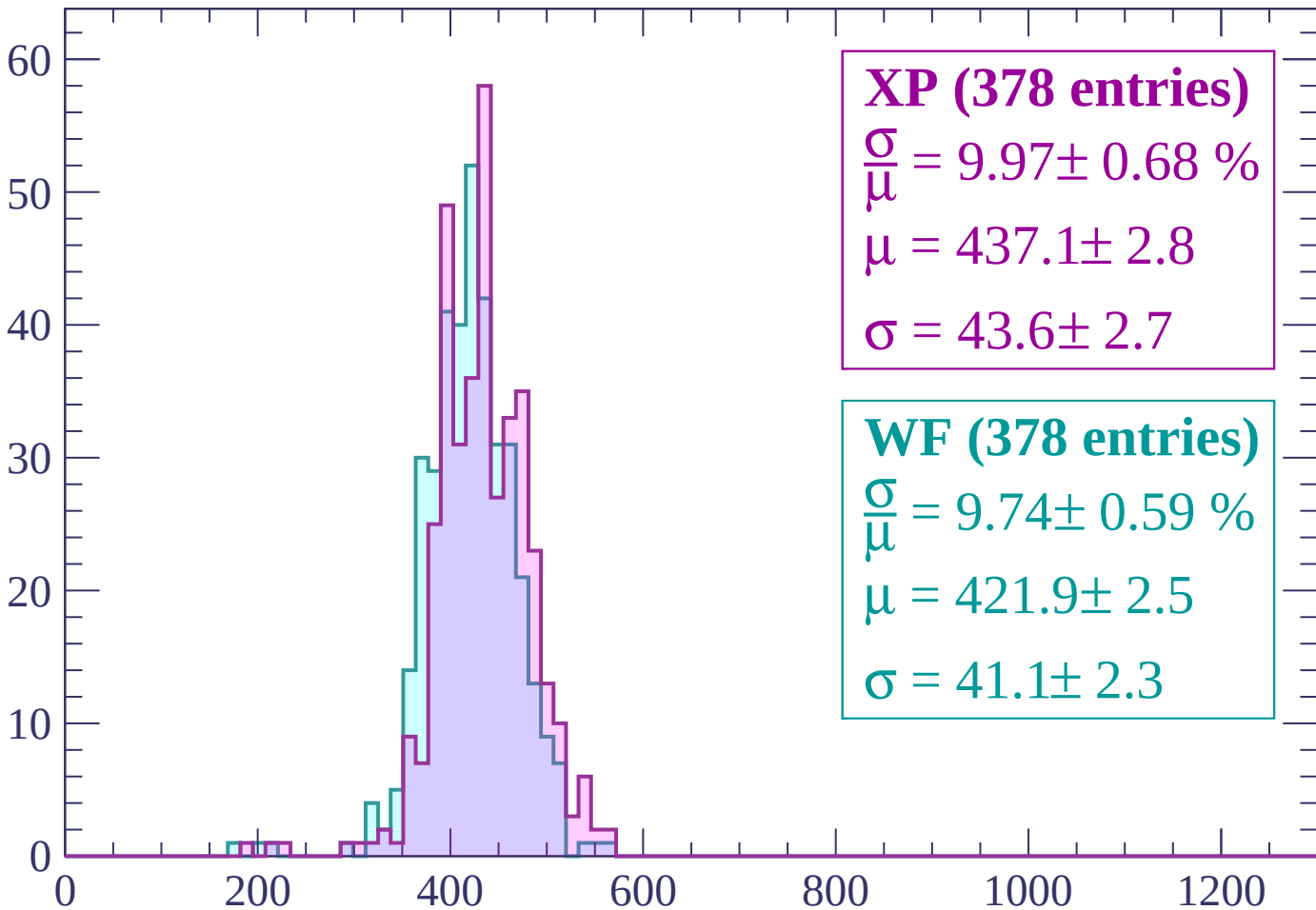
Energy loss | $780 < p < 840$

Count



Energy loss | $840 < p < 900$

Count



XP (378 entries)

$\frac{\sigma}{\mu} = 9.97 \pm 0.68 \%$

$\mu = 437.1 \pm 2.8$

$\sigma = 43.6 \pm 2.7$

WF (378 entries)

$\frac{\sigma}{\mu} = 9.74 \pm 0.59 \%$

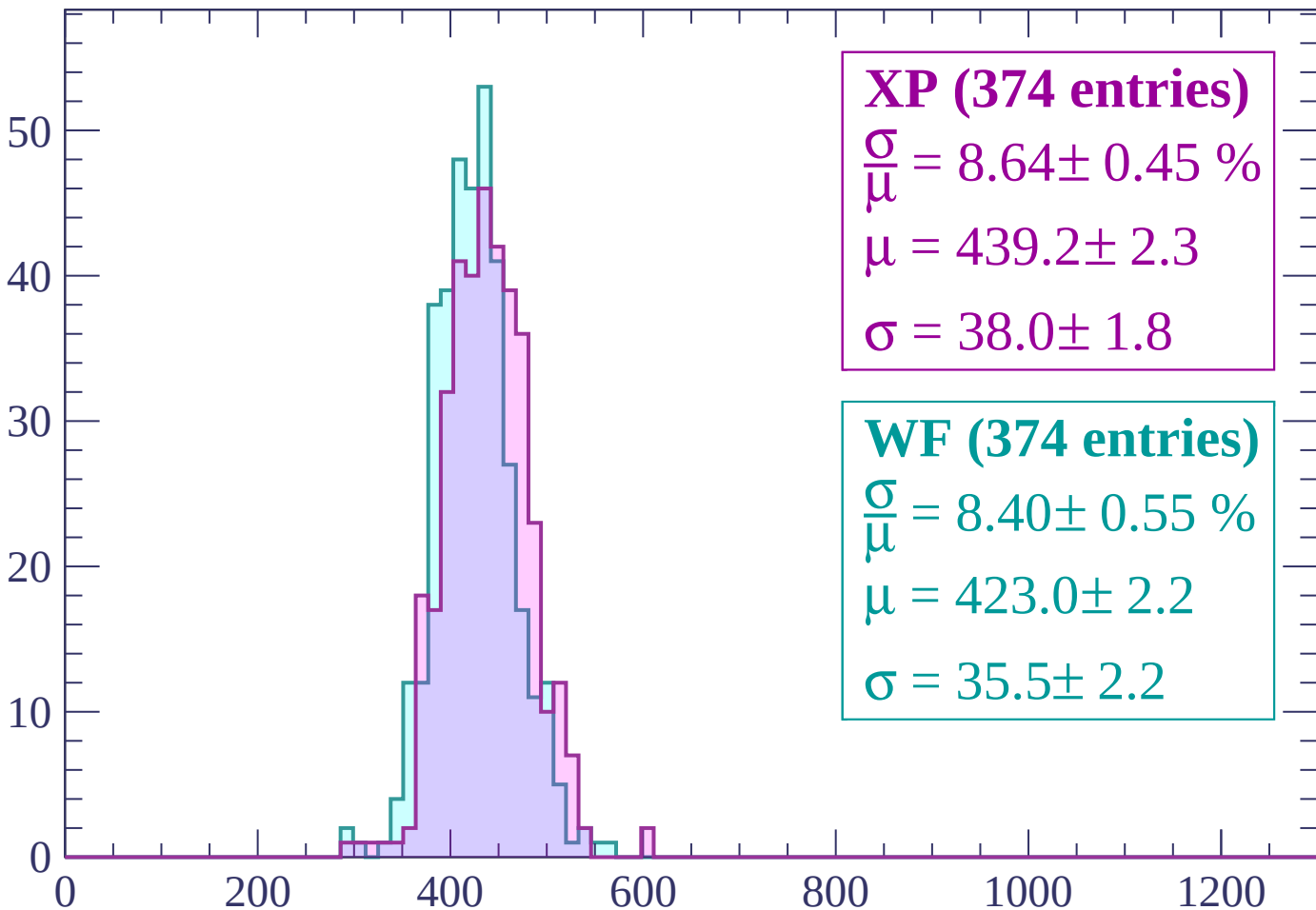
$\mu = 421.9 \pm 2.5$

$\sigma = 41.1 \pm 2.3$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (374 entries)

$\frac{\sigma}{\mu} = 8.64 \pm 0.45 \%$

$\mu = 439.2 \pm 2.3$

$\sigma = 38.0 \pm 1.8$

WF (374 entries)

$\frac{\sigma}{\mu} = 8.40 \pm 0.55 \%$

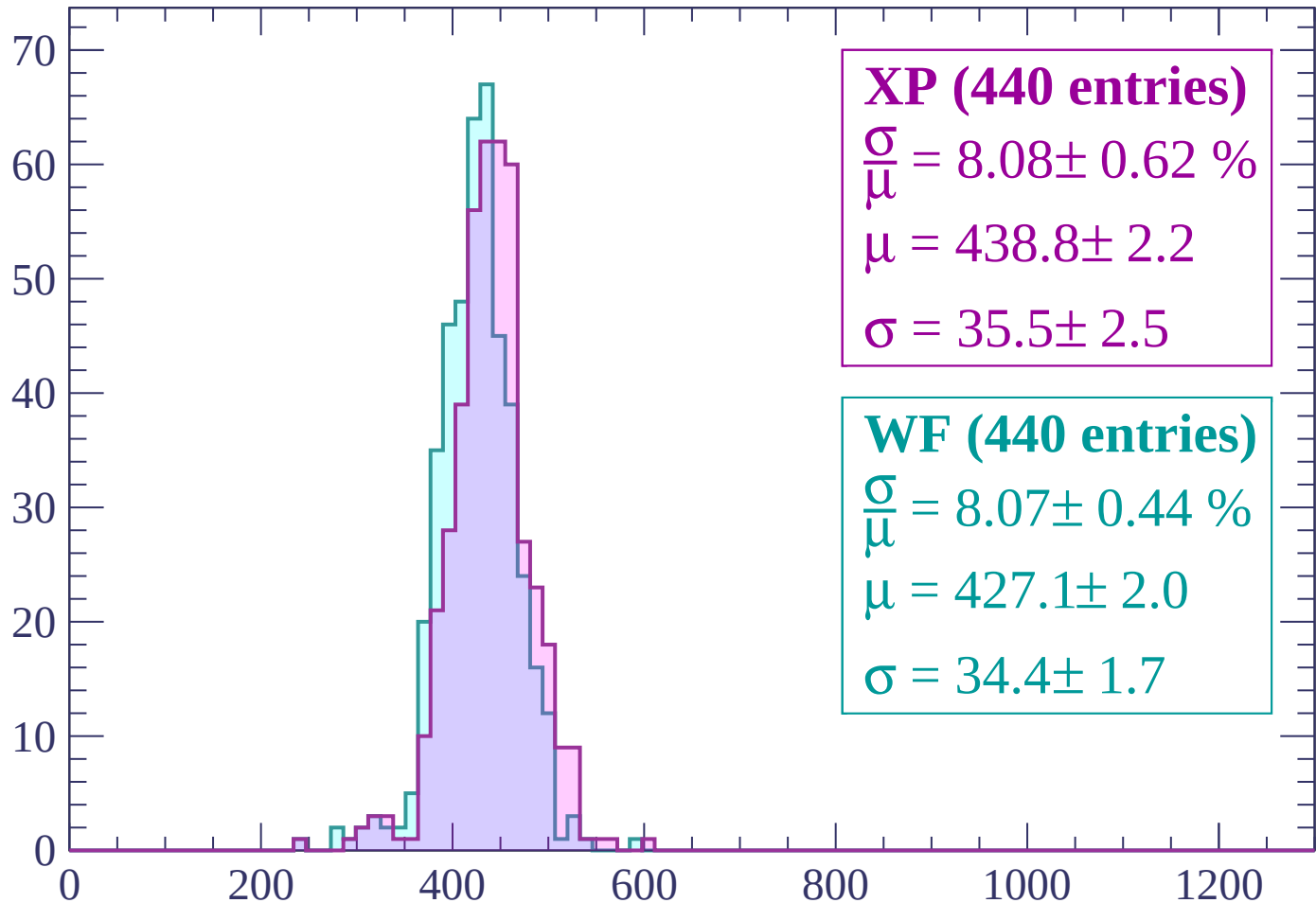
$\mu = 423.0 \pm 2.2$

$\sigma = 35.5 \pm 2.2$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count



XP (440 entries)

$$\frac{\sigma}{\mu} = 8.08 \pm 0.62 \%$$

$$\mu = 438.8 \pm 2.2$$

$$\sigma = 35.5 \pm 2.5$$

WF (440 entries)

$$\frac{\sigma}{\mu} = 8.07 \pm 0.44 \%$$

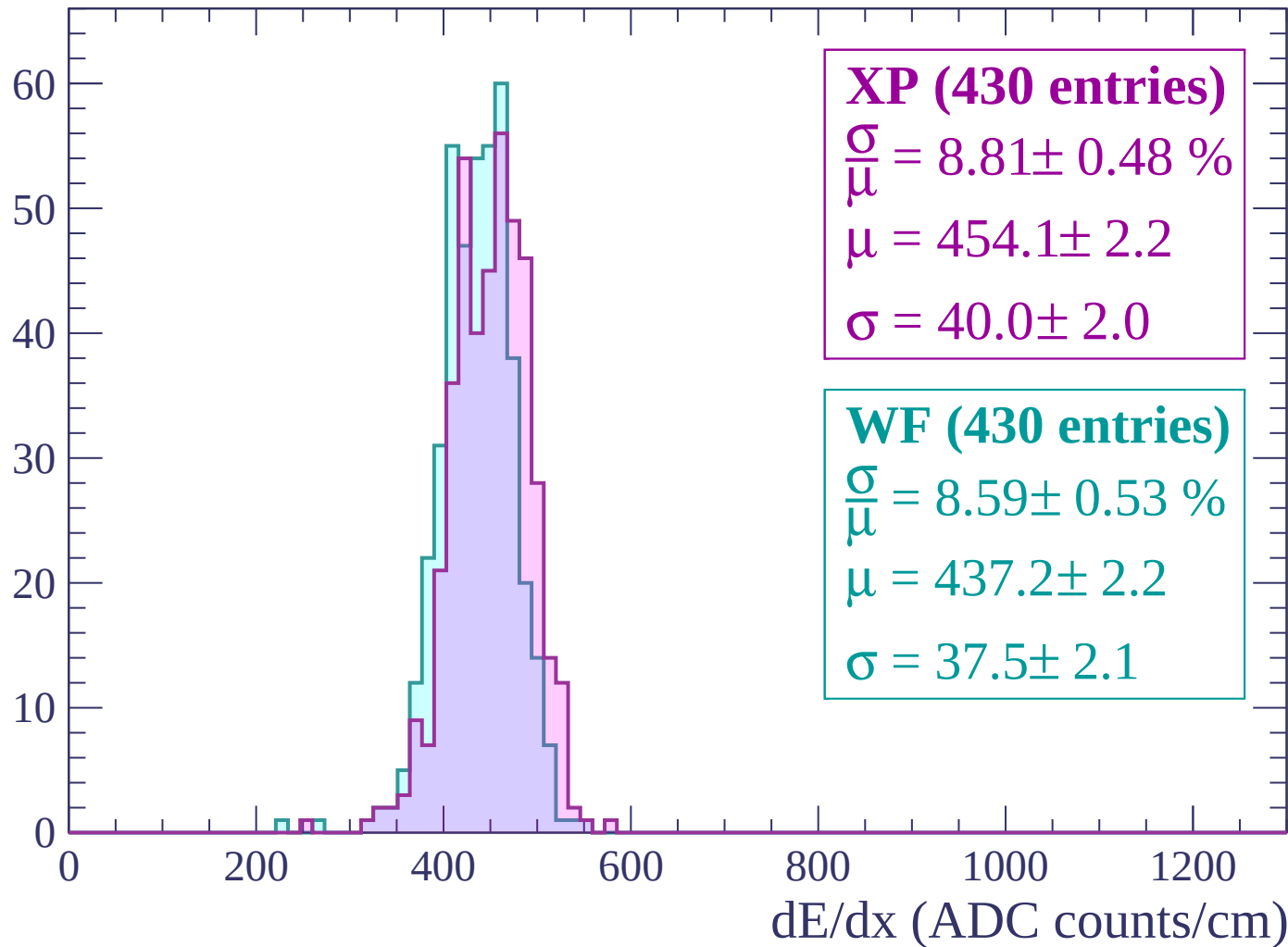
$$\mu = 427.1 \pm 2.0$$

$$\sigma = 34.4 \pm 1.7$$

dE/dx (ADC counts/cm)

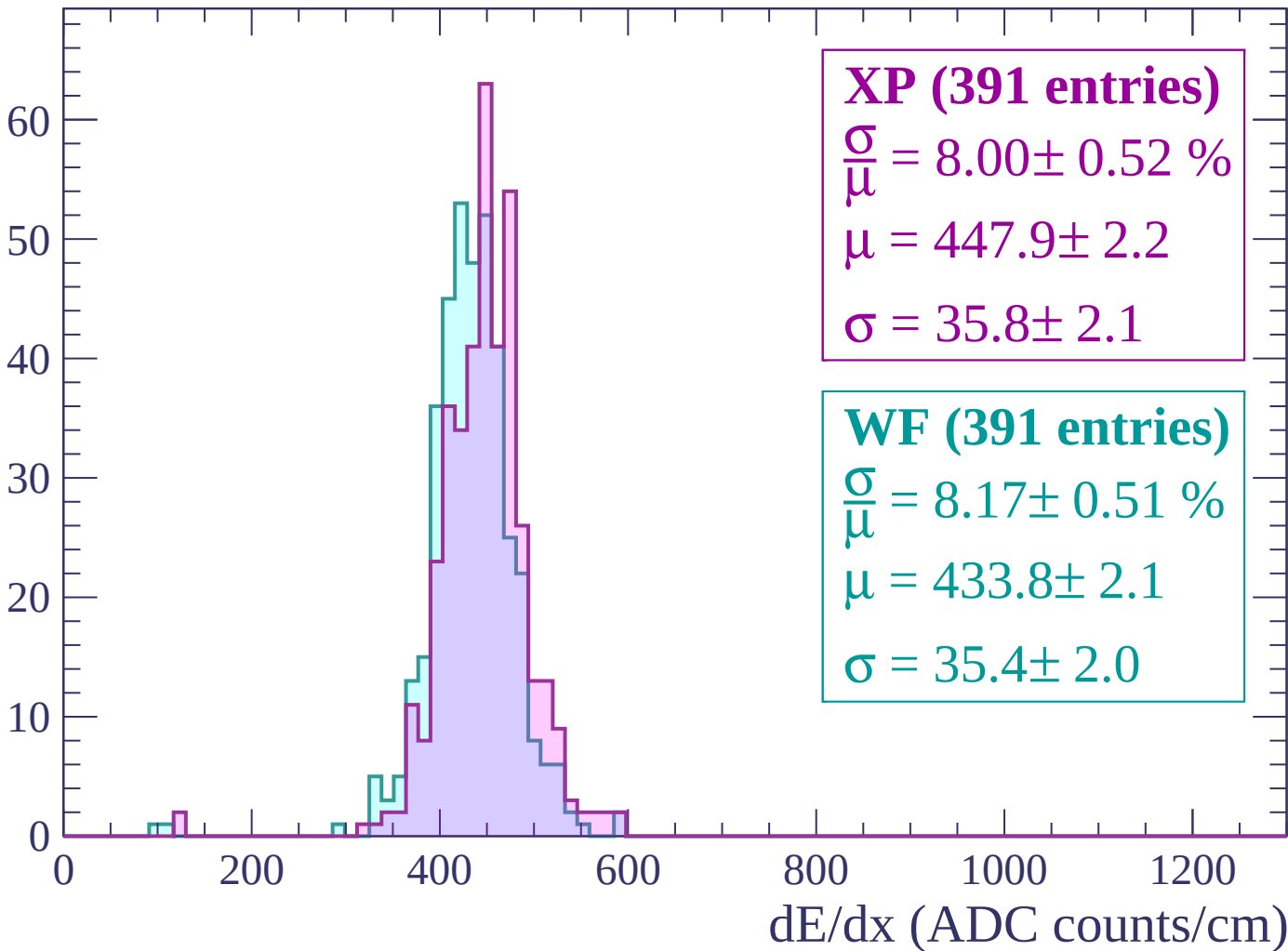
Energy loss | $1020 < p < 1080$

Count



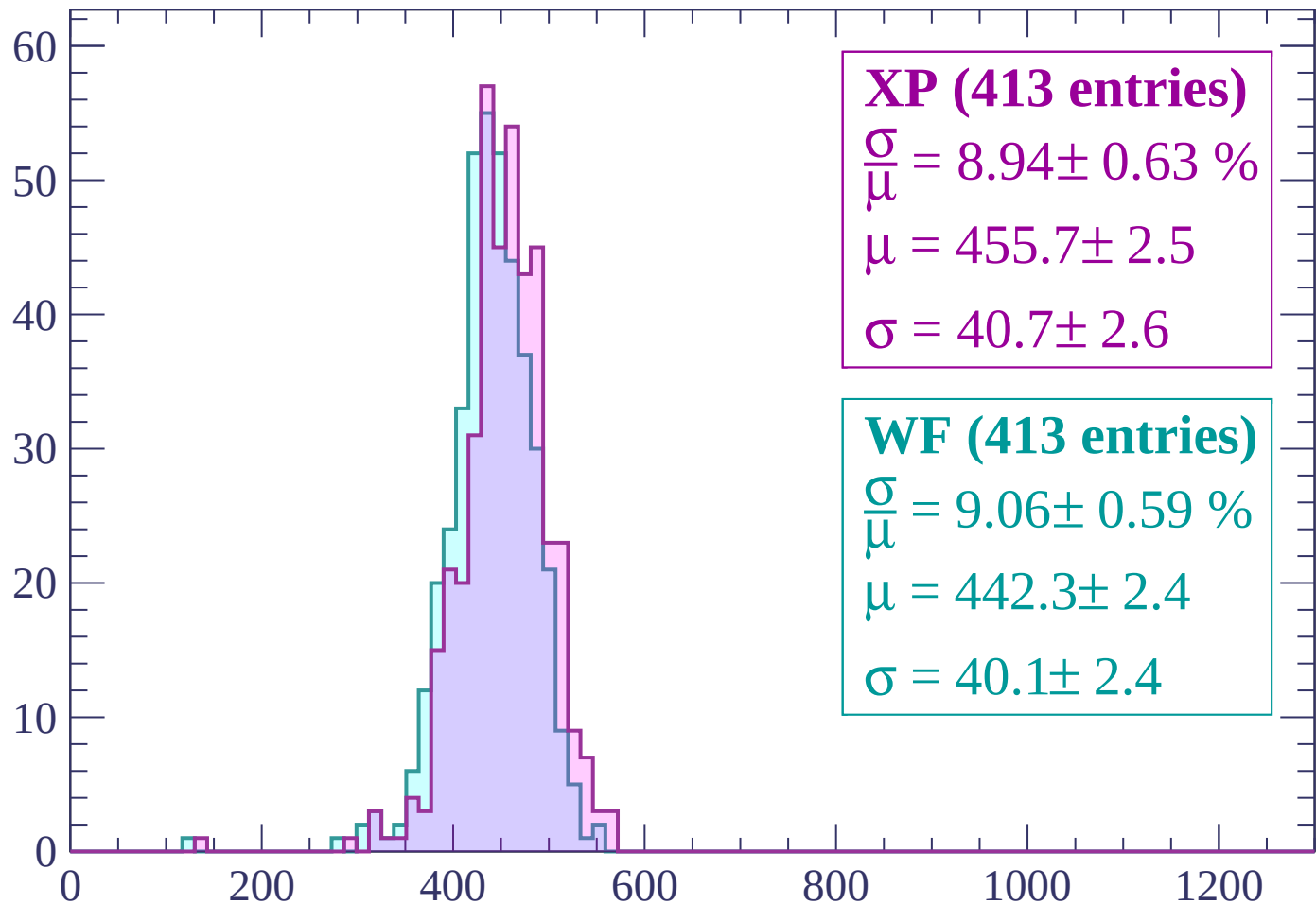
Energy loss | $1080 < p < 1140$

Count



Energy loss | $1140 < p < 1200$

Count



XP (413 entries)

$$\frac{\sigma}{\mu} = 8.94 \pm 0.63 \%$$

$$\mu = 455.7 \pm 2.5$$

$$\sigma = 40.7 \pm 2.6$$

WF (413 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.59 \%$$

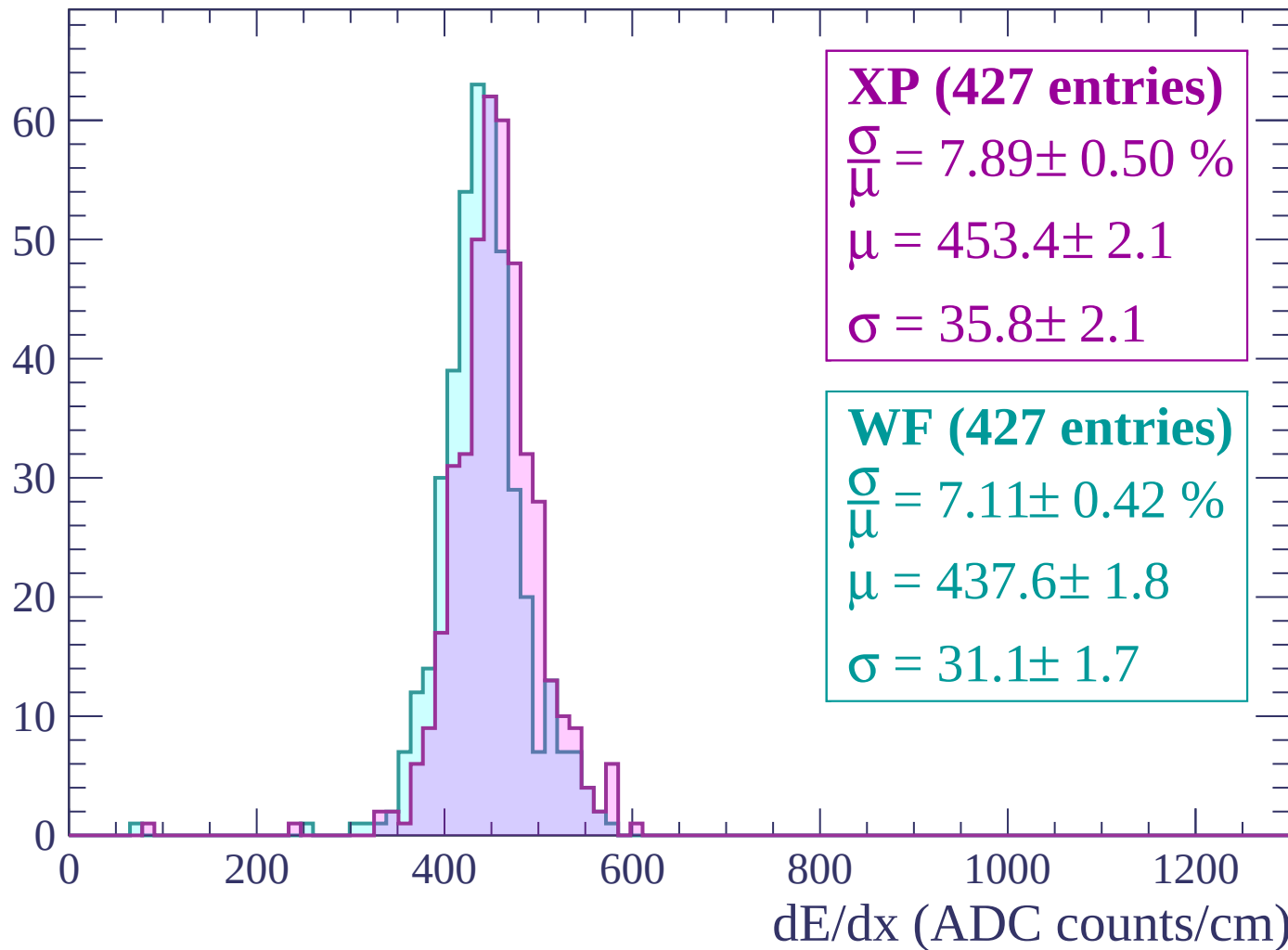
$$\mu = 442.3 \pm 2.4$$

$$\sigma = 40.1 \pm 2.4$$

dE/dx (ADC counts/cm)

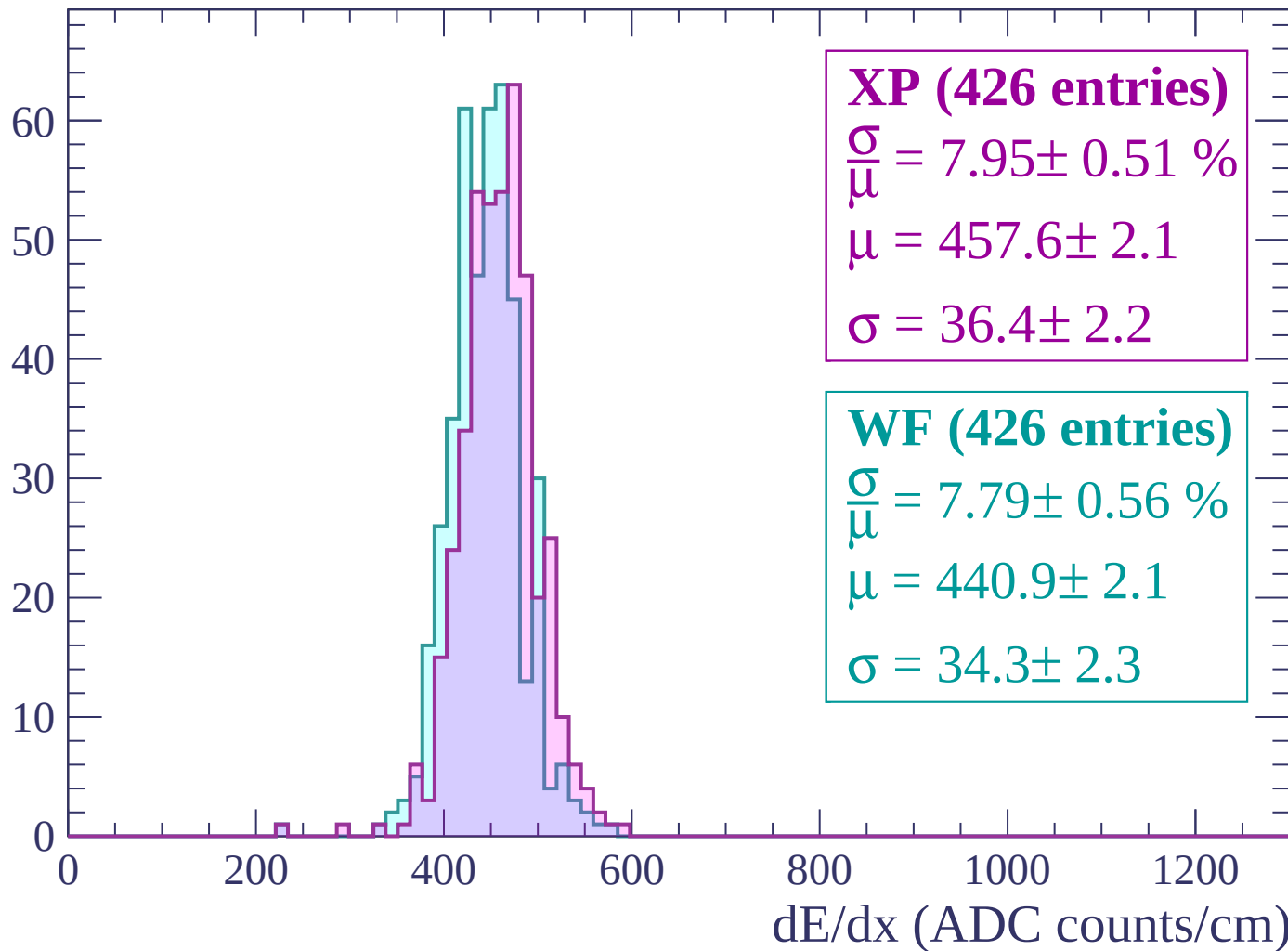
Energy loss | $1200 < p < 1260$

Count



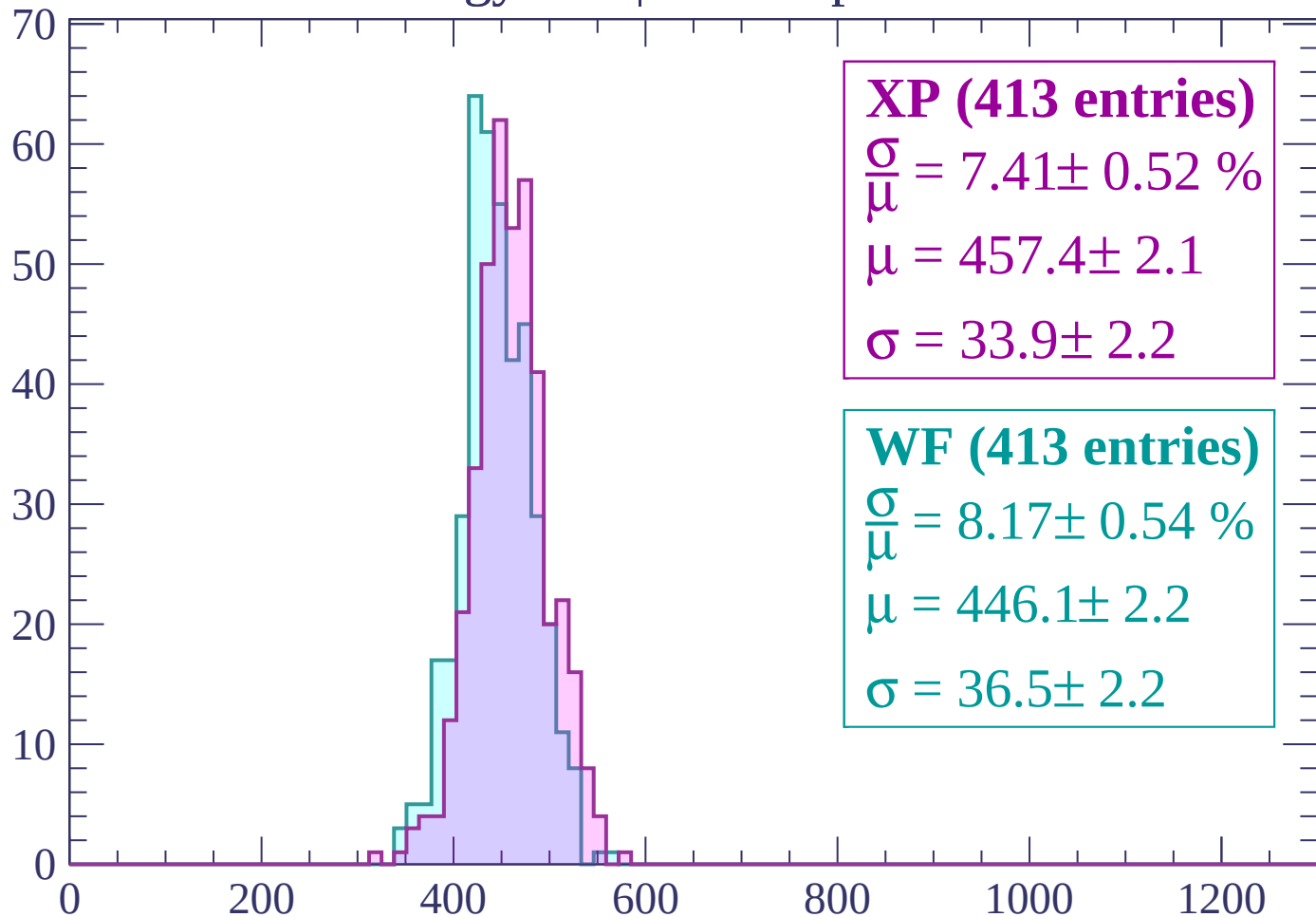
Energy loss | $1260 < p < 1320$

Count



Energy loss | $1320 < p < 1380$

Count



XP (413 entries)

$$\frac{\sigma}{\mu} = 7.41 \pm 0.52 \%$$

$$\mu = 457.4 \pm 2.1$$

$$\sigma = 33.9 \pm 2.2$$

WF (413 entries)

$$\frac{\sigma}{\mu} = 8.17 \pm 0.54 \%$$

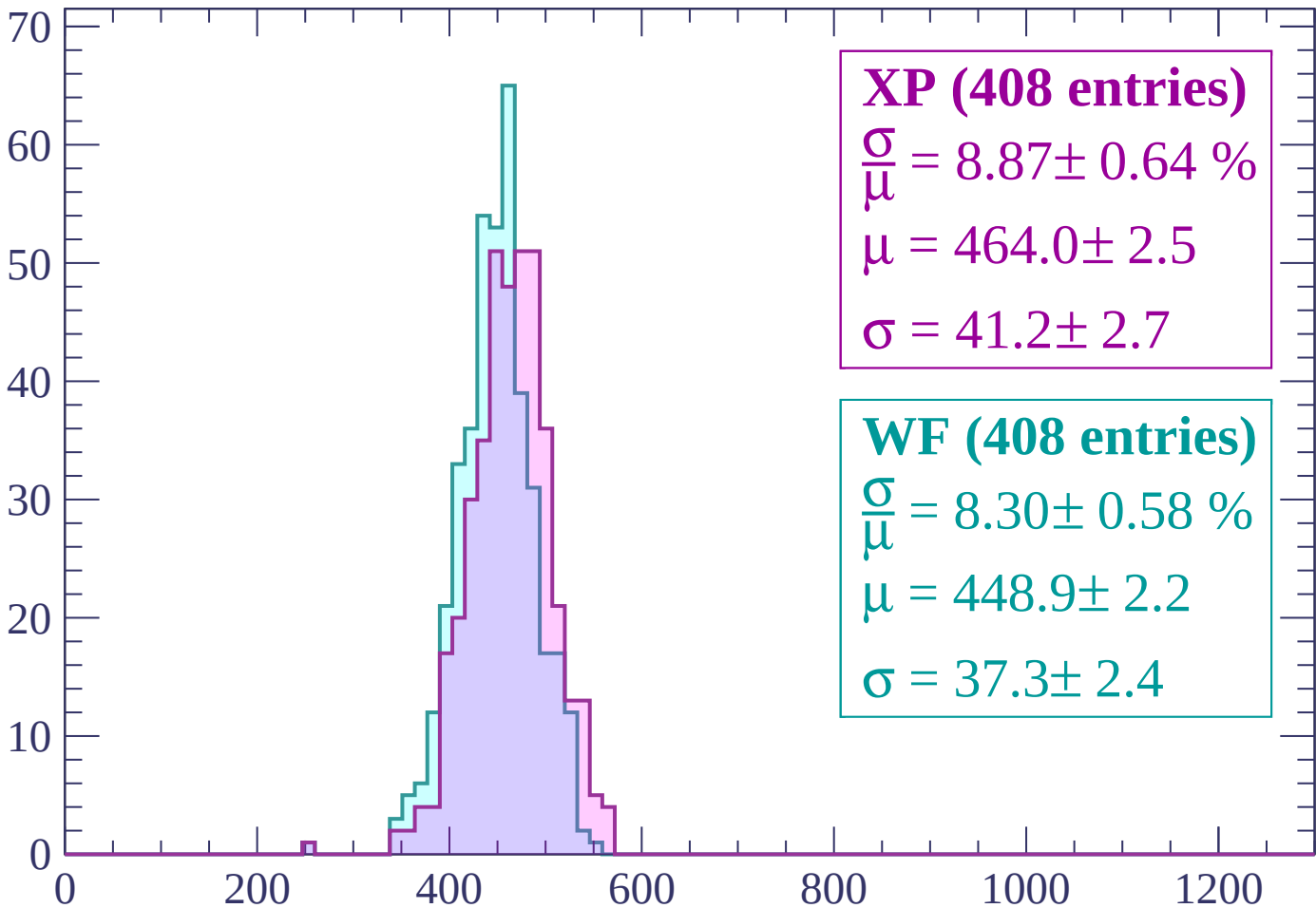
$$\mu = 446.1 \pm 2.2$$

$$\sigma = 36.5 \pm 2.2$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP (408 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.64 \%$$

$$\mu = 464.0 \pm 2.5$$

$$\sigma = 41.2 \pm 2.7$$

WF (408 entries)

$$\frac{\sigma}{\mu} = 8.30 \pm 0.58 \%$$

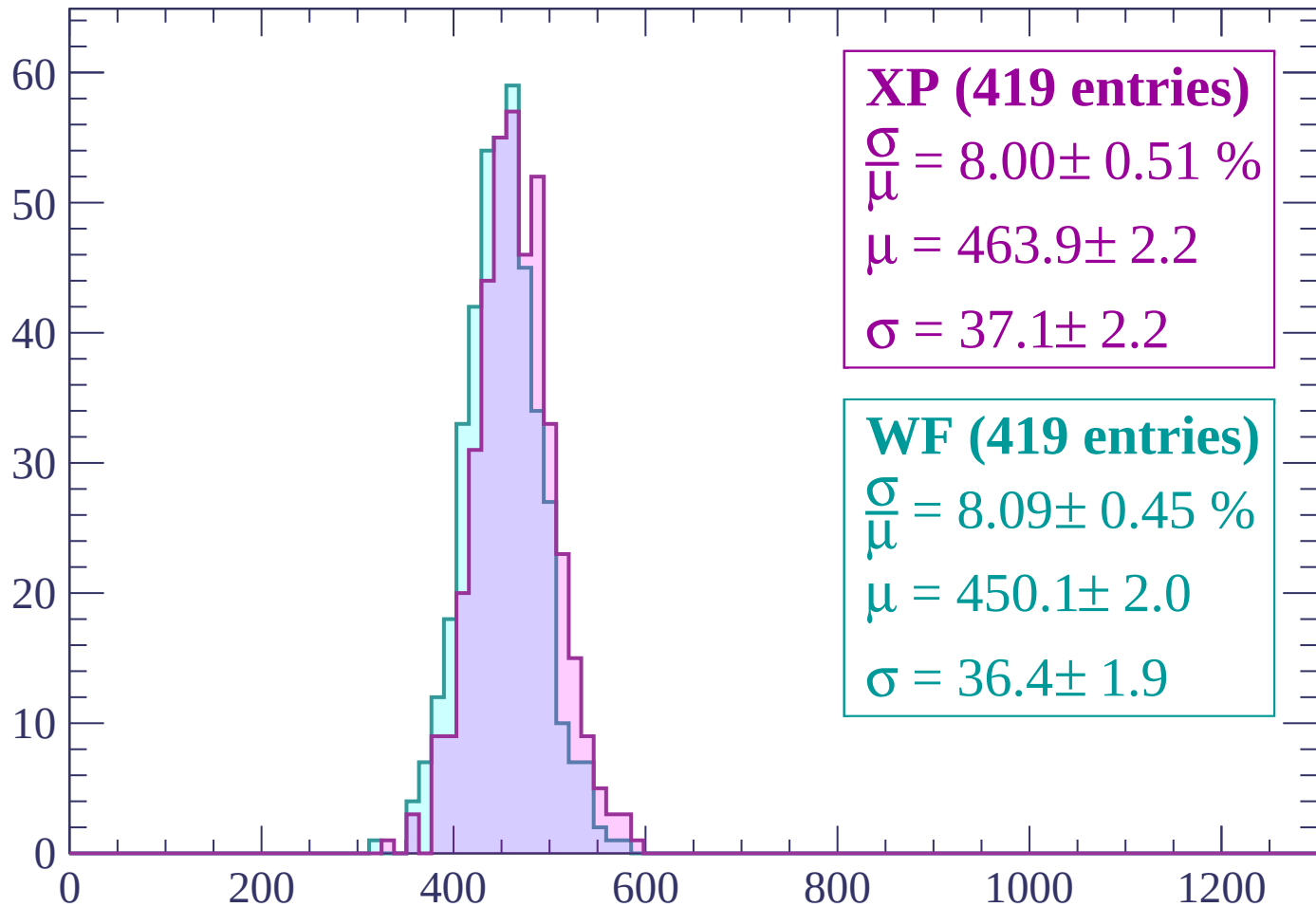
$$\mu = 448.9 \pm 2.2$$

$$\sigma = 37.3 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP (419 entries)

$\frac{\sigma}{\mu} = 8.00 \pm 0.51 \%$

$\mu = 463.9 \pm 2.2$

$\sigma = 37.1 \pm 2.2$

WF (419 entries)

$\frac{\sigma}{\mu} = 8.09 \pm 0.45 \%$

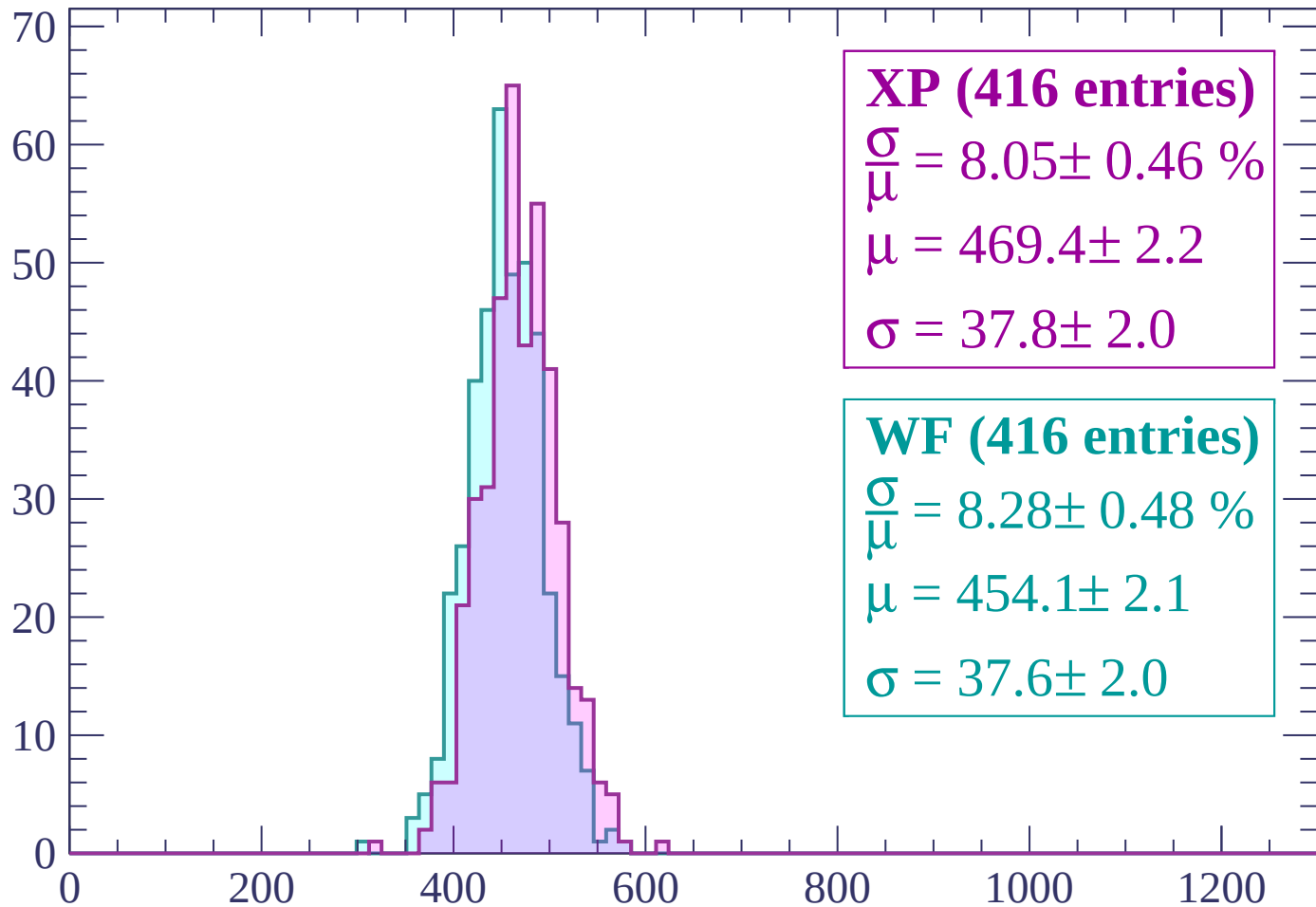
$\mu = 450.1 \pm 2.0$

$\sigma = 36.4 \pm 1.9$

dE/dx (ADC counts/cm)

Energy loss | $1500 < p < 1560$

Count



XP (416 entries)

$$\frac{\sigma}{\mu} = 8.05 \pm 0.46 \%$$

$$\mu = 469.4 \pm 2.2$$

$$\sigma = 37.8 \pm 2.0$$

WF (416 entries)

$$\frac{\sigma}{\mu} = 8.28 \pm 0.48 \%$$

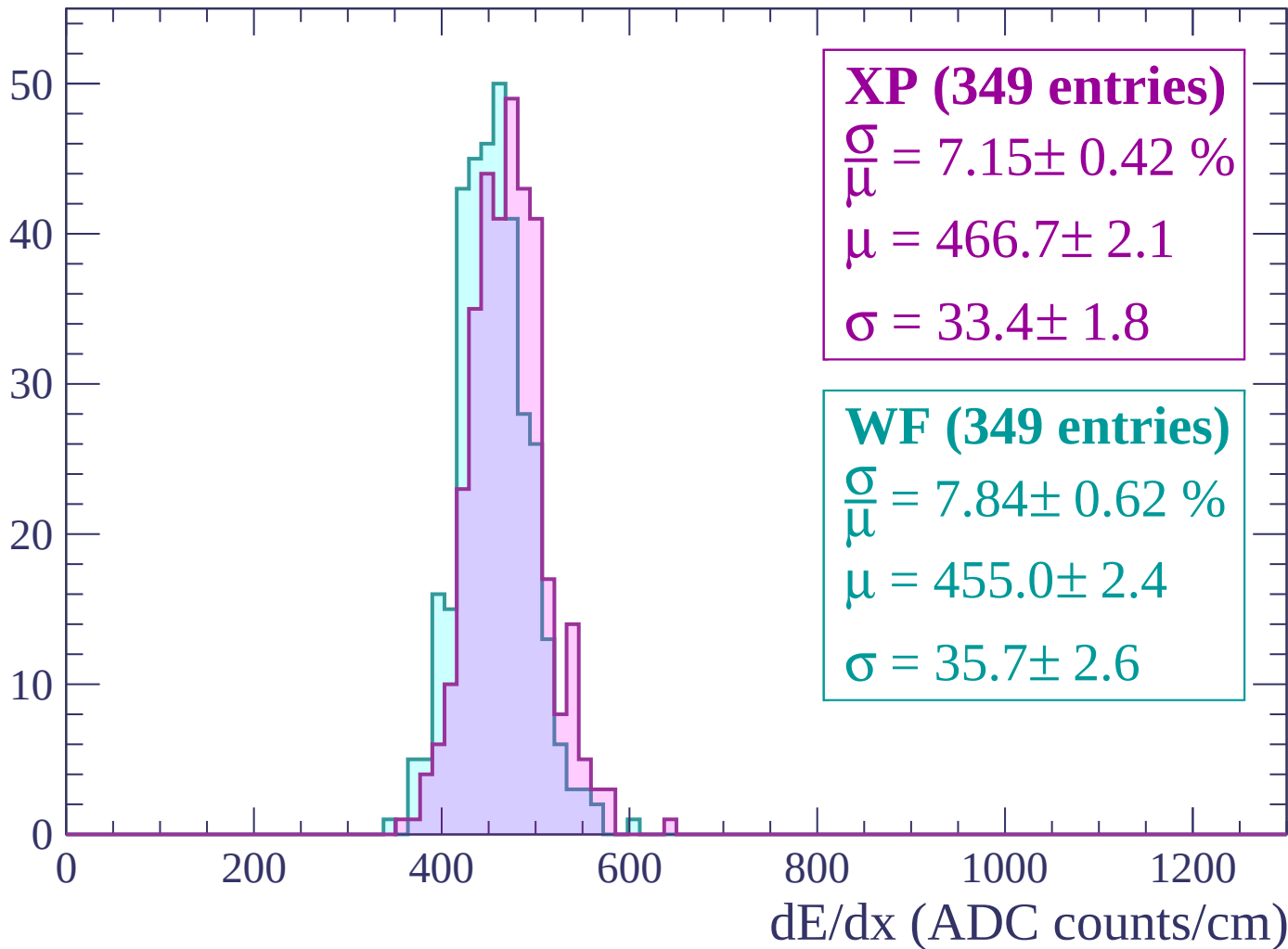
$$\mu = 454.1 \pm 2.1$$

$$\sigma = 37.6 \pm 2.0$$

dE/dx (ADC counts/cm)

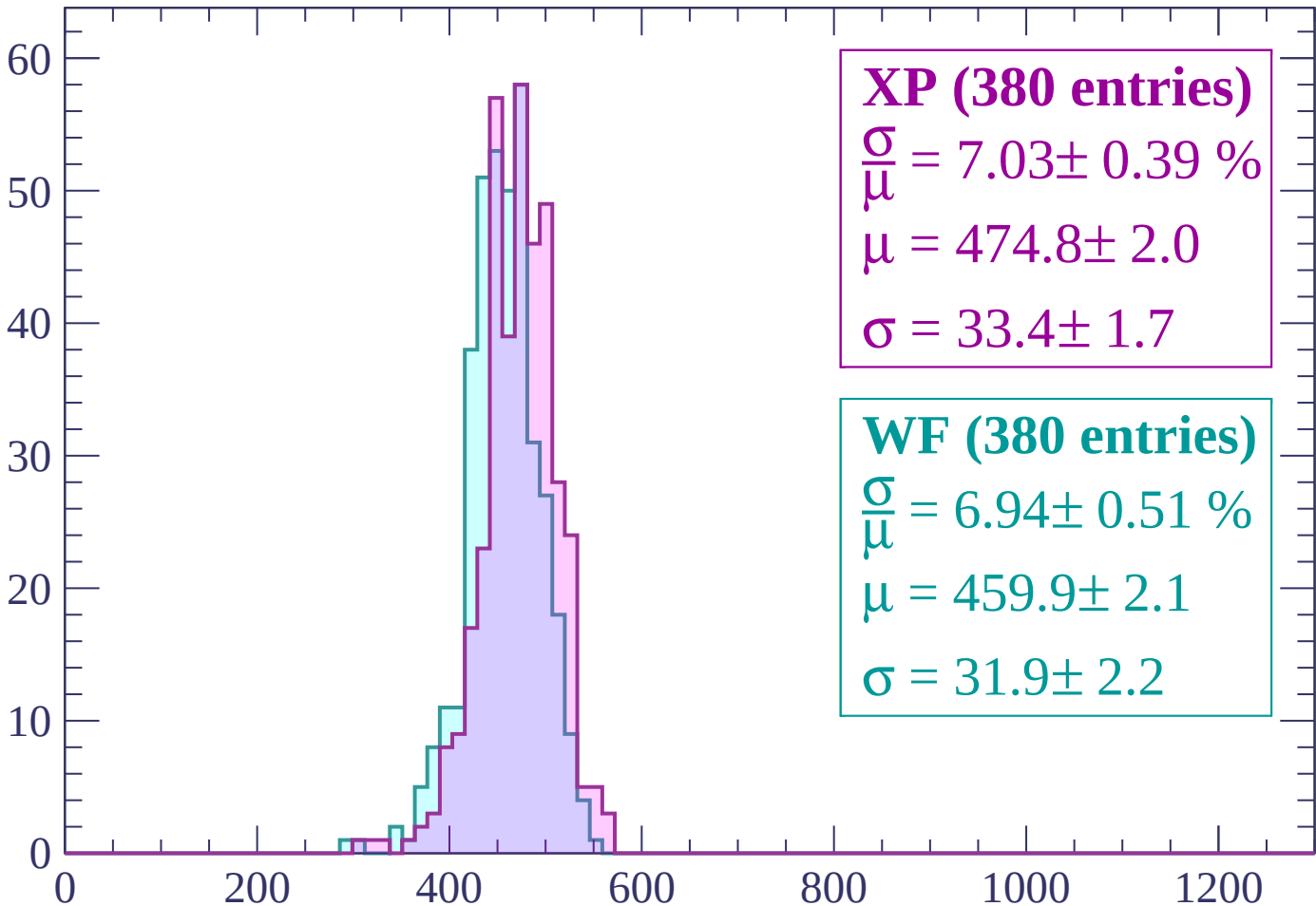
Energy loss | $1560 < p < 1620$

Count



Energy loss | $1620 < p < 1680$

Count



XP (380 entries)

$\frac{\sigma}{\mu} = 7.03 \pm 0.39 \%$

$\mu = 474.8 \pm 2.0$

$\sigma = 33.4 \pm 1.7$

WF (380 entries)

$\frac{\sigma}{\mu} = 6.94 \pm 0.51 \%$

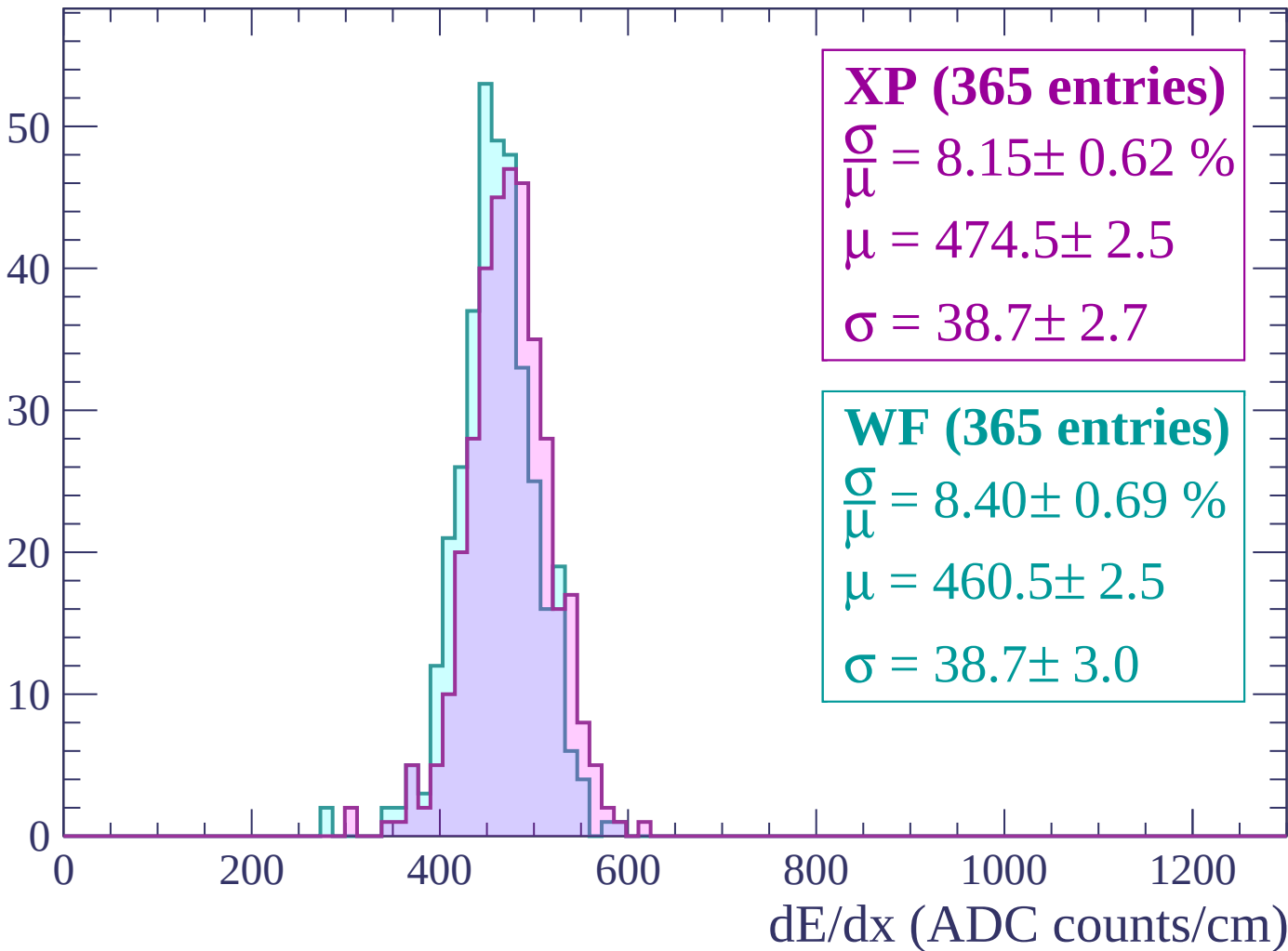
$\mu = 459.9 \pm 2.1$

$\sigma = 31.9 \pm 2.2$

dE/dx (ADC counts/cm)

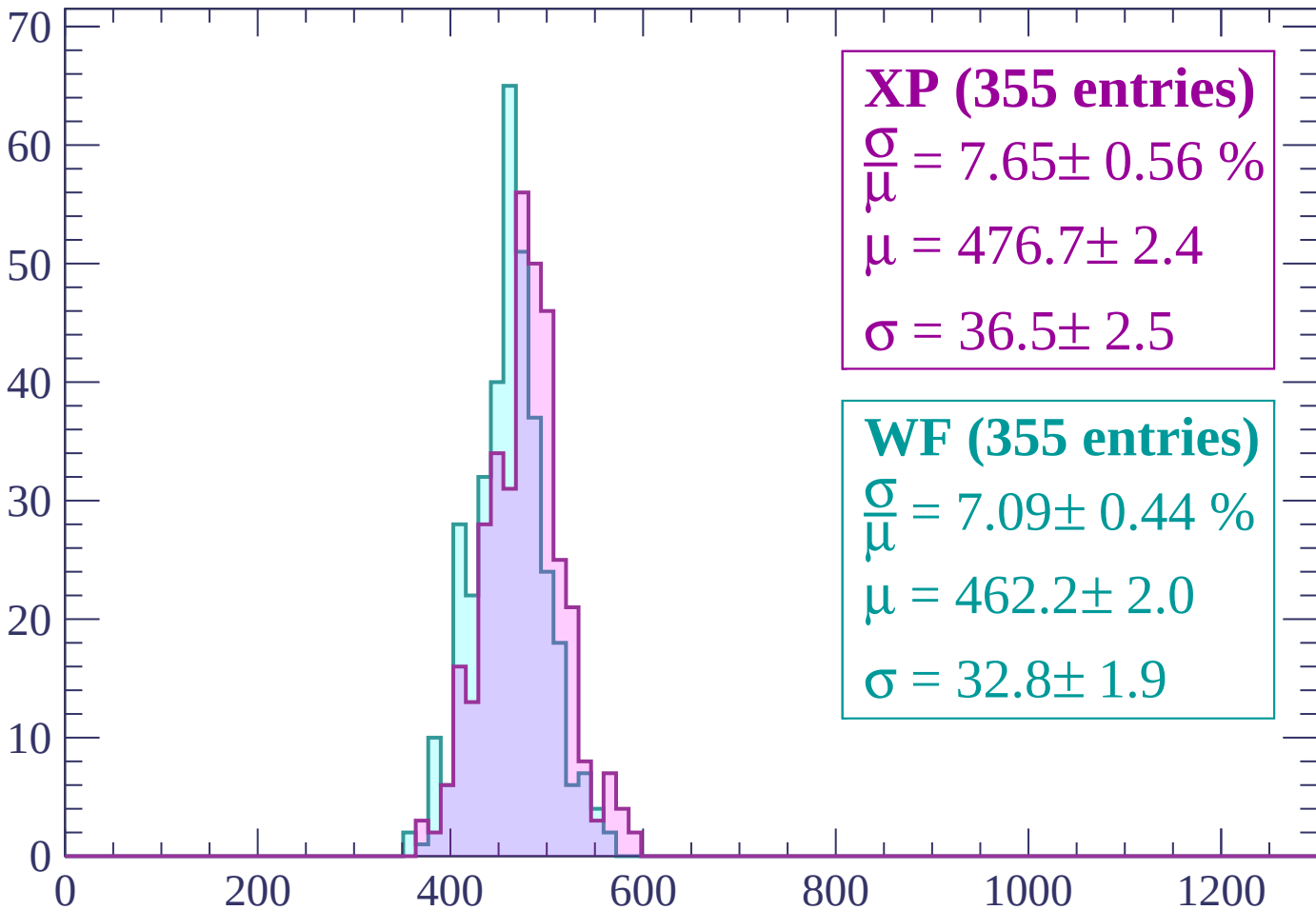
Energy loss | $1680 < p < 1740$

Count



Energy loss | $1740 < p < 1800$

Count



XP (355 entries)

$$\frac{\sigma}{\mu} = 7.65 \pm 0.56 \%$$

$$\mu = 476.7 \pm 2.4$$

$$\sigma = 36.5 \pm 2.5$$

WF (355 entries)

$$\frac{\sigma}{\mu} = 7.09 \pm 0.44 \%$$

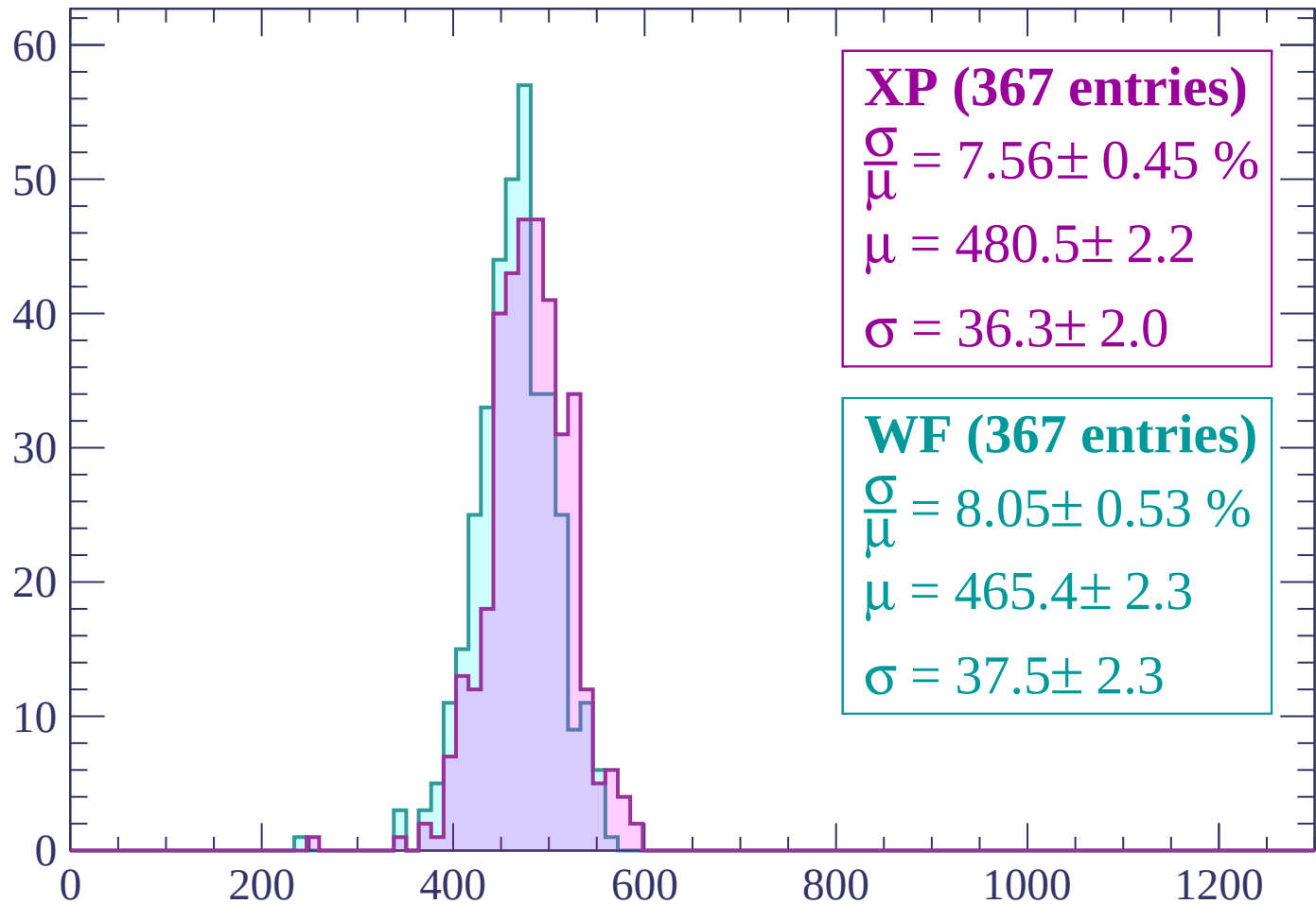
$$\mu = 462.2 \pm 2.0$$

$$\sigma = 32.8 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1860$

Count



XP (367 entries)

$\frac{\sigma}{\mu} = 7.56 \pm 0.45 \%$

$\mu = 480.5 \pm 2.2$

$\sigma = 36.3 \pm 2.0$

WF (367 entries)

$\frac{\sigma}{\mu} = 8.05 \pm 0.53 \%$

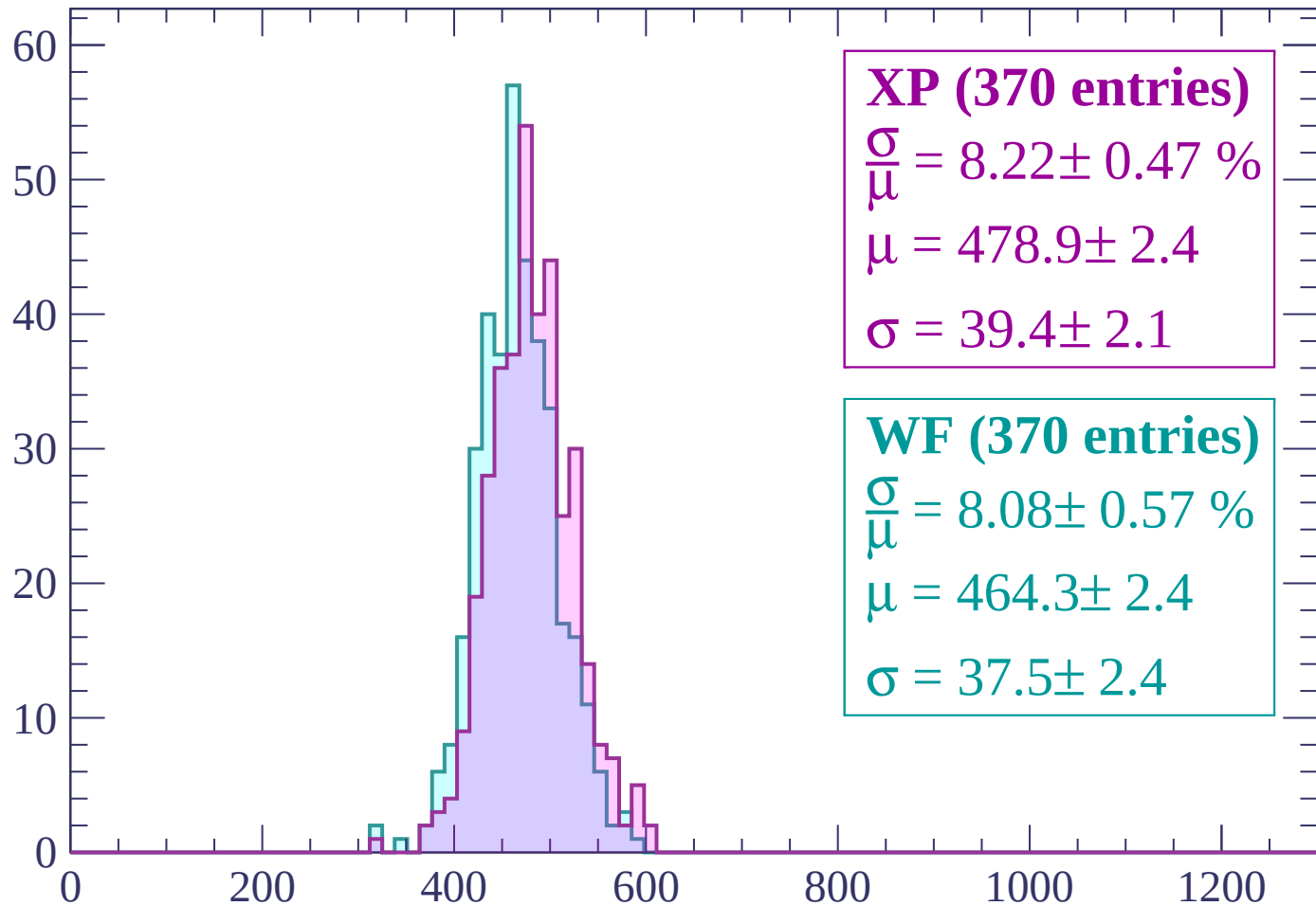
$\mu = 465.4 \pm 2.3$

$\sigma = 37.5 \pm 2.3$

dE/dx (ADC counts/cm)

Energy loss | $1860 < p < 1920$

Count



XP (370 entries)

$$\frac{\sigma}{\mu} = 8.22 \pm 0.47 \%$$

$$\mu = 478.9 \pm 2.4$$

$$\sigma = 39.4 \pm 2.1$$

WF (370 entries)

$$\frac{\sigma}{\mu} = 8.08 \pm 0.57 \%$$

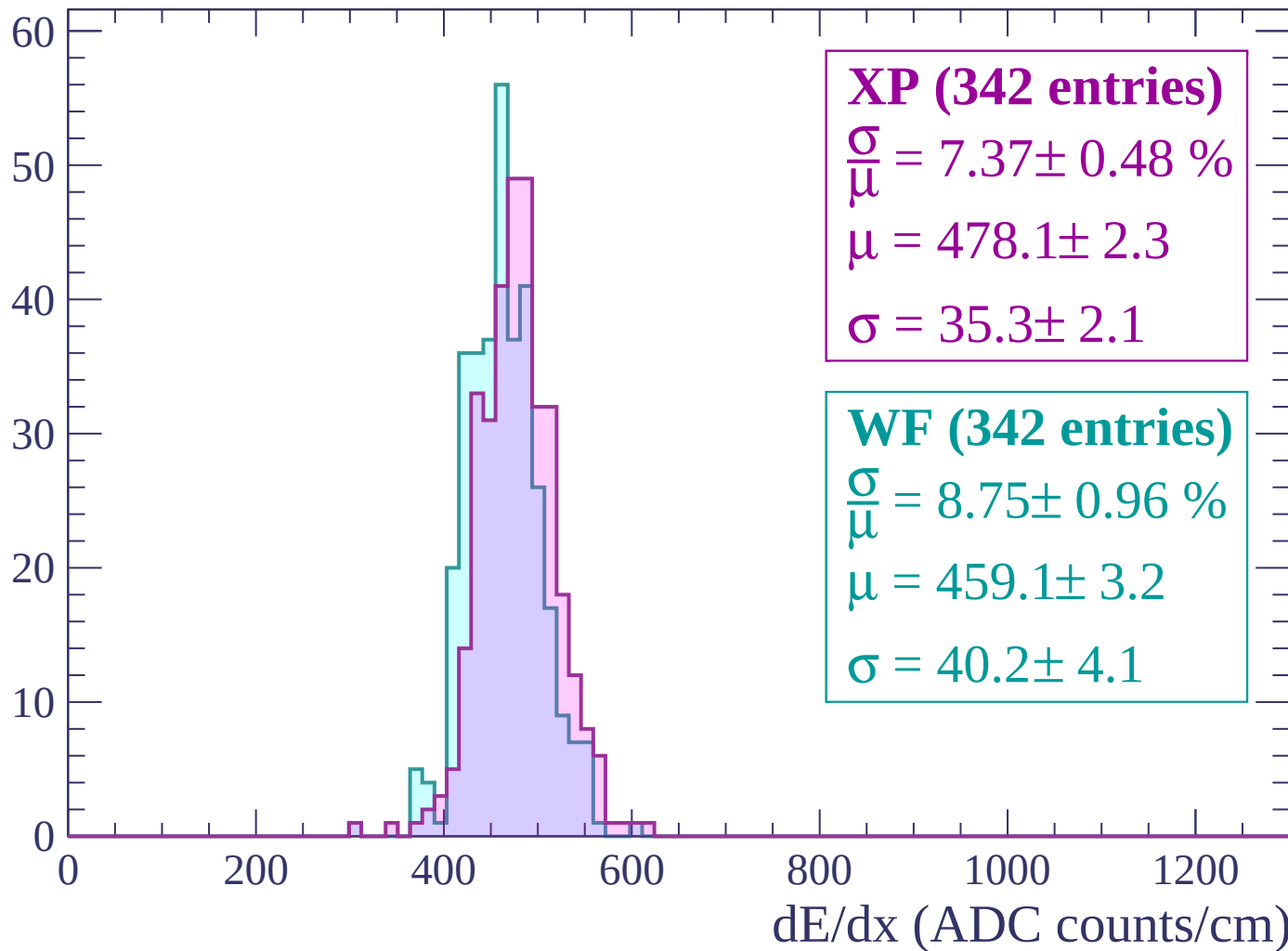
$$\mu = 464.3 \pm 2.4$$

$$\sigma = 37.5 \pm 2.4$$

dE/dx (ADC counts/cm)

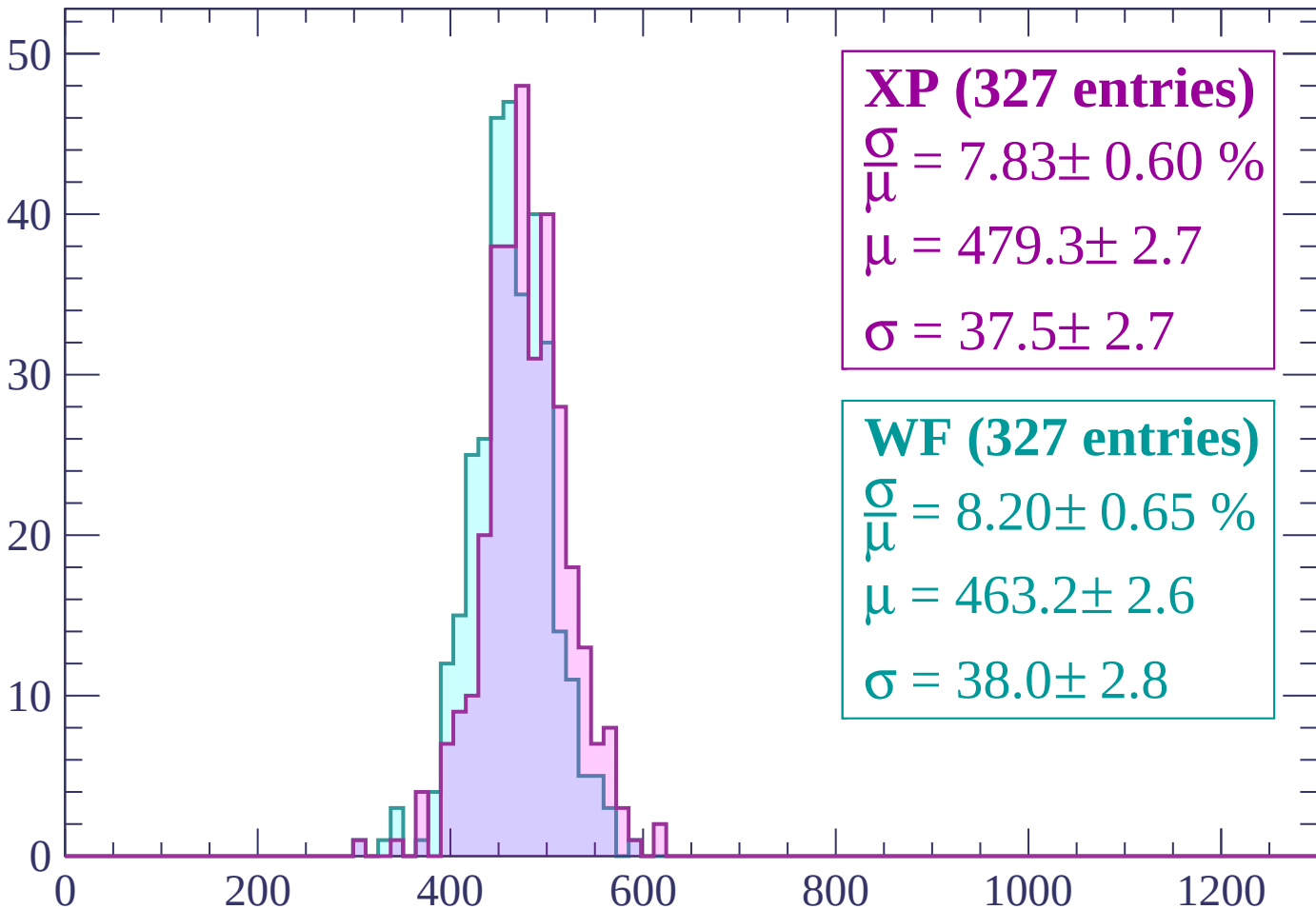
Energy loss | $1920 < p < 1980$

Count



Energy loss | $1980 < p < 2040$

Count



XP (327 entries)

$\frac{\sigma}{\mu} = 7.83 \pm 0.60 \%$

$\mu = 479.3 \pm 2.7$

$\sigma = 37.5 \pm 2.7$

WF (327 entries)

$\frac{\sigma}{\mu} = 8.20 \pm 0.65 \%$

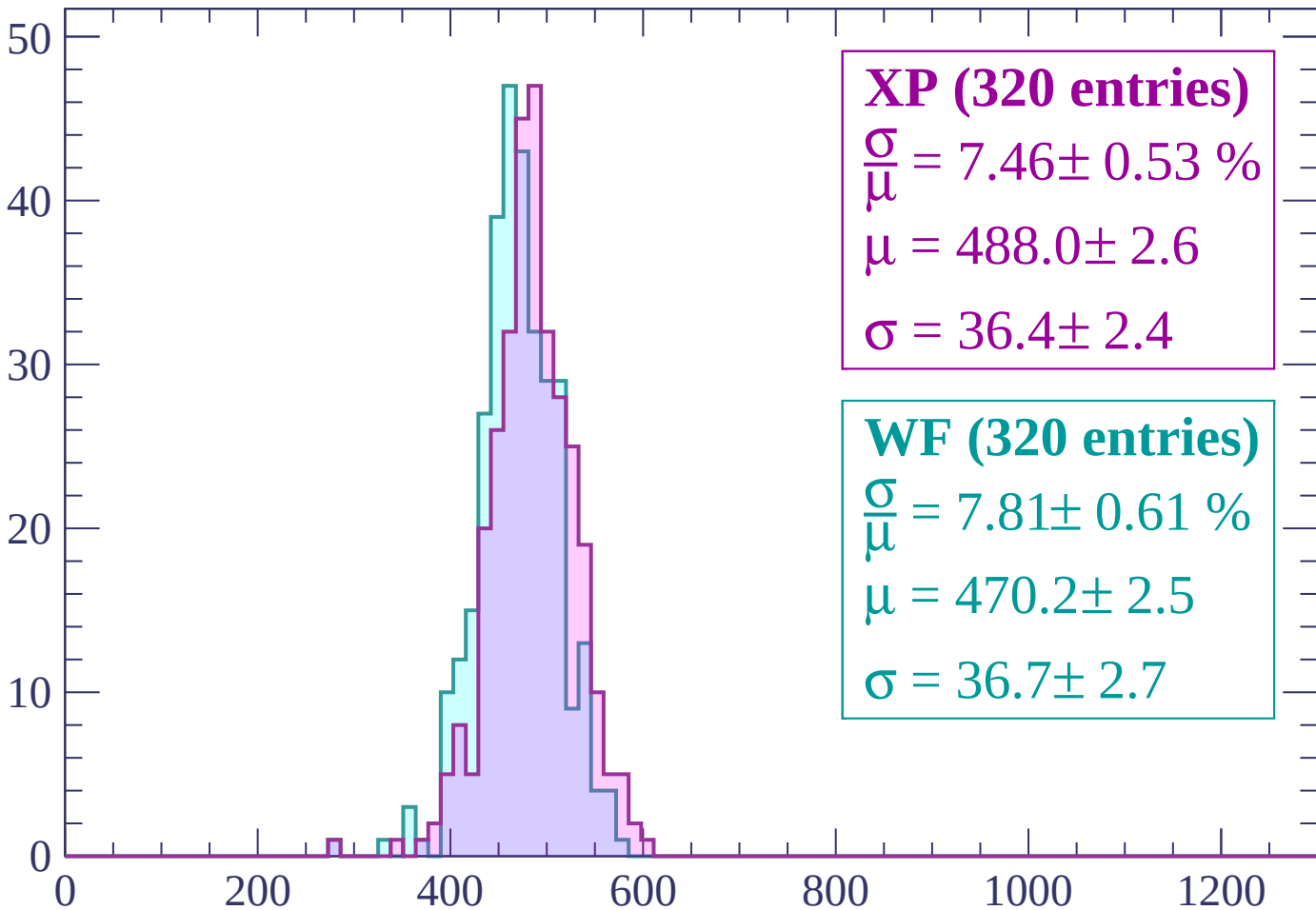
$\mu = 463.2 \pm 2.6$

$\sigma = 38.0 \pm 2.8$

dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

Count



XP (320 entries)

$\frac{\sigma}{\mu} = 7.46 \pm 0.53 \%$

$\mu = 488.0 \pm 2.6$

$\sigma = 36.4 \pm 2.4$

WF (320 entries)

$\frac{\sigma}{\mu} = 7.81 \pm 0.61 \%$

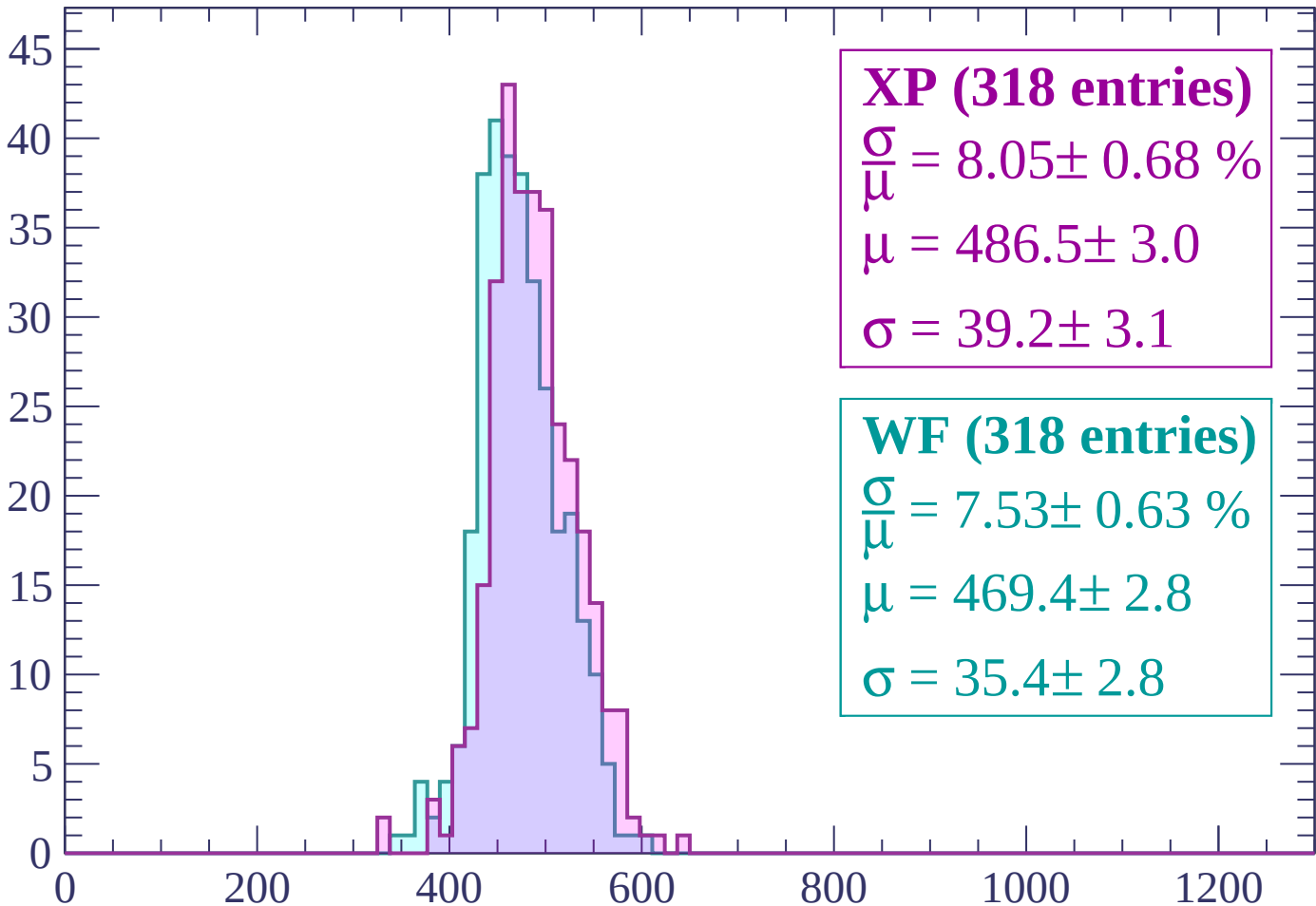
$\mu = 470.2 \pm 2.5$

$\sigma = 36.7 \pm 2.7$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP (318 entries)

$\frac{\sigma}{\mu} = 8.05 \pm 0.68 \%$

$\mu = 486.5 \pm 3.0$

$\sigma = 39.2 \pm 3.1$

WF (318 entries)

$\frac{\sigma}{\mu} = 7.53 \pm 0.63 \%$

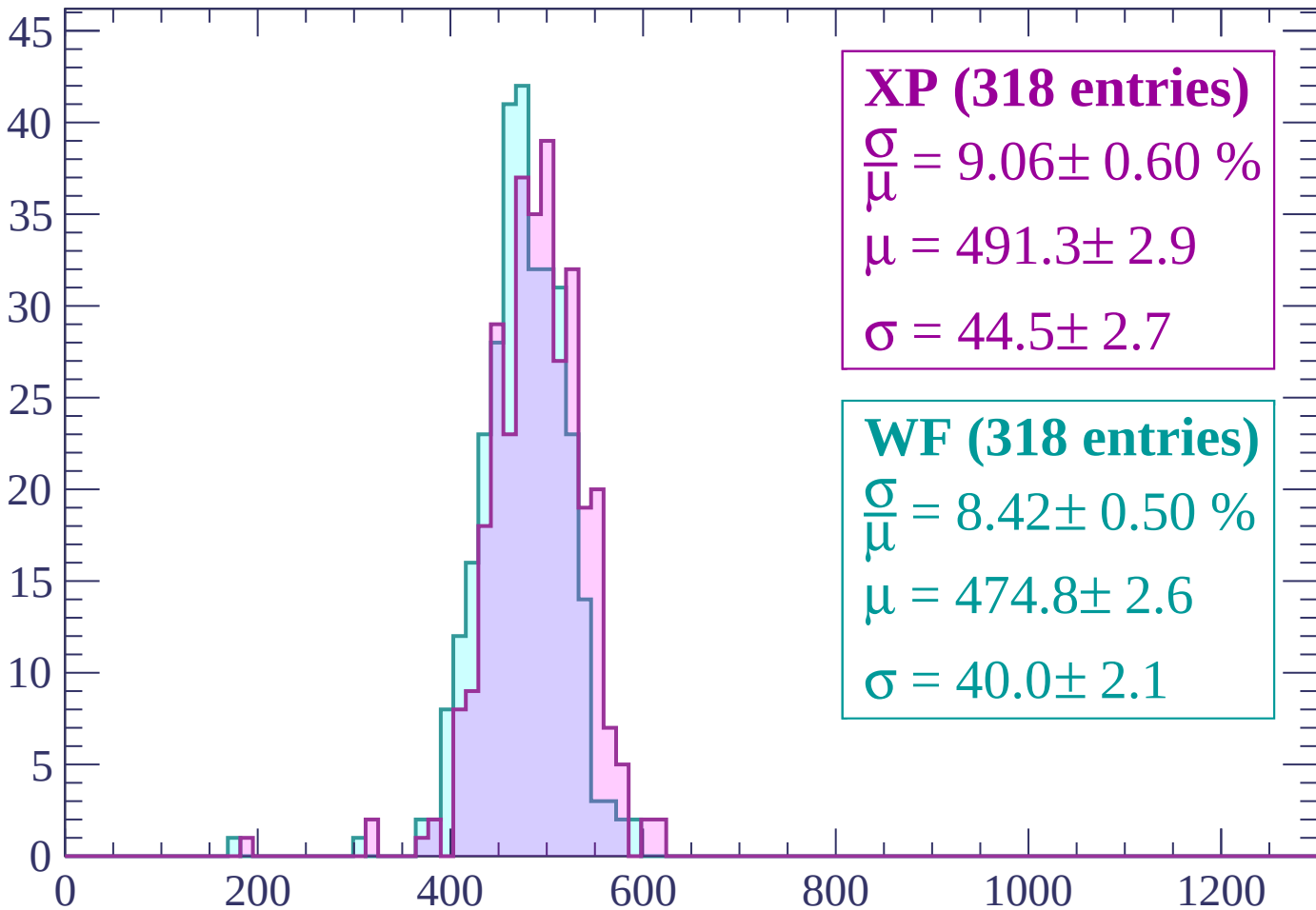
$\mu = 469.4 \pm 2.8$

$\sigma = 35.4 \pm 2.8$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP (318 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.60 \%$$

$$\mu = 491.3 \pm 2.9$$

$$\sigma = 44.5 \pm 2.7$$

WF (318 entries)

$$\frac{\sigma}{\mu} = 8.42 \pm 0.50 \%$$

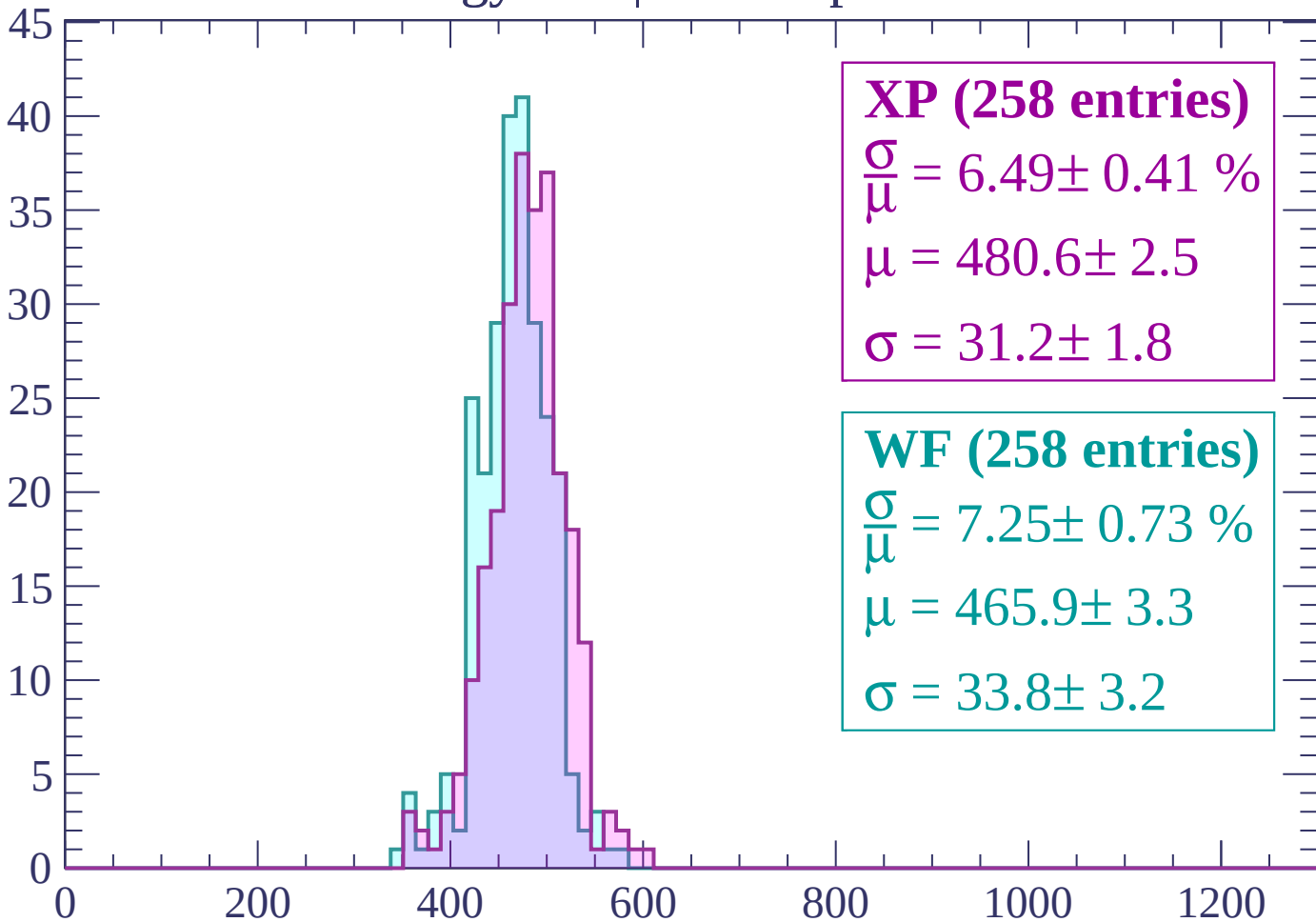
$$\mu = 474.8 \pm 2.6$$

$$\sigma = 40.0 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP (258 entries)

$\frac{\sigma}{\mu} = 6.49 \pm 0.41 \%$

$\mu = 480.6 \pm 2.5$

$\sigma = 31.2 \pm 1.8$

WF (258 entries)

$\frac{\sigma}{\mu} = 7.25 \pm 0.73 \%$

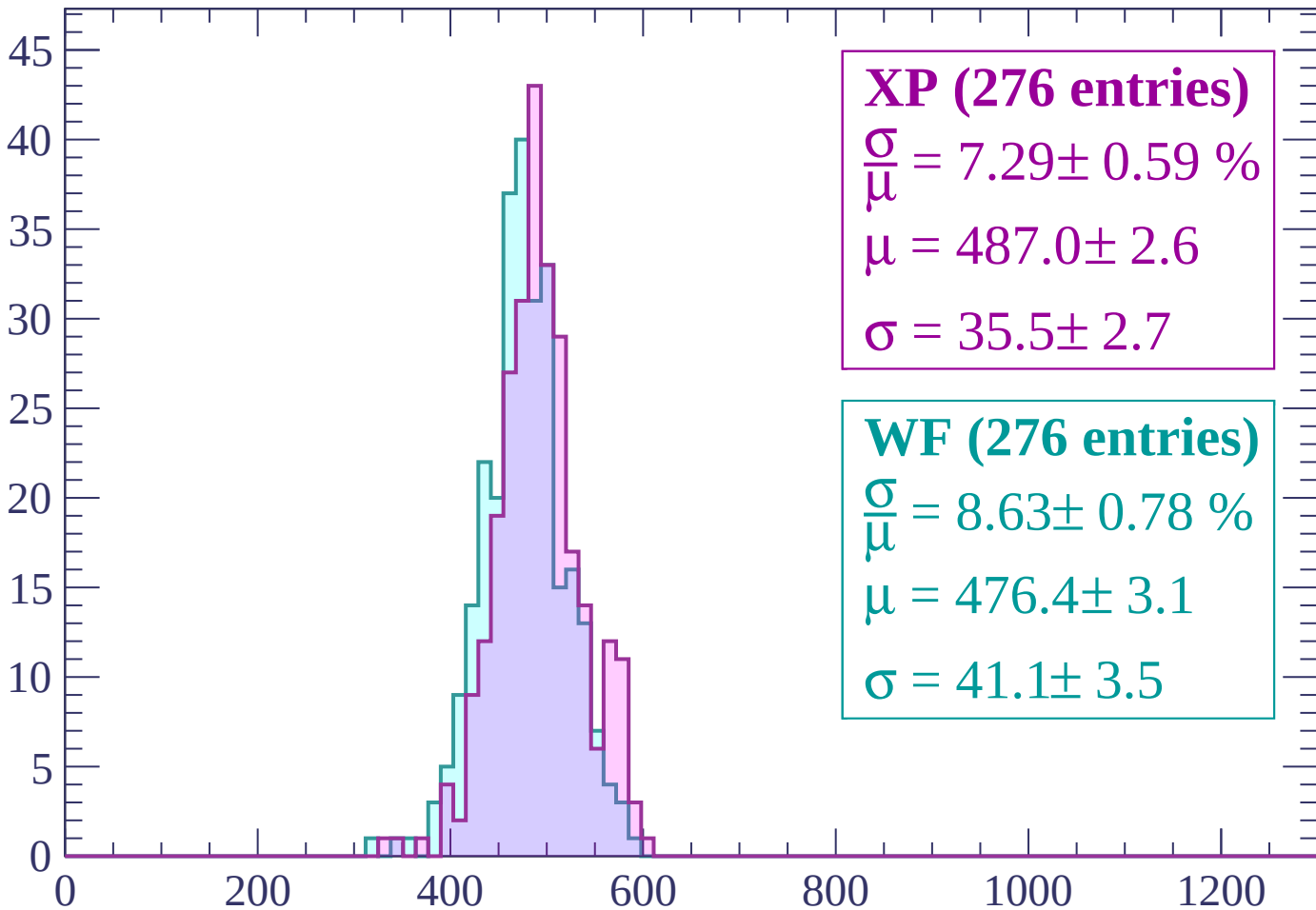
$\mu = 465.9 \pm 3.3$

$\sigma = 33.8 \pm 3.2$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

Count



XP (276 entries)

$\frac{\sigma}{\mu} = 7.29 \pm 0.59 \%$

$\mu = 487.0 \pm 2.6$

$\sigma = 35.5 \pm 2.7$

WF (276 entries)

$\frac{\sigma}{\mu} = 8.63 \pm 0.78 \%$

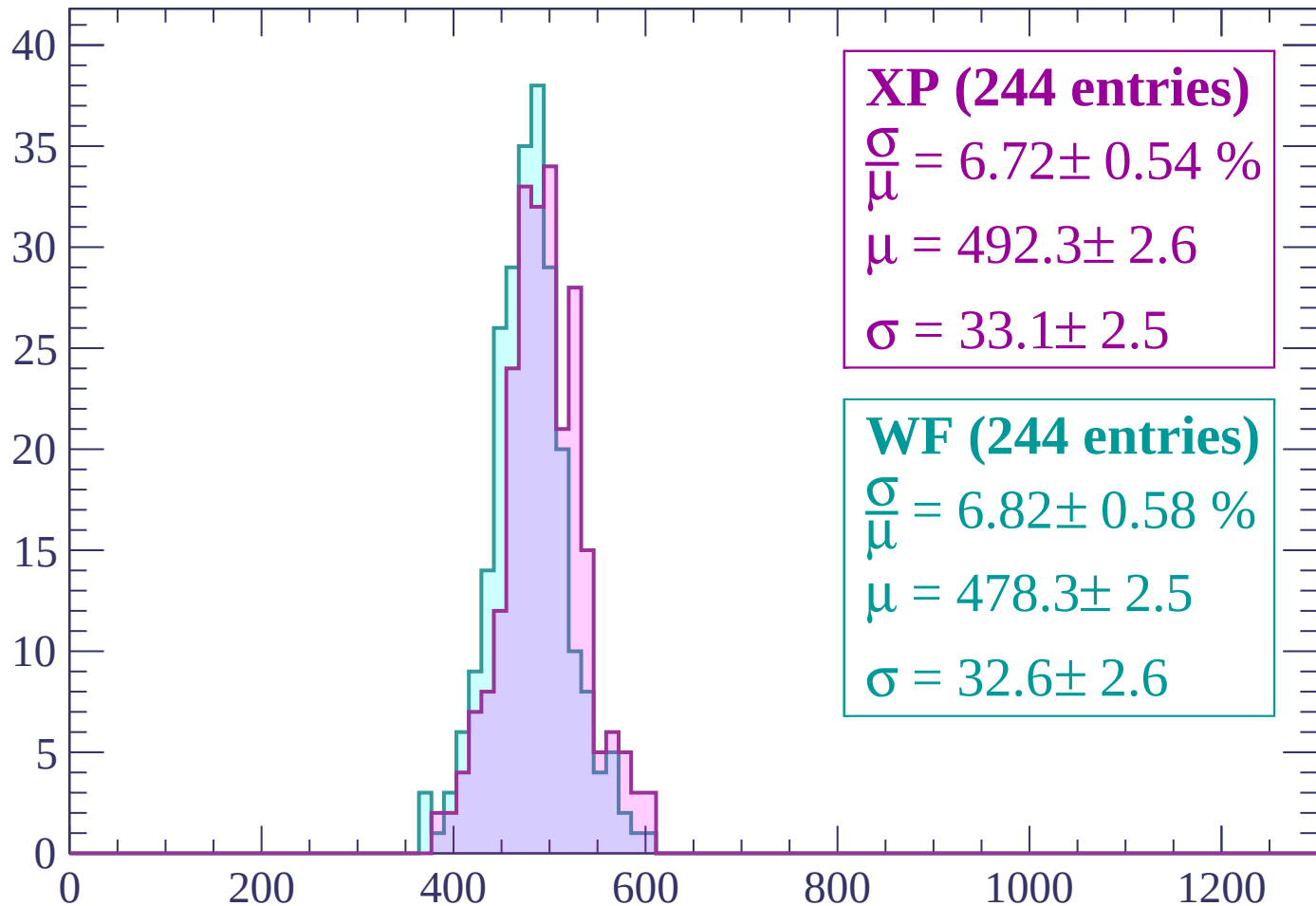
$\mu = 476.4 \pm 3.1$

$\sigma = 41.1 \pm 3.5$

dE/dx (ADC counts/cm)

Energy loss | $2340 < p < 2400$

Count



XP (244 entries)

$\frac{\sigma}{\mu} = 6.72 \pm 0.54 \%$

$\mu = 492.3 \pm 2.6$

$\sigma = 33.1 \pm 2.5$

WF (244 entries)

$\frac{\sigma}{\mu} = 6.82 \pm 0.58 \%$

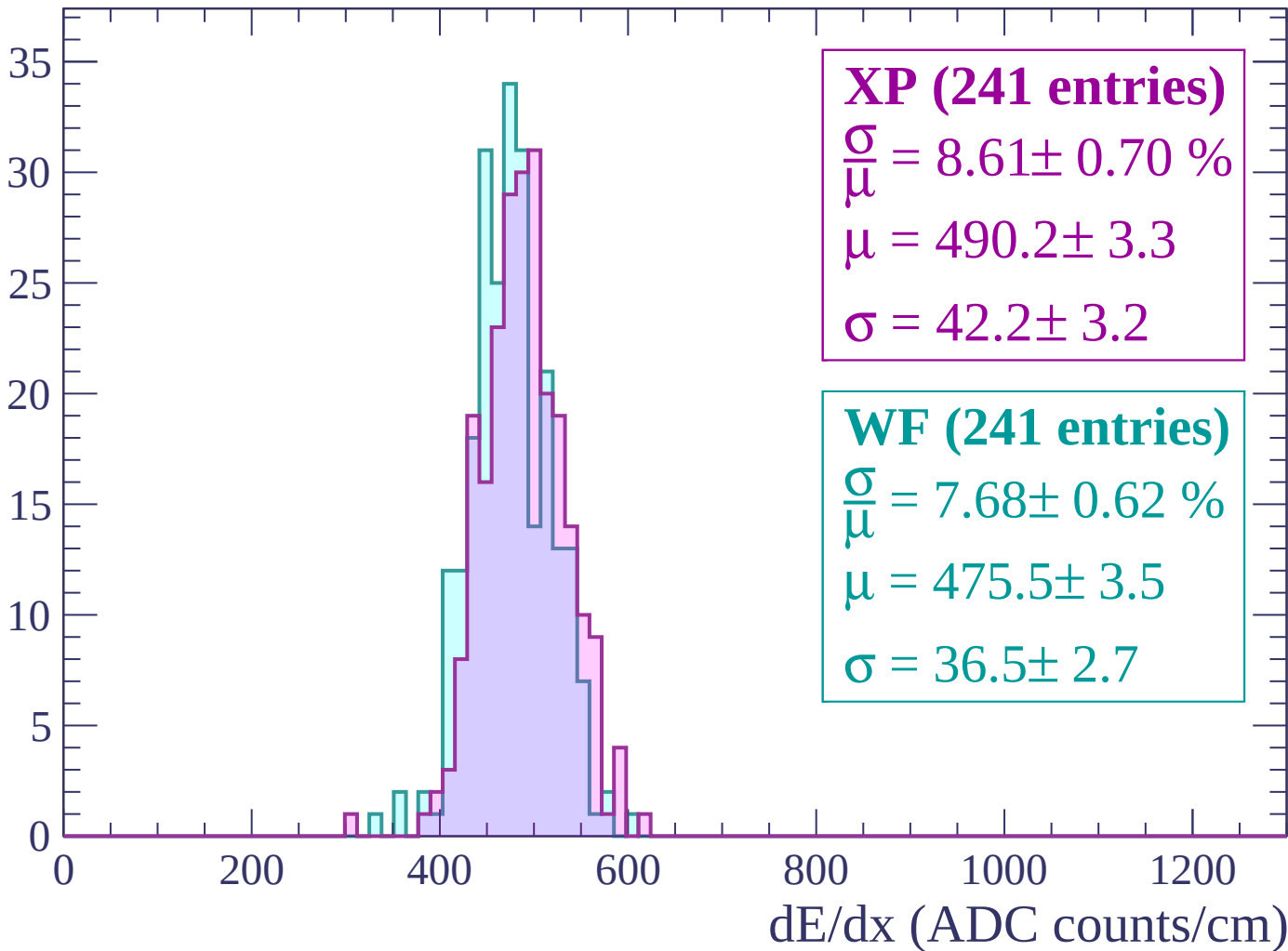
$\mu = 478.3 \pm 2.5$

$\sigma = 32.6 \pm 2.6$

dE/dx (ADC counts/cm)

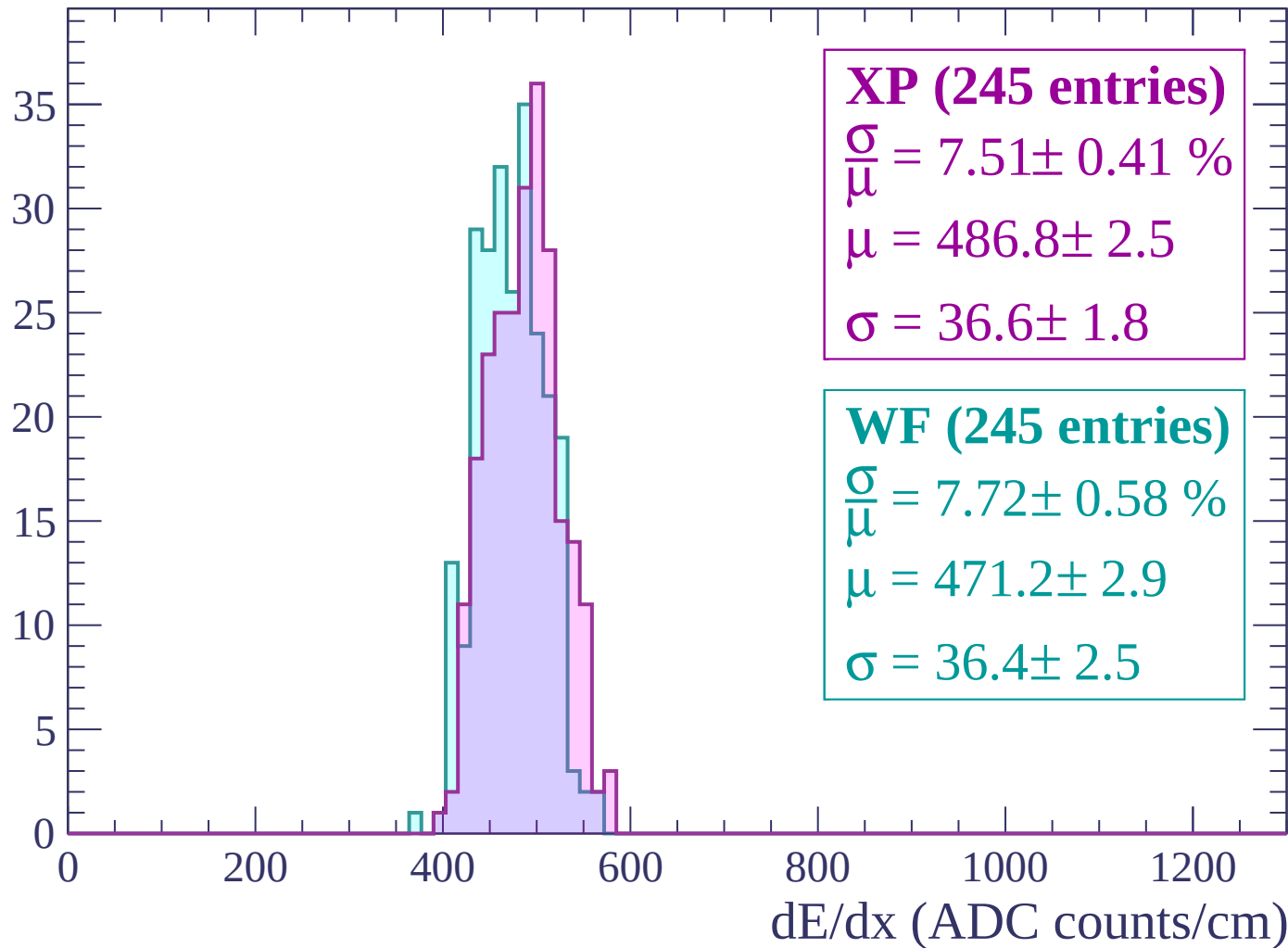
Energy loss | $2400 < p < 2460$

Count



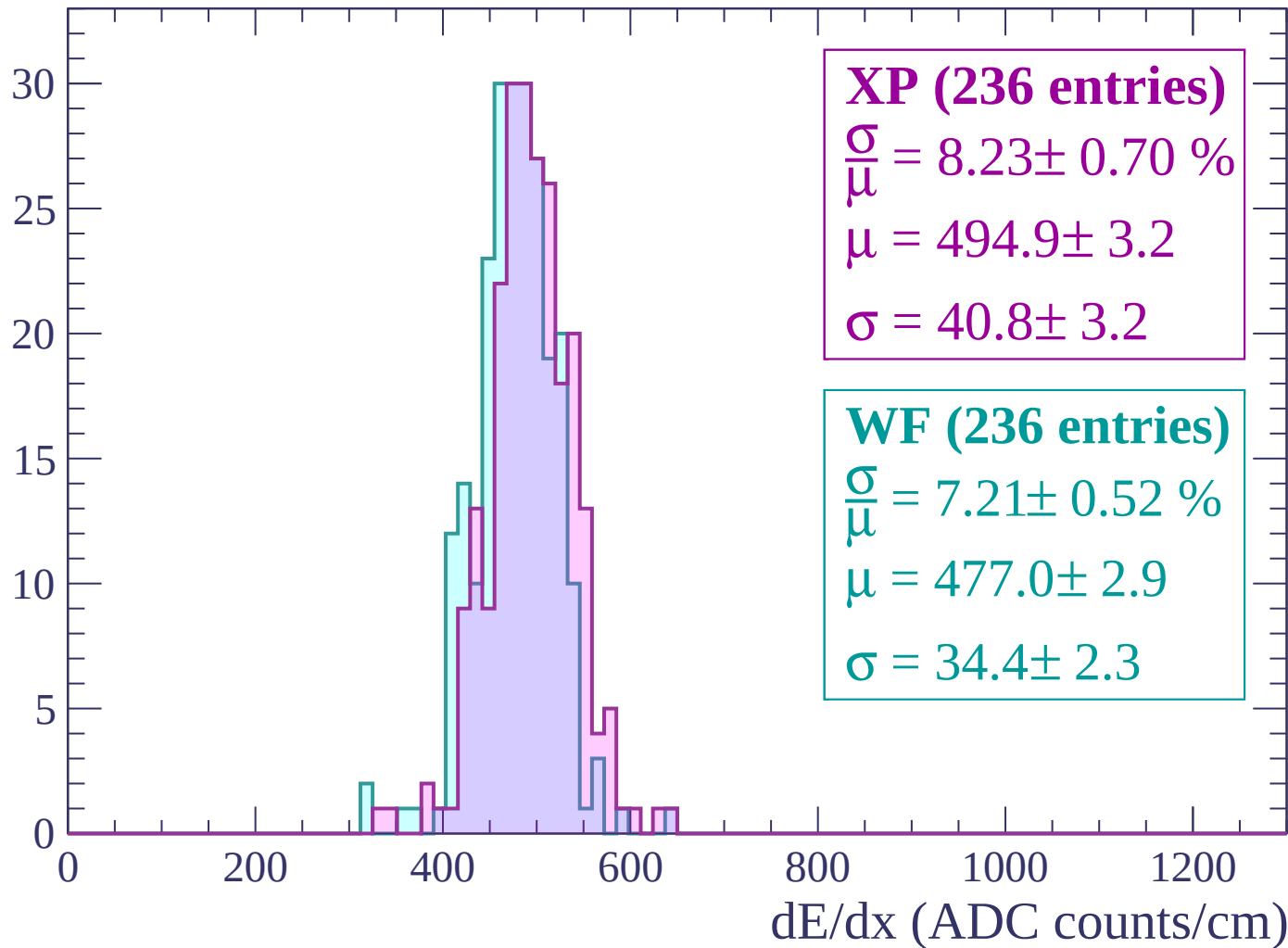
Energy loss | $2460 < p < 2520$

Count



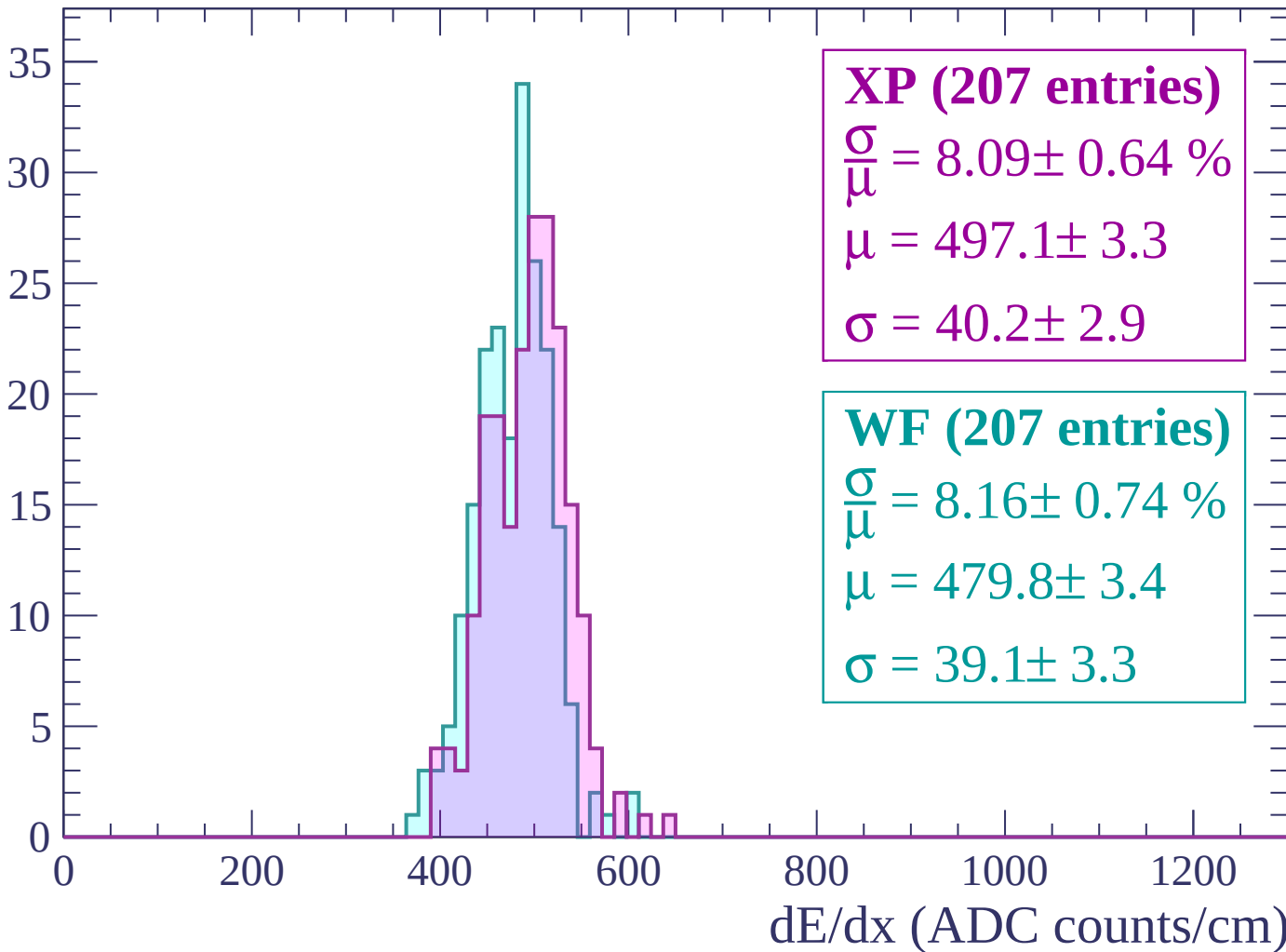
Energy loss | $2520 < p < 2580$

Count



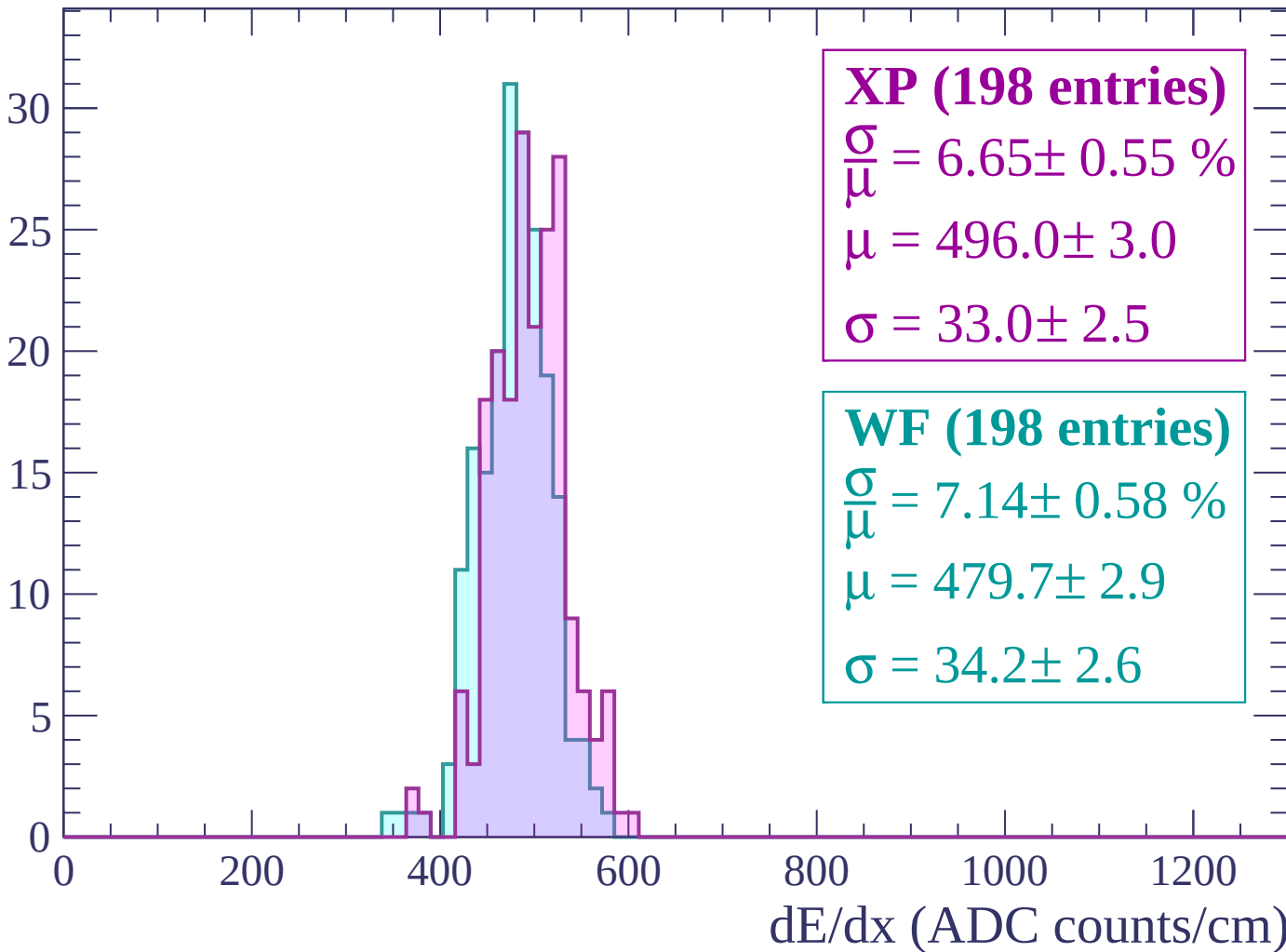
Energy loss | $2580 < p < 2640$

Count



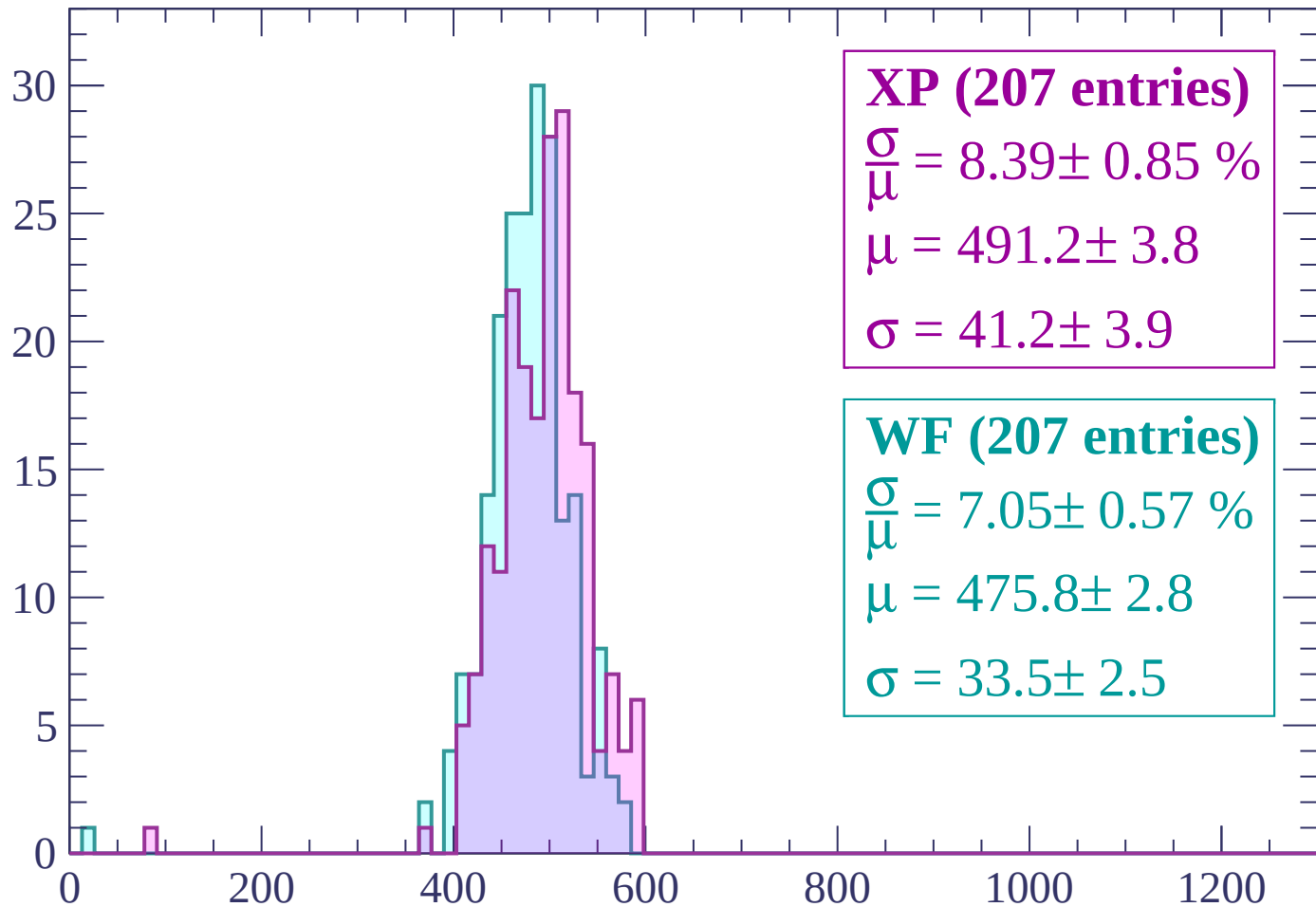
Energy loss | $2640 < p < 2700$

Count



Energy loss | $2700 < p < 2760$

Count



XP (207 entries)

$\frac{\sigma}{\mu} = 8.39 \pm 0.85 \%$

$\mu = 491.2 \pm 3.8$

$\sigma = 41.2 \pm 3.9$

WF (207 entries)

$\frac{\sigma}{\mu} = 7.05 \pm 0.57 \%$

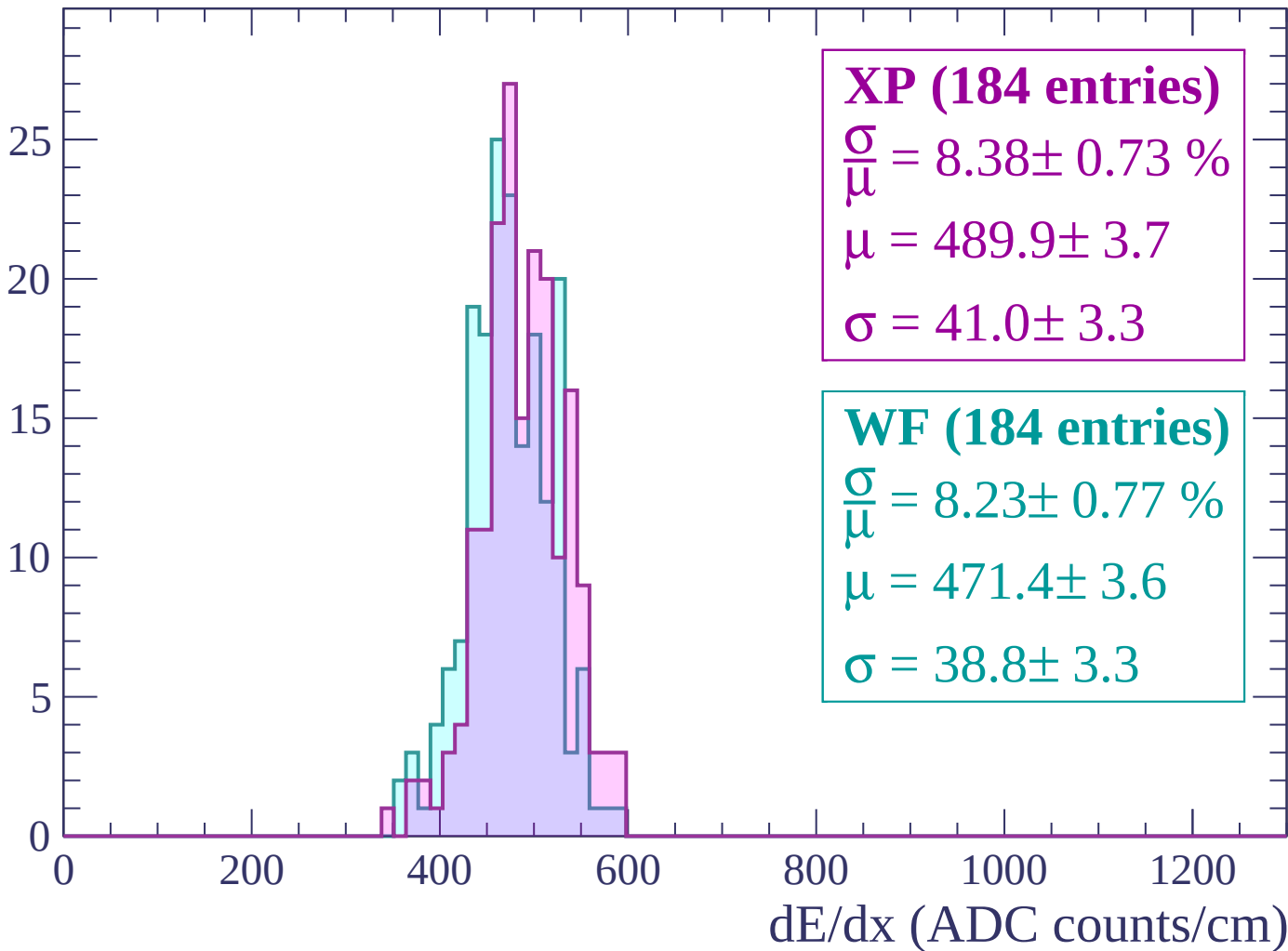
$\mu = 475.8 \pm 2.8$

$\sigma = 33.5 \pm 2.5$

dE/dx (ADC counts/cm)

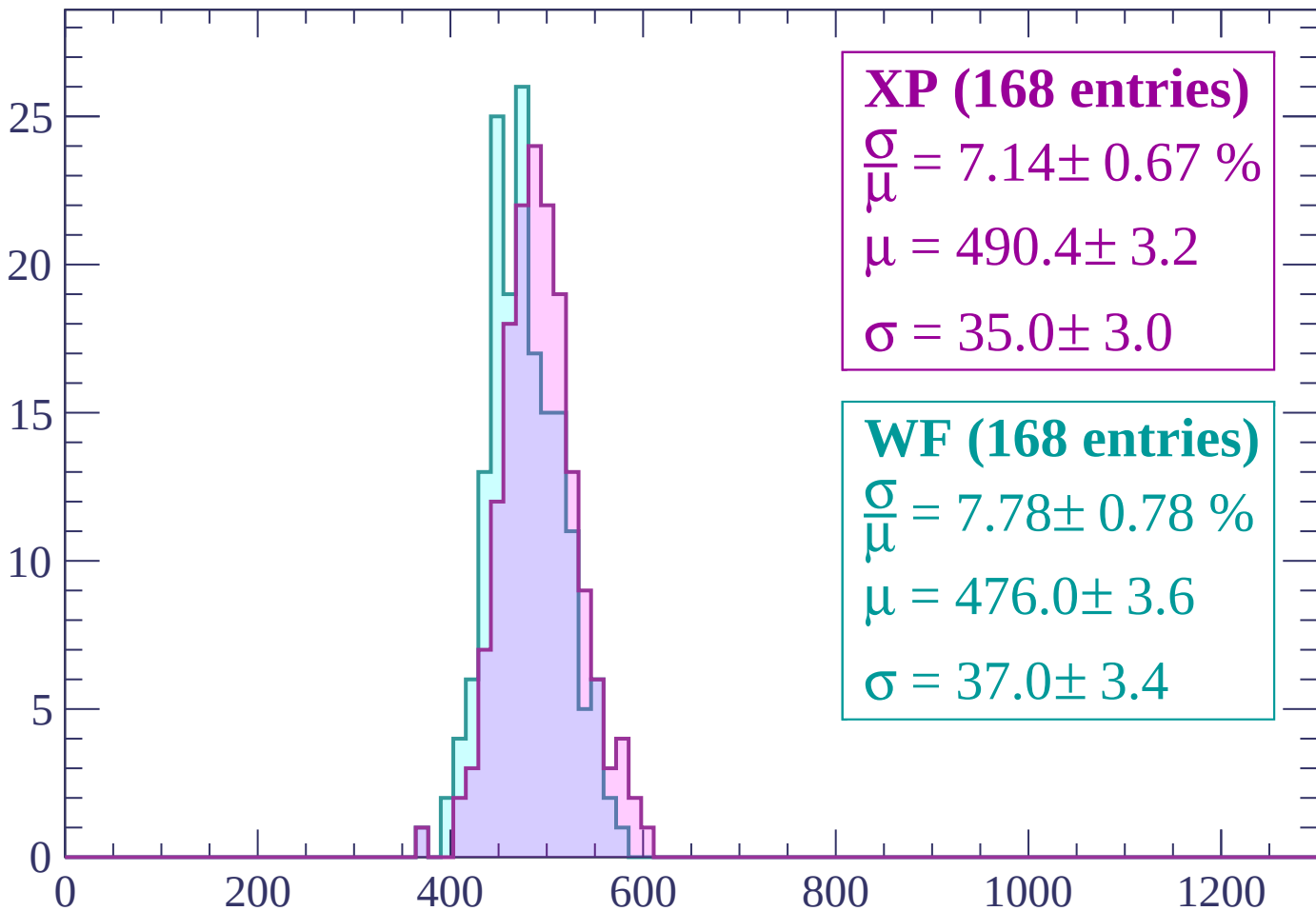
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP (168 entries)

$$\frac{\sigma}{\mu} = 7.14 \pm 0.67 \%$$

$$\mu = 490.4 \pm 3.2$$

$$\sigma = 35.0 \pm 3.0$$

WF (168 entries)

$$\frac{\sigma}{\mu} = 7.78 \pm 0.78 \%$$

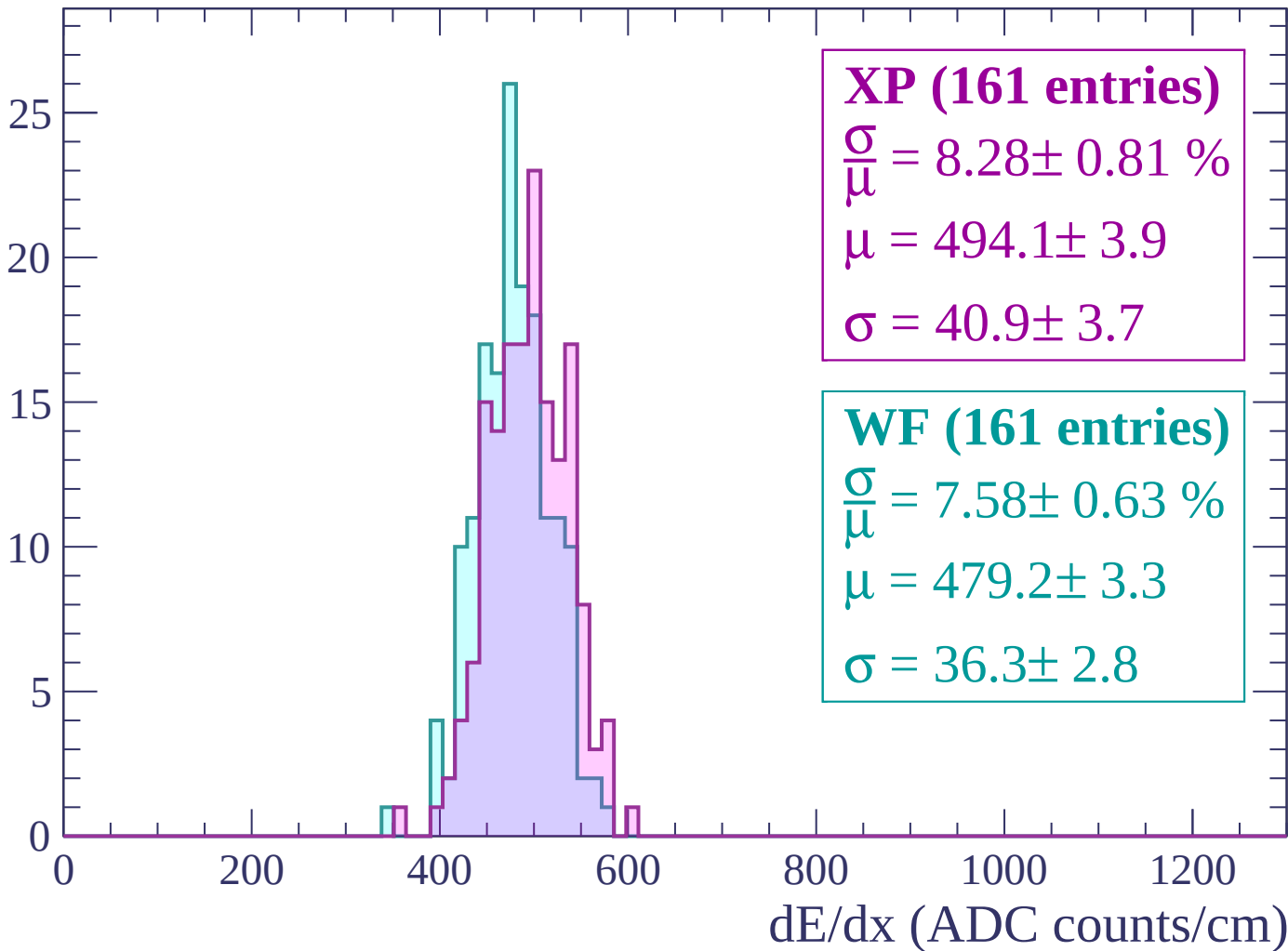
$$\mu = 476.0 \pm 3.6$$

$$\sigma = 37.0 \pm 3.4$$

dE/dx (ADC counts/cm)

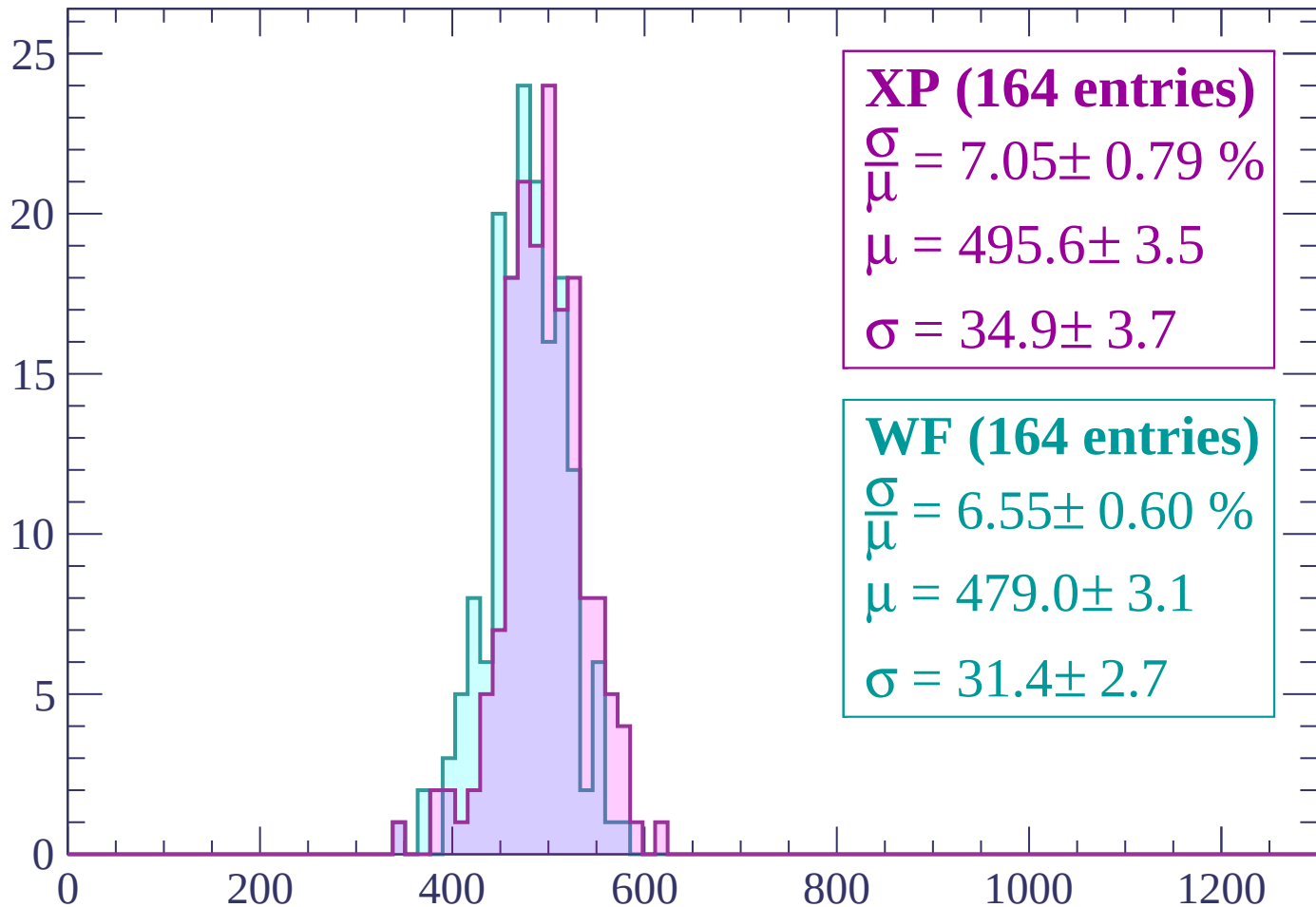
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



XP (164 entries)

$\frac{\sigma}{\mu} = 7.05 \pm 0.79 \%$

$\mu = 495.6 \pm 3.5$

$\sigma = 34.9 \pm 3.7$

WF (164 entries)

$\frac{\sigma}{\mu} = 6.55 \pm 0.60 \%$

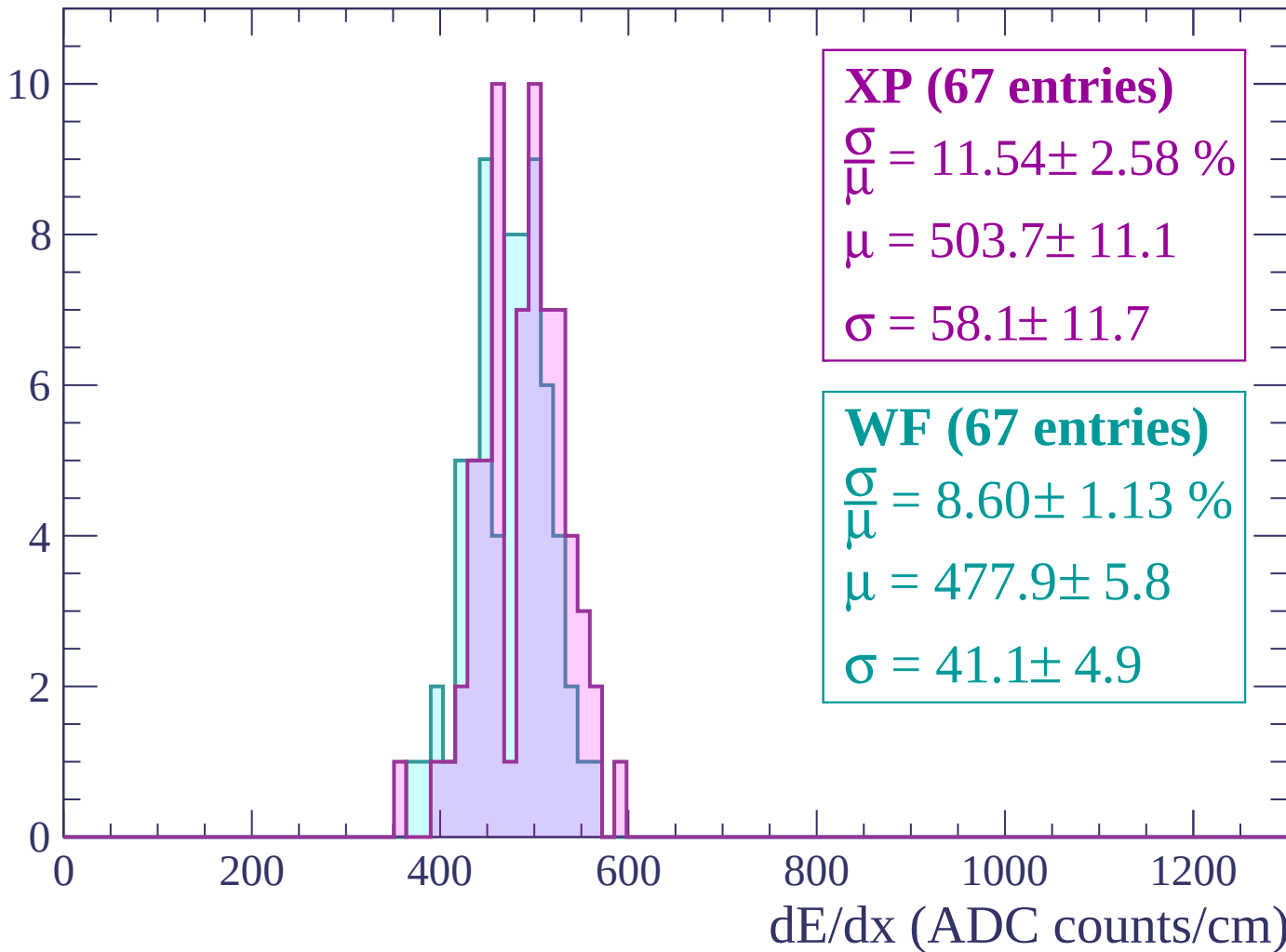
$\mu = 479.0 \pm 3.1$

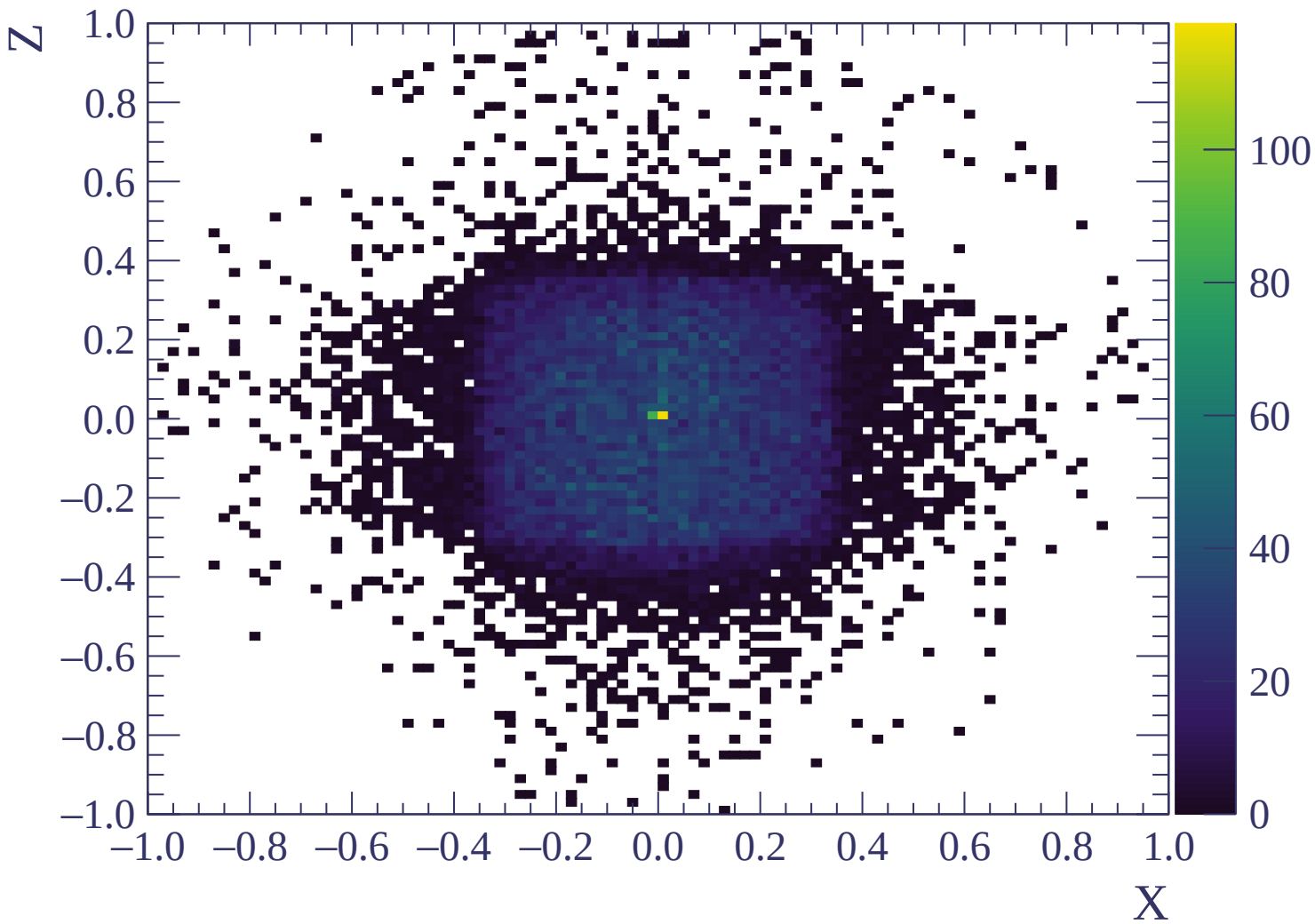
$\sigma = 31.4 \pm 2.7$

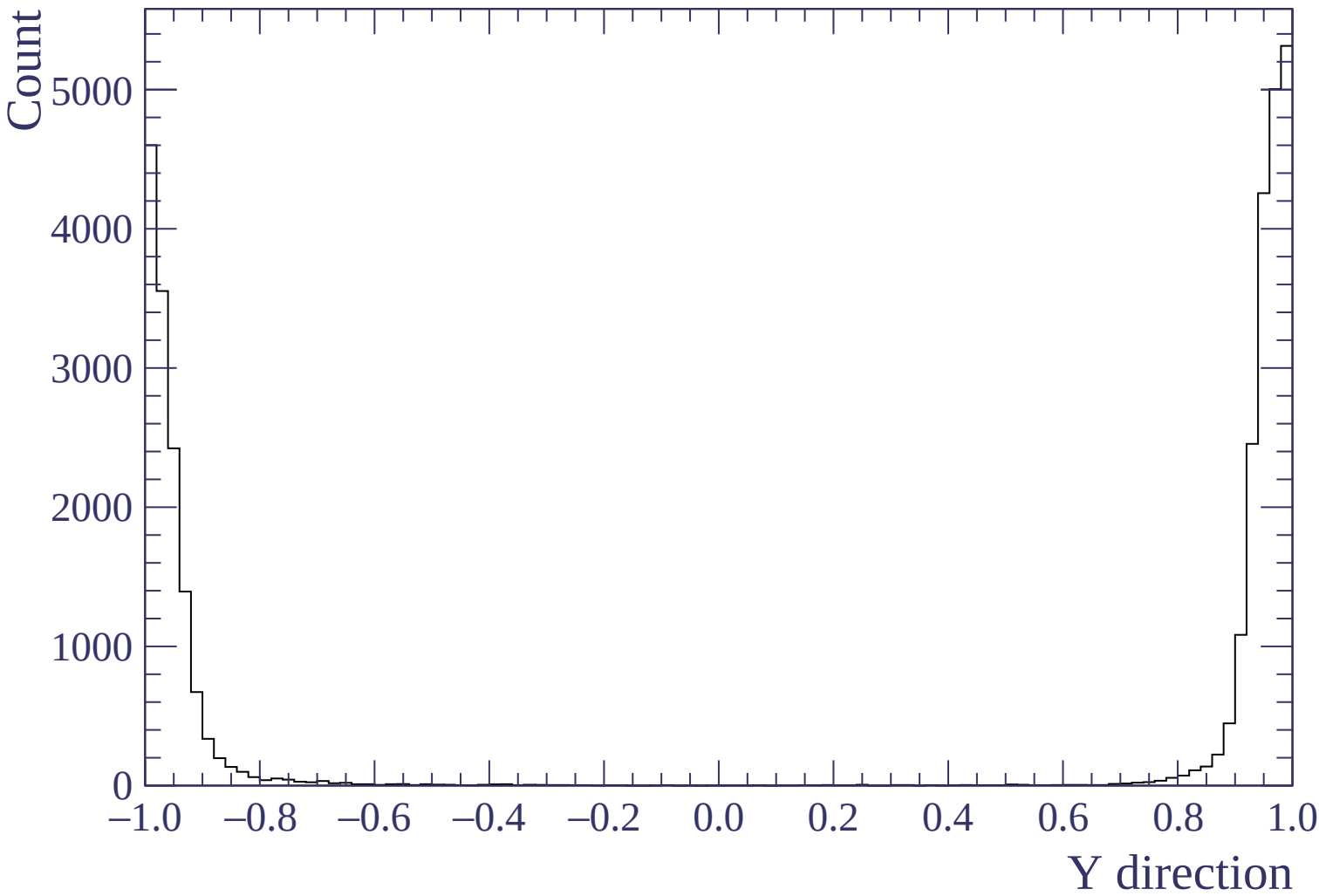
dE/dx (ADC counts/cm)

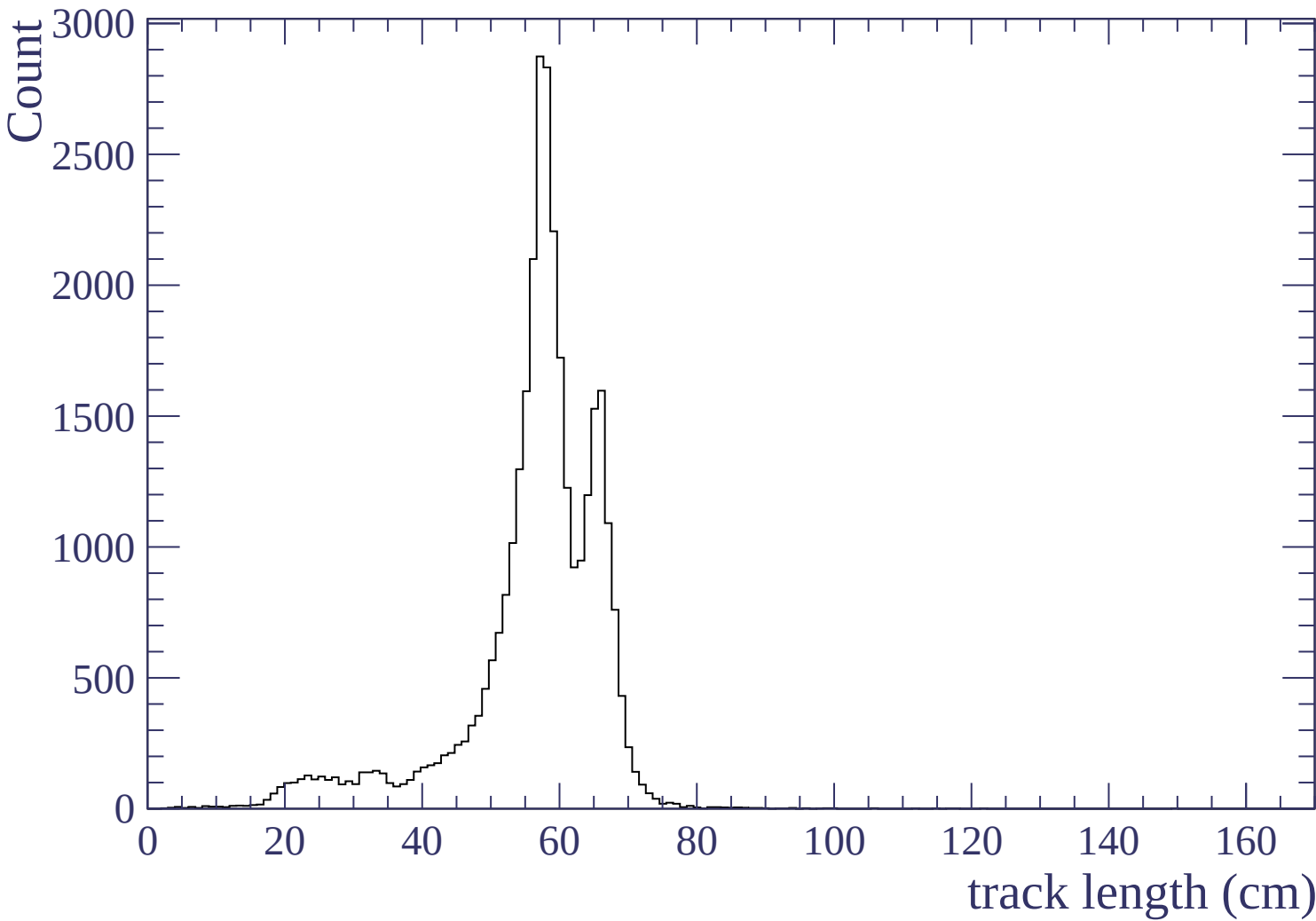
Energy loss | $3000 < p < 3060$

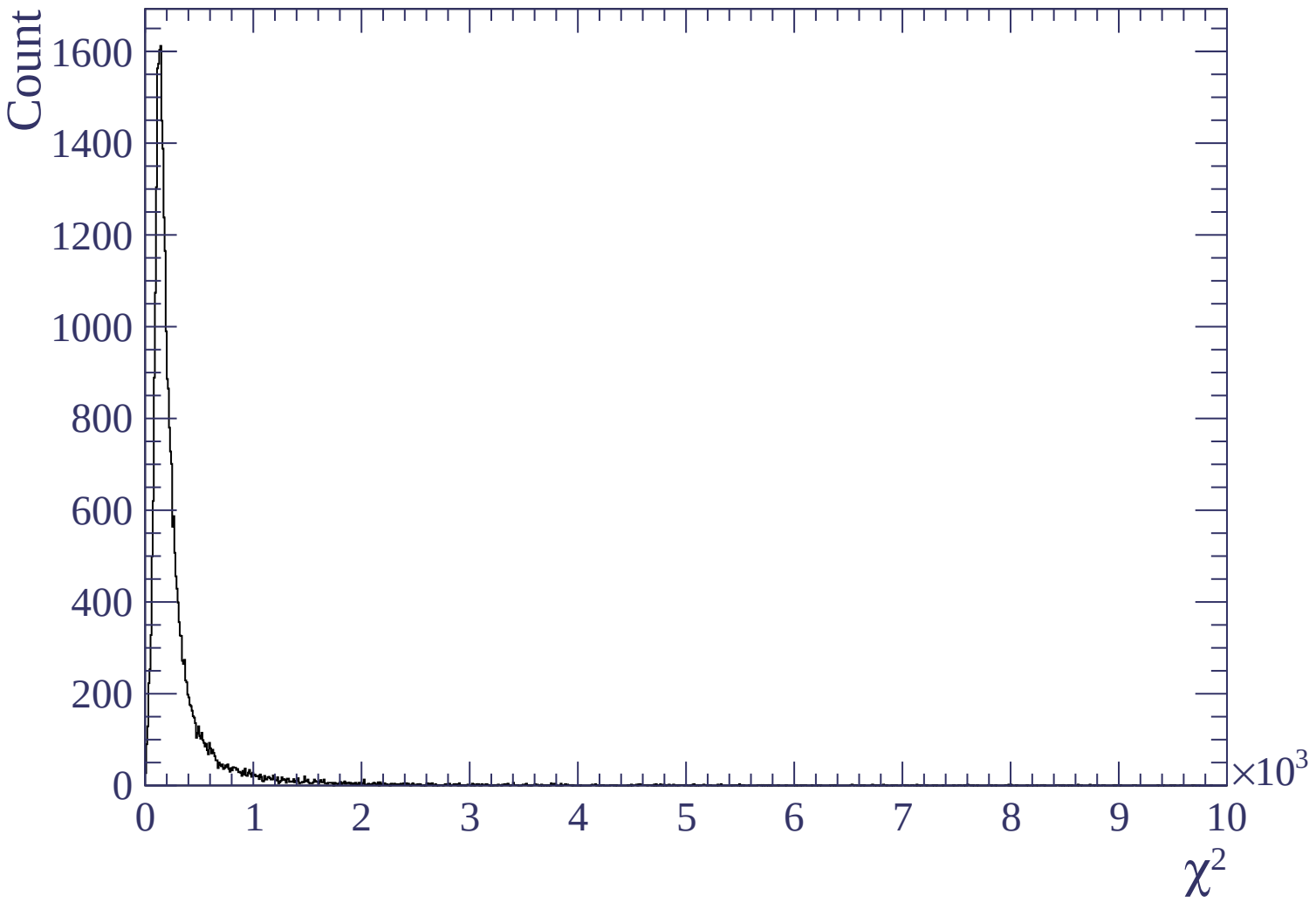
Count

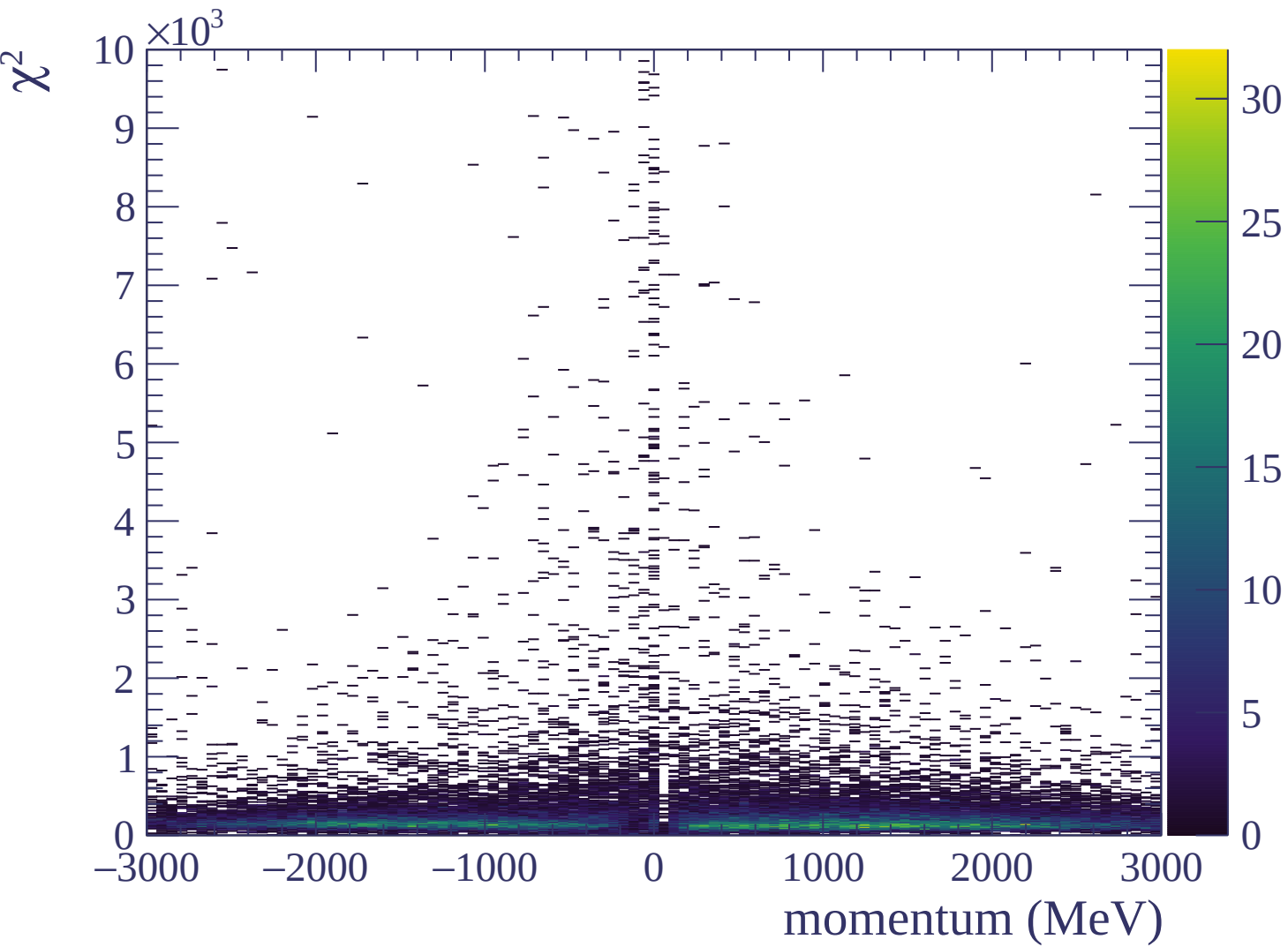












Count

- $T_{\min}$ (bottom cathode)
- $T_{\min}$ EP0
- $T_{\min}$ EP1
- $T_{\min}$ (top cathode)
- $T_{\min}$ EP2
- $T_{\min}$ EP3

50

40

30

20

10

0

0

100

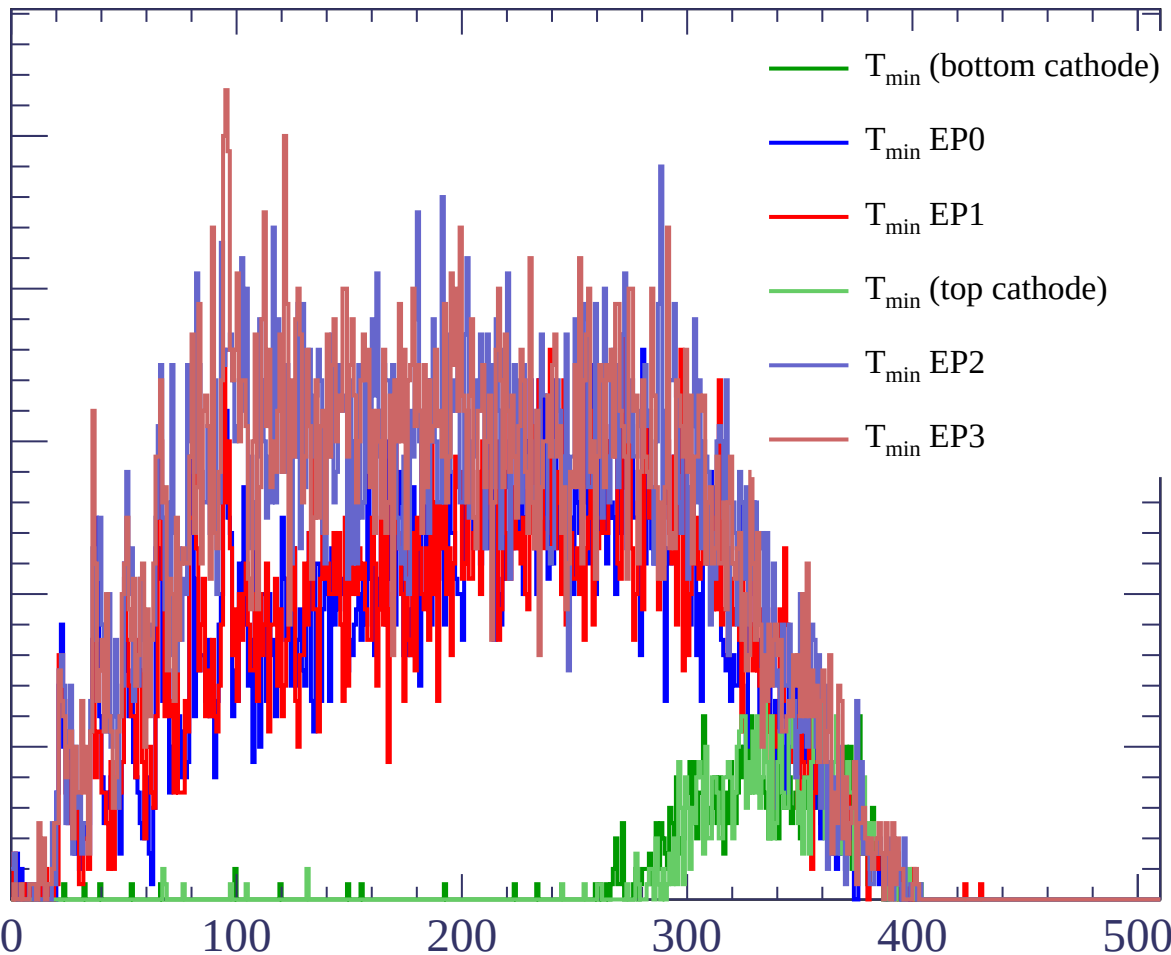
200

300

400

500

time bin



Count

